

Microscopic jamming in the negative spherical perceptron

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Abstract

We prove a sharp feasibility threshold and limiting gap and force laws for the spherical perceptron with margin -1 and quadratic penalty. The limits are taken successively in system size, inverse temperature, and density approaching the threshold from above. They agree for Gaussian coordinates and for the equal mixture of centered Gaussian coordinates with variances $1 - \varepsilon$ and $1 + \varepsilon$, for every sufficiently small fixed ε . The contact-removed gap cumulative law and the mean-one force cumulative law satisfy

$$G_J(u) = u^{1-\gamma+o(1)}, \quad F_J(s) = s^{1+\theta+o(1)},$$

as $u \downarrow 0$ and $s \downarrow 0$, with $\gamma = (2 + \theta)^{-1}$, $0.4126930 < \gamma < 0.4126934$, and $0.4231063 < \theta < 0.4231088$. A finite numerical certificate for these exponent intervals, together with its mathematical error bounds, is included.

1 The problem and the proof architecture

1.1 Background and contribution

Jamming links the loss of feasibility in a system of random constraints to singular distributions of small positive gaps and weak forces. The negative-margin spherical perceptron is a particularly simple setting in which this connection can be studied: its variables lie on a sphere, its constraints are linear inequalities, and its feasible set is nonconvex. Franz and Parisi [9] identified this model with the jamming universality class of high-dimensional spheres. Their analysis predicted isostaticity, the balance between contact constraints and degrees of freedom, together with the gap and force exponents. Their equations use a continuum of replica-overlap levels, called full replica-symmetry breaking.

The scaling mechanism and numerical values have an earlier origin in the full replica-symmetry-breaking solution for hard spheres developed by Charbonneau, Kurchan, Parisi, Urbani and Zamponi [5]. Franz, Parisi, Sevelev, Urbani and Zamponi [10] subsequently analyzed the spherical perceptron on both sides of jamming. In particular, they derived the zero-temperature scaling from the unsatisfiable side, the contact atom and normalized force distribution, and the relation between the critical exponents by matching the scaling regimes. Parisi and Zamponi [21] gave a direct analytic proof of that relation within the scaling equations. Their argument assumes the existence of the full replica-symmetry-breaking profile and the convergence and matching of its scaling expansion. Constructing and selecting the microscopic limit is a separate problem.

The relation between gap and force exponents also has a mechanical stability antecedent. Wyart [23] derived a stability inequality for sphere packings and predicted its saturation at marginal stability. Lerner, Düring and Wyart [14] distinguished the extended rearrangements governed by that inequality from localized buckling, which obeys a different bound. The force exponent in the full replica-symmetry-breaking hard-sphere solution is the extended-mode

exponent [5]. These mechanical arguments provide a conceptual comparison; the present proof derives its endpoint mass estimates from the microscopic equilibrium model.

The feasibility question also belongs to the older spherical storage-capacity problem, which asks how many random linear constraints admit a vector on a sphere. Gardner [11] and Gardner and Derrida [12] developed replica calculations for binary-pattern neural-network models; the latter identified instability of the replica-symmetric continuation at negative margins. For Gaussian constraints at a fixed negative margin, Stojnic [22, Section 2.3] proved a one-sided infeasibility bound. Montanari, Zhong and Zhou [16, Theorems 4.1–4.2] subsequently proved feasible and infeasible density regions whose bounds agree to leading order as the margin tends to $-\infty$. These bounds do not establish a sharp threshold at margin -1 or determine the microscopic gap and force laws studied here.

Here we establish the corresponding equilibrium statement directly for the random spherical model with quadratic penalty. The result includes a sharp feasibility threshold, the existence and identification of the ordered limiting gap and force laws, and their invariance under a fixed small independent Gaussian scale mixture in each pattern coordinate. It also identifies the critical susceptibility power and encloses the exponents with a finite certificate. The exponent values, relation, normalized force convention, and approach from the unsatisfiable side are thus consistent with the preceding physical predictions. The mathematical content is the passage from the microscopic model through these limits, including the selection and error bounds needed for the stated conclusions.

There are three linked difficulties. First, the spherical constraint prevents a direct convex-feasibility argument, while the quadratic penalty is unbounded. Second, vanishing energy alone does not imply exact satisfiability, and a weak critical limit need not preserve contact mass or force normalization. Third, writing down a stationary scaling profile does not show that the microscopic critical layer approaches it. The proof addresses these issues through a variational pressure formula and a separate feasibility repair, a convex Brownian martingale problem with endpoint control, and a contraction for all possible rescaled critical limits.

The pressure proof uses the interpolation scheme of Guerra [13] and the block-cavity organization of Aizenman, Sims and Starr [1], including its spherical implementation by Chen [8]. For the present row Hamiltonian, concavity of the row functional supplies the interpolation sign, while the lower bound requires a separate row expansion and identification of the empirical cavity covariance. Coordinate comparison uses the smooth-function form of Lindeberg replacement [6]; controlling spins concentrated on a few coordinates requires the additional entropy argument described below.

The hierarchical change-of-measure and cavity viewpoint follows the Ruelle probability-cascade framework of Bolthausen and Sznitman [3]. The Brownian formulation is related to the variational representation of Boué and Dupuis [4] and the stochastic-control treatment of the Parisi equation by Auffinger and Chen [2]. The passage from stochastic controls to a variational problem over Brownian martingales has a close precedent in Mourrat’s dual formulation of the Parisi functional [17]. The martingale representation and uniqueness results of Chen, Issa and Mourrat [7] further develop this approach for multi-species spin glasses. These works provide methodological precedents for the convex martingale problem used here. The present formulation requires a concave quadratic terminal penalty, nonnegative martingales with square-integrable terminal values, and control of a singular critical endpoint. The arguments below establish the corresponding identities, critical uniqueness, and scaling selection for the microscopic limits constructed here.

1.2 Model and observables

Fix $\alpha > 0$, let $M = \lfloor \alpha N \rfloor$, and let A be an $M \times N$ matrix with independent entries. In empirical observables containing M^{-1} , take N large enough that $M \geq 1$. Initially the entries

are standard normal. On

$$\mathbb{S}_N = \{x \in \mathbb{R}^N : \|x\|^2 = N\}$$

use normalized surface measure σ_N . For $y = x/\sqrt{N}$, define

$$g_\mu(x) = A_\mu y, \quad h_\mu(x) = 1 + g_\mu(x), \quad V(z) = \frac{1}{2}(-1 - z)_+^2, \quad H_N(x) = \sum_{\mu=1}^M V(g_\mu(x)). \quad (1.1)$$

Here $z_+ = \max(z, 0)$ and $z_- = \max(-z, 0)$. Since $\|y\| = 1$, the condition $H_N(x) = 0$ is exactly $Ay \geq -\mathbf{1}$ componentwise, the spherical storage-capacity problem at margin $\kappa = -1$. The realization is satisfiable (SAT) if H_N vanishes somewhere on $\mathbb{S}_N$ and is unsatisfiable (UNSAT) otherwise. At inverse temperature $\beta > 0$, write $\langle \cdot \rangle_{N,\beta,\alpha}$ for integration against

$$\frac{e^{-\beta H_N(x)} \sigma_N(dx)}{\int e^{-\beta H_N(x)} \sigma_N(dx)}.$$

Disorder expectation is $\mathbb{E}$; subscripts on the Gibbs brackets are omitted when unambiguous.

For $0 < \delta < u$, put

$$G_{N,\beta,\alpha}(\delta, u) = \mathbb{E} \left\langle \frac{1}{M} \sum_{\mu=1}^M \mathbf{1}\{\delta < h_\mu(x) \leq u\} \right\rangle. \quad (1.2)$$

Forces are defined on the violated constraints $C(x) = \{\mu : h_\mu(x) < 0\}$. When this set is nonempty, let

$$\bar{r}(x) = \frac{1}{|C(x)|} \sum_{\nu \in C(x)} (-h_\nu(x)), \quad f_\mu(x) = \frac{-h_\mu(x)}{\bar{r}(x)} \quad (\mu \in C(x)),$$

and set

$$F_{N,\beta,\alpha}(s) = \mathbb{E} \left\langle \frac{1}{|C(x)|} \sum_{\mu \in C(x)} \mathbf{1}\{f_\mu(x) \leq s\} \right\rangle, \quad s > 0. \quad (1.3)$$

The integrand is defined to be zero when $C(x) = \emptyset$.

All limits at jamming are taken from UNSAT in the order

$$\lim_{\alpha \downarrow \alpha_c} \lim_{\beta \rightarrow \infty} \lim_{N \rightarrow \infty}. \quad (1.4)$$

For gaps the lower cutoff is removed subsequently:

$$G_J(u) = \lim_{\delta \downarrow 0} \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \rightarrow \infty} \lim_{N \rightarrow \infty} G_{N,\beta,\alpha}(\delta, u), \quad (1.5)$$

$$F_J(s) = \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \rightarrow \infty} \lim_{N \rightarrow \infty} F_{N,\beta,\alpha}(s). \quad (1.6)$$

The limits $u \downarrow 0$ and $s \downarrow 0$ are taken last. This convention distinguishes positive gaps from contacts without assuming a limiting density.

The perturbations considered here have coordinate law

$$\nu_\varepsilon = \frac{1}{2}\mathcal{N}(0, 1 - \varepsilon) + \frac{1}{2}\mathcal{N}(0, 1 + \varepsilon), \quad 0 \leq \varepsilon < 1, \quad (1.7)$$

where the second argument is the variance. Coordinates and rows remain independent. This law is centered with variance one; for $\varepsilon > 0$ it is non-Gaussian, since its fourth moment is $3(1 + \varepsilon^2)$.

1.3 Main result

Theorem 1.1 (Microscopic criticality and mixture universality). *There exist $\alpha_c \in (0, \infty)$ and $\varepsilon_0 \in (0, 1)$ with the following properties for every fixed $\varepsilon \in [0, \varepsilon_0)$ and coordinate law ν_ε .*

- (i) *The SAT probability tends to one at every fixed $\alpha < \alpha_c$ and to zero at every fixed $\alpha > \alpha_c$.*
- (ii) *All the ordered limits in (1.5)–(1.6) exist for their indicated positive arguments. They are independent of ε .*
- (iii) *F_J is the cumulative distribution function on $(0, \infty)$ of a probability law of mean one. It has no atom at zero. There are exponents γ, θ such that*

$$G_J(u) = u^{1-\gamma+o(1)}, \quad F_J(s) = s^{1+\theta+o(1)},$$

as the corresponding argument decreases to zero, and

$$\gamma = \frac{1}{2+\theta}, \quad 0.4126930 < \gamma < 0.4126934, \quad 0.4231063 < \theta < 0.4231088.$$

The limiting full gap law has a contact atom of mass $1/\alpha_c$ (Lemma 8.10). Multiplying this contact fraction by the constraint density gives one contact per degree of freedom in the ordered critical equilibrium limit. This is the isostaticity predicted for the model; it also identifies the conditioning mass used to normalize forces.

1.4 The endpoint mechanism and the spectral characterization

The two exponents arise from different fluctuation scales with the same small probability mass. The proof constructs a positive, nonincreasing susceptibility $\chi : [0, 1) \rightarrow (0, 1]$, normalized by $\chi(0) = 1$ and vanishing at the critical endpoint $t = 1$. Here t is the variance time of the Brownian representation of the critical limit, rather than a microscopic dynamical time. Put

$$\tau = 1 - t, \quad w(t) = \int_t^1 \chi(v)^{-2} dv.$$

The remaining Brownian scales for gaps and residual forces are, respectively, $\sqrt{\tau}$ and $\sqrt{w(t)}$. Section 9 proves that $w(t) \asymp \tau/\chi(t)^2$ and that the strictly positive masses up to these two scales are both comparable to $\chi(t)$. Equivalently, for a fixed $c_f > 0$ accounting for mean force normalization,

$$G_J(\sqrt{\tau}) \asymp \chi(t), \quad F_J(c_f \sqrt{w(t)}) \asymp \chi(t).$$

Here $\asymp$ denotes bounds in both directions by fixed positive constants, uniformly as $t \uparrow 1$. These statements concern cumulative masses and require no assumption of a limiting density.

If $\chi(1 - \tau) = \tau^{a+o(1)}$ with $0 < a < 1/2$, the gap and force cutoffs have logarithmic powers $1/2$ and $(1 - 2a)/2$, while their masses have power a . Their cumulative powers are therefore $2a$ and $2a/(1 - 2a)$. The remaining task is to prove that the susceptibility has such a power and to select a ; a stationary scaling ansatz alone would not establish either assertion.

The selected value is characterized by a scalar boundary-value and spectral problem. For $a \in [0, 0.41]$, Section 10 constructs the unique stationary reference $M_a : \mathbb{R} \rightarrow \mathbb{R}$ in the class $-1 \leq M'_a \leq 0$, $M''_a > 0$, satisfying

$$\frac{1}{2}M''_a + \left(\frac{z}{2} + aM_a\right)M'_a + \left(a - \frac{1}{2}\right)M_a = 0, \tag{1.8}$$

with the bounds

$$\phi(z) - z\Phi(-z) \leq M_a(z) \leq \frac{\phi(z)}{\Phi(z)}.$$

Here ϕ and Φ are the standard normal density and distribution function. Put $B_a(z) = z/2 + aM_a(z)$ and let $\lambda(a)$ be the lowest eigenvalue of the Friedrichs realization on $L^2(\mathbb{R})$ of

$$\mathcal{H}_a = -\frac{1}{2} \frac{d^2}{dz^2} + \frac{1}{2}(B_a^2 + B_a'). \quad (1.9)$$

There is a unique root a_* of $\lambda(a) = a$ in

$$0.2936533 < a_* < 0.2936535. \quad (1.10)$$

The exponents in Theorem 1.1 are

$$\gamma = 1 - 2a_*, \quad \theta = \frac{2a_*}{1 - 2a_*} - 1. \quad (1.11)$$

The root enclosure and the contraction selecting it are proved with the finite certificate in Section 12 and Appendix A.

1.5 Architecture and conventions

Sections 2–3 construct the hierarchical functionals and the concavity estimates used in the Gaussian pressure formula, where pressure means the expected logarithm of the partition function divided by N . The hierarchy describes the possible laws of the overlap $N^{-1}x \cdot x'$ between replicas, meaning independent samples from a fixed-disorder Gibbs measure. The row functional records how adding one constraint changes the pressure; its concavity supplies the sign needed in the interpolation and cavity bounds of Section 4.

At fixed positive temperature, small row perturbations turn pressure information into an empirical row law. Adding $d\psi$ to the single-row exponent $-\beta V$ couples the pressure to the empirical average of a smooth compactly supported test ψ . An upper bound that touches the pressure at $d = 0$ for both signs of d pins down this average; a determining family of tests then identifies the row law. Section 5 compares that law with the coordinate-mixture model, including configurations whose mass is concentrated on a few coordinates. On a fixed density neighborhood of the threshold, the same sufficiently small fixed mixture is admissible at every fixed positive temperature, so the resulting identical functions pass through the prescribed later limits. For feasibility, Section 6 starts from subextensive minimum energy at a slightly higher density, deletes a fixed fraction of the rows, and repairs the remaining constraints exactly at the target density.

Sections 7 and 8 pass to zero temperature and then to the threshold. A susceptibility profile measures the response along the hierarchy; a nonnegative Brownian martingale represents the residual force. Joint convexity and a uniqueness argument on one Wiener space identify the critical profile. Strong endpoint convergence and small-cutoff estimates preserve the contact mass and normalized force law.

The final two stages address the critical singularity. Section 9 relates small-gap and small-force masses to the susceptibility and proves that its associated increasing order parameter has no constant interval sufficiently near the endpoint. Section 10 takes limits after translating logarithmic endpoint time. Every resulting solution is defined for all translated times and satisfies the same integral identities. Quantitative barriers first confine these solutions; a strict contraction then forces their susceptibility slope to be the selected scalar root. Section 12 certifies the finite inequalities used in that step. The principal outputs and their precise locations are:

Output	Proof location
Gaussian variational pressure and touching upper bounds for both signs of a compact row perturbation	Proposition 4.5; (4.11)
Equality of the thermodynamic row laws, and of the gap and normalized-force observables, at each fixed positive temperature for sufficiently small mixtures	Proposition 5.3
Exact satisfiability below the common threshold, obtained by repairing a system with subextensive energy	Theorem 6.7
A unique normalized physical critical limit, strong convergence through the endpoint, and the prescribed ordered gap and force laws	Proposition 8.6, Proposition 8.9, Theorem 8.12
Two-sided endpoint mass estimates and full increase support near the endpoint	Proposition 9.6; Proposition 9.9
The selected susceptibility slope, from entire-limit rigidity and the finite certificate	Proposition 10.4; Section 12
The cumulative exponents and their exact relation	Section 11

The mixture argument identifies the fixed-temperature functions; Corollary 6.8 transfers each subsequent Gaussian limit in the prescribed order. Sections 10 and 12 form one computer-assisted argument: the former proves the analytic implications of the quantitative inequalities, including the required tail bounds, and the latter verifies their finite enclosures. The short completion in Section 11 then combines the endpoint mass estimates with the selected slope.

We use C, c for finite positive constants whose permitted dependencies are stated locally. The notation $f \asymp g$ means two-sided bounds by such constants, uniform in the limit under discussion. Profiles are identified up to equality almost everywhere. A reference to a support concerns the support of the associated increasing measure, not the set where its cumulative distribution is positive.

The analytic prerequisites are Gaussian concentration and integration by parts, Brownian martingale representation, Brownian time change and reflection, Itô calculus, elementary parabolic comparison and interior regularity, the Dovbysh–Sudakov representation [19], the support-radius consequence of the Ghirlanda–Guerra identities [18], Panchenko’s ultrametricity theorem [20], and the Marchenko–Pastur limit [15]. Their precise uses are specified where they enter. All model-specific compactness, comparison, limiting, and numerical arguments are supplied below.

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2 Hierarchical Gaussian functionals and spherical entropy

The pressure calculation separates the contribution of a new constraint from that of new spin coordinates. We construct the corresponding row and spherical functionals on a common Gaussian hierarchy. For the row functional we identify its derivative with respect to the covariance profile. For the spherical functional we compute its limit and the entropy conjugate $E(q)$ used in the pressure formula.

The common hierarchy is a random family of leaf weights, together with Gaussian marks shared by leaves having a common ancestor. We first prove its change-of-measure and sampling rules. These rules identify which marks remain independent after a row is inserted. The continuity lemma then passes these conclusions through limits of replica arrays under explicit moment bounds. Throughout, a profile specifies only off-diagonal covariances; the diagonal is a separate parameter. This distinction determines the independent terminal Gaussian noise in a row insertion.

The finite cascade and its reweighting are the objects of Bolthausen–Sznitman’s abstract cavity method [3]. We give the marking calculation explicitly, including the genealogy and independent terminal noise used in the row insertions below.

2.1 Finite cascades and their change of measure

Let $\mathcal{Q}_Q$ be the set of nonnegative nondecreasing functions $q : [0, 1) \rightarrow [0, Q]$, identified up to Lebesgue-null sets. Write $q_* := \text{ess sup } q$. For a step profile, fix

$$0 = m_{-1} < m_0 < \cdots < m_k = 1, \quad q(s) = q_j \quad (m_{j-1} \leq s < m_j), \quad w_j := m_j - m_{j-1}.$$

Repeated values q_j are allowed. A cascade of depth k has leaves $\tau = (\tau_1, \dots, \tau_k) \in \mathbb{N}^k$. At every vertex of depth $j - 1$, independently, place a Poisson point process on $(0, \infty)$ with

intensity $m_{j-1}s^{-1-m_{j-1}}ds$; its points are the weights of the outgoing edges. Let W_τ be the product of the edge weights on the path to τ , and set

$$T = \sum_{\tau} W_\tau, \quad v_\tau = W_\tau/T.$$

A depth-zero cascade consists of one leaf of weight one. The following facts also prove that the normalizing sum is finite and positive.

Lemma 2.1 (Stable sums and the cascade change of measure). *For a fixed finite cascade, $0 < T < \infty$ almost surely and $\mathbb{E}|\log T|^2 < \infty$. Attach marks to its vertices, independently of the Poisson processes, with conditionally independent and identically distributed child marks given the ancestral marks. A terminal function X_τ of the path marks determines recursive functions by*

$$X_v = \frac{1}{m_{j-1}} \log \mathbb{E}[\exp(m_{j-1}X_{vi}) \mid \text{marks up to } v] \quad \text{at depth } j-1. \quad (2.1)$$

Suppose these moments are finite and $\mathbb{E}|X_\emptyset| < \infty$. Reweighting the leaf probabilities by e^{X_τ} preserves the law of their genealogy. Conditional on the transformed genealogy, the child-mark transition at depth j is multiplied by

$$\exp\{m_{j-1}(X_{vi} - X_v)\}. \quad (2.2)$$

Root marks retain their original distribution. Moreover,

$$\mathbb{E} \log \sum_{\tau} v_\tau e^{X_\tau} = \mathbb{E} X_\emptyset. \quad (2.3)$$

If the marks are independent families and X_τ is the sum of one function of each family, both the recursion and the tilted path law factor over the families, including conditional on the genealogy of finitely many replicas.

Proof. For $0 < m < 1$, a Poisson sum $S = \sum_i s_i Y_i$, with intensity $ms^{-1-m}ds$ and iid nonnegative marks satisfying $0 < \mathbb{E}Y^m < \infty$, has Laplace transform

$$\mathbb{E}e^{-tS} = \exp\{-\Gamma(1-m)\mathbb{E}(Y^m)t^m\}.$$

Indeed, the Poisson Laplace functional reduces the exponent to $m\mathbb{E}\int_0^\infty (1 - e^{-tsY})s^{-1-m}ds$; integration by parts evaluates this integral. The resulting positive stable random variable has finite positive moments of every order less than m and finite negative moments of every order. The latter follows directly from $x^{-a} = \Gamma(a)^{-1}\int_0^\infty t^{a-1}e^{-tx}dt$; the former follows by applying the corresponding integral formula to x^a , $0 < a < m$. Consequently $\mathbb{E}|\log S|^2 < \infty$. Starting at the last generation and using $m_0 < \dots < m_{k-1}$ proves these assertions for every subtree total, and hence for T .

Here is the change of measure at one vertex. Write $\mu(d\omega)$ for the conditional law of a child mark and put $c(\omega) = e^{X_{vi}-X_v}$. Under the map $(s, \omega) \mapsto (sc(\omega), \omega)$, the marked intensity becomes

$$m_{j-1}s^{-1-m_{j-1}}ds c(\omega)^{m_{j-1}}\mu(d\omega).$$

The second factor is a probability measure by (2.1). Thus the transformed weights have their original Poisson law and are independent of marks with transition (2.2).

We spell out why this yields independence of the *entire* transformed weight array from the marked tree. At depth two, write the root mark as B , a child mark as A_i , and its descendant marks as C_{ij} . Let s_i and t_{ij} be the edge processes of exponents m_0 and m_1 . Set

$$X_1(B, A) = m_1^{-1} \log \mathbb{E}_C e^{m_1 X_2(B, A, C)}, \quad X_0(B) = m_0^{-1} \log \mathbb{E}_A e^{m_0 X_1(B, A)}.$$

The lower transformation $t'_{ij} = t_{ij}e^{X_2(B, A_i, C_{ij}) - X_1(B, A_i)}$ has, conditional on (B, A_i) , the original unmarked Poisson law, denoted $\mathcal{P}_{m_1}$, independently of child marks with tilted transition $K_1(dC \mid B, A_i)$. In particular this law of the complete unmarked lower process is independent of A_i and B . Before the upper transformation a decorated child therefore has law

$$\mu_0(dA \mid B) \mathcal{P}_{m_1}(dt') \prod_j K_1(dC_j \mid B, A).$$

The upper map $s'_i = s_i e^{X_1(B, A_i) - X_0(B)}$ replaces only μ_0 by its tilted transition K_0 ; it leaves $\mathcal{P}_{m_1}$ unchanged. The marked-Poisson theorem makes the ordered s'_i independent of these iid decorated children. Thus, after both reorderings, all the transformed weights are independent of the hierarchical marked tree, conditional on B . The same argument inducts upward for any depth: replace $\mathcal{P}_{m_1}$ by the already transformed descendant weight-array law, which is independent of its ancestor mark. A sampled genealogy depends only on the complete weight array and independent sampling uniforms. This includes the descendant total masses used in selecting a child. Conditioning on that genealogy therefore does not tilt its marks again. Replicas sharing a vertex reuse the one mark already drawn there; no additional power depending on the number of replicas is introduced.

Reordering preserves a bijection of leaves. Under that correspondence, products of the edge multipliers telescope:

$$W'_\tau = W_\tau e^{X_\tau - X_\emptyset}, \quad \log \sum_\tau v_\tau e^{X_\tau} = X_\emptyset + \log T' - \log T.$$

Conditionally on root marks, T' and T have the same marginal law; they need not be independent. Their integrable logarithms therefore cancel in expectation. The root factor cancels in normalized leaf probabilities, so the root marks are not tilted. Finally, exponential moments of a sum of independent families factor at each recursion step, and so do the Radon–Nikodym factors in (2.2). This proves the last assertion. $\square$

For clarity, the genealogy used here can be constructed without appealing to any unproved identification of an overlap array. We record its sampling rule.

Lemma 2.2 (Sampling a cascade and the continuous rank array). *For two independent samples from a depth- k cascade, the probability that their leaves share exactly j edges is w_j , $0 \leq j \leq k$; sharing k edges includes selecting the same leaf. There exists a weakly exchangeable ultrametric array $(U_{ab})_{a,b \geq 1}$ with $U_{aa} = 1$ and U_{12} uniform on $[0, 1]$, such that every step-profile cascade covariance array has law*

$$C_{aa} = Q, \quad C_{ab} = q(U_{ab}) \quad (a \neq b).$$

All the finite-dimensional laws of this rank array are determined by its nested partition sampling rule.

Proof. We give the sampling computation, including the change caused by an ancestor. Let a Poisson process have intensity $c\beta s^{-1-\beta} ds$, $0 < \beta < 1$, and total S . Change its law by $S^\eta / \mathbb{E} S^\eta$, where $0 \leq \eta < \beta$. For a specified partition of n sampled indices into b blocks of sizes $n_1, \dots, n_b$, normalized Poisson weights give probability

$$\frac{\prod_{a=1}^{b-1} (a\beta - \eta)}{\prod_{a=1}^{n-1} (a - \eta)} \prod_{i=1}^b \prod_{a=1}^{n_i-1} (a - \beta). \quad (2.4)$$

Empty products are one. To verify this, insert

$$S^{\eta-n} = \frac{1}{\Gamma(n-\eta)} \int_0^\infty t^{n-\eta-1} e^{-tS} dt$$

into the expectation of the sum over distinct Poisson points. The Poisson factorial-moment formula gives one factor $c\beta\Gamma(n_i - \beta)t^{\beta-n_i}$ for each selected point and the Laplace factor $\exp\{-c\Gamma(1 - \beta)t^\beta\}$. The remaining integral, after $v = c\Gamma(1 - \beta)t^\beta$, is a gamma integral. Dividing by

$$\mathbb{E}S^\eta = [c\Gamma(1 - \beta)]^{\eta/\beta} \frac{\Gamma(1 - \eta/\beta)}{\Gamma(1 - \eta)}$$

(with value one at $\eta = 0$) gives (2.4). Its ratios show that an existing block of size l receives the next sample with probability $(l - \beta)/(n - \eta)$; a new block has probability $(b\beta - \eta)/(n - \eta)$.

At a cascade vertex, multiply each outgoing edge by the total mass of its child subtree. The Poisson mapping calculation in the preceding proof makes the resulting child masses a stable Poisson process and tilts the child subtree law by its total mass raised to the parent parameter. More explicitly, if the outgoing parameter is β , the masses $r_i = s_i T_i$ are independent of child-subtree marks whose law is multiplied by $T_i^\beta / \mathbb{E}T_i^\beta$. An incoming bias $(\sum_i r_i)^\eta$ changes only the mass process, so those child marks remain independent, also after conditioning on the choices of children. Thus (2.4) applies at the first level with $(\eta, \beta) = (0, m_0)$ and, inside a selected parent at later levels, with $(\eta, \beta) = (m_{j-2}, m_{j-1})$. For two samples the conditional joining probabilities are

$$1 - m_0, \quad \frac{1 - m_1}{1 - m_0}, \quad \dots, \quad \frac{1 - m_{k-1}}{1 - m_{k-2}}.$$

Their products give probability $1 - m_{j-1}$ of sharing the first j edges. Taking differences proves the first assertion.

More generally, multiply the conditional joining probabilities along the ancestral path to a cluster containing l of the first n samples at a level of parameter m . The denominators telescope, giving $(l - m)/n$. These probabilities, and differences between probabilities of joining a parent and joining its existing children, determine all ways the next sample can enter a finite nested partition. They also show that deleting intermediate levels leaves the same rule at the retained levels. Construct nested partitions at all dyadic parameters by these consistent finite-dimensional laws, taking the universal partition at parameter zero and the singleton partition at parameter one, and put $U_{ab} = \sup\{m : a, b \text{ belong to the same cluster at level } m\}$. The rule gives $\mathbb{P}(U_{12} \geq m) = 1 - m$ at dyadic m , hence uniformly on $[0, 1]$. Nesting gives ultrametricity. The predictive probabilities for a finite nested partition are differences of $(l - m)/n$, and hence its probability is continuous in the finitely many level parameters. Approximating arbitrary real levels by dyadic levels, and using that no U_{ab} has an atom at a prescribed level, therefore recovers the same sampling law at every real level. Applying a step function q now recovers the finite cascade covariance law. Set $q(1) = q_*$; nonnegative nondecreasing transforms of this nested array are positive semidefinite: for a step transform they are sums, with nonnegative coefficients, of block-indicator matrices, and the general case follows by limits. An additional nonnegative diagonal correction supplies any $Q \geq q_*$. $\square$

2.2 The row functional and tilted derivatives

Attach independent centered Gaussian increments of variances $q_0, q_1 - q_0, \dots, q_k - q_{k-1}$ to the common root and subsequent vertices, and denote their path sum by z_τ . Initially suppose $q_k = Q$ and that $u \in C^2(\mathbb{R})$ has bounded first and second derivatives. Define

$$\Phi_u(q; Q) = \mathbb{E} \log \sum_{\tau} v_{\tau} e^{u(z_{\tau})}.$$

When $q_k < Q$, replace $e^{u(z)}$ at a leaf by $\mathbb{E}_G e^{u(z + \sqrt{Q - q_k} G)}$, where $G \sim \mathcal{N}(0, 1)$. This is the *terminal convolution*; a residual Gaussian is drawn independently for each sampled replica, even if two replicas select the same leaf. We omit Q when the diagonal has been fixed.

For a finite profile put

$$f_k(z) = \log \mathbb{E}_G e^{u(z + \sqrt{Q - q_k} G)}, \quad f_{j-1}(z) = \frac{1}{m_{j-1}} \log \mathbb{E}_G e^{m_{j-1} f_j(z + \sqrt{q_j - q_{j-1}} G)}.$$

The terminal formula means $f_k = u$ when $q_k = Q$. Lemma 2.1 gives $\Phi_u(q; Q) = \mathbb{E}_G f_0(\sqrt{q_0} G)$. Let Z_j be the partial path sum under the tilted transitions (2.2), with the original Gaussian root law, and put $A_j = f'_j(Z_j)$. Differentiation under the Gaussian integrals gives

$$f'_{j-1}(z) = \mathbb{E}^{\text{tilt}}[f'_j(Z_j) \mid Z_{j-1} = z].$$

Thus (A_j) is a martingale. At the terminal convolution there is one further tilted transition, of parameter one, from f'_k to u' .

To pass from finite cascades to arbitrary covariance profiles, we use the following continuity principle. Its hypotheses concern replica arrays, so no convergence of random measures in a chosen representation is needed.

Lemma 2.3 (Replica-array continuity under a Gaussian tilt). *Let G_n be random probability measures. Suppose that the jointly sampled arrays of a finite list of uniformly bounded positive-semidefinite kernels, and any other bounded observables being retained, converge in law on every finite set of replicas. Conditional on these arrays, adjoin independent centered Gaussian fields with the prescribed kernels. Let X_n be the evaluation of one fixed continuous function on the retained single-replica variables and those fields, and assume, for every fixed $a > 0$,*

$$\sup_n \mathbb{E} G_n(e^{a|X_n|}) < \infty. \quad (2.5)$$

Then the expectations of $\log G_n(e^{X_n})$, and the expectations of bounded continuous functions of finitely many replicas sampled from $e^{X_n} G_n / G_n(e^{X_n})$, converge to the corresponding functionals of the limiting array. Independent auxiliary variables of a fixed law may be adjoined to each replica. The conclusion also holds for unbounded observables whose truncation errors vanish uniformly under the tilted law; in particular this follows for polynomial functions of the Gaussian fields under (2.5).

Proof. For a fixed number of replicas, the conditional characteristic function of the adjoined fields is $\exp(-t^T C t / 2)$. It is a bounded continuous function of the covariance matrix, including at singular matrices. Consequently the fields converge jointly with the arrays. The same argument with a product law adjoins the independent auxiliary variables.

Put $W = e^X$, $W_L = (W \vee L^{-1}) \wedge L$, $Z = G(W)$, and $Z_L = G(W_L)$. For every fixed $r \geq 1$, Jensen's inequality gives

$$Z^{-r} \leq G(e^{-rX}), \quad Z_L^{-r} \leq G(W_L^{-r}) \leq G(1 + e^{-rX}).$$

All these inverse moments are uniformly bounded in expectation. For arbitrary fixed $r \geq 1$ and $A > 0$, (2.5) and another application of Jensen give

$$\sup_n \mathbb{E} |Z - Z_L|^r \leq \sup_n \mathbb{E} G(|W - W_L|^r) = O(L^{-A}),$$

after using a sufficiently large moment in (2.5). The elementary inequality $|\log a - \log b| \leq |a - b|(a^{-1} + b^{-1})$ and Hölder's inequality show that replacing Z by Z_L changes its expected logarithm by a quantity tending to zero uniformly in n .

At fixed L , uniformly approximate $\log z$ on $[L^{-1}, L]$ by polynomials. Every expected polynomial of $G(W_L)$ is a finite linear combination of replica expectations, which converge

by the first paragraph. This proves the assertion for logarithms. For an r -replica observable B , its tilted expectation is

$$\mathbb{E} \frac{G_n^{\otimes r}(B \prod_{i=1}^r W_i)}{Z^r}.$$

Replace the W_i and Z by their clipped versions. Hölder's inequality, the inverse-moment bounds, and the displayed tail estimate make this error uniformly small for bounded B . At fixed L , approximate z^{-r} on $[L^{-1}, L]$ by polynomials and again use finitely many replicas. A further truncation proves the unbounded version whenever the stated uniform integrability holds. Polynomial Gaussian observables have moments of all orders uniformly, because the covariance kernels are bounded; Hölder's inequality with (2.5) and the inverse-moment bound then supplies this uniform integrability. $\square$

Lemma 2.4 (Derivative profile and Gaussian interpolation). *For a step profile define $D_q(s) = \mathbb{E} A_j^2$ on $[m_{j-1}, m_j]$. This is nonnegative and nondecreasing, satisfies $D_q \leq \|u'\|_\infty^2$, and is constant across adjacent equal values of q . The diagonal quantity $D_{q,d} := \mathbb{E}^{\text{tilt}} u'(Z)^2$ is at least the final off-diagonal value. For two profiles with common diagonal Q ,*

$$\frac{d}{dt} \Phi_u((1-t)q + tr; Q) = -\frac{1}{2} \int_0^1 (r-q)(s) D_{(1-t)q+tr}(s) ds. \quad (2.6)$$

For arbitrary $q, r \in \mathcal{Q}_Q$ this identity holds for $0 < t < 1$ and as the appropriate one-sided derivative at the endpoints. Moreover,

$$|\Phi_u(q; Q) - \Phi_u(r; Q)| \leq \frac{1}{2} \|u'\|_\infty^2 \|q - r\|_1, \quad q \mapsto D_q \text{ is continuous into } L^1[0, 1]. \quad (2.7)$$

For a general profile, D_q is constant almost everywhere on every interval on which q is constant almost everywhere.

Proof. We first prove the finite-profile identity, then use replica-array continuity to construct its extension and pass the derivative formula to the limit. The square of a bounded martingale is a submartingale. A zero-variance increment has identical incoming and outgoing derivative, proving all the monotonicity and plateau assertions. Given that two replicas share exactly j levels, their later marks are independent under the tilted transitions. Their terminal derivatives therefore have product expectation $\mathbb{E} A_j^2$. The shared path itself has the single-path tilted law, because transformed weights and marks are independent, also conditional on the genealogy.

Put q and r on a common finite partition, first with no residual convolution. Gaussian covariance differentiation gives

$$\frac{d}{dt} \Phi_u(q_t; Q) = \frac{1}{2} \mathbb{E} \left\langle [u''(z_1) + u'(z_1)^2] \dot{C}_{11} - u'(z_1) u'(z_2) \dot{C}_{12} \right\rangle_t.$$

The diagonal is fixed, so $\dot{C}_{11} = 0$; off the diagonal $\dot{C}_{12} = (r-q)(U_{12})$. Lemmas 2.1 and 2.2, followed by the conditional product calculation, give (2.6). Bounded derivatives justify Gaussian integration by parts; one may truncate the number of leaves and then pass by dominated convergence and the exponential moments of the Gaussian marks. Integrating the derivative proves the Lipschitz bound for finite profiles.

Use the common rank array of Lemma 2.2. If bounded profiles $q_n \rightarrow q$ in L^1 , then $q_n(U_{ab}) \rightarrow q(U_{ab})$ in probability for every $a \neq b$; a union bound handles any finite subarray. Keep diagonal Q fixed. Since u has at most linear growth, (2.5) holds uniformly for $X = u(z)$, as the base single-replica field has law $\mathcal{N}(0, Q)$. Lemma 2.3 therefore constructs the limiting

functional and tilted replica expectations independently of the step approximation. If an approximating profile is raised to Q on a last interval of length ε , its final recursion is

$$\frac{1}{1-\varepsilon} \log \mathbb{E} e^{(1-\varepsilon)u(z+\sqrt{Q-q_k}G)}.$$

As $\varepsilon \downarrow 0$ this is exactly the terminal convolution. The limiting covariance array has an independent residual Gaussian for each replica. This verifies both descriptions of the limit.

The D_{q_n} are bounded monotone functions and hence precompact in L^1 . The two-replica identity can be tested against the full rank U_{12} : refine a step profile at the endpoints of any step test function, using zero Gaussian increments on the added levels inside each plateau. Lemma 2.1 preserves the entire refined genealogy, while the derivative martingale is unchanged across those zero increments. The finite shared-level calculation thus proves the identity for step test functions, and uniform approximation proves it for continuous ones. For each continuous $\varphi : [0, 1] \rightarrow \mathbb{R}$, this identity and Lemma 2.3 give convergence of

$$\int_0^1 \varphi(s) D_{q_n}(s) ds = \mathbb{E} \langle \varphi(U_{12}) u'(z_1) u'(z_2) \rangle_{q_n}.$$

These limits identify every L^1 subsequential limit, proving convergence of D_{q_n} and defining D_q . The same argument proves continuity for any convergent sequence of general profiles. On an interval where q is constant, choose step approximations that have that value on every compact subinterval. Their derivative profiles are constant there, so the limit is constant almost everywhere there.

Pass to the limit in the integrated finite-profile version of (2.6), using boundedness and L^1 continuity of D . For fixed q, r , the resulting integrand is continuous in t . The fundamental theorem of calculus gives the claimed derivative, including one-sided endpoint derivatives, and completes the proof. $\square$

The moment assumption in Lemma 2.3 is a hypothesis, not a consequence of convergence of overlaps. Above it follows from bounded slope and Gaussian exponential moments; the later cavity applications verify the same bound before inserting capped rows. Potentials with the negative quadratic tail of the perceptron are handled by the cap-removal estimates in Section 3.

2.3 The spherical functional

Let σ_N be normalized surface measure on $\mathbb{S}_N = \{x \in \mathbb{R}^N : \|x\|^2 = N\}$. Take independent Gaussian fields $z_i(\tau)$, $1 \leq i \leq N$, each with off-diagonal profile p and diagonal $p_d \geq p_*$. Define

$$S_N(p; p_d) = \frac{1}{N} \mathbb{E} \log \sum_{\tau} v_{\tau} \int_{\mathbb{S}_N} e^{\sum_{i=1}^N x_i z_i(\tau)} d\sigma_N(x), \quad \mathcal{S}_N(p) = S_N(p; p_d) - \frac{p_d}{2}.$$

As above, a residual diagonal is integrated separately for each replica. It contributes $\exp\{N(p_d - p_*)/2\}$ inside each spin integral, because $\|x\|^2 = N$. Thus $\mathcal{S}_N$ does not depend on the diagonal convention. Gaussian interpolation, followed by the cascade change of measure for the spin-integrated marks, gives

$$|\mathcal{S}_N(p) - \mathcal{S}_N(\tilde{p})| \leq \frac{1}{2} \|p - \tilde{p}\|_1. \quad (2.8)$$

Indeed the diagonal term is removed by $p_d/2$ and the off-diagonal derivative is $-\frac{1}{2} \mathbb{E} \langle R_{12}(\tilde{p} - p)(U_{12}) \rangle$, where $R_{12} = N^{-1} x^1 \cdot x^2$ and $|R_{12}| \leq 1$. The tilted marginal law of U_{12} remains uniform by Lemma 2.1. The lemma applies here because the entire vector of Gaussian

increments can be taken as the mark at each vertex; no coordinate factorization is needed for this interpolation. For general bounded profiles, $\mathcal{S}_N$ is defined by its unique L^1 -continuous extension from step profiles, whose existence follows from (2.8). Lemma 2.3, applied at fixed N , identifies this extension with the Gaussian functional of the continuous rank array.

Proposition 2.5 (Explicit spherical entropy). *The limit $\mathcal{S}(p) = \lim_{N \rightarrow \infty} \mathcal{S}_N(p)$ exists and is uniform over all nonnegative nondecreasing profiles bounded by any fixed constant. For a finite-step profile without residual diagonal, put*

$$\Delta p_j = p_j - p_{j-1}, \quad I = \sum_{j=1}^k m_{j-1} \Delta p_j, \quad \lambda_j = b - \sum_{i=j+1}^k m_{i-1} \Delta p_i \quad (0 \leq j \leq k).$$

Thus $\lambda_k = b$, $\lambda_0 = b - I$, and $\lambda_{j-1} = \lambda_j - m_{j-1} \Delta p_j$. Define, for $b > I$,

$$X(b) = -\frac{1}{2} \log b + \frac{p_0}{2\lambda_0} + \sum_{j=1}^k \frac{1}{2m_{j-1}} \log \frac{\lambda_j}{\lambda_{j-1}}. \quad (2.9)$$

There is exactly one $b > I$ satisfying

$$-2X'(b) = \frac{1}{b} + \frac{p_0}{\lambda_0^2} + \sum_{j=1}^k \frac{p_j - p_{j-1}}{\lambda_j \lambda_{j-1}} = 1. \quad (2.10)$$

At this value of b ,

$$\mathcal{S}(p) = X(b) + \frac{b}{2} - \frac{1}{2} - \frac{p_k}{2}. \quad (2.11)$$

If $p \leq B$, its saddle point satisfies $1 \leq b < 2(B+1)$.

Proof. The calculation has four steps. We evaluate a Gaussian spin integral, choose its quadratic multiplier so that one disorder-averaged spin has variance one, and compare the resulting typical shell with the sphere. The common Lipschitz bound then makes the limit uniform over bounded profiles.

First replace the spherical integral by

$$Z_N^G(b) = \sum_{\tau} v_{\tau} \int_{\mathbb{R}^N} \exp\{x \cdot z(\tau) - b\|x\|^2/2\} \frac{dx}{(2\pi)^{N/2}}.$$

The terminal logarithm factors into a sum of $-\frac{1}{2} \log b + z_i^2/(2b)$. The elementary Gaussian integral

$$\frac{1}{m} \log \mathbb{E} \exp \left\{ m \left(c + \frac{(z + \sqrt{d}G)^2}{2\lambda} \right) \right\} = c + \frac{1}{2m} \log \frac{\lambda}{\lambda - md} + \frac{z^2}{2(\lambda - md)}$$

is valid when $\lambda > md$. Iterating it and averaging the root Gaussian gives (2.9), with $\mathbb{E} \log Z_N^G(b) = NX(b)$. The condition $b > I$ ensures every required Gaussian exponential moment is finite.

Differentiate this explicit expression to get (2.10). Its left-hand side is strictly decreasing and continuous on (I, ∞) and tends to zero at infinity. At $b \downarrow I$ it diverges: if $p_0 > 0$, use p_0/λ_0^2 ; otherwise use the summand corresponding to the first positive increment; if every $p_j = 0$, use $1/b$. Hence the solution is unique. The equation also gives $b \geq 1$. If $p_k \leq B$, then $I \leq B$; at $b = 2(B+1)$ its left-hand side is at most

$$\frac{1}{2(B+1)} + \frac{B}{(B+2)^2} \leq \frac{5}{8} < 1.$$

Thus the solution is below $2(B + 1)$.

We now justify replacing the Gaussian spin integral by the sphere. For *one* spin vector sampled from $Z_N^G(b)$ and then averaged over the disorder, the coordinates are iid. Indeed the terminal function is a sum of independent coordinate functions; Lemma 2.1 makes every path transition factor by coordinate, and the independent root Gaussian marks are not tilted. Conditional on the terminal fields, the spins are independent Gaussians of mean z_i/b and variance $1/b$. Integrating the factorized tilted path law therefore preserves independence and identical distribution. Their common law can be computed directly. Write $f_j(z) = c_j + z^2/(2\lambda_j)$ for the quadratic recursion. Completing the square in its tilted transition gives

$$Z_j \mid Z_{j-1} = z \sim \mathcal{N}\left(\frac{\lambda_j}{\lambda_{j-1}}z, \frac{\Delta p_j \lambda_j}{\lambda_{j-1}}\right). \quad (2.12)$$

Consequently $Y_j = Z_j/\lambda_j$ is a centered Gaussian martingale with independent increments of variances $\Delta p_j/(\lambda_j \lambda_{j-1})$, starting with variance p_0/λ_0^2 . The terminal spin is Y_k plus an independent $\mathcal{N}(0, 1/b)$ variable, so its variance is

$$\frac{p_0}{\lambda_0^2} + \sum_{j=1}^k \frac{\Delta p_j}{\lambda_j \lambda_{j-1}} + \frac{1}{b} = 1$$

by (2.10). Thus the averaged single-vector law is exactly $\mathcal{N}(0, I_N)$. This assertion concerns a single disorder-averaged sample; it does not assert independence at fixed disorder or between replicas.

Let G_N^G denote the x -marginal of the normalized Gibbs measure associated with the partition function $Z_N^G(b)$. Fix $0 < \delta < 1$ and write $A_{N,\delta} = \{|\|x\|^2/N - 1| \leq \delta\}$. The law of large numbers gives $\mathbb{E}G_N^G(A_{N,\delta}^c) \rightarrow 0$, and Markov's inequality gives $G_N^G(A_{N,\delta}) \rightarrow 1$ in disorder probability. To pass to logarithms, let $Z_{N,\delta}^G$ be the integral restricted to this shell. Lemma 2.1 expresses $\log Z_N^G$ as an iid coordinate sum at the root plus a difference of two log total cascade masses. More explicitly the root shift is $\sum_{i=1}^N [c_0 + Z_{0i}^2/(2\lambda_0)]$, where

$$c_0 = -\frac{1}{2} \log b + \sum_{j=1}^k \frac{1}{2m_{j-1}} \log \frac{\lambda_j}{\lambda_{j-1}}.$$

At each vertex the edge multiplier has m th moment exactly one by (2.1), also when the marks comprise N coordinates. Hence the transformed weight processes have exactly their original intensities. Conditionally on root marks, T' has the same N -independent marginal law as T ; all scale changes have been included in the root shift. Its quadratic Gaussian summands have fixed second moments, and Lemma 2.1 bounds the second moments of the two log masses. Therefore $\mathbb{E}|\log Z_N^G|^2 = O(N^2)$, with a constant allowed to depend on the fixed step profile. The restricted logarithm has a deterministic lower bound $-C_{b,\delta}N$ for large N : at any radius the sphere average of $e^{x \cdot z}$ is at least one by Jensen's inequality, and the Gaussian volume of the shell is at least $e^{-C_{b,\delta}N}$. Consequently

$$0 \leq \frac{1}{N} \log \frac{Z_N^G}{Z_{N,\delta}^G} \leq \frac{1}{N} \log Z_N^G + C_{b,\delta}$$

is uniformly integrable. The left side tends to zero in probability, so

$$\mathbb{E} \log Z_{N,\delta}^G = NX(b) + o(N). \quad (2.13)$$

For completeness the radial comparison includes the normalization of surface measure. With $x = r\hat{x}$, $\hat{x} \in \mathbb{S}_N$, polar coordinates give

$$\frac{dx}{(2\pi)^{N/2}} = c_N r^{N-1} dr d\sigma_N(\hat{x}), \quad c_N = \frac{2(N/2)^{N/2}}{\Gamma(N/2)}, \quad \frac{1}{N} \log c_N \rightarrow \frac{1}{2}.$$

For every fixed field the sphere integral of $e^{\widehat{r\cdot x}\cdot z}$ is nondecreasing in $r \geq 0$, by symmetry and monotonicity of the hyperbolic cosine. On $\sqrt{1-\delta} \leq r \leq \sqrt{1+\delta}$, the logarithm of the remaining radial factor, divided by N , is $1/2 - b/2 + O_b(\delta) + o(1)$. Its integral over this interval has the same estimate. Sandwich the spin integral between the values at the two shell endpoints. Scaling a field by r scales its covariance by r^2 , and (2.8), including the diagonal correction, gives

$$|S_N(r^2 p; r^2 p_k) - S_N(p; p_k)| \leq B|r^2 - 1|.$$

Together with (2.13), the radial sandwich shows $S_N(p; p_k) = X(b) + b/2 - 1/2 + O_{b,B}(\delta) + o(1)$. First let $N \rightarrow \infty$, then $\delta \downarrow 0$, to obtain (2.11).

Finally, nonnegative nondecreasing profiles bounded by B form a compact set in L^1 , and finite-step profiles are dense in it. The common Lipschitz estimate (2.8) transfers convergence from a finite net of step profiles to uniform convergence on this compact set. This proves the proposition for general bounded profiles. $\square$

2.4 Concavity and the entropy conjugate

For a trial overlap q , the pressure calculation in Section 4 will couple the row functional to the spherical functional through $p_q = \alpha D_q$. Its two bounds first lead to the expression

$$\alpha \Phi_u(q) + \mathcal{S}(p_q) + \frac{1}{2} \int_0^1 q(s) p_q(s) \, ds.$$

To identify the last two terms, we compute the spherical response. At a step profile p , the differential of $\mathcal{S}$ is represented by $-r/2$, where r is a nondecreasing profile strictly below one. For general p we construct a supergradient of this form. The inverse relation between p and the selected r will both prove concavity and identify when the last two terms equal the entropy conjugate $E(q)$.

For any nonnegative nondecreasing v with $v_* < 1$, define its overlap distribution and integrated distribution by

$$m_v(t) = \mathbb{P}(v(U) \leq t), \quad \chi_v(t) = \int_t^1 m_v(a) \, da, \quad U \sim \text{Unif}[0, 1].$$

The continuous form of the inverse map will be

$$\mathcal{I}(v)(s) = \int_0^{v(s)} \frac{dt}{\chi_v(t)^2}. \quad (2.14)$$

Its denominator is positive on the integration range, since $\chi_v(t) \geq 1 - v_*$ for $t \leq v_*$. The next proposition proves the inverse relation for the selected supergradient.

Proposition 2.6 (Concavity and supergradients). *The functional $\mathcal{S}$ is concave on bounded nonnegative nondecreasing profiles. At a finite-step profile its differential is*

$$d\mathcal{S}(p) = -\frac{1}{2} \sum_{j=0}^k w_j r_j \, dp_j, \quad r_j = \frac{p_0}{\lambda_0^2} + \sum_{i=1}^j \frac{p_i - p_{i-1}}{\lambda_i \lambda_{i-1}}, \quad r_k = 1 - \frac{1}{b}. \quad (2.15)$$

For every general bounded profile p there exists a nonnegative nondecreasing profile r , with $r_* < 1$, such that

$$\mathcal{S}(\tilde{p}) \leq \mathcal{S}(p) - \frac{1}{2} \int_0^1 r(s)(\tilde{p} - p)(s) \, ds \quad (2.16)$$

for every bounded nonnegative nondecreasing $\tilde{p}$. The profile r can be chosen so that $p = \mathcal{I}(r)$ almost everywhere. If $p \leq B$, r may be chosen with $r_* \leq 1 - [2(B+1)]^{-1}$.

Proof. All the derivatives can first be taken with $p_0 > 0$ and strictly positive increments, and then extended to boundary points by continuity. Differentiate (2.9) and (2.11). The coefficient of db is $X'(b) + 1/2 = 0$ by (2.10). Substituting $d\lambda_j = db - \sum_{i>j} m_{i-1}(dp_i - dp_{i-1})$ and collecting coefficients of dp_j gives (2.15). The last equality there is precisely (2.10).

To check the Hessian sign without a sign convention ambiguity, invert this gradient map. Write $\chi_j = 1/\lambda_j$. The differences of (2.15) and the recursion for λ give

$$r_j - r_{j-1} = \Delta p_j \chi_j \chi_{j-1}, \quad \chi_{j-1} - \chi_j = m_{j-1}(r_j - r_{j-1}), \quad \chi_k = 1 - r_k.$$

Telescoping yields the inverse formulas

$$\begin{aligned} \chi_j &= 1 - m_j r_j - \sum_{i>j} w_i r_i, \\ p_0 &= r_0 / \chi_0^2, \quad p_j - p_{j-1} = \frac{\chi_j^{-1} - \chi_{j-1}^{-1}}{m_{j-1}} = \frac{r_j - r_{j-1}}{\chi_j \chi_{j-1}}. \end{aligned} \tag{2.17}$$

Conversely every $0 \leq r_0 \leq \dots \leq r_k < 1$ gives positive χ_j and a nonnegative nondecreasing p by these formulas; putting $b = 1/\chi_k$ recovers (2.10). Thus they are genuine inverse relations.

Here is the full quadratic-form calculation. Put

$$T_j = \sum_{i \geq j} w_i dr_i, \quad T_{k+1} = 0, \quad L_j = m_j dr_j + T_{j+1} = -d\chi_j; \quad L_0 = T_0.$$

Differentiating (2.17) gives

$$dp_0 = \frac{dr_0}{\chi_0^2} + \frac{2r_0 T_0}{\chi_0^3}, \quad d(\Delta p_j) = \frac{1}{m_{j-1}} \left(\frac{L_j}{\chi_j^2} - \frac{L_{j-1}}{\chi_{j-1}^2} \right).$$

Insert these into $\sum_j w_j dr_j dp_j = T_0 dp_0 + \sum_{j \geq 1} T_j d(\Delta p_j)$. Summation by parts gives exactly

$$\sum_{j=0}^k w_j dr_j dp_j = \frac{m_0 (dr_0)^2 - T_1^2 / m_0}{\chi_0^2} + \frac{2r_0 T_0^2}{\chi_0^3} + \sum_{j=1}^k \frac{w_j L_j^2}{m_{j-1} m_j \chi_j^2}. \tag{2.18}$$

Since $\chi_j \leq \chi_0$, replace χ_j^{-2} in the last, nonnegative sum by χ_0^{-2} . The identity

$$\frac{w_j L_j^2}{m_{j-1} m_j} = w_j (dr_j)^2 + \frac{T_j^2}{m_{j-1}} - \frac{T_{j+1}^2}{m_j}$$

then telescopes and proves

$$\sum_j w_j dr_j dp_j \geq \frac{1}{\chi_0^2} \sum_j w_j (dr_j)^2 \geq 0. \tag{2.19}$$

For $k = 0$ the same calculation has $m_0 = 1$ and empty sums. Along a line in p , (2.15) and (2.19) give $d^2 S \leq 0$. Continuity proves concavity on the boundary and, by step approximation, on the full domain.

For general $p \leq B$, choose step profiles $p_n \leq B$ converging to p . Their gradients have nonnegative nondecreasing representatives r_n and, by Proposition 2.5,

$$(r_n)_* = 1 - 1/b_n \leq 1 - [2(B+1)]^{-1}.$$

Pass to an L^1 convergent subsequence. The finite-step tangent inequality holds against every step competitor after refining to a common partition. Letting $n \rightarrow \infty$, and then

approximating an arbitrary bounded competitor, gives (2.16). The displayed uniform bound passes to the limit.

This same construction gives the asserted inverse relation. For a step profile v , χ_v is constant below its lowest value and affine of slope $-m_{j-1}$ between successive values. Integrating χ_v^{-2} therefore gives v_0/χ_0^2 and increments $(v_j - v_{j-1})/(\chi_j \chi_{j-1})$. Thus the finite inverse formulas are exactly $p_n = \mathcal{I}(r_n)$. To pass to the limit, Fubini gives

$$\chi_v(t) = 1 - t - \int_0^1 (v(s) - t)_+ ds, \quad \sup_t |\chi_v(t) - \chi_{\tilde{v}}(t)| \leq \|v - \tilde{v}\|_1. \quad (2.20)$$

For $v, \tilde{v} \leq Q < 1$, their integrated distributions are at least $1 - Q$ on $[0, Q]$. Comparing the integrands and the upper integration limits in (2.14) gives

$$\|\mathcal{I}(v) - \mathcal{I}(\tilde{v})\|_1 \leq C_Q \|v - \tilde{v}\|_1.$$

Consequently $p = \mathcal{I}(r)$. This identifies the supergradient obtained from the step approximations; no assertion about uniqueness of all supergradients on the monotone cone is needed. $\square$

Proposition 2.7 (Entropy conjugacy). *Let q be nonnegative and nondecreasing with $q_* < 1$, and use χ_q defined above. Then the following supremum is attained:*

$$\begin{aligned} E(q) &:= \sup_{\substack{p \geq 0 \text{ bounded} \\ p \text{ nondecreasing}}} \left\{ \mathcal{S}(p) + \frac{1}{2} \int_0^1 q(s)p(s) ds \right\} \\ &= \frac{1}{2} \left\{ \log(1 - q_*) + \int_0^{q_*} \frac{dt}{\chi_q(t)} \right\}, \quad \chi_q(t) = \int_t^1 m_q(v) dv. \end{aligned} \quad (2.21)$$

An optimizer may be chosen with $p_* \leq q_*/(1 - q_*)^2$. Equivalently,

$$E(q) = \frac{1}{2} \int_0^1 \left\{ \frac{1}{\chi_q(t)} - \frac{1}{1 - t} \right\} dt, \quad (2.22)$$

where the integrand is identically zero for $t > q_*$. For profiles $q, \tilde{q} \leq Q < 1$,

$$|E(q) - E(\tilde{q})| \leq \frac{Q}{2(1 - Q)^2} \|q - \tilde{q}\|_1. \quad (2.23)$$

Proof. For a step profile q , set $r = q$ in (2.17) and obtain p . The tangent inequality in Proposition 2.6 gives, for every competitor $\tilde{p}$,

$$\mathcal{S}(\tilde{p}) + \frac{1}{2} \int q\tilde{p} \leq \mathcal{S}(p) + \frac{1}{2} \int qp;$$

thus p attains the supremum. In these inverse formulas $\chi_j = \chi_q(q_j)$, $b = 1/(1 - q_k)$, and

$$p_0 = \frac{q_0}{\chi_0^2}, \quad \Delta p_j = \frac{q_j - q_{j-1}}{\chi_j \chi_{j-1}}.$$

The algebra in the value can be made explicit. Abel summation and the inverse formulas give

$$\begin{aligned} p_k - \sum_j w_j q_j p_j &= p_0 \chi_0 + \sum_{j=1}^k \Delta p_j (\chi_{j-1} + m_{j-1} q_{j-1}) \\ &= \frac{q_0}{\chi_0} + \sum_{j=1}^k \left(\frac{q_j}{\chi_j} - \frac{q_{j-1}}{\chi_{j-1}} \right) = \frac{q_k}{\chi_k} = b - 1. \end{aligned}$$

Substitution into (2.9) and (2.11) therefore yields

$$2E(q) = \log(1 - q_k) + \frac{q_0}{\chi_0} + \sum_{j=1}^k \frac{1}{m_{j-1}} \log \frac{\chi_{j-1}}{\chi_j}.$$

On $[0, q_0]$, $\chi_q = \chi_0$; on each interval $[q_{j-1}, q_j]$ it is affine of slope $-m_{j-1}$. Integrating its reciprocal on these intervals gives exactly (2.21), also when an interval has zero length.

For passage to general profiles, fix $Q < 1$ with $q \leq Q$ and approximate by step profiles $q_n \leq Q$ in L^1 . Their inverse profiles satisfy

$$(p_n)_* = \frac{(q_n)_0}{(\chi_n)_0^2} + \sum_j \frac{(q_n)_j - (q_n)_{j-1}}{(\chi_n)_j(\chi_n)_{j-1}} \leq \frac{Q}{(1 - Q)^2},$$

because every $\chi_n \geq 1 - Q$. Compactness gives a subsequence $p_n \rightarrow p$ in L^1 . The finite-step tangent inequalities, first against step competitors and then against arbitrary bounded competitors, pass to the limit by the Lipschitz continuity of $\mathcal{S}$ and the uniform bounds. They prove attainment for q and show that its value is the limit of the finite-step values. This argument does not interchange an uncontrolled supremum with a limit.

To identify the limiting expression, use (2.20). For $q \leq Q$, $\chi_q(t) \geq 1 - Q$ on $[0, Q]$, and $\chi_q(t) = 1 - t$ on $[Q, 1]$. Consequently the right side of (2.22) is continuous under these approximations, with the explicit bound (2.23). It agrees with (2.21) because $\int_0^{q_*} (1 - t)^{-1} dt = -\log(1 - q_*)$. Approximating with $Q = q_*$ gives the stated optimizer bound; when $q_* = 0$, $p = 0$ suffices. This completes the proof. $\square$

3 Concavity and stability of the row functional

The pressure argument needs three properties of the row functional at a fixed diagonal covariance Q . Ordinary concavity in the off-diagonal profile q supplies the interpolation sign. Strict concavity makes the derivative profile D_q strongly monotone; this will force a minimizing overlap to equal the spherical response constructed in the preceding section. Finally, concavity must persist under both signs of a small terminal perturbation, so that pressure bounds identify empirical row laws.

We first derive a Hessian identity for smooth profiles and terminals. For a concave terminal it is a sum of nonpositive terms. The hinge has enough curvature on one fixed interval to make the bound uniformly negative. We then control the change in that Hessian under a compact perturbation and remove the terminal and profile regularizations.

For the hinge terminal $u = -\beta V$, two perturbation regimes are used below. Corollary 3.6 permits arbitrary $\psi \in C_c^\infty(\mathbb{R})$ in $u + d\psi$, with $|d|$ small at each fixed β . Corollary 3.7 treats $-\beta[V + w\psi(1 + z)]$ with $\psi \in C_c^\infty((-\infty, 0))$; convexity of the loss gives an admissible $|w|$ independent of β .

3.1 The parabolic representation and its regularity

Suppose initially that $q \in C^\infty([0, 1])$, $q(0) = 0$, $q(1) = Q$, and $\dot{q} > 0$. The Gaussian recursion defining the row functional is equivalent to

$$f_s = -\frac{\dot{q}(s)}{2}(f_{zz} + sf_z^2), \quad f(1, z) = u(z), \quad \Phi_u(q; Q) = f(0, 0). \quad (3.1)$$

Here and below a dot denotes differentiation in rank time s . In variance time $t = q(s)$, write $F(t, z) = f(q^{-1}(t), z)$ and $m(t) = q^{-1}(t)$. The equation becomes

$$F_t = -\frac{1}{2}(F_{zz} + m(t)F_z^2), \quad F(Q, z) = u(z). \quad (3.2)$$

When m is constant, equal to $a \in [0, 1]$, on an interval of length T , its backward solution operator is

$$\mathcal{T}_{a,T}v(z) = \begin{cases} a^{-1} \log \mathbb{E} \exp\{av(z + \sqrt{T}G)\}, & a > 0, \\ \mathbb{E}v(z + \sqrt{T}G), & a = 0, \end{cases} \quad G \sim \mathcal{N}(0, 1).$$

Consequently step functions m reproduce exactly the finite Gaussian recursion, including its initial heat interval and terminal convolution.

We record the analytic class in which subsequent differentiations are made. Assume that u is smooth, all its derivatives of order at least two are bounded, and, for fixed constants $A, C \geq 0$, $B \in \mathbb{R}$, and $\kappa \geq 0$,

$$-A(1 + z^2) \leq u(z) \leq B, \quad -C \leq u''(z) \leq \kappa, \quad \kappa Q < 1. \quad (3.3)$$

Constants in the following bounds depend only on these constants and Q , unless a dependence on a higher derivative of u is explicitly indicated.

Lemma 3.1 (Parabolic bounds and differentiability). *Equation (3.2) has a unique solution of at most quadratic growth with a spatial derivative of at most linear growth. For continuous $m : [0, Q] \rightarrow [0, 1]$, the solution is classical and satisfies*

$$-C \leq F_{zz}(t, z) \leq \frac{\kappa}{1 - \kappa(Q - t)}, \quad |F_z(t, z)| \leq C_1(1 + |z|). \quad (3.4)$$

For a fixed smooth terminal, each spatial derivative of order at least two is bounded. The corresponding bounds are uniform over $m \in [0, 1]$. For a smooth profile q with $\dot{q} > 0$ and a smooth direction η with $\eta(0) = \eta(1) = 0$, the solution of (3.1) is twice differentiable in the parameter ε at $q + \varepsilon\eta$.

Proof. Construction and curvature bounds. For a step coefficient m , use the displayed Gaussian operators. Their integrals exist since u is bounded above. Jensen's inequality and iteration give

$$-A(1 + z^2 + Q - t) \leq F(t, z) \leq B.$$

The differentiated equation, with $\mathcal{D} = \partial_t + \frac{1}{2}\partial_{zz} + mF_z\partial_z$, is

$$\mathcal{D}F_z = 0, \quad \mathcal{D}F_{zz} = -mF_{zz}^2.$$

Backward parabolic comparison, or forward comparison after the change of time $\tau = Q - t$, bounds the second derivative below by $-C$ and above by the solution of $K' = K^2$, $K(0) = \kappa$. These comparisons can first be made for bounded-slope terminals and on bounded spatial intervals, with quadratic barriers, and then passed to expanding intervals. Here is an explicit bounded-slope approximation preserving the constants needed for that passage. Keep u on $[-R, R]$. If $p = u'(R) \leq 0$, continue affinely to the right. If $p > 0$, continue with derivative $p - C(z - R)$ until that derivative reaches zero, and then continue constantly. The maximum of this continuation is $u(R) + p^2/(2C) \leq B$: the inequality follows by applying $u(R + h) \geq u(R) + ph - Ch^2/2$ at $h = p/C$ to the original terminal. On the left, continue affinely if $u'(-R) \geq 0$; otherwise continue with second derivative $-C$ until the derivative reaches zero, and then constantly. The same argument bounds its maximum by B . If $C = 0$, the original terminal is convex and bounded above, hence constant, so no continuation is needed. The resulting $C^{1,1}$ function has curvature in $[-C, \kappa]$ distributionally. Its value and derivative at zero agree with those of u , so the lower curvature bound supplies a common negative quadratic envelope on the whole line. Convolution with a compact nonnegative mollifier preserves both curvature bounds and the upper bound; let its width tend to zero as

$R \rightarrow \infty$. The preceding quadratic barriers and Gaussian integrability now permit passage to the original terminal. The two-sided bound on F near $z = 0$, together with the bound on F_{zz} , bounds $F_z(t, 0)$; integration of F_{zz} then gives the linear-growth bound in (3.4).

Higher spatial derivatives. These estimates do not require bounds on derivatives of m . For example,

$$\mathcal{D}F_{zzz} = -3mF_{zz}F_{zzz}.$$

The diffusion representation bounds this derivative by $\|u'''\|_\infty \exp(3KQ)$, where K bounds $|F_{zz}|$. Each further spatial differentiation gives a linear equation whose zeroth-order coefficient is bounded and whose source contains only already bounded derivatives. Induction gives the assertion.

General coefficients. Approximate a continuous m uniformly by step functions. If F and F' are the two step solutions, their difference solves a linear backward equation with diffusion coefficient $1/2$, drift $m(F_z + F'_z)/2$, and source $-(m - m')(F'_z)^2/2$. Its drift is globally Lipschitz and has linear growth. The diffusion representation, with the terminal difference zero, gives on each compact spatial interval

$$|F - F'| \leq C_R \|m - m'\|_\infty.$$

The spatial derivative bounds and the equation give local compactness also for the derivatives. Thus the limit solves (3.2); the same difference equation proves uniqueness. This constructs the claimed classical solution and identifies it with the Gaussian recursion. The argument also works with step approximations of bounded measurable m , using convergence in measure and dominated convergence in the diffusion representation whenever needed below.

Dependence on the covariance profile. Work in rank time and keep $\dot{q}$ positive in a neighborhood of the parameter under consideration. Subtracting two equations gives a linear equation with drift $\dot{q}_\varepsilon s(f_{\varepsilon,z} + f_z)/2$ and source $-(\dot{q}_\varepsilon - \dot{q})(f_{zz} + sf_z^2)/2$. Its diffusion representation and its spatially differentiated equations bound the difference and each fixed number of its spatial derivatives by $C\|\dot{q}_\varepsilon - \dot{q}\|_\infty(1 + z^2)$. To see that differentiation does not increase this polynomial degree, keep the linear-growth drift in the transport operator: every spatial derivative of that drift of order at least one is bounded. The commutator terms are therefore bounded coefficients times derivatives already estimated, while the source and all its fixed-order spatial derivatives have at most quadratic growth. Induction and the diffusion moment bounds preserve the same quadratic envelope. The constant here may depend on the fixed profile and on finitely many terminal derivatives. Applying the same bounds to difference quotients, and then to the difference between those quotients and their linearized equations, proves first and second parameter differentiability. This also justifies differentiation under the diffusion expectations below. Indeed the diffusions have moments of every fixed order, and stopping at $|Z| = R$ followed by $R \rightarrow \infty$ removes all spatial localizations. $\square$

3.2 An exact Hessian identity

Return to rank time. Put $a = f_z$, $b = f_{zz}$, and let Z solve

$$dZ_s = \sqrt{\dot{q}(s)} dW_s + \dot{q}(s)sa(s, Z_s) ds, \quad Z_0 = 0. \quad (3.5)$$

Its generator, including the time derivative, is $\mathcal{D} = \partial_s + \dot{q}\partial_{zz}/2 + \dot{q}sa\partial_z$. All unqualified expectations in the rest of this subsection evaluate the displayed functions along Z_s . Differentiation of (3.1) gives

$$\mathcal{D}a = 0, \quad \mathcal{D}b = -\dot{q}sb^2, \quad U = b + sa^2, \quad \mathcal{D}U = a^2.$$

Thus $a(s, Z_s)$ is precisely the derivative martingale of the tilted Gaussian recursion.

Let $v = \partial_\varepsilon f_{q+\varepsilon\eta}|_{\varepsilon=0}$ and $w = \partial_\varepsilon^2 f_{q+\varepsilon\eta}|_{\varepsilon=0}$. Their terminal values vanish, and

$$\begin{aligned}\mathcal{D}v &= -\dot{\eta}U/2, \\ \mathcal{D}w &= -\dot{\eta}(v_{zz} + 2sav_z) - \dot{q}sv_z^2.\end{aligned}\tag{3.6}$$

Since η vanishes at both endpoints, integration by parts in $\mathbb{E}U(s, Z_s)$ gives

$$v(0, 0) = \frac{1}{2} \int_0^1 \dot{\eta}(s) \mathbb{E}U(s, Z_s) ds = -\frac{1}{2} \int_0^1 \eta(s) \mathbb{E}a(s, Z_s)^2 ds.$$

This is the differential formula (2.6) in the smooth setting.

For the second derivative, the term $\dot{\eta}$ in the equation for v would obscure the sign. Adding $\eta U/2$ cancels that term and leaves $\mathcal{D}(v + \eta U/2) = \eta a^2/2$. The spatial derivative of this modified variation is the useful unknown in the Hessian identity.

Lemma 3.2 (Hessian identity). *Set $\bar{v} = v + \eta U/2$ and $l = \bar{v}_z$. Then*

$$\begin{aligned}w(0, 0) &= \mathbb{E} \int_0^1 \left(\dot{q}sl^2 - 2\eta al - \frac{1}{2}\eta^2 b^2 \right) ds, \\ \mathcal{D}l &= \eta ab - \dot{q}sb, \quad l(1, \cdot) = 0.\end{aligned}\tag{3.7}$$

Proof. For $Mg = g_{zz} + 2sag_z$, direct differentiation gives

$$\mathcal{D}(Mg) = M(\mathcal{D}g) + (2a - \dot{q}sU_z)g_z.$$

Integrate the first term on the right side of the second equation in (3.6) by parts, using $\frac{d}{ds} \mathbb{E}Mg(s, Z_s) = \mathbb{E}\mathcal{D}(Mg)(s, Z_s)$. Substitute $v_z = l - \eta U_z/2$ and complete the square. Apart from $\dot{q}sl^2 - 2\eta al$, the remaining integrand is

$$-\frac{1}{4}\dot{q}s\eta^2 U_z^2 + \eta^2 a U_z + \frac{1}{2}\eta\dot{\eta} MU.$$

The second identity

$$\mathcal{D}(MU) = 2b^2 + 4aU_z - \dot{q}sU_z^2$$

and one further integration by parts turn this integrand into $-\eta^2 b^2/2$. There are no boundary terms because $\eta(0) = \eta(1) = 0$. Finally $\mathcal{D}\bar{v} = \eta a^2/2$ and spatial differentiation, taking account of the derivative $\dot{q}sb$ of the drift, give the equation for l . $\square$

Proposition 3.3 (Concavity). *For every concave terminal in the class (3.3), $q \mapsto \Phi_u(q; Q)$ is concave on smooth strictly increasing profiles. For every concave globally Lipschitz terminal, which need not be smooth or bounded above, and for the quadratic hinge $u = -\beta V$, it is concave on the full convex set of nonnegative nondecreasing profiles bounded by Q .*

Proof. First suppose $u'' \leq -\rho < 0$. The stochastic flow in (3.5), started from z at time s , has spatial Jacobian $J_{s,r} = \exp\{\int_s^r \dot{q}(t)tb(t, Z_t) dt\}$. Differentiating the martingale representation $a(s, z) = \mathbb{E}_{s,z}u'(Z_1)$ gives

$$b(s, z) = \mathbb{E}_{s,z} \left[u''(Z_1) \exp \left\{ \int_s^1 \dot{q}(t)tb(t, Z_t) dt \right\} \right].\tag{3.8}$$

Thus $k = -b$ is bounded and bounded away from zero. The product and quotient rules for $\mathcal{D}$ give

$$\mathcal{D}(l^2/k) = -2\eta al + \dot{q}sl^2 + \frac{\dot{q}}{k}(l_z - lk_z/k)^2.$$

Since $l(1, \cdot) = 0$, substitution into (3.7) yields the exact identity

$$\begin{aligned} w(0, 0) = & -\frac{l(0, 0)^2}{k(0, 0)} - \mathbb{E} \int_0^1 \frac{\dot{q}}{k} (l_z - lk_z/k)^2 ds \\ & - \frac{1}{2} \int_0^1 \eta(s)^2 \mathbb{E} b(s, Z_s)^2 ds \leq 0. \end{aligned} \quad (3.9)$$

For a merely concave u , apply this argument to $u - \lambda z^2$ and let $\lambda \downarrow 0$. The parabolic comparison and moment bounds in Lemma 3.1 give convergence on every smooth profile and its segments. Concavity passes to that limit. Finally approximate two arbitrary profiles and their segment together by smooth increasing profiles with the prescribed endpoint values. The L^1 continuity in (2.6) proves this extension for Lipschitz terminals already in the analytic class.

To cover a general concave globally L -Lipschitz terminal, convolve it with a compact nonnegative mollifier of width δ . The resulting u_δ is smooth and concave, has bounded derivatives of every order at least two, and satisfies $\|u_\delta - u\|_\infty \leq L\delta$. For each $\lambda > 0$, $u_\delta - \lambda z^2$ belongs to (3.3), so concavity holds for smooth profiles. Let $\lambda \downarrow 0$ on each of the three profiles of a fixed chord. This limit needs no bound on $\sup u_\delta$ uniform in λ . Indeed, for $0 < \lambda \leq 1$, Jensen gives a common negative quadratic lower bound for the solutions, while $u_\delta(z) - \lambda z^2 \leq u_\delta(0) + L|z|$ and $\mathcal{T}_{a,T}v \leq \mathcal{T}_{1,T}v$ for $a \in [0, 1]$ give the upper bound

$$F_{\delta,\lambda}(t, z) \leq u_\delta(0) + L|z| + \log \mathbb{E} \exp\{L\sqrt{Q}|G|\}.$$

The curvature bounds, hence the linear-growth drift and its diffusion moments, are uniform for $\lambda \leq 1$ at fixed δ . The linear equation comparing two such solutions therefore bounds their difference by $C_\delta|\lambda - \lambda'|(1 + z^2)$. It identifies the limit with the Gaussian recursion for u_δ ; alternatively the same conclusion follows by monotone convergence of its terminal weights, with the displayed upper and lower integrable bounds. Thus concavity holds for u_δ on smooth profiles. Profile continuity from (2.6) extends it to all profiles, and the 1-Lipschitz dependence of Φ on the terminal uniform norm lets $\delta \downarrow 0$. The quadratic hinge and its other required limits are treated below. $\square$

3.3 Strict concavity uniform over profiles

Fix $\beta > 0$. Let u_L be the function $-\beta V$ on $[-L, \infty)$, extended by its tangent line at $-L$ on $(-\infty, -L)$, where $L > 4$. Convolve with a nonnegative smooth mollifier of total mass one and support in $[-\delta, \delta]$, $0 < \delta \leq 1/4$. Denote the resulting terminal by $u_{L,\delta}$. For the present argument one may also subtract λz^2 , $0 < \lambda \leq 1$. All these terminals have uniform quadratic-growth bounds, and

$$-\beta - 2 \leq u'' \leq 0, \quad \int_{\mathbb{R}} |u'''(z)| dz \leq 2\beta, \quad u'' \leq -\beta \quad \text{on } I = [-3, -2]. \quad (3.10)$$

The added quadratic is used only to justify division by $-b$; it will always be removed before an argument that requires a monotone terminal.

Lemma 3.4 (Diffusion density bounds). *Let X solve $dX_t = dB_t + A(t, X_t)dt$, with $|\partial_z A| \leq K$ and $|A(t, 0)| \leq K$. Over an interval of length $T > 0$, its transition density is bounded above by $e^{KT}/\sqrt{2\pi T}$. Moreover, when $X_0 = 0$, $Q \in [1/2, 2]$, and K is fixed, the probability $\mathbb{P}(X_Q \in I)$ is bounded below by a positive constant depending only on K and the nondegenerate bounded interval I .*

Proof. Condition on the Brownian bridge and write the Brownian path over the interval as $b_t + (t/T)\xi$, where $\xi \sim \mathcal{N}(0, T)$ is independent of the bridge. The endpoint $X_T = \mathcal{F}(\xi)$ is an increasing function of ξ . Differentiation of the integral equation gives

$$\mathcal{F}'(\xi) = \frac{1}{T} \int_0^T \exp \left\{ \int_r^T \partial_z A(v, X_v) dv \right\} dr \in [e^{-KT}, e^{KT}].$$

The conditional change-of-variables formula proves the upper density bound, which remains true after averaging over the bridge. For the lower bound, restrict the bridge to an event on which its supremum is at most a fixed constant. Such an event has uniformly positive probability for $Q \in [1/2, 2]$, by Brownian scaling. Gronwall's inequality then bounds $|\mathcal{F}(0)|$ uniformly. The inverse image of I is an interval of length at least $e^{-2K}|I|$ contained in a fixed bounded interval. Gaussian densities with variance $Q \in [1/2, 2]$ have a positive common lower bound there. Multiplication by the bridge-event probability proves the claim. $\square$

Proposition 3.5 (Uniform strictness and stability). *There is $c_\beta > 0$ such that, for every $Q \in [1/2, 2]$, every regularized base terminal just described, every smooth strictly increasing profile, and every smooth endpoint-zero direction,*

$$\frac{d^2}{d\varepsilon^2} \Phi_u(q + \varepsilon\eta; Q)|_{\varepsilon=0} \leq -c_\beta \|\eta\|_2^2. \quad (3.11)$$

For each $\psi \in C_c^\infty(\mathbb{R})$ there is $d_0 = d_0(\beta, \psi) > 0$, independent of L, δ, λ, Q , and the profile, such that (3.11) holds with $c_\beta/2$ for the terminal $u + d\psi$ whenever $|d| \leq d_0$.

Proof. In variance time the drift is $m(t)F_z(t, z)$. Its Lipschitz and growth bounds are uniform by Lemma 3.1. Thus Lemma 3.4 gives $\mathbb{P}(Z_1 \in I) \geq p_\beta > 0$, where Z_1 denotes the rank-time endpoint. From (3.8), the bound on $|b|$, and (3.10), for every s ,

$$\mathbb{E}[-b(s, Z_s)] \geq \beta e^{-KQ} \mathbb{P}(Z_1 \in I) \geq \beta e^{-2K} p_\beta.$$

Jensen's inequality supplies a positive uniform lower bound on $\mathbb{E}b(s, Z_s)^2$. Equation (3.9) proves (3.11). The lower bound does not depend on λ , so it survives $\lambda \downarrow 0$.

To prove stability, we compare slopes, then curvatures, then the modified variation l under a common Brownian coupling. These three comparisons control every term in the Hessian identity. The curvature comparison uses the total variation of u'' , so its constants remain uniform as the mollifier is removed.

Slope comparison. A tilde will denote the terminal $u + d\psi$. Choose $|d|$ initially so that $|d|\|\psi''\|_\infty < 1/4$; then its positive curvature bound is strictly less than $1/(2Q)$ for $Q \in [1/2, 2]$, and all solution and drift bounds remain uniform. In this proof only, use variance-time notation $a = F_z$, $b = F_{zz}$ and its tilde analogue. At the same spatial point,

$$\tilde{\mathcal{D}}(\tilde{a} - a) = -mb(\tilde{a} - a), \quad (\tilde{a} - a)(Q, \cdot) = d\psi'.$$

The diffusion representation implies

$$\sup_{t,z} |\tilde{a} - a| \leq C|d|. \quad (3.12)$$

Couple both diffusions using the same Brownian motion. Starting from a common position at time t , Gronwall gives $|\tilde{X}_r - X_r| \leq C|d|(r - t)$ for $t \leq r \leq Q$.

Curvature comparison. We use estimates that depend on the total variation of u'' , and not on $\|u'''\|_\infty$. The equation $\mathcal{D}b_z = -3mbb_z$ and its diffusion representation give

$$\int_{\mathbb{R}} |b_z(t, z)| dz \leq C, \quad \sup_z |b_z(t, z)| \leq \frac{C}{\sqrt{Q-t}} \quad (t < Q). \quad (3.13)$$

For the first bound integrate the representation over starting positions; the pathwise spatial flow has Jacobian between e^{-KQ} and e^{KQ} . For the second use the upper density bound of Lemma 3.4 and $\int |u'''| \leq 2\beta$.

Compare the flow representations (3.8) for b and $\tilde{b}$, started from the same point at time t . The terminal perturbation contributes at most $C|d|$. For the displacement in the base terminal, use the elementary inequality

$$|u''(y) - u''(x)| \leq \int_{x-h}^{x+h} |u'''(v)| dv \quad (|y - x| \leq h).$$

With $h = C|d|(Q - t)$ and the transition-density bound, its expectation is at most $C|d|\sqrt{Q - t}$. The difference of the two exponential Jacobians is at most a constant times the integral of the difference of their curvatures along the paths. Putting $A(t) = \sup_z |\tilde{b}(t, z) - b(t, z)|$ and using (3.13) therefore gives

$$A(t) \leq C|d| + C \int_t^Q \left(A(r) + \frac{|d|(r - t)}{\sqrt{Q - r}} \right) dr.$$

The singularity is integrable, and backward Gronwall yields

$$\sup_{t,z} |\tilde{b}(t, z) - b(t, z)| \leq C|d|. \quad (3.14)$$

For paths coupled from the root, their displacement at variance time t is at most $C|d|t$. The same elementary inequality, now applied to $b(t, \cdot)$, and its total variation and the density bound at time t , give

$$\mathbb{E}|b(t, \tilde{X}_t) - b(t, X_t)| \leq C|d|\sqrt{t}.$$

At $t = 0$ the positional difference is zero. Combining this estimate with (3.14) and the uniform boundedness of both curvatures gives, for every fixed $p \geq 1$,

$$\sup_t \|\tilde{b}(t, \tilde{X}_t) - b(t, X_t)\|_{L^p} \leq C_p |d|^{1/p}. \quad (3.15)$$

The coupled slopes differ pointwise by $C|d|$ and have uniformly bounded moments of every fixed order.

Comparison of the modified variations. Return to rank time and write $a_s = a(s, Z_s)$, $b_s = b(s, Z_s)$ and $l_s = l(s, Z_s)$. On the common Brownian filtration $\mathcal{F}_s$, the second equation of (3.7) gives

$$l_s = -\mathbb{E} \left[\int_s^1 \eta(r) a_r b_r \exp \left\{ \int_s^r \dot{q}(v) v b_v dv \right\} dr \middle| \mathcal{F}_s \right]. \quad (3.16)$$

This same filtration is used for the tilde representation; no comparison of unrelated conditional laws is involved. Both exponential factors are uniformly bounded since $\int_0^1 \dot{q} ds = Q \leq 2$. Conditional expectation is a contraction in L^p , so

$$\sup_s \|l_s\|_{L^p} + \sup_s \|\tilde{l}_s\|_{L^p} \leq C_p \|\eta\|_1.$$

Hölder's inequality, (3.15) with $p = 4$, the slope estimates, and Minkowski's inequality for the exponential integrals give

$$\sup_s \|\tilde{l}_s - l_s\|_{L^2} \leq C|d|^{1/4} \|\eta\|_1. \quad (3.17)$$

For example, the difference of the exponential factors has L^4 norm at most $C \int_s^r \dot{q}(v) \|\tilde{b}_v - b_v\|_{L^4} dv \leq C|d|^{1/4}$; multiplying by $a_r b_r$, whose L^4 norm is bounded, proves the relevant L^2 bound.

Hessian comparison. Compare the three terms of (3.7). The $\dot{q}sl^2$ term changes by at most $C|d|^{1/4}Q\|\eta\|_1^2$; the $2\eta al$ term by at most $C|d|^{1/4}\|\eta\|_1^2$; and the $\eta^2 b^2/2$ term by at most $C|d|\|\eta\|_2^2$. Since $\|\eta\|_1 \leq \|\eta\|_2$,

$$|\tilde{w}(0,0) - w(0,0)| \leq C|d|^{1/4}\|\eta\|_2^2. \quad (3.18)$$

Choosing d_0 so that $Cd_0^{1/4} \leq c_\beta/2$ proves the proposition. $\square$

3.4 Removing regularizations and general profiles

We now pass the estimates to the terminals and profiles used in the pressure formula. The resulting statement is the row-functional input for both minimizer identification and compact row perturbations.

Corollary 3.6 (The uncapped row functional). *Fix $\beta > 0$ and $\psi \in C_c^\infty(\mathbb{R})$. There are constants $c_\beta > 0$ and $d_0 = d_0(\beta, \psi) > 0$ such that, for $u_d = -\beta V + d\psi$, $|d| \leq d_0$, $Q \in [1/2, 2]$, and all $q, r \in \mathcal{Q}_Q$,*

$$\Phi_{u_d}((1-t)q + tr; Q) \geq (1-t)\Phi_{u_d}(q; Q) + t\Phi_{u_d}(r; Q) + \frac{c_\beta}{4}t(1-t)\|r - q\|_2^2, \quad 0 \leq t \leq 1. \quad (3.19)$$

When $d = 0$, the last coefficient can be replaced by $c_\beta/2$. The derivative formula (2.6), including its one-sided endpoint interpretation, holds with the terminal u_d and fixed diagonal Q . Its nonnegative nondecreasing derivative profile satisfies $\sup_{q, Q, d} \|D_q^{u_d}\|_\infty \leq C_{\beta, \psi}$ in these ranges and depends continuously on q in L^1 at fixed Q, d . The functional values and derivative profiles are limits of those for $u_{L, \delta} + d\psi$, uniformly over q, Q, d , with derivative convergence in $L^1(ds)$. Here $\delta \downarrow 0$ at each fixed cap before $L \rightarrow \infty$.

Proof. The profile inequalities above are uniform in $\dot{q}$: their constants use only $Q = \int \dot{q}$. To approximate two arbitrary profiles q, r , smooth their monotone extensions, add a small increasing linear function, and insert short ramps at 0 and 1 to impose the values 0, Q . These approximations can be chosen to remain in $[0, Q]$ and converge in L^1 , hence also in L^2 ; their convex combinations approximate the whole segment. For a fixed capped and mollified terminal, first let $\lambda \downarrow 0$ on these smooth segments. Then (2.6) passes the strong concavity inequalities to all profiles. Integrating the Hessian bound gives (3.19) for each capped and mollified terminal, with coefficient $c_\beta/2$ in the unperturbed case and $c_\beta/4$ in the perturbed case.

We justify the remaining terminal limits uniformly in the profile. If μ is any probability measure on $\mathbb{R}$, w is nonnegative and increasing with $0 < \int w d\mu < \infty$, and T is nonnegative and decreasing, then

$$\frac{\int Tw d\mu}{\int w d\mu} \leq \int T d\mu. \quad (3.20)$$

First take T bounded. The covariance of T and w equals half the integral of $(T(x) - T(y))(w(x) - w(y))$ against $\mu \otimes \mu$, and is nonpositive. Apply this to $T \wedge K$ and let $K \rightarrow \infty$ to obtain the general case by monotone convergence. To apply this inequality, condition first on the cavity measure and the new Gaussian field, and let μ be the pushforward of the cavity measure under that field. Insertion of an increasing terminal u replaces μ by its tilt with weight $w = e^u$. Only after applying the inequality do we average over the independent field and the cavity randomness. Before insertion, one sampled field value then has law $\mathcal{N}(0, Q)$, because its diagonal covariance is Q . For the RPC row functional the same construction includes any independent residual Gaussian in the base space. Adding $d\psi$ increases the right side of (3.20) by at most the factor $\exp(2|d|\|\psi\|_\infty)$.

The tangent cap satisfies the exact identity

$$0 \leq u_L(z) + \beta V(z) = \frac{\beta}{2}(z + L)_-^2.$$

Every terminal between u_L and $-\beta V$ is increasing. Interpolating between them and using (3.20) proves

$$\sup_q |\Phi_{u_L+d\psi}(q; Q) - \Phi_{-\beta V+d\psi}(q; Q)| \leq e^{2|d|\|\psi\|_\infty} \mathbb{E} \frac{\beta}{2} (\sqrt{Q}G + L)_-^2 \longrightarrow 0. \quad (3.21)$$

The convergence is also uniform for $Q \in [1/2, 2]$. For each fixed L , mollification converges uniformly in terminal value, since u_L is Lipschitz; the row functional is 1-Lipschitz for the uniform norm on its terminal. Thus the limits are taken in the order $\lambda \downarrow 0$, $\delta \downarrow 0$, and $L \rightarrow \infty$. Equation (3.19), including its perturbed form, survives them.

The derivative functional also converges uniformly in the profile. Here are the necessary uniform-integrability details. The derivatives of $u_L + d\psi$ agree with that of u_d outside $(-\infty, -L)$ and their absolute values there are bounded by a constant times $1 + |z|$. Equation (3.20), applied to higher negative-tail moments, bounds their tilted moments of every fixed order uniformly. Let ν_L denote a row measure tilted by $u_L + d\psi$ and let ν be the corresponding uncapped measure. Both measures are defined on the same marked base space: the cascade leaf, its full Gaussian field, and an independent residual Gaussian integrated inside the terminal weight, when present. Include the common refined rank genealogy, inserting zero-variance levels as in Section 2, or its equivalent auxiliary rank variables. In a two-replica expectation the continuous rank is then a common measurable observable even when both replicas select the same leaf of the original finite cascade. Thus comparison of the full product measures also controls every bounded rank test. Their unnormalized density ratio is $e^{-\beta V - u_L} \leq 1$ and equals one on $[-L, \infty)$. If $\varepsilon_L = \nu_L((-\infty, -L))$, normalization and the triangle inequality give

$$\|\nu - \nu_L\|_{\text{TV},1} \leq 4\varepsilon_L,$$

where $\|\cdot\|_{\text{TV},1}$ is total variation without the factor $1/2$. The averaged right side tends to zero uniformly by (3.20). The same is true for two replicas. Test the product of the two terminal derivatives against any bounded function of rank, first truncating the derivatives and then using their uniform higher moments. The two-replica formula for D_q therefore gives uniform $L^1(ds)$ convergence of the capped derivative profiles to $D_q^{u_d}$. Mollification is handled similarly at each fixed cap. In particular $D_q^{u_d}$ is continuous in q in L^1 , by the corresponding capped assertion and this uniform approximation. Moreover its L^∞ norm is uniformly bounded: the derivative martingale and the terminal second-moment bound give $D_q^{u_d}(s) \leq \mathbb{E}_{\text{tilted}}(u_d')^2 \leq C_{\beta,\psi}$. Thus the integrated derivative formula (2.6) and its one-sided derivative interpretation extend to u_d . $\square$

For $u = -\beta V$, write $p_q = \alpha D_q$, $\alpha > 0$. Adding the two tangent inequalities furnished by the unperturbed version of (3.19), and using the gradient $-D_q/2$, proves

$$\int_0^1 (r - q)(p_r - p_q) ds \geq c_{\alpha,\beta} \|r - q\|_2^2, \quad c_{\alpha,\beta} = 2\alpha c_\beta > 0. \quad (3.22)$$

All these assertions hold at the fixed diagonal $Q = 1$ used below; the uniform neighborhood $Q \in [1/2, 2]$ is available when varying a diagonal in a cavity calculation.

Corollary 3.7 (Convex perturbations of the loss). *Let*

$$W(z) = cV(z) + \chi(z), \quad c > 0, \quad \chi \in C_c^\infty((-\infty, -1)),$$

and suppose that $W \geq 0$, $W' \leq 0$, and $W'' \geq 0$ distributionally. For every fixed $\beta > 0$, the terminal $u = -\beta W$ has a concave row functional on all profiles with fixed diagonal $Q \in [1/2, 2]$. The derivative formula (2.6), continuity of D_q in L^1 , and the uniform tangent-cap limits hold for this terminal. In particular, the Gaussian interpolation upper bound derived for smooth concave caps applies to $-\beta W$ after removal of the cap.

Proof. The terminal is concave and increasing. Its derivative is Lipschitz, has at most linear growth, and vanishes on $[-1, \infty)$. Choose L so large that $-L < \inf \text{supp } \chi$, with the condition omitted when $\chi = 0$, and extend u tangentially below $-L$. The resulting terminal u_L is concave, increasing, and globally Lipschitz. Proposition 3.3 applies after mollification and its removal. Because the cutoff lies below the entire support of χ , its error is exactly

$$u_L(z) - u(z) = \frac{\beta c}{2}(z + L)_-^2.$$

Every intermediate terminal is increasing, so (3.20) bounds the difference of the row functionals by $\frac{\beta c}{2}\mathbb{E}(\sqrt{Q}G + L)_-^2$, uniformly in the profile. The same comparison applies to a row insertion in the microscopic Gaussian model. The derivatives agree outside the cutoff region and are bounded by $C_{\beta, W}(1 + |z|)$ within it. Higher Gaussian negative-tail moments and the marked-space total-variation comparison above prove uniform convergence of D_q and its differential formula. This proves all assertions, including passage of the interpolation bound to the uncapped loss. $\square$

If $\psi \in C_c^\infty((-\infty, 0))$ has support in $[-R, -a]$, $a > 0$, then $W_w(z) = V(z) + w\psi(1 + z)$ satisfies its assumptions for both signs of all sufficiently small *fixed* w . For example it suffices to require

$$|w||\psi''|_\infty \leq \frac{1}{2}, \quad |w||\psi'|_\infty \leq \frac{a}{2}, \quad |w||\psi|_\infty \leq \frac{a^2}{4}.$$

On the perturbation support these inequalities give $W_w'' \geq 1/2$, $W_w' \leq -a/2$, and $W_w \geq a^2/4$; outside it the loss equals V . Hence the concavity assertion applies to $-\beta W_w$ at every β , without requiring $|\beta w| \leq d_0$. The scaled loss $(1 + w)V$, $w > -1$, is covered by $\chi = 0$, $c = 1 + w$.

4 The Gaussian pressure

For each row terminal u used below, define the finite-system Gaussian pressure by

$$P_N(u) = \frac{1}{N} \mathbb{E} \log \int_{\mathbb{S}_N} \exp \left\{ \sum_{\mu=1}^M u(g_\mu(x)) \right\} \sigma_N(dx), \quad M = \lfloor \alpha N \rfloor.$$

Thus $P_N(-\beta V)$ is the pressure of the model at inverse temperature β .

We prove, for $\alpha, \beta > 0$, the Gaussian pressure formula

$$\lim_{N \rightarrow \infty} P_N(-\beta V) = \min_{q_* < 1} \{ \alpha \Phi_{-\beta V}(q) + E(q) \}.$$

We first prove matching bounds for smooth caps of the row potential. The upper bound inserts one independent row into a limiting overlap array and uses concavity of its row functional. The lower bound adds a fixed number of spin coordinates. Their Gaussian fields have an empirical covariance built from row derivatives; deleting and reinserting two rows identifies that covariance and its fluctuations. These two arguments use the same cascade representation, established below from a vanishing perturbation and the Ghirlanda–Guerra identities. After removing the cap, the spin inverse relation turns the common row–spin expression into the displayed entropy formula.

Throughout this section a profile has diagonal 1, unless another diagonal is explicitly specified. We write $\langle \cdot \rangle$ for the Gibbs expectation of the system currently under consideration, and include all its randomness in $\mathbb{E}$. The perturbation coefficients introduced below are held fixed inside $\mathbb{E}$; their product-uniform average is denoted by $\mathbb{E}_c$.

We first work with a smooth concave tangent cap u having bounded first three derivatives and at most linear growth. The concentration and cavity arguments need only these regularity bounds. The interpolation upper bound additionally uses profile concavity from Section 3; it therefore also applies to the small terminal perturbations constructed there.

4.1 Perturbations and overlap arrays

The perturbation will identify every subsequential overlap law needed in the bounds, without requiring convergence of the unperturbed overlap law. An overlap array is viewed as an element of the compact product space of its entries. Convergence of its law means convergence of every finite subarray. For spins $x^\ell \in \sqrt{N}\mathbb{S}^{N-1}$ put $R_{\ell\ell'} = N^{-1}x^\ell \cdot x^{\ell'}$. When a finite trial cascade is present, let $Q_{\ell\ell'}$ be its covariance overlap, including $Q_{\ell\ell} = 1$.

Lemma 4.1 (A vanishing perturbation). *Fix $\rho \in (1/4, 1/2)$ and put $s_N = N^\rho$. There is an independent centered Gaussian perturbation H_N^{pert} with the following properties.*

- (i) *Its addition changes the expected pressure by at most Cs_N^2/N .*
- (ii) *For the sphere-only system, every subsequential overlap-array limit obtained at almost every perturbation parameter satisfies the Ghirlanda–Guerra identities. With a finite trial cascade, the corresponding identities hold for every continuous function of the pair (R, Q) :*

$$\mathbb{E}\langle f\psi(R_{1,n+1}, Q_{1,n+1}) \rangle = \frac{1}{n}\mathbb{E}\langle f \rangle \mathbb{E}\langle \psi(R_{12}, Q_{12}) \rangle + \frac{1}{n} \sum_{\ell=2}^n \mathbb{E}\langle f\psi(R_{1\ell}, Q_{1\ell}) \rangle. \quad (4.1)$$

Here $n \geq 2$ and f is a bounded measurable function of the first n joint overlaps. For the sphere-only system, omit Q .

- (iii) *The extraction can be made simultaneously for any prescribed countable collection of systems with finitely many rows removed, and for almost every time in a bounded interpolation parameter interval.*

In (ii)–(iii) the precise extraction convention is this: from any dimension subsequence one can first choose a further subsequence on which all the identity errors vanish for almost every coefficient vector, and then, for each such vector, take any further convergent array subsequence. No assertion is made that averaging limiting arrays over coefficient vectors preserves these identities.

Proof. Enumerate pairs (j, λ) , where $j \geq 1$ and λ belongs to a countable dense subset of $[0, \infty)$ containing 0. Let $h_{j,\lambda}$ be independent unit-variance Gaussian fields with covariance

$$K_{j,\lambda}((x, \tau), (x', \tau')) = \left(\frac{R(x, x') + \lambda Q(\tau, \tau')}{1 + \lambda} \right)^j.$$

These are positive semidefinite kernels: $R + \lambda Q$ is a Gram kernel and integer powers preserve positive semidefiniteness by the Schur product identity. The binomial expansion separates powers of R and Q . Each power of Q has nonnegative increments along the cascade, so

the fields can be realized using independent child fields conditional on their ancestor fields. Choose positive weights $w_{j,\lambda}$ with $\sum w_{j,\lambda}^2 j < \infty$, and set

$$H_N^{\text{pert}} = s_N \sum_{j,\lambda} w_{j,\lambda} c_{j,\lambda} h_{j,\lambda}, \quad c_{j,\lambda} \in [1, 2].$$

In the sphere-only construction use just covariance kernels R^j . The Gaussian covariance metric is continuous on each sphere, and the fields are taken in their separable versions. Gaussian interpolation bounds the pressure change by Cs_N^2/N .

Here is the concentration estimate needed for the perturbation argument:

$$\mathbb{E}|\log Z - \mathbb{E}\log Z| \leq C(\sqrt{N} + s_N). \quad (4.2)$$

Without a cascade, Gaussian concentration gives this directly. Indeed the gradient of $\sum_{\mu \leq M} u(g_\mu(x))$ with respect to the row entries has squared norm at most $M\|u'\|_\infty^2$, uniformly on the sphere. Any linear Gaussian spin terms have squared reproducing-kernel norm $O(N)$, and the perturbation has squared norm $O(s_N^2)$.

For a fixed finite cascade apply Lemma 2.1 to the spin-integrated leaf partition marks. Conditional on shared root Gaussian fields, the log partition is a recursive root shift plus the difference of two log cascade masses. The latter have uniformly bounded second moments, with a constant depending on the fixed cascade. Recursive logarithmic moments preserve a uniform Lipschitz constant: if $|F(g, z) - F(g', z)| \leq L\|g - g'\|$ for every child mark z , the same bound holds after applying $m^{-1} \log \mathbb{E}_z e^{mF}$. Thus the root shift has Lipschitz constant $C(\sqrt{N} + s_N)$ in its shared Gaussian fields. All child moments needed here are finite, by bounded slope and Gaussian moments; at fixed dimension one can also bound them by the supremum of the corresponding Gaussian sphere process. This proves (4.2), uniformly in interpolation time and after deletion of a fixed number of rows.

For completeness we derive the identities rather than invoke a separate perturbation theorem. Fix one field h with coefficient swc , where $s = s_N$ and $w > 0$ is fixed, and put $F(c) = \log Z(c)$. Then

$$F'(c) = sw\langle h \rangle, \quad F''(c) = (sw)^2 \langle (h - \langle h \rangle)^2 \rangle.$$

Gaussian integration by parts gives

$$\mathbb{E}\langle h \rangle = swc \mathbb{E}\langle K_{11} - K_{12} \rangle.$$

Consequently integration over $c \in [1, 2]$ bounds the mean thermal variance of h by a constant. The resulting integrated thermal absolute deviation is $O(1) = o(s)$.

For the disorder deviation, the convex difference-quotient inequalities for F and $\mathbb{E}F$, at distance $d \in (0, 1/4)$, give

$$\int_1^2 \mathbb{E}|\langle h \rangle - \mathbb{E}\langle h \rangle| dc \leq C_w \left\{ \frac{\sqrt{N} + s}{sd} + sd \right\}.$$

To justify the second term, integrate the increments of the monotone function $(\mathbb{E}F)'$ over the enlarged interval $[1 - d, 2 + d]$; their total is at most $C_w ds^2$, before division by sw . The concentration bound applies on this enlarged interval as well. Taking $d = N^{1/4-\rho}$ makes the last display $o(s)$. In Gaussian integration by parts against a bounded replica test f , one may therefore replace $h(x^1, \tau^1)$ by its disorder mean at an error $o(s)$ after coefficient averaging. Division by swc gives a vanishing integrated absolute error in (4.1), first with $\psi = K$.

For each total degree j , the polynomials $(R + \lambda Q)^j$ at $j + 1$ distinct values of λ span the homogeneous polynomials of degree j in (R, Q) . Polynomial approximation gives all

continuous ψ and all continuous finite-array tests; a monotone-class argument gives measurable f . Choose countable determining families. The sum of their integrated errors, with summable weights, tends to zero. A subsequence with summable weighted errors has pointwise vanishing errors for almost every coefficient vector. Including interpolation time and a countable list of deleted-row systems in this extraction proves (iii). Compactness of the array space then gives the asserted limiting identities. $\square$

The only general ultrametricity theorem used here is the following array formulation of Panchenko's theorem [20, Theorem 1]. A weakly exchangeable positive semidefinite random array with fixed diagonal, whose off-diagonal entries satisfy the Ghirlanda–Guerra identities for all bounded measurable pair tests, is almost surely ultrametric:

$$R_{23} \geq \min\{R_{12}, R_{13}\}.$$

The equivalent random-measure formulation concerns independent samples from a random probability measure on a separable Hilbert ball. Precisely, the Gram-array representation [19, Proposition 1] says that a weakly exchangeable positive semidefinite array with diagonal 1 has a realization

$$R_{\ell\ell'} = v^\ell \cdot v^{\ell'} + \mathbf{1}_{\{\ell=\ell'\}}(1 - \|v^\ell\|^2),$$

where, conditional on a random probability measure on the unit Hilbert ball, the vectors v^ℓ are independent samples from that measure. To account for the latent diagonals, let $q^\dagger$ be the supremum of the one-overlap support. The directing measure is almost surely supported on $\|v\|^2 = q^\dagger$; this is also [18, Theorem 2(a)]. Here is the short argument needed for this fact. If an overlap set B has probability $c > 0$, GG implies that the probability of avoiding B in all n overlaps with replica 1 equals $\prod_{j=1}^n (1 - c/j)$, which tends to zero. Conditional independent sampling then shows that almost every directing point has positive mass of partners with overlap in B . Apply this to $B = (t, q^\dagger]$, for every rational $t < q^\dagger$. Since two directing points almost surely have inner product at most $q^\dagger$, continuity and separability imply $\|v\|^2 \leq q^\dagger$ throughout the support: a sufficiently small ball around a counterexample would give larger inner products with positive probability. Cauchy–Schwarz now excludes $\|v\|^2 < q^\dagger$, since such a point could not have partners with overlaps arbitrarily close to $q^\dagger$. Thus the latent diagonals are deterministic, and the random-measure GG identities follow from the array identities. Panchenko's ultrametricity theorem applies. Adding a constant Gram kernel and rescaling allows this reasoning for any bounded fixed diagonal before positivity is known.

Lemma 4.2 (Synchronization and identification). *Suppose the joint array (R, Q) is a subsequential limit from Lemma 4.1. Then $R \geq 0$ off the diagonal. There are nonnegative nondecreasing profiles r, q , on a common canonical rank array $(U_{\ell\ell'})$, such that*

$$(R_{\ell\ell'}, Q_{\ell\ell'}) \stackrel{\text{law}}{=} (r(U_{\ell\ell'}), q(U_{\ell\ell'})), \quad \ell \neq \ell'.$$

If the finite trial cascade has prescribed profile q , this is that same profile. In the sphere-only case the array is the RPC array determined by its one-overlap marginal.

Proof. Apply ultrametricity separately to R , Q , and $R + Q$, using (4.1). First, if $\mathbb{P}(R_{12} \leq a) = m > 0$ for some $a < 0$, ultrametricity makes $R_{ij} > a$ an equivalence relation on distinct replicas. On the event that n sampled replicas occupy distinct classes, GG gives probability $(1 - m)/n$ for the next replica to join any specified class. These joining events are disjoint, so the probability of avoiding them all is m . Induction gives positive probability of arbitrarily many pairwise overlaps at most a . Positive semidefiniteness forbids this, since on that event the sum of the entries of the $n \times n$ Gram matrix is at most $n + n(n - 1)a < 0$ for sufficiently large n .

Let μ be the law of (R_{12}, Q_{12}) . If two points in its support were strictly incomparable coordinatewise, choose disjoint neighborhoods A, B with $R_A < R_B$ and $Q_A > Q_B$. GG with two old replicas gives

$$\mathbb{P}((R_{12}, Q_{12}) \in A, (R_{13}, Q_{13}) \in B) = \frac{1}{2}\mu(A)\mu(B) > 0.$$

On this event scalar ultrametricity forces $R_{23} = R_{12}$ and $Q_{23} = Q_{13}$. Thus $R_{23} + Q_{23}$ is strictly smaller than both $R_{12} + Q_{12}$ and $R_{13} + Q_{13}$, contradicting ultrametricity of the sum. The support of μ is therefore a chain. Each coordinate is a nondecreasing function of the sum on that support, and the pair has comonotone quantiles.

It remains to identify the whole array, since a pair marginal alone would otherwise be insufficient. At any threshold with lower mass m , GG says, conditional on the old array, that the new replica joins an existing ultrametric class containing l old replicas with probability $(l - m)/n$. For finitely many thresholds, the classes form a nested partition. These joining probabilities, and their differences for a parent and its children, determine the distribution of the new branch. Induction in the number of replicas determines the law of the discretized array. By Lemma 2.2 it is exactly the finite RPC sampling law. Refining thresholds at continuity points identifies the full scalar array. Apply this to the sum and then to its two coordinate functions to obtain the common rank representation.

Finally, at finite N the cascade genealogy retains its prescribed law after the spin-integrated tilt: all Gaussian fields in the model, including the perturbation, have conditional child independence. This is precisely Lemma 2.1. Hence the limiting second marginal is the prescribed trial profile. $\square$

Lemma 4.3 (Deleting and reinserting finitely many rows). *Consider any convergent deleted-row overlap-array subsequence satisfying the preceding lemmas. Reinsert finitely many independent Gaussian rows, with potentials of bounded slope and with covariances given by continuous functions of the available overlap kernels. Suppose those covariances are represented by nondecreasing profiles on the common rank array. Then:*

- (i) *the entire common-rank genealogy retains its law after insertion;*
- (ii) *conditional on this genealogy, separate inserted row-mark families are independent and have their individually tilted RPC laws;*
- (iii) *for a row of covariance profile C , the conditional mean of the product of two terminal derivatives is $D_C(U_{12})$, with diagonal mean $D_{C,d}$.*

The conclusions include convergence of tilted polynomial-growth row observables whenever their requisite moments are uniformly integrable. In particular any convergent rowful array and any further limit with a fixed finite set of its rows removed have the same profile.

Proof. Conditionally on the deleted-row Gibbs measure, the new row values sampled on finitely many replicas are centered Gaussian vectors with the displayed covariance arrays. They therefore converge jointly with the base arrays. For every fixed $a < \infty$, their bounded-slope potentials obey $\mathbb{E}\langle e^{a|u(g)|} \rangle \leq C_a$ before insertion. The replica truncation argument of Lemma 2.3 consequently passes both the finite-row tilt and its observables to the limit. For a finite RPC, claims (i)–(iii) are the marking transformation and additive-mark factorization of Lemma 2.1, together with the derivative-martingale definition of D_C . In particular, the marking transformation makes the transformed weight tree independent of the separate row-mark families. Sampling replicas from those weights therefore preserves their conditional factorization given the genealogy. This is the depth-two calculation in the proof of Lemma 2.1,

applied to the sum of the inserted-row potentials; it uses no independence between a stable total and its normalized jumps.

For the passage to general profiles, retain the canonical rank array itself in the replica convergence, even on plateaus of the covariance profile. The following conditional-kernel version of the truncation argument makes the factorization passage precise. For a bounded continuous observable F of the row marks at the first n replicas, its conditional expectation given their ranks is approximated in L^1 of the fixed rank law by the finite-replica polynomial approximants to the truncated tilt numerator and denominator. The truncation error is uniform over bounded profiles, by the exponential moment bounds. At finite profile, invariance of the tilted rank marginal lets us interpret these approximants as conditional kernels relative to the same canonical rank law. A polynomial approximant uses a fixed finite number of additional replicas. Integrating its Gaussian marks gives a bounded function K_j of all the ranks of those replicas. Under $C_j \rightarrow C$ in L^1 , the covariance entries converge almost surely along a subsequence; Gaussian covariance continuity and dominated convergence give $K_j \rightarrow K$ in L^1 of the fixed rank law. Averaging the additional ranks conditional on the first n ranks preserves this convergence, since

$$\|\mathbb{E}[K_j - K \mid (U_{ab})_{a,b \leq n}]\|_1 \leq \|K_j - K\|_1.$$

The conditional law of those additional ranks is independent of the covariance profile. Thus the exact bounded conditional kernels converge in L^1 . Products of such kernels also converge in L^1 . The finite-profile factorization identities against any bounded cylinder test of the ranks therefore pass to the limit. A monotone-class argument extends them to the entire countable genealogy. This proves (i)–(iii) without presuming that conditional independence is closed under weak convergence.

In particular, for two inserted rows A, B with profile C_j and for any bounded cylinder test H of the full rank array, the identity needed below is

$$\mathbb{E}\langle H a_A^1 a_A^2 a_B^1 a_B^2 \rangle = \mathbb{E}[H D_{C_j}(U_{12})^2].$$

Its limit also follows directly from $D_{C_j} \rightarrow D_C$ in L^1 and bounded slope. Single-replica analogues have constant right-hand factors. Polynomial observables follow by truncation and the same exponential moment bounds. The last assertion follows by first taking a joint subsequence of the rowful and deleted-row laws, then representing the former as the tilt of the latter. Claim (i) identifies their complete overlap-array laws. $\square$

4.2 The interpolation upper bound

The comparison follows Guerra's interpolation scheme [13, §4]: we connect the interacting system to a hierarchical trial system and control the sign of the overlap remainder. Here that sign follows from the row-profile concavity proved in Section 3. The synchronized cavity limits used for this nonlinear spherical model are proved here; the SK bound itself is not an input for the present model.

Fix a finite trial profile q reaching diagonal 1, and put $p = p_q = \alpha D_q$, with diagonal $p_d = \alpha D_{q,d}$. Since D_q is constant on plateaus of q , the off-diagonal field p is a measurable function of the trial overlap Q itself. Take independent fields $z_\mu(\tau)$, $z_i^p(\tau)$, and $Y(\tau)$ of covariance profiles q , p , and qp , respectively. On the product of sphere and cascade interpolate with exponent

$$\sum_{\mu \leq M} u(\sqrt{t} g_\mu(x) + \sqrt{1-t} z_\mu(\tau)) + \sqrt{1-t} \sum_{i=1}^N x_i z_i^p(\tau) + \sqrt{tN} Y(\tau) + H_N^{\text{pert}}. \quad (4.3)$$

Write $\varphi_N(t)$ for its normalized expected log partition, including coefficient averaging. For an individual row put $a_\mu = u'(\sqrt{t}g_\mu + \sqrt{1-t}z_\mu)$. Gaussian covariance differentiation cancels all diagonal terms and gives

$$\varphi'_N(t) = -\frac{M}{2N}\mathbb{E}_c\mathbb{E}\langle(R_{12}-Q_{12})a_\mu^1a_\mu^2\rangle + \frac{1}{2}\mathbb{E}_c\mathbb{E}\langle(R_{12}-Q_{12})p(U_{12}^\tau)\rangle. \quad (4.4)$$

Here row exchangeability permits any fixed μ , and U^τ denotes the trial cascade rank. The last term is the combined derivative of the linear spin and compensation fields: their covariance derivative is $N(Q-R)p$, and their diagonal derivatives cancel.

To control a subsequential limit at almost every t and coefficient vector, remove row μ . Lemmas 4.1 and 4.2 give comonotone cavity profiles (r, q) . The independent removed row has covariance profile $C = tr + (1-t)q$ and diagonal 1. Lemma 4.3 identifies the derivative limit as

$$-\frac{\alpha}{2}\int_0^1(r-q)D_C ds + \frac{1}{2}\int_0^1(r-q)p ds.$$

Concavity of Φ_u and (2.6) imply, for $t > 0$,

$$\int_0^1(r-q)(D_C - D_q) ds \geq 0.$$

The limiting derivative is therefore nonpositive. For fixed bounded-slope u , (4.4) is uniformly bounded. Starting from any subsequence realizing the upper limit of the integrated derivative, use the almost-everywhere extraction in Lemma 4.1, and at each such parameter-time pair use compactness to examine every further array limit. Every resulting derivative limit is nonpositive. Reverse Fatou gives $\limsup_N \int_0^1 \varphi'_N(t) dt \leq 0$.

After removing the negligible perturbation, the endpoint $t = 0$ separates into $M\Phi_u(q) + NS_N(p)$ by additive-mark factorization. At $t = 1$ the physical sphere partition separates from the cascade compensation, whose expected logarithm is

$$\frac{N}{2}\int_0^1 s d(qp)(s) = \frac{N}{2}\left(p_d - \int_0^1 qp ds\right).$$

Thus, with $\mathcal{S} = \lim_N(S_N - p_d/2)$,

$$\limsup_{N \rightarrow \infty} P_N(u) \leq \mathcal{P}_u(q) := \alpha\Phi_u(q) + \mathcal{S}(p_q) + \frac{1}{2}\int_0^1 qp_q ds. \quad (4.5)$$

Profile continuity extends this to every admissible q . The argument uses profile concavity rather than concavity of the terminal itself, so it also proves (4.5) for the small terminal perturbations of Section 3.

4.3 Identification of the empirical cavity fields

Adding a spin coordinate couples its fresh Gaussian column to the vector of row derivatives. Its conditional covariance between two replicas is therefore a row average, rather than their original spin overlap. The quadratic expansion also produces a scalar shift: the variance of the new columns gives a term u'' , while normalization of the old projections gives $-g_\mu u'$. We identify both quantities before comparing dimensions. In a sphere-only system write $a_\mu(x) = u'(g_\mu(x))$ and

$$p_{\ell\ell'}^N = \frac{1}{N} \sum_{\mu \leq M_N} a_\mu(x^\ell) a_\mu(x^{\ell'}), \quad b_N(x) = \frac{1}{N} \sum_{\mu \leq M_N} [u''(g_\mu(x)) - g_\mu(x) a_\mu(x)].$$

The two-row deletion argument below identifies the first quantity as a function of the limiting spin overlap and proves that the second has a deterministic limit under the joint disorder and replica law.

Lemma 4.4 (Empirical cavity coefficients). *Take a subsequence on which the base overlap array converges to profile q and the GG errors for the base, one-row-deleted, and two-row-deleted systems vanish. In every joint array limit,*

$$\begin{aligned} p_{\ell\ell'} &= \alpha D_q(U_{\ell\ell'}), & \ell \neq \ell', \\ p_d &= \alpha D_{q,d}, & b_N(x^\ell) \longrightarrow b = -p_d + \int_0^1 q p_q \, ds. \end{aligned} \quad (4.6)$$

In particular, $\mathbb{E}(|b_N(x^\ell) - b|^2) \rightarrow 0$ along this subsequence.

Proof. For a continuous function F of one overlap, remove a specified row and reinsert it using Lemma 4.3. Row exchangeability gives

$$\lim_N \mathbb{E} \langle p_{12}^N F(R_{12}) \rangle = \alpha \int_0^1 D_q(s) F(q(s)) \, ds.$$

On expanding $(p_{12}^N)^2$, the terms with equal row indices have total size $O(N^{-1})$, since u' is bounded. For the distinct indices, delete both rows and use conditional factorization of their tilted mark families. Hence

$$\lim_N \mathbb{E} \langle (p_{12}^N)^2 \rangle = \alpha^2 \int_0^1 D_q(s)^2 \, ds.$$

The function D_q is constant almost everywhere on each plateau of q . Thus there is a bounded Borel function d_q such that $D_q(s) = d_q(q(s))$ almost everywhere. Approximate d_q in L^2 of the overlap marginal by continuous functions. The preceding two displays imply that any joint limit p_{12} satisfies $\mathbb{E}(p_{12} - \alpha d_q(R_{12}))^2 = 0$. Exchangeability gives all off-diagonal entries simultaneously.

The same one-row and two-row calculation for the single-replica mark $a_\mu(x)^2$ identifies the diagonal limit. With only one sampled replica there is no variable pair genealogy, so its conditional mean is the scalar $D_{q,d}$. Apply the calculation also to $u''(g_\mu) - g_\mu u'(g_\mu)$. This observable has at most linear growth. Its moments after inserting one or two rows are uniformly bounded by reciprocal Jensen, Gaussian exponential moments, and bounded slope; this justifies truncation in the preceding argument and gives a deterministic L^2 limit for b_N .

To compute that limit, Gaussian integration by parts in a single row gives

$$\mathbb{E} \langle g_\mu u'(g_\mu) \rangle = \mathbb{E} \langle u''(g_\mu) + u'(g_\mu)^2 \rangle - \mathbb{E} \langle R_{12} u'(g_\mu^1) u'(g_\mu^2) \rangle.$$

Multiplication by M_N/N and the already identified row limits give exactly $b = -p_d + \int q p_q$. All deletions and limits take place at a fixed admissible coefficient vector; there is no mixture-of-profiles step. $\square$

4.4 The dimension-increment lower bound

We compare systems of dimensions N and $N + m$, using the block-cavity organization of Aizenman–Sims–Starr [1, Theorem 1(ii), equations (27)–(33)] and its spherical implementation by Chen [8, Theorem 2.1]. For this row Hamiltonian, the preceding lemma supplies the empirical covariance identification; the expansion below supplies the corresponding insertion and normalization terms. The spherical-shell comparison, nonlinear and fresh-row estimates, and telescoping argument without superadditivity are supplied locally below. Throughout the cavity expansion, m and the shell width δ are fixed while $N \rightarrow \infty$. The preceding lemma will identify the limiting insertion of these m coordinates. Only after that limit do we let $m \rightarrow \infty$ to recover the spherical functional, and finally let $\delta \downarrow 0$.

Sphere coordinates and the cavity expansion. Let A_N be the expected log partition including the sphere-only perturbation and the coefficient average. Fix a positive integer m , put $k = M_{N+m} - M_N = \alpha m + O(1)$, and fix $\delta \in (0, 1)$. Use coordinates

$$X = \left(\sqrt{\frac{N + m - \|\varepsilon\|^2}{N}} x, \varepsilon \right), \quad x \in \sqrt{N} \mathbb{S}^{N-1}, \quad \varepsilon \in \mathbb{R}^m,$$

and restrict ε to $\mathcal{A}_{m,\delta} = \{m(1-\delta) \leq \|\varepsilon\|^2 \leq m(1+\delta)\}$. Write $G_m \sim \mathcal{N}(0, I_m)$ for a standard Gaussian vector. At fixed m, δ the marginal surface density of ε converges relatively uniformly on this compact shell to standard Gaussian density. The angular law of x is independent uniform surface measure.

The $N + m$ perturbation can be replaced here by the same N -spin perturbation used in A_N . For any two shell configurations their full overlaps differ from the old overlaps by $O_m(N^{-1})$. Since $|r^j - r'^j| \leq j|r - r'|$ on $[-1, 1]$, covariance interpolation and $\sum w_j^2 j < \infty$ bound the change of expected log partition by $O_m(N^{2\rho-1}) = o(1)$. This includes the change from s_{N+m} to s_N .

Let $B_{\mu i}$ be the new independent Gaussian columns. For each old row, its new argument is

$$\sqrt{1 - \frac{\|\varepsilon\|^2}{N + m}} g_\mu(x) + \frac{B_\mu \cdot \varepsilon}{\sqrt{N + m}}.$$

Taylor expansion of its potential around $g_\mu(x)$ yields, after summing rows,

$$\sum_{i=1}^m \varepsilon_i z_i(x) + \frac{1}{2} \|\varepsilon\|^2 b_N(x) + T(x, \varepsilon) + o(1), \quad z_i(x) = \frac{1}{\sqrt{N}} \sum_{\mu \leq M_N} a_\mu(x) B_{\mu i}, \quad (4.7)$$

where

$$T(x, \varepsilon) = \frac{1}{2N} \sum_{\mu \leq M_N} u''(g_\mu(x)) [(B_\mu \cdot \varepsilon)^2 - \|\varepsilon\|^2].$$

Writing $\text{Rem}_N(x, \varepsilon)$ for the $o(1)$ term in (4.7), we have, for fixed m, δ ,

$$\mathbb{E} \sup_{\substack{x \in \mathbb{S}_N \\ \varepsilon \in \mathcal{A}_{m,\delta}}} |\text{Rem}_N(x, \varepsilon)| \rightarrow 0.$$

To check this directly, use $\sum_\mu g_\mu(x)^2 \leq \|A\|_{\text{op}}^2$ and $\mathbb{E} \|A\|_{\text{op}}^j = O(N^{j/2})$, together with $\mathbb{E} \sum_\mu \|B_\mu\|^j = O_m(N)$ for $j \leq 3$. For the cubic remainder, write the argument increment as $d_\mu = c_N g_\mu + B_\mu \cdot \varepsilon / \sqrt{N + m}$, where $c_N = -\|\varepsilon\|^2 / (2N) + O_m(N^{-2})$. Then

$$\sum_\mu |d_\mu|^3 \leq C_m \left(N^{-3} \|A\|_{\text{op}}^3 + N^{-3/2} \sum_\mu \|B_\mu\|^3 \right).$$

This uses $|a + b|^3 \leq 4(|a|^3 + |b|^3)$, so no mixed fourth moment is needed. The quadratic mixed term is at most

$$C_m N^{-3/2} \|A\|_{\text{op}} \left(\sum_\mu \|B_\mu\|^2 \right)^{1/2},$$

whose mean is $O_m(N^{-1/2})$. The shrinkage-square term is bounded by $C_m N^{-2} \|A\|_{\text{op}}^2 = O_{L^1, m}(N^{-1})$, and denominator replacements use only the first two displayed column moments. These estimates prove the claimed uniform remainder. The operator-norm estimate follows from finite bilinear sphere nets and Gaussian tail bounds. The k fresh rows can similarly be replaced by independent old-dimensional fields $g_a^{\text{new}}(x)$: their new-column contribution is $O_m(N^{-1/2})$ and their shrinkage error is bounded by $O_m(N^{-1})$ times a row norm.

The centered quadratic term. For the lower bound we may discard T at expected cost $o(1)$, although its supremum need not be small. Here is the required estimate. Condition the Gaussian shell measure to be a probability, and let $\langle \cdot \rangle_0$ include it and the old Gibbs measure. Put

$$W(x, \varepsilon) = \varepsilon \cdot z(x) + \frac{1}{2} \|\varepsilon\|^2 b_N(x) + \sum_{a=1}^k u(g_a^{\text{new}}(x)).$$

Jensen's inequality gives

$$\log \langle e^{W+T} \rangle_0 - \log \langle e^W \rangle_0 \geq -\langle |T| \rangle_W, \quad \mathbb{E} \langle |T| \rangle_W \leq \mathbb{E} \langle |T_1| e^{W_1 - W_2} \rangle_0.$$

The second inequality uses $\langle e^W \rangle_0^{-1} \leq \langle e^{-W} \rangle_0$. Conditional on the old disorder and (x, ε) , the centered Gaussian squares in T are independent across rows; hence $\mathbb{E} \langle T_1^2 \rangle_0 = O_m(N^{-1})$. Furthermore $\mathbb{E} \langle e^{a(W_1 - W_2)} \rangle_0$ is bounded for every fixed a, m, δ . Indeed the Gaussian fields z_i have covariance bounded by $(M_N/N) \|u'\|_\infty^2$, fresh row potentials have linear growth, and

$$|b_N(x)| \leq C \left(1 + \frac{\|A\|_{\text{op}}}{\sqrt{N}} \right).$$

Cauchy-Schwarz proves the desired $o(1)$ bound. Here is the lower bound needed later for coefficient averaging. For fixed coefficients put $F_N(c) = \mathbb{E} \log Z_{N+m}(c) - \mathbb{E} \log Z_N(c)$. All the preceding sphere-density, perturbation, Taylor, and fresh-row errors are $o(1)$ uniformly in c , with m, δ fixed. Before tilting, $\mathbb{E} \langle z_i \rangle_0 = 0$ and $\mathbb{E} \langle T \rangle_0 = 0$, while the fresh-row potential means are finite constants and $\mathbb{E} \langle b_N \rangle_0 \geq -C$. Jensen therefore gives

$$F_N(c) \geq \log \mathbb{P}(G_m \in \mathcal{A}_{m,\delta}) - C_{m,\delta} - o(1),$$

uniformly in c . This is a bound on the actual dimension increment, not only on its limiting RPC expression.

The limiting insertion. At fixed m , the row count k has at most two possible values; pass first to a dimension subsequence on which it is constant. Now fix an admissible coefficient vector and extract a base-array limit. Conditionally on the old disorder, the z_i are independent centered Gaussian fields with covariance array p^N . Lemma 4.4 identifies that array and b_N , while the fresh rows have covariance array R . To handle the unbounded retained observable b_N , set $K_N = C(1 + \|A\|_{\text{op}}/\sqrt{N})$. Clipping b_N to $[-L, L]$ changes W uniformly in (x, ε) by at most $C_m K_N \mathbf{1}_{\{K_N > L\}}$. Thus the expected log integral changes by at most $C_m \mathbb{E}[K_N \mathbf{1}_{\{K_N > L\}}]$, which tends to zero uniformly in N and c . Apply Lemma 2.3 with this bounded clipping, then let $L \rightarrow \infty$. Together with the exponential bounds for the Gaussian terms, this passes the logarithmic increment to its RPC limit. The independent fresh-row contributions separate, giving $k\Phi_u(q)$. If p has a diagonal residual, its independent Gaussian convolution contributes precisely half that variance times $\|\varepsilon\|^2$. Thus the remaining integral is the radial Gaussian version of the spin functional with covariance $p = p_q$ and scalar shift b .

On the shell, write $\varepsilon = r\omega$, where $\omega \in \sqrt{m}S^{m-1}$ and $r^2 \in [1 - \delta, 1 + \delta]$. Let $Z_{\text{sph}}(r)$ be the cascade sum of the spherical exponential integrals at radius r , including any residual-diagonal factor, and let Z_{shell} be the corresponding Gaussian shell integral including the scalar shift $b\|\varepsilon\|^2/2$. For each fixed field realization, $Z_{\text{sph}}(r)$ increases with $r \geq 0$ by symmetry of ω and nonnegativity of the residual variance. Thus, with $r_- = \sqrt{1 - \delta}$,

$$Z_{\text{shell}} \geq \mathbb{P}(G_m \in \mathcal{A}_{m,\delta}) \exp\{mb/2 - |b|m\delta/2\} Z_{\text{sph}}(r_-).$$

The expected logarithm of the last factor is $mS_m((1 - \delta)p; (1 - \delta)p_d)$. Covariance interpolation changes it to $mS_m(p; p_d)$ at cost at most $C\delta m$, uniformly over bounded profiles p . Adding

the fresh-row contribution $k\Phi_u(q)$ and using $|k - \alpha m| \leq 1$ and the uniform bound on $|\Phi_u(q)|$ gives

$$\liminf_{N \rightarrow \infty} (A_{N+m} - A_N) \geq \inf_q \{ \alpha m \Phi_u(q) + m S_m(p_q) + m b(q)/2 \} - C(1 + \delta m) + \log \mathbb{P}(G_m \in \mathcal{A}_{m,\delta}), \quad (4.8)$$

with C independent of m, δ . The uniformity follows from boundedness of $p_q, b(q)$, and $\Phi_u(q)$. More explicitly, take a subsequence realizing the left lower limit; extract GG errors almost everywhere in the coefficients for the base and both row deletions; then take pointwise array subsequences. Every resulting functional is bounded below by the displayed infimum. The uniform finite lower bound on increments permits Fatou in the coefficients. This establishes (4.8) without requiring a common profile for different coefficient vectors or different dimensions.

From increments to pressure. The insertion bound still contains the finite-dimensional spin functional S_m . Telescoping the dimension increments and then sending m to infinity will replace it by $\mathcal{S}$, giving the trial functional in the upper bound. To carry this out, for any real sequence (A_N) with the present linear bounds, telescoping separately in the m residue classes gives

$$\liminf_{N \rightarrow \infty} \frac{A_N}{N} \geq \liminf_{N \rightarrow \infty} \frac{A_{N+m} - A_N}{m}.$$

Indeed, for any number below the right side, every sufficiently late increment is at least m times that number, and summing proves the inequality. Substitute $b = -p_d + \int q p_q$, use the uniform convergence $S_m - p_d/2 \rightarrow \mathcal{S}$ of Proposition 2.5, and divide the error in (4.8) by m . For fixed δ , $\mathbb{P}(G_m \in \mathcal{A}_{m,\delta}) \rightarrow 1$ as $m \rightarrow \infty$. Sending first $m \rightarrow \infty$ and then $\delta \downarrow 0$ proves

$$\liminf_{N \rightarrow \infty} P_N(u) \geq \inf_q \mathcal{P}_u(q).$$

Together with (4.5), this proves the pressure formula for smooth concave caps.

4.5 Removing the cap and identifying the minimizing profile

The uniform tail comparison of Section 3 removes the cap. For clarity, the mechanism is row insertion: conditional on a cavity spin, a fresh Gaussian projection is standard normal; insertion with an increasing potential decreases expectations of decreasing nonnegative tail functions. The error between a tangent cap below $-L$ and $u = -\beta(-1 - z)_+^2/2$ is bounded by a Gaussian negative-tail quadratic expectation tending to zero with L . A bounded terminal perturbation only multiplies this bound by $e^{2|d|\|\psi\|_\infty}$. The same estimate applies to the RPC row functional and to its tilted derivative moments. Consequently Φ_u, D_q , and $\mathcal{P}_u$ have uniform cap limits on the compact space of monotone profiles, with its L^1 topology. We obtain

$$P(u) := \lim_{N \rightarrow \infty} P_N(u) = \min_q \mathcal{P}_u(q).$$

Continuity and profile compactness give the stated minimum.

Proposition 4.5 (Entropy form of the pressure). *For Gaussian rows, $u = -\beta V$, and $\alpha > 0$,*

$$P(u) = \min_{q: q_* < 1} \{ \alpha \Phi_u(q) + E(q) \}, \quad q_* := \text{ess sup } q. \quad (4.9)$$

Every minimizer of $\mathcal{P}_u$ has $q_ < 1$ and satisfies the spin inverse relations of Section 2 with $p = p_q$.*

Proof. Let q minimize $\mathcal{P}_u$. By Proposition 2.6, $\mathcal{S}$ has at p_q a supergradient $-r/2$, where r is a nonnegative nondecreasing profile bounded strictly below 1 and $p_q = \mathcal{I}(r)$ for the inverse map (2.14). Put $q_t = q + t(r - q)$ and $p_t = p_{q_t}$. The supporting inequality for the concave row functional and the supergradient inequality for $\mathcal{S}$ give

$$\begin{aligned}\alpha[\Phi_u(q_t) - \Phi_u(q)] &\leq -\frac{t}{2} \int (r - q)p_q, \\ \mathcal{S}(p_t) - \mathcal{S}(p_q) &\leq -\frac{1}{2} \int r(p_t - p_q).\end{aligned}$$

Adding the exact difference of the coupling terms yields

$$\mathcal{P}_u(q_t) - \mathcal{P}_u(q) \leq -\frac{1-t}{2} \int (r - q)(p_t - p_q).$$

By (3.22), for $0 < t < 1$ the integral on the right is at least $c_{\alpha,\beta}t\|r - q\|_2^2$. Minimality therefore forces $r = q$. In particular $q_* < 1$ and $p_q = \mathcal{I}(q)$, proving the asserted spin inverse relation for the original profile p_q . The equality of this spin supergradient with $-q/2$ means that p_q attains the entropy conjugate:

$$E(q) = \mathcal{S}(p_q) + \frac{1}{2} \int qp_q.$$

For any competing profile q' with $q'_* < 1$, the definition of E gives $\mathcal{P}_u(q') \leq \alpha\Phi_u(q') + E(q')$. Equality at the minimizing q proves (4.9). $\square$

One useful endpoint estimate follows directly from minimality. Let a_0 be the root first derivative before the initial Gaussian variance is added, and recall $\chi_q(0) = \int_0^1 (1 - q(s)) ds$. Then

$$\alpha a_0^2 \chi_q(0) \leq \frac{1}{\chi_q(0)} - 1. \quad (4.10)$$

Indeed, for $a \geq 0$ small, set $q^{(a)} = a + (1 - a)q$. Formula (2.21) gives the exact entropy change

$$E(q^{(a)}) - E(q) = \frac{1}{2} \log(1 - a) + \frac{a}{2(1 - a)\chi_q(0)}.$$

The row derivative from (2.6) is $-\alpha \int (1 - q)D_q/2 \leq -\alpha a_0^2 \chi_q(0)/2$, because the derivative martingale gives $D_q \geq a_0^2$. The nonnegative right derivative of the minimized objective is thus at most half the right side of (4.10) minus half its left side, proving the claim.

4.6 Touching supports for row observables

The pressure formula concerns the base quadratic loss. To identify row observables, we also need an upper bound for perturbed terminals that agrees with the base pressure at zero perturbation. No pressure formula for the perturbed model is required.

Fix $\psi \in C_c^\infty(\mathbb{R})$. For $|d|$ sufficiently small, Corollary 3.6 and its regularized proof give profile concavity for $u + d\psi$ and all its approximations. The interpolation upper bound applies at any base minimizer q . With $p = \alpha D_{q,u+d\psi}$, the entropy conjugate bounds $\mathcal{S}(p) + \int qp/2 \leq E(q)$. After removing the cap this gives the upper support inequality

$$\limsup_{N \rightarrow \infty} P_N(u + d\psi) \leq \alpha \Phi_{u+d\psi}(q) + E(q). \quad (4.11)$$

The permissible size of d may depend on β and ψ , as required for the terminal perturbation argument.

Corollary 4.6 (Upper bound for convex loss perturbations). *Let $\psi \in C_c^\infty((-\infty, 0))$. There is $w_0 > 0$, independent of β , such that for both signs of w with $|w| < w_0$ the loss $L_w(z) = V(z) + w\psi(1+z)$ is nonnegative, convex, and nonincreasing. For every fixed $\beta > 0$ and every fixed profile q with $q_* < 1$,*

$$\limsup_{N \rightarrow \infty} P_N(-\beta L_w) \leq \alpha \Phi_{-\beta L_w}(q) + E(q). \quad (4.12)$$

More generally this upper bound holds for any nonpositive, concave, nondecreasing terminal of at most quadratic growth which agrees with a negative quadratic hinge outside a compact set.

Proof. The support of $\psi(1+z)$ is a compact subset of $z < -1$, where $V'' = 1$, $V > 0$, and $V' < 0$. Taking $|w|$ sufficiently small preserves all three asserted inequalities, including at the boundary of that support. The threshold for w is a property of the loss and does not depend on β . For the fixed concave terminal $-\beta L_w$, take tangent caps and concave mollifications. Ordinary concave-terminal profile concavity from Section 3 gives the interpolation upper bound for each such cap. The entropy conjugate bounds its spin terms by $E(q)$, so the right side is simply $\alpha \Phi(q) + E(q)$ at this fixed trial. Mollification errors vanish uniformly in terminal value for a fixed cap. Beyond the compact perturbation support the cap error is the original negative Gaussian quadratic-tail error, controlled by monotone row insertion both microscopically and in the cascade. Sending the cap to infinity proves the displayed bound. Only pressure and row-functional values are passed to the limit; no derivative-profile uniformity in w and no complete pressure formula for the perturbed loss are required. $\square$

5 Universality at fixed temperature

Write $\nu_\varepsilon = \frac{1}{2}\mathcal{N}(0, 1 - \varepsilon) + \frac{1}{2}\mathcal{N}(0, 1 + \varepsilon)$, and let $P_{N,\varepsilon}(w)$ denote the pressure when the entries of A have law ν_ε . Thus $P_{N,0} = P_N$. Surface measure on each sphere is normalized to have mass one. We use unit spins $y = x/\sqrt{N}$ when comparing matrix entries. Fix $0 < B_0 < \infty$ and $u(z) = -\frac{\beta}{2}(-1 - z)_+^2$.

Theorem 5.1 (Pressure and upper supports). *There is $\varepsilon_0(B_0) > 0$, independent of $\beta > 0$, with the following property. If $0 < \alpha \leq B_0$ and $0 \leq \varepsilon < \varepsilon_0(B_0)$, then*

$$\lim_{N \rightarrow \infty} P_{N,\varepsilon}(u) = P(u).$$

If q is any Gaussian minimizer in (4.9) and $\psi \in C_c^\infty(\mathbb{R})$, then, for all sufficiently small $|d|$,

$$\limsup_{N \rightarrow \infty} P_{N,\varepsilon}(u + d\psi) \leq \alpha \Phi_{u+d\psi}(q; 1) + E(q).$$

The allowed size of d can depend on β, ψ and B_0 .

The principal issue is that a unit spin may place a positive fraction of its squared norm on a few coordinates. Moment matching controls the remaining coordinates, but gives no vanishing error for these large ones. We therefore bound separately the slices on which their values are fixed. A slice with squared norm t in the large coordinates loses spherical entropy. Its rows acquire a random variance change of size at most εt . The decisive estimate compares both effects on the same scale, $t/\chi_q(0)$, with constants independent of temperature. This permits one choice of $\varepsilon_0(B_0)$ before any later zero-temperature limit. After proving the pressure bounds, we use their derivatives in d to identify the empirical row law and the normalized-force observables.

5.1 Concentration, truncation, and replacement

We record the quantitative elementary estimates needed below. Constants in this subsection may depend on a fixed smooth capped potential, on β , and on B_0 , but are uniform in N and $0 \leq \varepsilon \leq 1/2$. Every entry is subgaussian with variance proxy at most $3/2$. Applying the exponential Markov inequality to a fixed bilinear form and taking $1/4$ -nets of the two unit spheres gives

$$\mathbb{P}\{\|A\|_{\text{op}} > K\sqrt{N}\} \leq 2\exp(C(M+N) - cK^2N). \quad (5.1)$$

Indeed the nets have at most 9^{M+N} pairs, each bilinear form has a subgaussian tail, and their maximum controls the operator norm within an absolute factor. Integrating this estimate also gives every fixed moment bound for $\|A\|_{\text{op}}/\sqrt{N}$.

A mixture entry is a Lipschitz function of a standard Gaussian. To see this, let $T = F_\varepsilon^{-1} \circ \Phi$, where F_ε is the mixture CDF and Φ the standard Gaussian CDF, and put $\sigma_+ = \sqrt{1+\varepsilon}$. For $z \geq 0$, $0 \leq T(z) \leq \sigma_+ z$, and the mixture density f_ε satisfies

$$f_\varepsilon(T(z)) \geq \frac{1}{2\sigma_+} \phi(T(z)/\sigma_+) \geq \frac{\phi(z)}{2\sigma_+}.$$

Consequently $T' \leq 2\sigma_+$; symmetry gives the same bound on the negative half-line. Product Gaussian concentration therefore applies to a Lipschitz function of the entries with only this absolute change in its Lipschitz constant.

For a potential w with bounded first three derivatives and any nonrandom nonzero finite measure λ supported on unit spins, set

$$L(A) = \log \int \exp \left\{ \sum_{\mu=1}^M w(A_\mu y) \right\} \lambda(dy).$$

The total mass of λ need not be one. Differentiation gives $\|\nabla_A L\|_F \leq \|w'\|_\infty \sqrt{M}$. If $|y_i| \leq b_N$ on the integration domain, differentiating three times in $A_{\mu i}$ gives

$$\begin{aligned} \partial_{\mu i}^3 L &= \langle w'''(g_\mu) y_i^3 \rangle + 3 \text{Cov}(w''(g_\mu) y_i^2, w'(g_\mu) y_i) \\ &\quad + \left\langle (w'(g_\mu) y_i - \langle w'(g_\mu) y_i \rangle)^3 \right\rangle. \end{aligned}$$

Its absolute value is at most $C_w b_N^3$. Taylor's theorem with integral remainder, replacing entries one at a time, thus shows that two entry laws with matching first two moments and uniformly bounded third absolute moments satisfy

$$|\mathbb{E}L(A) - \mathbb{E}L(\tilde{A})| \leq C_w M N b_N^3. \quad (5.2)$$

This is the smooth-function Lindeberg replacement argument [6, Theorem 1.1]. The coordinate bound b_N is what makes its error small; the localization argument below handles spins for which that bound fails. The estimate is unchanged by adding arbitrary, fixed row biases, and also holds conditionally on such biases.

Here and below the hinge can be capped and smoothed as in Section 3. More explicitly, writing $r = (-1 - z)_+$, replace $-\beta r^2/2$ for $r > L$ by its tangent $-\beta Lr + \beta L^2/2$. The difference from u is $\frac{\beta}{2}(r - L)_+^2$. Mollification with a symmetric kernel, followed by a constant of size $O(\beta\delta^2)$, gives a smooth upper approximation with bounded first three derivatives and arbitrarily small smoothing error. The pressure error of the cap tends to zero uniformly in N , including on any fixed restricted spin domain. For completeness, insert the rows one at a time. For a fixed row and any cavity measure, weighting by a decreasing function of r decreases the mean of any increasing function of r . This is the identity

$$2 \text{Cov}(a(R), b(R)) = \mathbb{E}[(a(R) - a(R'))(b(R) - b(R'))] \leq 0$$

for increasing a , decreasing b , and independent copies R, R' . Interpolate between the cap and the hinge and apply this observation to the cap error. Before row insertion, the fresh projection conditional on the spin is Gaussian with a random variance in $[1 - \varepsilon, 1 + \varepsilon]$. The expected error is therefore bounded by $\frac{\beta}{2} \sup_{v \in [1/2, 3/2]} \mathbb{E}[(-1 - \sqrt{v}Z - L)_+^2]$ per row, which tends to zero. Addition of the bounded test $d\psi$ changes this bound by at most $e^{2d\|\psi\|_\infty}$. For the row functional, interpolate the terminal potentials in the same way. The derivative is the tilted expectation of the cap error. Every interpolating base terminal has nonnegative first derivative, so its tilted diffusion has nonnegative drift; driven by the same Brownian motion, its endpoint is at least the Brownian endpoint. Its residual therefore has no larger upper tail than the fresh Gaussian residual. The same Gaussian tail bound applies, uniformly over profiles and diagonals in $[1/2, 3/2]$. The bounded-terminal-perturbation comparison proved below supplies the factor $e^{2d\|\psi\|_\infty}$ for row functionals as well.

Choose once and for all

$$\frac{1}{3} < a < \frac{1}{2}, \quad \mathcal{D}_N = \{y \in S^{N-1} : \max_i |y_i| \leq N^{-a}\}. \quad (5.3)$$

For Gaussian disorder, rotation invariance implies that the *disorder-averaged* Gibbs spin is uniform on the sphere. Spherical coordinate tails and a union bound give $\mathbb{E}\langle \mathbf{1}_{\mathcal{D}_N^c} \rangle \leq CN \exp(-cN^{1-2a})$. In particular the probability that the Gibbs mass of $\mathcal{D}_N$ is below $1/2$ is exponentially small on this scale. On the complementary event the loss in log partition is at most $\log 2$. On the exceptional event, Jensen's inequality on the restricted surface measure bounds the restricted log partition below by $\log \sigma_N(\mathcal{D}_N) - C_\beta(M + \|A\|_{\text{op}}^2)$. The full log partition has the same polynomial moment bound. By Cauchy–Schwarz and (5.1), the expected restriction loss is $o(N)$. This argument applies to the hinge and to its caps. For a fixed smooth cap, (5.2) costs $O(N^{2-3a}) = o(N)$ on $\mathcal{D}_N$. Removing the cap afterwards proves

$$\liminf_{N \rightarrow \infty} P_{N,\varepsilon}(u) \geq P(u). \quad (5.4)$$

5.2 Reduction of the upper bound to a localized slice

Fix a smooth upper cap w of $u + d\psi$. For a unit spin define its large-coordinate set $I(y) = \{i : |y_i| > N^{-a}\}$. Its cardinality is at most N^{2a} . The number of possible sets is at most $\exp(CN^{2a} \log N) = \exp(o(N))$. Fix such a set I of size k , write $c = y_I$, $t = \|c\|^2$, and set $n = N - k$. Conditional on c , the remaining coordinates have the form $\sqrt{1-t}z$ with $z \in S^{n-1}$. For $k > 0$ the marginal density of c is

$$a_{N,k}(1-t)^{(n-2)/2} \mathbf{1}_{\{t < 1\}}, \quad a_{N,k} = \frac{\Gamma(N/2)}{\pi^{k/2} \Gamma(n/2)}.$$

Stirling's formula gives $|\log a_{N,k}| = O(k \log N) = o(N)$. For $k = 0$ there is one slice, $c = 0, t = 0$.

Fix $\eta > 0$. The total contribution of $t > 1 - \eta$, after summing over I , has normalized log at most $\frac{1}{2} \log \eta + B_0 \sup w + o(1)$, since ball volumes and density prefactors contribute $o(N)$. The lower bound $P(u) \geq -C\beta B_0$ follows from Jensen's inequality, and the desired upper-support value is at least $P(u) - \alpha|d|\|\psi\|_\infty$. Hence η can be chosen, at the fixed temperature and test perturbation under consideration, so that this part lies strictly below that value. Dependence of η on β will have no bearing on ε_0 .

For $t \leq 1 - \eta$, the original restriction on the remaining coordinates implies $|z_i| \leq \eta^{-1/2} N^{-a}$. Enlarge their integration domain to this fixed restriction, independently of c . Cover the c -ball by a Euclidean N^{-2} -net, retaining centers with $\|c\|^2 \leq 1 - \eta/2$. Across all I the number of centers is still $\exp(O(N^{2a} \log N)) = \exp(o(N))$. On $\|A\|_{\text{op}} \leq K\sqrt{N}$, changing a center by h

changes the row arguments in Euclidean norm by at most $C_\eta K \sqrt{N} \|h\|$. Cauchy–Schwarz over the rows shows that the log integral changes by at most $C_{w,\eta,K} N \|h\|$. The logarithm of the marginal density changes by at most $C_\eta N \|h\|$ as well. Both errors are $o(N)$ on these nets.

For each center, the restricted log integral is $C_w \sqrt{N}$ -Lipschitz in the matrix. Gaussian concentration through T and a union bound make this uniform over all centers. Explicitly, put $\delta_N = N^{-(1-2a)/4}$. The probability that any centered log integral exceeds $N\delta_N$ in absolute value is at most

$$2 \exp\{CN^{2a} \log N - cN^{(1+2a)/2}\} \longrightarrow 0.$$

Conditional on the localized columns, replace the remaining entries by Gaussians using (5.2); their coefficients are bounded by $C_\eta N^{-a}$, so the error is $o(N)$ uniformly over the centers. Average this conditional inequality over the localized columns *before* maximizing over centers. The maximum is therefore over deterministic, fully averaged expected log partitions; no maximum of conditional expectations is taken. Then enlarge the remaining spin domain to the full sphere. Consequently it suffices to bound, for every deterministic sequence of centers, the unrestricted hybrid expected log partition plus $\frac{N}{2} \log(1-t)$. There is no compactness assertion uniform in an unspecified growing collection here: if the required bound failed for the maximum, one could choose a maximizing center for each N and apply the following subsequence argument to that deterministic sequence. The passage from probability to expected pressure is legitimate because $N^{-1} \log Z$ is bounded above for these potentials, and its negative part has uniformly bounded moments by Jensen and (5.1).

For each localized row, condition on independent fair signs s_i in the variance-mixture representation. Its bias is Gaussian of variance $t + \varepsilon D$, where

$$D = \sum_{i \in I} c_i^2 s_i, \quad |D| \leq t. \quad (5.5)$$

These marks and biases are independent from row to row and from the Gaussian remaining columns. Along a subsequence, t converges and the laws of D converge weakly to a probability measure supported on $[-t, t]$. We use t, D for these limits. The next calculation is valid for any such law; in particular it does not need a lower bound on the number of large coordinates.

5.3 The marked-row interpolation

Here we give the extension of the Gaussian upper interpolation from Section 4, including the terms caused by the localized coordinates. For two remaining spins with overlap R , put $R_f = t + (1-t)R$. Conditional on the row mark D , the actual row field has covariance $R_f + \varepsilon D$ and diagonal $v = 1 + \varepsilon D$. Its common Gaussian variance $t + \varepsilon D$ is nonnegative. For a finite trial profile q reaching diagonal 1, use a row trial field of profile vq and diagonal v , and put

$$p = \alpha \mathbb{E}_D D_{vq}, \quad p_d = \alpha \mathbb{E}_D D_{vq,d},$$

where the row derivative profile is calculated with terminal w . The spin field must have covariance $NR_f p$. To realize it, write $x = \sqrt{n} z$ for the remaining spin and take independent cascade fields $z_0^p, z_1^p, \dots, z_n^p$, each with profile p and diagonal p_d . Use

$$\sqrt{Nt} z_0^p(\tau) + \sqrt{\frac{N(1-t)}{n}} \sum_{i=1}^n x_i z_i^p(\tau).$$

The first term is common to all remaining spins and supplies covariance Ntp ; the second supplies $N(1-t)Rp$. Add a compensation field with covariance Nqp . At interpolation time $s \in [0, 1]$, using independent fields conditional on the marks, each row input is $\sqrt{s}$ times

the actual field plus $\sqrt{1-s}$ times the trial field; the spin and compensation fields enter the exponent with factors $\sqrt{1-s}$ and $\sqrt{s}$, respectively, while H_N^{pert} is unchanged. At finite N one may use the current mark law and M/N in this definition; continuity of the finite cascade recursion gives the displayed limits. Since $n/N \rightarrow 1$, the difference between the n -spin and N -spin normalizations vanishes.

Here are the fluctuation and deletion details needed to apply the cavity argument of Section 4. All limits in this paragraph are taken at a fixed finite trial grid and a fixed cap. Keep s fixed, and let Q be the covariance of two cascade leaves. Changing one mark from D to D' changes that row's covariance by

$$\Delta C = \varepsilon(D' - D)[s + (1-s)Q], \quad \Delta C_{\text{diag}} = \varepsilon(D' - D).$$

The profile p is defined from the deterministic mark law, rather than the realized marks, and is held fixed in this comparison. Gaussian covariance differentiation of the expected log partition, conditional on all marks, gives the change-rate bound

$$\frac{\varepsilon|D' - D|}{2} (\|w''\|_\infty + 2\|w'\|_\infty^2).$$

Indeed its two terms are the diagonal expectation of $w'' + (w')^2$ and the off-diagonal expectation of $[s + (1-s)Q]w'_1w'_2$; the latter coefficient lies in $[0, 1]$. The bound is $O(1)$ uniformly in the other marks, N , and s . Conditional on the marks, the independent row fields give a Gaussian Lipschitz constant at most $\|w'\|_\infty \sqrt{3M/2}$; spin and compensation fields add $O(\sqrt{N})$ and the perturbation adds $O(s_N)$. The finite-cascade log-mass estimate from Section 4 handles the weight randomness. Bounded differences for the marks therefore imply an unconditional expected fluctuation of $O(\sqrt{N} + s_N)$, also after any fixed number of row deletions.

Apply the GG argument after averaging the marks as well as the Gaussian disorder. There is no assertion of GG for every quenched mark vector. Deleting row 1 removes its row field and mark but leaves the deterministic p , the spin and compensation fields, and the perturbation fixed. Thus at each N the joint law of the removed mark and deleted-row replica array is exactly $\rho_N \otimes \text{Law}(\text{deleted-row array})$. At fixed almost-everywhere perturbation parameters and interpolation time, take deterministic subsequences of these distributions. Every limit retains this product structure. The independent mark may therefore be adjoined to the limiting cavity array and integrated after Gaussian row reinsertion. This argument does not select a subsequence depending on the realized removed mark. Bounded derivatives at the fixed cap justify the extra integration by dominated convergence. The positivity and comonotone-profile conclusions of the Gaussian cavity argument now apply. Finite-grid limits are taken before profile approximation, so no concentration bound uniform in cascade depth is required.

To make the cancellation explicit, let Q denote the overlap of two cascade leaves, and let a_μ^ℓ be w' evaluated at the interpolating field of row μ in replica ℓ . Gaussian integration by parts gives the following exact derivative of the normalized expected log partition at fixed perturbation coefficients (with $\mathbb{E}$ averaging the row marks, Gaussian fields, and cascade weights):

$$-\frac{M}{2N} \mathbb{E} \left\langle [(R_f - Q) + \varepsilon D_1(1 - Q)] a_1^1 a_1^2 \right\rangle + \frac{1}{2} \mathbb{E} \langle (R_f - Q) p(Q) \rangle.$$

Here $p(Q)$ means the value of the spin covariance at the same cascade rank; repeated covariance levels can be refined without changing the fields. The diagonal differences are zero in both terms.

Let r be the limiting remaining-spin cavity profile. It is nonnegative and comonotone with the prescribed cascade profile q . Thus $r_f = t + (1-t)r$ satisfies $r_f + \varepsilon D \geq (1-\varepsilon)t \geq 0$. The interpolating conditional profile is $C_D(s) = s(r_f + \varepsilon D) + (1-s)vq$, with fixed diagonal v .

Row concavity, with the small perturbation allowed by Section 3, bounds the row derivative by

$$-\frac{\alpha}{2} \int_0^1 \mathbb{E}_D[(r_f - q) + \varepsilon D(1 - q))D_{vq}] da.$$

The spin and compensation derivatives cancel exactly the term $-\frac{1}{2} \int (r_f - q)p$. The remaining upper bound is $-\frac{\alpha\varepsilon}{2} \int (1 - q)\mathbb{E}_D[DD_{vq}]$. As in the Gaussian proof, this pointwise subsequential derivative bound integrates by Fatou after the perturbation parameters are averaged; the fixed-cap derivatives are bounded, and the perturbation contributes $o(N)$.

At the trial endpoint the row contribution is $\alpha\mathbb{E}_D\Phi_w(vq; v)$. The common spin field contributes $t(p_d - \int p)/2$. The remaining-spin field contributes, at finite N , $(n/N)S_n((N/n)(1 - t)p)$; by uniform convergence on bounded profiles, this converges to $\mathcal{S}((1 - t)p) + (1 - t)p_d/2$. The second term restores the diagonal contribution removed in the definition of $\mathcal{S}$. Compensation subtracts $(p_d - \int qp)/2$. Using $|D| \leq t$, $D_{vq} \geq 0$ and the definition of p , the sum of these terms, the residual derivative bound, and the slice density is at most

$$\alpha\mathbb{E}_D\Phi_w(vq; v) + \frac{1}{2} \log(1 - t) + \mathcal{S}((1 - t)p) + \frac{1}{2} \int_0^1 (q_e - t)p, \quad q_e = \varepsilon t + (1 - \varepsilon)tq. \quad (5.6)$$

In this cancellation the restored diagonal term $(1 - t)p_d/2$ combines with the common-field contribution and the compensation, leaving no diagonal term. This also checks the normalization of the common field.

We now fix q to be a Gaussian base minimizer, so that $q_* < 1$. This includes every minimizer of (4.9): the conjugate bound gives $\mathcal{P}_u(q) \leq \alpha\Phi_u(q) + E(q) = P(u)$, whereas the pressure formula gives $\mathcal{P}_u(q) \geq P(u)$. Thus such a q is also a minimizer of $\mathcal{P}_u$ and inherits its root estimates. At a fixed cap first pass from finite trial profiles to this q by profile continuity. Since $p \geq 0$, clipping $(q_e - t)/(1 - t)$ below at zero only increases its pairing with p . The spin conjugate (2.21) therefore bounds the non-row terms by

$$\begin{aligned} & \frac{1}{2} \log(1 - t) + E\left(\max\left\{0, \frac{q_e - t}{1 - t}\right\}\right) \\ &= E(q_e) - \frac{1}{2} \int_0^t \frac{dz}{\chi_{q_e}(z)} \\ &\leq E(q) + \frac{1}{2} \log(1 - \varepsilon t) - \frac{(1 - \varepsilon)t}{2(1 - \varepsilon t)\chi_q(0)}. \end{aligned} \quad (5.7)$$

Here is a direct check, including clipping. If $\hat{q} = (q_e - t)_+/(1 - t)$, then for $z \geq 0$

$$m_{\hat{q}}(z) = m_{q_e}(t + (1 - t)z), \quad \chi_{\hat{q}}(z) = \frac{\chi_{q_e}(t + (1 - t)z)}{1 - t}.$$

Substitution into (2.21) proves the equality. When t exceeds the essential supremum of q_e , use $\chi_{q_e}(z) = 1 - z$ above that supremum; the same equality reduces to $E(\hat{q}) = E(0) = 0$. For $b = \varepsilon t$, the affine change $q_e = b + (1 - b)q$ gives

$$E(q_e) = E(q) + \frac{1}{2} \log(1 - b) + \frac{b}{2(1 - b)\chi_q(0)}, \quad \chi_{q_e}(0) = (1 - b)\chi_q(0).$$

Since χ_{q_e} is nonincreasing, $\int_0^t dz/\chi_{q_e}(z) \geq t/\chi_{q_e}(0)$, proving the inequality. Crucially, the resulting estimate contains no p . We can now remove the cap, using its uniform row-functional error over $v \in [1/2, 3/2]$.

5.4 A variance estimate uniform in temperature

The entropy estimate (5.7) gives a loss of at least $t/(4\chi_q(0))$ when $\varepsilon \leq 1/2$. We now bound the possible row gain by $C(B_0)\varepsilon t/\chi_q(0)$ with no temperature dependence in $C(B_0)$. The needed residual estimate is a property of the single-row diffusion.

Lemma 5.2 (Residual moments under a decrease of variance). *Let q be a nonnegative nondecreasing profile with diagonal 1, and let $u(z) = -\beta(-1-z)_+^2/2$, $\beta > 0$. Put $a_0 = f_z(0,0)$ for its row solution before the root Gaussian variance. Write $\mathbb{E}_{u,v}$ for expectation under the tilted endpoint law for profile vq and diagonal v . For $1/2 \leq v \leq 1$,*

$$\sqrt{v}\beta\mathbb{E}_{u,v}(-1-g)_+ \leq a_0, \quad \mathbb{E}_{u,v}(-1-g)_+^2 \leq \sqrt{\pi/2}\mathbb{E}_{u,v}(-1-g)_+. \quad (5.8)$$

Proof. For the first inequality put the variance scale into the terminal, so the Gaussian variance profile is q on the fixed time interval $[0,1]$. Let A_v and A_1 be the derivative solutions for the terminals $u(\sqrt{v}z)$ and $u(z)$, respectively. Their difference $H = A_v - A_1$ satisfies

$$H_t + \frac{1}{2}H_{zz} + m_q A_v H_z + m_q (A_1)_z H = 0, \quad H(1, z) = \sqrt{v}\beta(-1-\sqrt{v}z)_+ - \beta(-1-z)_+ \leq 0.$$

The zeroth-order coefficient lies in $[-\beta, 0]$, and the drift is Lipschitz with at most linear growth. The maximum principle gives $H \leq 0$. To handle the hinge, convolve the two terminal derivative functions with the same nonnegative mollifier, preserving their order, and pass to the limit. When $q(0) > 0$, $m_q = 0$ before that root variance, so the same comparison includes the initial heat interval. The derivative martingale at $(0,0)$ identifies $A_v(0,0)$ with the expectation on the left of (5.8).

For the second inequality use variance time, writing the tilted diffusion as

$$dZ_s = dB_s + m(s)a(s, Z_s)ds, \quad Z_0 = 0, \quad 0 \leq s \leq v.$$

The base derivative $a = f_z$ is nonnegative, nonincreasing, and convex in position. Nonnegativity and monotonicity follow from the terminal and parabolic comparison. Convexity follows by differentiating once more: $c = f_{zzz}$ has nonnegative terminal value (a positive measure for the hinge) and solves the linear equation $(\partial_s + \frac{1}{2}\partial_{zz} + ma\partial_z)c = -3mf_{zz}c$. Smooth concave approximations with convex first derivative and no far cap justify the claim; their diffusion limits give the hinge. Thus the drift is nonnegative, nonincreasing, and convex.

Condition on the Brownian bridge $\omega_s = B_s - (s/v)B_v$ and write $b = B_v \sim \mathcal{N}(0, v)$, independent of the bridge. The endpoint map $T_\omega(b) = Z_v$ is increasing, convex, and 1-Lipschitz. Indeed its first variation J_s solves $J'_s = 1/v + d_z(s, Z_s)J_s$, $J_0 = 0$, for $d = ma$; hence $0 \leq J_s \leq s/v$ and, more precisely,

$$T'_\omega(b) = \frac{1}{v} \int_0^v \exp \left\{ \int_r^v d_z(s, Z_s) ds \right\} dr \geq \frac{1 - e^{-\beta v}}{\beta v} > 0.$$

Its second variation solves $K'_s = d_z K_s + d_{zz} J_s^2$, $K_0 = 0$, and is nonnegative. These deterministic integral-equation variations remain valid for every continuous bridge path by smooth approximation. Also $T_\omega(b) \geq b$ by nonnegative drift. The strictly positive lower slope bound makes T_ω onto, so it has a unique crossing $b_0 \leq -1$ of -1 . Set $R(s) = -1 - T_\omega(b_0 - s)$, $s \geq 0$. Then R is increasing and concave, $R(0) = 0$, and $0 \leq R(s) \leq s$. Consequently $R(s)/s$ is nonincreasing. Under the Gaussian left-tail measure with density proportional to $\exp(-(b_0 - s)^2/(2v))\mathbf{1}_{s \geq 0}$, decreasing-likelihood-ratio comparison gives

$$\frac{\mathbb{E}[R(s)^2]}{\mathbb{E}[R(s)]} \leq \frac{\mathbb{E}[sR(s)]}{\mathbb{E}[R(s)]} \leq \frac{\mathbb{E}[s^2]}{\mathbb{E}[s]} \leq \sqrt{\pi/2}.$$

The middle inequality follows from the same covariance identity used above, now relative to the probability density proportional to $s \exp(-(b_0 - s)^2/(2v))$. Relative to the density proportional to $se^{-s^2/2}$, the logarithmic derivative of its likelihood ratio is $b_0/v - (1/v - 1)s \leq 0$. The last ratio is therefore at most $\int_0^\infty s^2 e^{-s^2/2} ds / \int_0^\infty se^{-s^2/2} ds = \sqrt{\pi/2}$. Multiply by the conditional first moment and integrate over the bridge to obtain the second assertion of (5.8). The lower slope bound is used only for the crossing and approximation; its dependence on β does not enter the moment constant. Mollified drifts retain all three shape properties and the common Lipschitz bound β . Approximate a general monotone m by steps in L^1 ; the linear PDE difference equation and uniform diffusion moments give local convergence of solutions, and the flow maps converge on compact endpoint sets. Finally $(-1 - Z_v)_+ \leq (-1 - B_v)_+$ supplies uniform integrability for all residual moments. These observations justify the hinge and coefficient limits in the bridge argument. $\square$

Bounded terminal tests. A bounded terminal perturbation changes a tilted endpoint expectation of a nonnegative observable by at most the factor $e^{2|d|\|\psi\|_\infty}$. On the same realized cascade and Gaussian field, the normalized Gibbs measure with terminal $u + d\psi$ has density

$$\frac{e^{d\psi(g)}}{\langle e^{d\psi(g)} \rangle_u} \in [e^{-2|d|\|\psi\|_\infty}, e^{2|d|\|\psi\|_\infty}]$$

relative to the measure with terminal u . Average this pointwise bound over the cascade and field. The endpoint expectation here is precisely the PDE tilted expectation: for a bounded continuous terminal direction h , differentiating $\Phi_{u+\lambda h}$ at zero gives $\mathbb{E}\langle h(g) \rangle_u$ directly, while differentiating the recursion gives the following linear equation, now denoting the fixed diagonal by Q :

$$J_t + \frac{1}{2}J_{zz} + mf_z J_z = 0, \quad J(Q, z) = h(z).$$

Its diffusion representation identifies the same expectation with the tilted endpoint law. The initial $m = 0$ heat interval includes common root variance. For residual diagonal variance, augment the base measure at each leaf by the residual Gaussian variable and integrate it inside that leaf weight. Each replica draws a fresh residual variable, even when leaf labels coincide, so this convention adds no off-diagonal variance. The terminal $m = 1$ interval is exactly this convolution. Profile approximation preserves the equality for bounded tests and the likelihood-ratio bound; monotone truncation extends the latter to nonnegative observables. In particular there is no factor depending on cascade depth.

Comparison with the entropy loss. Put $w = u + d\psi$ after removing the cap. Differentiating the variance scale in the terminal gives

$$\partial_v \Phi_w(vq; v) = \frac{1}{2v} \mathbb{E}_{w,v}[gw'(g)].$$

This identity follows first by differentiating the finite cascade recursion and then by the uniform quadratic moment bounds. If $r = (-1 - g)_+$, then $gw'(g) = -\beta(r^2 + r)$. For $v \geq 1$ this part of the derivative is nonpositive. For $1/2 \leq v \leq 1$, the preceding perturbation bound and (5.8) show that its absolute value is at most $C_0 a_0$, provided $|d|\|\psi\|_\infty \leq 1$, where $C_0 = e^2 \sqrt{2}(1 + \sqrt{\pi/2})$. The test part is bounded by $|d| \sup_z |z\psi'(z)|/(2v)$. Impose also $|d| \sup_z |z\psi'(z)| \leq 1$. Since $|v - 1| \leq \varepsilon t$,

$$\alpha(\Phi_w(vq; v) - \Phi_w(q; 1)) \leq \varepsilon t(C_0 \alpha a_0 + \alpha) \leq \frac{(C_0 \sqrt{B_0} + B_0) \varepsilon t}{\chi_q(0)}.$$

The final inequality uses (4.10), which implies $\alpha a_0 \leq \sqrt{\alpha}/\chi_q(0)$, and $\chi_q(0) \leq 1$. The constant C_0 is absolute: the dependence on β has disappeared. For $\varepsilon \leq 1/2$ the negative last term in

(5.7) has magnitude at least $t/(4\chi_q(0))$, and its logarithmic term is nonpositive. Choose, for example, any

$$0 < \varepsilon_0(B_0) < \min \left\{ \frac{1}{2}, \frac{1}{8(C_0\sqrt{B_0} + B_0)} \right\}.$$

The entropy loss then dominates the row improvement in every slice. To specify the quantifiers, choose this $\varepsilon_0(B_0)$ before choosing β , a minimizer, a test perturbation, or a cap. For each fixed β and ψ , Proposition 3.5 and Corollary 3.6 allow both signs of d with

$$|d| \leq \min \left\{ d_{\text{stab}}(\beta, \psi), \frac{1}{1 + \|\psi\|_\infty}, \frac{1}{1 + \sup_z |z\psi'(z)|} \right\}.$$

The stability radius is uniform in the trial profile, the diagonal in $[1/2, 2]$, and the cap and mollifier. The thermodynamic limit is taken at each fixed cap and finite trial grid; profile approximation follows, then removal of mollification at fixed cap, and finally removal of the cap. The entropy factor $1/\chi_q(0)$ may depend on β but appears on both sides of the comparison and cancels. No convergence rate uniform in β is required. The same fixed mixture law therefore remains valid throughout the later ordered limits. Equations (5.6)–(5.8) prove the upper support in Theorem 5.1. Combined with (5.4) at $d = 0$, they prove its pressure assertion.

5.5 The inner thermodynamic observables

Let $\mu_{\beta,\alpha}$ be the Gaussian tilted endpoint law associated with a minimizer q . The following argument proves, in particular, that this law does not depend on the choice of minimizer. Define the random empirical projection measure

$$L_N = \frac{1}{M} \sum_{\mu=1}^M \delta_{A_\mu y}.$$

Convergence in probability below refers to the joint law of disorder and one Gibbs sample.

Proposition 5.3 (Empirical row law and force normalization). *For either the Gaussian law or any mixture in Theorem 5.1, at fixed $\beta > 0$ and $0 < \alpha \leq B_0$, L_N converges weakly in probability to $\mu_{\beta,\alpha}$. This measure is atomless, assigns positive mass to $(-\infty, -1)$, and has finite moments of its negative part. The gap and normalized-force observables in the problem have inner $N \rightarrow \infty$ limits at every fixed argument, and those limits agree for all these entry laws.*

Proof. First, the normalized random log partition concentrates for the uncapped potentials $u + d\psi$. On the convex operator-norm ball $\|A\|_{\text{op}} \leq K\sqrt{N}$, differentiation gives

$$\|\nabla_A \log Z\|_F \leq \beta(\|A\|_{\text{op}} + \sqrt{M}) + |d|\|\psi'\|_\infty \sqrt{M} = O(\sqrt{N}).$$

Extend its restriction from this ball with the same Lipschitz constant and use Gaussian concentration, through T for mixture entries. The complement has probability exponentially small in N for a large fixed K , by (5.1). Jensen's lower bound $\log Z \geq -C_\beta(M + \|A\|_{\text{op}}^2) - M|d|\|\psi\|_\infty$ and the corresponding upper bound show that the extension changes its expectation by $o(N)$. Hence $N^{-1} \log Z - P_{N,\varepsilon}(u + d\psi)$ converges to zero in probability.

For a fixed test ψ , put $S(d) = \alpha \Phi_{u+d\psi}(q; 1) + E(q)$. Theorem 5.1 gives $S(0) = P(u)$ and $S'(0) = \alpha \int \psi d\mu_{\beta,\alpha}$. If $b > S'(0)$, choose a sufficiently small $d > 0$ and $c > 0$ such that $S(d) - S(0) < db - 2c$. The expected upper support and base pressure limit, together with the preceding concentration, imply with disorder probability tending to one that

$$\log Z(u + d\psi) - \log Z(u) \leq N(db - c).$$

This uses only a limsup bound at d ; a thermodynamic pressure limit for the perturbed terminal has not been assumed. Exponential Markov under the Gibbs measure gives

$$\left\langle \mathbf{1}_{\{N^{-1} \sum_{\mu} \psi(g_{\mu}) \geq b\}} \right\rangle \leq \exp\{\log Z(u + d\psi) - \log Z(u) - Ndb\}.$$

On that event the right side is at most e^{-Nc} . The same argument with $d < 0$ gives the lower bound. Therefore $\int \psi dL_N \rightarrow \int \psi d\mu_{\beta,\alpha}$ in joint probability for every $\psi \in C_c^\infty$. A countable determining class, together with tightness obtained by smooth cutoffs whose $\mu_{\beta,\alpha}$ -masses tend to one, proves weak convergence. The same limits for every base minimizer prove their equality of row marginals.

Here are direct bounds ensuring that threshold tests and normalization are permitted. Remove one row and let $\langle \cdot \rangle_0$ be its cavity measure. Its reweighting denominator is $D = \langle e^{u(g)} \rangle_0 \leq 1$, and Jensen gives $D \geq \exp\langle u(g) \rangle_0$. Since the row is independent of this measure, $\mathbb{E}[-\langle u(g) \rangle_0] \leq C_\beta$. Thus $\mathbb{P}(D < e^{-K}) \leq C_\beta/K$. For an interval J , conditional on a unit cavity spin the fresh projection has density bounded by an absolute constant, and $e^u \leq 1$. Consequently its reweighted marginal satisfies

$$\mathbb{E}\langle \mathbf{1}_{\{g \in J\}} \rangle \leq Ce^K |J| + C_\beta/K.$$

This estimate is uniform in N and passes to the deterministic limiting law by continuous tests. Sending $|J| \downarrow 0$ and then $K \rightarrow \infty$ proves that the law has no atoms. Conversely, for a fixed nonempty interval $J \Subset (-\infty, -1)$, $D \leq 1$ gives

$$\mathbb{E}\langle \mathbf{1}_{\{g \in J\}} \rangle \geq \inf_{g \in J} e^{u(g)} \inf_{v \in [1/2, 3/2]} \mathbb{P}(\sqrt{v}Z \in J) > 0.$$

Hence its limiting negative-gap probability is positive. Finally the single-row covariance inequality used above gives, for every fixed $p > 0$,

$$\mathbb{E}\langle (-1 - g)_+^p \rangle \leq \sup_{v \in [1/2, 3/2]} \mathbb{E}(-1 - \sqrt{v}Z)_+^p < \infty.$$

Using an exponent larger than p gives uniform integrability, including for empirical averages. Set

$$p_- = \mu_{\beta,\alpha}((-\infty, -1)), \quad m_- = \int (-1 - g)_+ \mu_{\beta,\alpha}(dg), \quad \bar{r} = m_-/p_-.$$

Both p_- and m_- are strictly positive. Weak convergence, atomlessness, and uniform integrability imply in joint probability

$$\frac{|C(y)|}{M} \rightarrow p_-, \quad \frac{1}{M} \sum_{\mu \in C(y)} (-1 - g_\mu) \rightarrow m_-.$$

For the second assertion, the second-moment bound gives $\mathbb{E} \int r \mathbf{1}_{\{r > K\}} dL_N \leq C/K$, where $r = (-1 - g)_+$. Markov's inequality makes this truncation error small in joint probability. The bounded continuous truncated residual converges by weak convergence, and then $K \rightarrow \infty$ gives the assertion. Thus the configuration-dependent conditional mean residual converges to $\bar{r}$. For every $s > 0$, sandwich its random threshold between $s(\bar{r} - \delta)$ and $s(\bar{r} + \delta)$, apply weak convergence to these fixed thresholds, and send $\delta \downarrow 0$. Atomlessness at $g = -1 - s\bar{r}$ justifies the last step. Since the observables lie in $[0, 1]$, convergence in probability also gives convergence of their expectations. Explicitly,

$$\begin{aligned} \lim_{N \rightarrow \infty} G_{N,\beta,\alpha}(\delta, u_+) &= \mu_{\beta,\alpha}((\delta - 1, u_+ - 1]), \quad 0 < \delta < u_+, \\ \lim_{N \rightarrow \infty} F_{N,\beta,\alpha}(s) &= \frac{\mu_{\beta,\alpha}(\{-1 - s\bar{r} \leq g < -1\})}{p_-}. \end{aligned}$$

The convention when $C(y)$ is empty is immaterial, since its probability tends to zero. These formulas prove the proposition with exactly the force normalization specified in the problem. $\square$

6 Ground energy and the sharp satisfiability threshold

This section separates an energetic issue from a geometric one. The pressure formula determines when the minimum energy is subextensive. A uniform repair lemma then turns subextensive energy into exact satisfiability after deleting a fixed fraction of the constraints. There are two geometric obstacles. Solving for the violated rows can create violations in other rows, so the repair enlarges the set of rows whose projections it controls. The repaired vector may then have norm less than one, so a second displacement restores its norm without spoiling those controlled projections. The negative margin permits radial contraction when the repaired vector instead has norm at least one.

Write $A \in \mathbb{R}^{M \times N}$ for the pattern matrix, and use unit-spin coordinates $y = x/\sqrt{N}$. Set

$$r_A(y) = (-\mathbf{1} - Ay)_+, \quad H_A(y) = \frac{1}{2} \|r_A(y)\|_2^2, \quad R(A) = \min_{\|y\|_2=1} \|r_A(y)\|_2.$$

All vector norms without a subscript are Euclidean, and $\|A\|$ denotes the operator norm between Euclidean spaces. Let $P_{N,\alpha}^\nu(\beta) = N^{-1} \mathbb{E} \log \int e^{-\beta H_A(y)} d\sigma(y)$, where $M = \lfloor \alpha N \rfloor$, σ is normalized surface measure on the unit sphere, and ν is the entry law. We use $\nu_0 = \mathcal{N}(0, 1)$ and the variance mixtures ν_ε from the introduction. The pressure and universality results of Sections 4 and 5 are the only input from the variational analysis: the Gaussian pressure has a limit at every fixed (α, β) ; for every finite B_0 , the same limit holds for $0 \leq \varepsilon < \varepsilon_0(B_0)$ and $0 < \alpha \leq B_0$.

We will repeatedly use a uniform matrix-norm estimate. For $\varepsilon \leq 1/2$, an entry satisfies $\mathbb{E} e^{tA_{\mu i}} \leq e^{3t^2/4}$. Independence gives the same bound for $v^\top Ay$ when $\|v\| = \|y\| = 1$. Nets of radius $1/4$ on the two spheres have at most 9^{M+N} pairs, and approximating both arguments gives $\|A\| \leq 2 \max |v^\top Ay|$ on these nets. Thus

$$\mathbb{P}(\|A\| > 2t) \leq 2 \cdot 9^{M+N} e^{-t^2/3}, \quad t > 0. \quad (6.1)$$

In particular $\mathbb{E} \|A\|^2 \leq C(M + N)$, and sufficiently large constant multiples of $\sqrt{M} + \sqrt{N}$ bound $\|A\|$ with exponential error. The case $M = 0$ is immediate.

6.1 The ground-energy limit

Lemma 6.1 (Uniform approximation by the pressure). *For every finite B_0 there is a constant $C = C(B_0)$ such that, for $N \geq 2$, $0 \leq M \leq B_0 N$, $0 \leq \varepsilon \leq 1/2$, $\beta > 0$, and $0 < \delta < 1/2$,*

$$0 \leq -\frac{P_{N,\alpha}^{\nu_\varepsilon}(\beta)}{\beta} - \frac{1}{N} \mathbb{E} \min_{\|y\|=1} H_A(y) \leq C\delta + \frac{1}{\beta} \log \frac{C}{\delta}. \quad (6.2)$$

Consequently the Gaussian limit

$$e(\alpha) = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \min_{\|y\|=1} H_A(y) = \lim_{\beta \rightarrow \infty} -\frac{P_\alpha^{\nu_0}(\beta)}{\beta}$$

exists. On every range where the fixed-temperature pressures agree, the mixture ground-energy limit also exists and equals $e(\alpha)$. Moreover, e is nondecreasing and 1-Lipschitz, and $e(\alpha) = 0$ for $0 \leq \alpha \leq 1$.

Proof. The positive-part map is 1-Lipschitz, so for unit y, y' ,

$$\|r_A(y) - r_A(y')\| \leq \|A\| \|y - y'\|, \quad \|r_A(y)\| \leq \sqrt{M} + \|A\|.$$

It follows that H_A has Lipschitz constant at most $L_A = \|A\|(\sqrt{M} + \|A\|)$ on the sphere. A spherical cap of Euclidean radius δ about a minimizer has surface probability at least $(c\delta)^N$, for an absolute $c > 0$. For completeness, rotate its center to the last coordinate vector and parametrize the upper hemisphere by $z \mapsto (z, \sqrt{1 - \|z\|^2})$. The ball $\|z\| \leq \delta/2$ maps inside the cap, and its surface Jacobian is at least one. The ratio of the $(N-1)$ -ball volume to the $(N-1)$ -sphere area is bounded below by c_1^N , as follows directly from their gamma-function formulas; this proves the stated bound. Integrating over this cap and also using $H_A \geq \min H_A$ gives the deterministic estimate

$$0 \leq -\frac{1}{\beta N} \log \int e^{-\beta H_A} d\sigma - \frac{\min H_A}{N} \leq \frac{L_A \delta}{N} + \frac{1}{\beta} \log \frac{C}{\delta}.$$

Equation (6.1) gives $\mathbb{E}\|A\|^2 \leq C(B_0)N$; hence $\mathbb{E}L_A/N \leq C(B_0)$. This proves (6.2).

At fixed N , the quantity $-P_{N,\alpha}^{\nu_0}(\beta)/\beta$ is nonincreasing in β and nonnegative. Indeed, $\beta \mapsto \log \int e^{-\beta H_A} d\sigma$ is convex and vanishes at zero. Its limiting pressure inherits these properties, so its zero-temperature limit exists. First let $N \rightarrow \infty$ in the inequality, then let $\beta \rightarrow \infty$ with, for example, $\delta = \beta^{-1/2}$. The liminf and limsup of the expected ground energy coincide with this limit. The identical argument applies to any law with the same fixed-temperature pressures.

Couple matrices of different row counts by appending independent rows. Appending rows cannot decrease the minimum energy. Conversely, evaluate the new rows at a measurable approximate minimizer of the old matrix and then let its approximation error tend to zero. For every fixed unit y , a fresh projection $g = A_\mu y$ has mean zero and variance one, and

$$\mathbb{E}V(g) \leq \frac{1}{2}\mathbb{E}(1+g)^2 = 1.$$

Thus the expected minimum energy increases by at most one per appended row. After dividing by N and passing to the limit this proves the monotonicity and Lipschitz assertions. Approximate minimizers can be selected measurably by taking the first adequate point in a fixed countable dense subset of the sphere, since H_A is continuous. If $M < N$, the row kernel contains a unit vector, at which every gap is one. Hence $e(\alpha) = 0$ for $\alpha < 1$; continuity gives $e(1) = 0$. $\square$

Lemma 6.2 (Positive energy at a finite density). *There is an absolute finite $B > 1$ such that the Gaussian ground energy satisfies $e(B) > 0$. The same finite-dimensional lower bound used to prove this assertion holds uniformly for $0 \leq \varepsilon \leq 1/2$.*

Proof. Conditioned on the variance marks of a mixture matrix, a fixed unit-spin projection is Gaussian with variance in $[1/2, 3/2]$. Thus

$$\mathbb{P}(A_\mu y < -2) \geq p, \quad p = \mathbb{P}(Z < -2\sqrt{2}) > 0, \quad Z \sim \mathcal{N}(0, 1).$$

At a deterministic unit y , row independence and the elementary Chernoff bound give

$$\mathbb{P}(\#\{\mu : A_\mu y < -2\} < pM/2) \leq e^{-c_p M}.$$

Outside this event, $\|r_A(y)\| \geq \sqrt{pM/2}$. There is an absolute K_0 such that $\|A\| \leq K_0(\sqrt{N} + \sqrt{M})$ with probability tending to one, uniformly in these entry laws. Choose a fixed $\delta > 0$ so small that $2K_0\delta \leq \frac{1}{2}\sqrt{p/2}$, and take a δ -net of the unit sphere of cardinality at most

$(1 + 2/\delta)^N$. For $M \geq N$, the Lipschitz bound for r_A transfers the residual lower bound on the net to

$$R(A) \geq \frac{1}{2}\sqrt{pM/2}.$$

The probability that a net point fails is at most $\exp\{N \log(1 + 2/\delta) - c_p M\}$. Choosing B sufficiently large and setting $M = \lfloor BN \rfloor$ makes this probability tend to zero. Therefore $\min H_A/N \geq pM/(16N)$ with probability tending to one, and $e(B) \geq pB/16 > 0$. $\square$

Fix henceforth $B_0 > B$ and reduce the pressure-comparison constant, if necessary, so that $\varepsilon_0(B_0) \leq 1/2$. Define

$$\alpha_J = \sup\{\alpha \geq 0 : e(\alpha) = 0\}. \quad (6.3)$$

The preceding lemmas show that $1 \leq \alpha_J < B < B_0$. By continuity and monotonicity, $e(\alpha) = 0$ for $\alpha \leq \alpha_J$ and $e(\alpha) > 0$ for $\alpha > \alpha_J$. It remains to show that vanishing energy density yields exact satisfiability at each strictly smaller density.

6.2 Uniform geometry of the rows

The repair needs a uniformly bounded right inverse for every set of slightly fewer than N rows. It also needs a bound on how many rows can be close to contact at an arbitrary spin. The next lemma supplies both facts before the spin is selected from the random matrix. This uniformity is what permits us to apply it at a ground minimizer.

Lemma 6.3 (Uniform row estimates). *Fix $B_0 < \infty$ and $0 < \rho < 1/8$. There are constants $K < \infty$, $c > 0$, $u_0 > 0$, $a > 0$, and $C < \infty$, depending only on (B_0, ρ) , such that, for all sufficiently large N , all $M \leq B_0 N$, and every $0 \leq \varepsilon \leq 1/2$, the following hold simultaneously with probability at least $1 - Ce^{-aN}$:*

1. $\|A\| \leq K\sqrt{N}$;
2. for every row set I with $1 \leq |I| \leq (1 - \rho)N$,

$$\|A_I^\top v\| \geq c\sqrt{N} \|v\| \quad \text{for all } v \in \mathbb{R}^{|I|}; \quad (6.4)$$

3. every unit y satisfies

$$\#\{\mu : |1 + A_\mu y| \leq u_0\} \leq (1 + \rho/2)N. \quad (6.5)$$

Proof. Choose $K = K(B_0)$ large enough in (6.1) that the operator-norm failure probability is exponentially small, and work on this event for the two net transfers below.

For a fixed I , $k = |I|$, and fixed $v \in \mathbb{R}^k$ with $\|v\| = 1$, the coordinates of $A_I^\top v$, conditional on all variance marks, are independent centered Gaussians with variances at least $1/2$. Their joint density is at most $\pi^{-N/2}$. The Euclidean ball-volume formula therefore gives

$$\mathbb{P}(\|A_I^\top v\| \leq 2c\sqrt{N}) \leq (C_1 c)^N.$$

Use a (c/K) -net of the unit sphere in $\mathbb{R}^k$, of cardinality at most $(3K/c)^k$, taking $c < K$. If (6.4) fails while $\|A\| \leq K\sqrt{N}$, some net point has image norm at most $2c\sqrt{N}$. Summing over dimensions and row sets bounds the failure probability by

$$N 2^{B_0 N} (3K/c)^{(1-\rho)N} (C_1 c)^N.$$

Its logarithm divided by N is bounded by a constant depending on (B_0, K) plus $\rho \log c + o(1)$. Choose c sufficiently small to make this bound negative.

For the gap-count estimate take a net of radius $\eta = u_0\sqrt{\rho}/(4K)$. If y and its net representative y_0 satisfy $\|y - y_0\| \leq \eta$, then

$$\#\{\mu : |A_\mu(y - y_0)| > u_0\} \leq \frac{\|A(y - y_0)\|^2}{u_0^2} \leq \rho N/16.$$

If (6.5) fails, more than $(1 + 7\rho/16)N$ rows at y_0 therefore satisfy $|1 + A_\mu y_0| \leq 2u_0$; in particular at least $\lceil(1 + \rho/4)N\rceil$ do. For fixed y_0 , each row projection has density at most $1/\sqrt{\pi}$, conditionally on its variance marks and also unconditionally. Hence its probability of this interval is at most $C_2 u_0$. Row independence, a union bound over row subsets, and then a union bound over the spin net give, for $C_2 u_0 < 1$,

$$\mathbb{P}(\text{gap-count failure, } \|A\| \leq K\sqrt{N}) \leq (1 + 8K/(u_0\sqrt{\rho}))^N 2^{B_0 N} (C_2 u_0)^{(1+\rho/4)N}.$$

Its logarithm divided by N is at most a constant depending on (B_0, K, ρ) plus $(\rho/4)\log u_0$. Choose u_0 sufficiently small. These choices establish all three assertions with exponential error. $\square$

The following elementary fact supplies a direction with uniformly small row projections. It does not require any randomness.

Lemma 6.4 (A direction inside finitely many slabs). *Let W be a D -dimensional Euclidean space, $D \geq 1$, and $L : W \rightarrow \mathbb{R}^m$ a linear map. There exists a unit $v \in W$ such that*

$$\|Lv\|_\infty \leq \frac{\|L\|}{\sqrt{D}}. \quad (6.6)$$

Proof. If L has a nonzero kernel, take a unit kernel vector. Otherwise the set $Q = \{w \in W : \|Lw\|_\infty \leq 1\}$ is compact and has nonempty interior. Let w_* maximize $\|w\|$ over Q . The active linear functionals, those satisfying $|(Lw_*)_j| = 1$, span W^* . Indeed, if they had a common nonzero annihilator h , both $w_* + th$ and $w_* - th$ would remain in Q for sufficiently small $t > 0$: active inequalities would be unchanged and the finitely many inactive inequalities have positive slack. But at least one of these two points has squared norm at least $\|w_*\|^2 + t^2\|h\|^2$, a contradiction. There are consequently at least D distinct active rows, and

$$D \leq \|Lw_*\|^2 \leq \|L\|^2 \|w_*\|^2.$$

The vector $v = w_*/\|w_*\|$ satisfies the required bound. $\square$

6.3 Repair after deleting constraints

Lemma 6.5 (Deterministic repair). *Fix $K, c, u_0 > 0$ and $0 < \rho < 1/8$. There exists $d_* = d_*(K, c, u_0, \rho) > 0$ such that the following holds for all sufficiently large N . Suppose A satisfies the first two assertions of Lemma 6.3, and some unit y satisfies*

$$\|r_A(y)\| \leq d\sqrt{N}, \quad S = \{\mu : 1 + A_\mu y \leq u_0\}, \quad |S| \leq (1 - 2\rho)N, \quad 0 < d \leq d_*.$$

Then there is a unit $\hat{y}$ with $1 + A_\mu \hat{y} \geq 0$ for every row μ .

Proof. We first correct the deficient gaps in the ambient vector space, keeping all uncontrolled rows a fixed distance above zero. We then return the corrected vector to the unit sphere.

Correct the gaps. Choose $d_* > 0$ sufficiently small that, for $d \leq d_*$,

$$\frac{d}{c} \leq \frac{1}{2}, \quad L_d := \frac{16K^2 d^2}{c^2 u_0^2} \leq \frac{\rho}{4}, \quad K\sqrt{\frac{4d}{c\rho}} \leq \frac{u_0}{2}. \quad (6.7)$$

Write $h = \mathbf{1} + Ay$ and $r = (-h)_+ = r_A(y)$; these vectors remain fixed throughout the construction. For each imposed set $T \supseteq S$, consider the equations $A_S z = r_S$ and $A_{T \setminus S} z = 0$. When they are solvable, let z_T be their minimum-norm solution. Whenever $|T| \leq (1 - \rho)N$, the surjection bound gives existence and

$$\|z_T\| \leq \frac{\|r_S\|}{c\sqrt{N}} \leq \frac{d}{c}. \quad (6.8)$$

The empty-set case is understood as $z_\emptyset = 0$.

Start at $T = S$. If

$$J = \{j \notin T : |A_j z_T| > u_0/4\}$$

is nonempty, add all of J to T , impose $A_J z = 0$, and solve again. Here is a count that verifies that every solve is permitted. If z' is the new solution, then $z' - z_T \in \ker A_T$ while z_T belongs to the row span of A_T . Thus

$$\|z'\|^2 - \|z_T\|^2 = \|z' - z_T\|^2 \geq \frac{\|A_J z_T\|^2}{K^2 N} > \frac{|J|u_0^2}{16K^2 N}.$$

Summing over completed additions and using (6.8) shows that they contain at most $L_d N$ rows altogether. The next, not yet imposed, set also has at most $L_d N$ rows, since

$$|J|u_0^2/16 < \|A_J z_T\|^2 \leq K^2 N \|z_T\|^2 \leq K^2 N (d/c)^2.$$

Consequently, before every proposed solve,

$$|T \cup J| \leq (1 - 2\rho)N + 2L_d N \leq (1 - 3\rho/2)N < (1 - \rho)N.$$

This proves the induction without assuming existence of the proposed solution. Each nonterminal step adds at least one row, so the procedure terminates after finitely many steps. At termination write T for the imposed set and $z = z_T$. In particular $|T| \leq (1 - \rho)N$ and $\|z\| \leq d/c$.

Put $y' = y + z$. On S its gaps are $1 + A_S y' = h_S + (-h_S)_+ = (h_S)_+ \geq 0$. On $T \setminus S$ the correction has zero image and the original gaps exceed u_0 . Every remaining row has original gap greater than u_0 and $|A_j z| \leq u_0/4$, so its new gap is at least $3u_0/4$. Thus y' is feasible in the ambient space.

Restore the norm. If $n = \|y'\| \geq 1$, set $\hat{y} = y'/n$. Each gap becomes

$$1 + A_\mu \hat{y} = (1 - n^{-1}) + n^{-1}(1 + A_\mu y') \geq 0.$$

If $n < 1$, take $W = \ker A_T \cap (y')^\perp$, whose dimension satisfies $D \geq N - |T| - 1 \geq \rho N - 1 \geq \rho N/2$ for large N . Apply Lemma 6.4 to the remaining rows restricted to W . There is a unit $v \in W$ such that

$$\|A_{T^c} v\|_\infty \leq K\sqrt{N/D} \leq K\sqrt{2/\rho}.$$

Here the infinity norm of an empty vector is zero. Set $\hat{y} = y' + \sqrt{1 - n^2} v$. Orthogonality gives $\|\hat{y}\| = 1$, and imposed gaps are unchanged. Since $n \geq 1 - \|z\| \geq 1 - d/c$, we have $1 - n^2 \leq 2d/c$. The absolute change of every remaining gap is at most $K\sqrt{4d/(c\rho)} \leq u_0/2$, leaving it at least $u_0/4$. This completes the repair. $\square$

Proposition 6.6 (Subextensive parent energy implies satisfiability). *Let $0 < \alpha < \alpha' \leq B_0$, and let the entries have any one of the laws ν_ε , $0 \leq \varepsilon \leq 1/2$. If*

$$\frac{1}{N} \mathbb{E} \min H_{\lfloor \alpha' N \rfloor \times N} \longrightarrow 0,$$

then a matrix with $\lfloor \alpha N \rfloor$ rows is satisfiable with probability tending to one.

Proof. Deleting rows serves one precise purpose: it reduces the number of small or negative gaps from at most slightly more than N to strictly fewer than N . The remaining rows can then be repaired using the surplus dimension in Lemma 6.5.

The constants are chosen in the following order. First fix $\eta \in (0, 1 - \alpha/\alpha')$, so that for all sufficiently large N the retained row count $m = \lfloor \alpha N \rfloor$ is at most $(1 - \eta)M$, where $M = \lfloor \alpha' N \rfloor$. Next choose $0 < \rho < \min(1/8, \eta/8)$ and obtain K, c, u_0 from Lemma 6.3 for the parent matrix. Finally fix $d > 0$ satisfying (6.7) and $d^2/u_0^2 \leq \rho/2$.

Choose a measurable approximate minimizer y of the parent matrix, with energy at most $\min H_A + N^{-1}$. Markov's inequality and the hypothesis show that $\|r_A(y)\| \leq d\sqrt{N}$ with probability tending to one. On the uniform geometric event, the set of problematic parent rows $S_{\text{par}} = \{\mu : 1 + A_\mu y \leq u_0\}$ has size at most $(1 + \rho)N$. Indeed, the rows with $|1 + A_\mu y| \leq u_0$ number at most $(1 + \rho/2)N$, while those with $1 + A_\mu y < -u_0$ number at most $\|r_A(y)\|^2/u_0^2 \leq \rho N/2$.

Independently of the parent matrix, retain a uniformly chosen set I of exactly m rows. Conditional on the parent and y , the random variable $X = |I \cap S_{\text{par}}|$ is hypergeometric, with

$$\mathbb{E}[X \mid A, y] \leq (1 - \eta)(1 + \rho)N \leq (1 - 7\rho)N.$$

Writing X as a sum of inclusion indicators gives, with $k = |S_{\text{par}}|$ and $r = m/M$,

$$\text{Var}(X \mid A, y) = kr(1 - r) \frac{M - k}{M - 1} \leq \frac{k}{4} \leq \frac{(1 + \rho)N}{4}.$$

Here the covariance of two distinct inclusion indicators is $-r(1 - r)/(M - 1)$; the formula is also immediate when $k = 0$. Chebyshev's inequality therefore gives

$$\mathbb{P}\{X > (1 - 2\rho)N \mid A, y\} \leq \frac{1 + \rho}{100\rho^2 N}.$$

Thus $X \leq (1 - 2\rho)N$ with probability tending to one, uniformly over the parent on the stated event.

The retained matrix inherits the parent's operator-norm and surjection bounds, and its residual at y is no larger than the parent residual. Lemma 6.5 makes it exactly feasible on the unit sphere. Finally, an independently chosen subset of rows of an iid matrix, listed in increasing order, has precisely the law of an iid $m \times N$ matrix. This proves the assertion for the desired row count. $\square$

6.4 The common threshold and subsequent ordered limits

Theorem 6.7 (Sharp common satisfiability threshold). *For Gaussian entries and, with the choice of B_0 above, for every fixed $0 \leq \varepsilon < \varepsilon_0(B_0)$, the number α_J in (6.3) satisfies $1 \leq \alpha_J < \infty$ and*

$$\lim_{N \rightarrow \infty} \mathbb{P}(\text{SAT}) = \begin{cases} 1, & 0 < \alpha < \alpha_J, \\ 0, & \alpha > \alpha_J. \end{cases}$$

In particular the threshold itself agrees for all these laws. No assertion at $\alpha = \alpha_J$ is needed here.

Proof. If $\alpha < \alpha_J$, choose a fixed $\alpha' \in (\alpha, \alpha_J)$. The expected parent ground energy is $o(N)$ for both laws by Lemma 6.1 and pressure universality. Proposition 6.6 proves satisfiability with probability tending to one.

For $\alpha_J < \alpha \leq B_0$, the same ground-energy limit is strictly positive. The map $A \mapsto R(A)$ is 1-Lipschitz in Frobenius norm: for every fixed unit y ,

$$\left| \|r_A(y)\| - \|r_{A'}(y)\| \right| \leq \|(A - A')y\| \leq \|A - A'\|_F,$$

and taking minima preserves this bound. Gaussian concentration gives $\text{Var } R(A) \leq C$ for Gaussian entries. For mixture entries use the coordinatewise Gaussian quantile representation from Section 5, whose Lipschitz constant is at most $2\sqrt{1+\varepsilon}$; the same variance bound follows with an absolute constant for $\varepsilon \leq 1/2$. Since

$$\frac{\mathbb{E}R(A)^2}{N} = 2 \frac{\mathbb{E} \min H_A}{N} \longrightarrow 2e(\alpha),$$

we obtain $\mathbb{E}R(A)/\sqrt{N} \rightarrow \sqrt{2e(\alpha)} > 0$ by subtracting the bounded variance. Chebyshev's inequality now implies

$$\mathbb{P}(\text{SAT}) = \mathbb{P}(R(A) = 0) \leq \frac{\text{Var } R(A)}{(\mathbb{E}R(A))^2} \longrightarrow 0.$$

For larger densities, retain the first $\lfloor \alpha_* N \rfloor$ rows at any fixed $\alpha_* \in (\alpha_J, B_0)$ and use monotonicity of satisfiability. $\square$

Corollary 6.8 (Transfer of the remaining ordered limits). *Fix $0 \leq \varepsilon < \varepsilon_0(B_0)$. On a full interval above the common threshold, the thermodynamic gap and normalized-force observables at every fixed positive temperature and fixed argument agree with their Gaussian counterparts. Consequently each further ordered limit in the problem exists for the mixture law if and only if it exists for the Gaussian law, and their values agree whenever it exists.*

Proof. The fixed-temperature row-marginal result in Section 5 gives equality of the inner $N \rightarrow \infty$ limits for every fixed β , argument, and $\alpha \in (\alpha_J, B_0]$. Limits of these identical functions, taken successively in the prescribed order, have identical existence and values. This assertion makes no exchange of limits; existence of the Gaussian zero-temperature and jamming limits is established in the following sections. $\square$

7 Zero temperature and mechanical bounds

The pressure formula supplies variational minimizers at each positive temperature. We now extract their zero-temperature limits and obtain the mechanical estimates that will control the subsequent approach to the threshold. There are two endpoints to distinguish: the diffusion immediately before the concentrated terminal layer, and the physical row gap after that layer. We first derive their relation. A supporting functional then identifies the negative part of the physical law with ground-minimizer residuals. Finally the contact count, spectral estimate, and small-gap bounds give control uniform as the density approaches the threshold. Section 8 will use that control to preserve mass and moments in the critical limit.

Throughout this section the rows are Gaussian and $\alpha_J < \alpha \leq B_0$ is fixed. Write $e = e(\alpha) > 0$. Constants with a subscript e may depend on this fixed density; constants called uniform are uniform on a fixed bounded interval of UNSAT densities adjacent to α_J . All assertions about a random matrix are in probability as $N \rightarrow \infty$, at fixed values of every cutoff occurring in the assertion. No uniqueness of the limiting variational parameters is assumed in this section.

7.1 Compactness of the variational parameters

Let q_β be any minimizer in (4.9). Denote its variance-time row solution by f_β ; thus, in the notation of (3.2),

$$(f_\beta)_t = -\frac{1}{2}((f_\beta)_{zz} + m_{q_\beta}(t)(f_\beta)_z^2), \quad f_\beta(1, z) = -\beta V(z), \quad f_\beta(0, 0) = \Phi_{-\beta V}(q_\beta).$$

Put

$$g_\beta = f_\beta/\beta, \quad \gamma_\beta(t) = \beta m_{q_\beta}(t), \quad \bar{\chi}_\beta(t) = \beta \chi_{q_\beta}(t), \quad q_{\beta,*} = \text{ess sup } q_\beta.$$

Thus $g_{\beta,z} \geq 0$, $-1 \leq g_{\beta,zz} \leq 0$, and $\gamma_\beta = \beta$ on $(q_{\beta,*}, 1)$. The next lemma shows that this last interval shrinks while its mass $\beta(1 - q_{\beta,*})$ stays bounded away from zero. The limiting scaled measure therefore retains an endpoint mass, denoted by Δ below; mass from just below $q_{\beta,*}$ can also contribute to it.

Lemma 7.1 (UNSAT compactness bounds). *For all sufficiently large β ,*

$$0 < c \leq \bar{\chi}_\beta(0) \leq C_e, \quad 0 < c_e \leq \beta(1 - q_{\beta,*}) \leq C_e. \quad (7.1)$$

Consequently every sequence $\beta \rightarrow \infty$ has a subsequence for which

$$\gamma_\beta(t) dt \implies \gamma(t) dt + \Delta \delta_1, \quad \bar{\chi}(t) = \Delta + \int_t^1 \gamma(v) dv, \quad L = \bar{\chi}(0),$$

where $\gamma \geq 0$ is nondecreasing and integrable and $\Delta \geq c_e$. The convergence of γ_β is in $L^1([0, T])$ for every $T < 1$, and $\bar{\chi}_\beta(t) \rightarrow \bar{\chi}(t)$ for $t < 1$.

Here γ_β and $\gamma(t)$ are variational profiles; the unindexed scalar γ in Theorem 1.1 denotes the gap exponent.

Proof. The entropy $E(q)$ is nonnegative. Since $P(-\beta V)/\beta \rightarrow -e$, at a minimizer $\alpha g_\beta(0, 0) \leq -e/2$ for large β . Jensen's inequality in the row recursion gives $g_\beta(0, z) \geq -\mathbb{E}V(z + G)$, where $G \sim \mathcal{N}(0, 1)$. Choose $R_e > 0$ so that $\mathbb{E}V(R_e + G) < e/(4\alpha)$. Concavity then implies $g_{\beta,z}(0, 0) \geq e/(4\alpha R_e)$. After dividing (4.10) by the appropriate powers of β ,

$$\alpha g_{\beta,z}(0, 0)^2 \bar{\chi}_\beta(0)^2 \leq 1,$$

which proves the upper bound for $\bar{\chi}_\beta(0)$. Conversely $E(q_\beta) \leq C\beta$, since the pressure is nonpositive and $\Phi_{-\beta V} \geq -\beta \mathbb{E}V(G)$. The integral version of (2.21), restricted to $[0, 1/2]$, gives

$$E(q_\beta) \geq \frac{1}{4\chi_{q_\beta}(0)} - \frac{\log 2}{2},$$

and hence the lower bound.

We shall also use the following consequences of the spin inverse relations:

$$\frac{1}{1 - q_{\beta,*}} = \frac{1}{\chi_{q_\beta}(0)} + \int_0^1 s dp_{q_\beta}(s), \quad q_{\beta,*} \leq \chi_{q_\beta}(0)^2 p_{q_\beta}(1-). \quad (7.2)$$

These follow directly from the continuous inverse relation (2.14). For brevity write $q = q_\beta$, $p = p_q$, and $\chi = \chi_q$. Stieltjes integration by parts and Fubini's theorem give

$$p(1-) = \int_0^{q_*} \frac{dt}{\chi(t)^2}, \quad \int_0^1 s dp(s) = p(1-) - \int_0^1 p(s) ds = \int_0^{q_*} \frac{m_q(t)}{\chi(t)^2} dt.$$

Since $\chi' = -m_q$ and $\chi(q_*) = 1 - q_*$, the last integral is $(1 - q_*)^{-1} - \chi(0)^{-1}$. Also $\chi(t) \leq \chi(0)$ in the first integral, so $p(1-) \geq q_*/\chi(0)^2$. This proves both relations, including general profiles and their top limits.

Let Z_t solve $dZ_t = m_{q_\beta}(t) f_{\beta,z}(t, Z_t) dt + dB_t$, $Z_0 = 0$, and put $\mathcal{P}_\beta(t) = \alpha \mathbb{E} f_{\beta,z}(t, Z_t)^2$. The derivative martingale and Itô's isometry imply $\mathcal{P}'_\beta(t) = \alpha \mathbb{E} f_{\beta,zz}(t, Z_t)^2 \leq \alpha \beta^2$, and

$p_{q_\beta}(s) = \mathcal{P}_\beta(q_\beta(s))$. The Stieltjes chain rule, including jumps by integration over the intervening variance interval, yields

$$\int s \, dp_{q_\beta}(s) \leq \alpha \beta^2 \int s \, dq_\beta(s) \leq \alpha \beta^2 \chi_{q_\beta}(0).$$

Thus the first relation in (7.2) gives

$$\frac{1}{\beta(1 - q_{\beta,*})} \leq \frac{1}{\bar{\chi}_\beta(0)} + \alpha \bar{\chi}_\beta(0).$$

The reverse endpoint bound follows from $1 - q_{\beta,*} \leq \chi_{q_\beta}(0)$.

Finally, bounded mass and monotonicity give subsequential weak compactness of $\gamma_\beta \, dt$. On every compact interval before 1, monotonicity bounds the densities uniformly and gives L^1 compactness. Any singular mass is therefore at 1. The terminal interval has mass $\beta(1 - q_{\beta,*}) \geq c_e$, so the endpoint atom has mass at least c_e . $\square$

Proposition 7.2 (Zero-temperature variational principle). *For every subsequential limit in Lemma 7.1, let g be the solution of*

$$g_t = -\frac{1}{2}(g_{zz} + \gamma(t)g_z^2), \quad g(1, z) = -\frac{V(z)}{1 + \Delta}.$$

It satisfies

$$\mathcal{E}_\alpha(\gamma, \Delta) := \alpha g(0, 0) + \frac{1}{2} \int_0^1 \frac{dt}{\bar{\chi}(t)} = -e, \quad (7.3)$$

and minimizes this functional over nonnegative nondecreasing integrable γ and $\Delta > 0$. Here solutions have quadratic growth, nonnegative first derivative, and second derivative in $[-1, 0]$; an integrable time coefficient is interpreted by approximation in L^1 .

Proof. We give the endpoint comparison explicitly. For $t < 1$, set

$$U_\beta(t, z) = \sup_{w \in \mathbb{R}} \left\{ -V(w) - \frac{(z - w)^2}{2\bar{\chi}_\beta(t)} \right\} = -\frac{V(z)}{1 + \bar{\chi}_\beta(t)}.$$

At $t = 1$ use the continuous extension $U_\beta(1, z) = -V(z)$. This solves $U_t = -\gamma_\beta U_z^2/2$ and has curvature in $[-1, 0]$. Subtraction from the viscous equation for g_β produces a linear equation with drift $\gamma_\beta(g_{\beta,z} + U_{\beta,z})/2$ and a source of absolute value at most $1/2$. Its diffusion representation, or comparison after smoothing the hinge, gives

$$\sup_z |g_\beta(t, z) - U_\beta(t, z)| \leq (1 - t)/2, \quad \sup_z |g_{\beta,z}(t, z) - U_{\beta,z}(t, z)| \leq C\sqrt{1 - t}. \quad (7.4)$$

For the gradient bound use the value bound and the common Lipschitz bound on the two gradients, comparing difference quotients with step $\sqrt{1 - t}$. The identical comparison holds for g , with $\bar{\chi}$ in place of $\bar{\chi}_\beta$.

On $[0, T]$, subtraction of two viscous equations again gives a linear equation. Its source is bounded by the coefficient difference times $C(1 + z^2)$, and its drift has a spatial Lipschitz constant bounded by an integrable function with uniformly bounded integral. The associated diffusions have uniform moments of every fixed order, by Gronwall's inequality. The diffusion representation therefore proves stability under L^1 coefficient convergence and local uniform terminal convergence with a common quadratic growth bound. Applying this at $T < 1$, using (7.4), and then sending $T \uparrow 1$, proves local uniform convergence of values and gradients on compact time intervals before 1. The same argument constructs, and proves uniqueness of, the stated solution for an integrable γ .

For the entropy, (2.21) gives

$$\frac{E(q_\beta)}{\beta} = \frac{1}{2} \int_0^{q_{\beta,*}} \frac{dt}{\bar{\chi}_\beta(t)} + \frac{\log(1 - q_{\beta,*})}{2\beta}.$$

On this integration interval the denominator is at least $\beta(1 - q_{\beta,*}) \geq c_e$. Dominated convergence applies, and the logarithmic term is $O_e(\log \beta/\beta)$. This proves (7.3).

For an arbitrary trial pair, use the finite-temperature CDF $m_\beta(t) = \min\{\gamma(t)/\beta, 1\}$ until $1 - \Delta/\beta$, and set $m_\beta = 1$ thereafter. This is nondecreasing and its scaled measure converges to $\gamma dt + \Delta\delta_1$; integrability of γ ensures that its mass on the shrinking terminal interval vanishes. The comparisons just proved show convergence of its row term and entropy. The finite-temperature minimum is at most this trial value, so $-e \leq \mathcal{E}_\alpha(\gamma, \Delta)$. $\square$

7.2 Endpoint laws and the negative marginal

Proposition 7.3 (Subsequential endpoint law). *Let $Z_0 = 0$ and $dZ_t = \gamma(t)g_z(t, Z_t) dt + dB_t$, and put $H = 1 + Z_1$. Along the subsequence defining (γ, Δ) , the tilted row gap converges in distribution to*

$$h^* = \begin{cases} H, & H \geq 0, \\ H/(1 + \Delta), & H < 0. \end{cases} \quad (7.5)$$

Every fixed moment of the negative part converges. Both H and h^ are atomless and assign positive probability to every nonempty open interval. This statement is subsequential; uniqueness is addressed separately below and in Section 8.*

Proof. On each constant-CDF interval the row tilt is the Gaussian Doob transform. Composing these transforms, then approximating a general CDF, identifies its physical gap with $1 + Z_1^\beta$, where

$$Z_t^\beta = B_t + \int_0^t \gamma_\beta(v)g_{\beta,z}(v, Z_v^\beta) dv.$$

Couple Z^β and Z with the same Brownian motion. Before a fixed time $T < 1$, the coefficient and gradient convergence above and Gronwall's inequality imply convergence in expected supremum norm. On $[T, 1]$, replace the gradient by $U_{\beta,z}$. The error has expected size at most $C_e\sqrt{1-T}$, by (7.4) and the uniform bound on $\int \gamma_\beta$; the Brownian increment has the same bound. The resulting deterministic flow fixes positive gaps and multiplies a negative gap between times T and t by $(1 + \bar{\chi}_\beta(t))/(1 + \bar{\chi}_\beta(T))$. First send $\beta \rightarrow \infty$, and then $T \uparrow 1$. The limiting integrable drift on $[T, 1]$ has vanishing total mass, whereas the total mass concentrating near time 1 in $\gamma_\beta(t) dt$ is Δ . Since $\bar{\chi}_\beta(1) = 0$ and $\bar{\chi}_\beta(T) \rightarrow \bar{\chi}(T)$, the displayed flow factor tends to $(1 + \Delta)^{-1}$ as $\beta \rightarrow \infty$ and then $T \uparrow 1$. This proves (7.5). Since the drift is nonnegative, the negative part of the gap is bounded by $(1 + B_1)_-$; this gives uniform integrability of every moment of the negative part.

For atomlessness, condition on the Brownian bridge $\tilde{B}_t = B_t - tB_1$. The map from $b = B_1$ to Z_1 is increasing and has slope between $\exp(-\int \gamma)$ and 1. Indeed its variation satisfies $J'(t) = 1 + \gamma(t)g_{zz}(t, Z_t)J(t)$, $J(0) = 0$. Approximation proves the same bounds when derivatives exist only almost everywhere. The conditional endpoint law is thus the pushforward of a Gaussian under a globally bi-Lipschitz map. That map is onto $\mathbb{R}$, so the law has a bounded density and assigns positive probability to every nonempty open interval. The piecewise linear map in (7.5) preserves these properties. $\square$

The variable H is the pre-layer gap; h^* is the gap sampled by the microscopic observable. In particular their signs agree, but a negative gap is reduced by the factor $1 + \Delta$. Define

$$p = \mathbb{P}(h^* < 0) = \mathbb{P}(H < 0), \quad \mu = \mathbb{E}(h^*)_-, \quad z_c = \alpha p.$$

The parameter z_c will be identified with the number of active constraints per spin coordinate. That identification requires a small-residual estimate: convergence against tests supported away from zero alone would miss constraints whose negative gaps tend to zero.

Proposition 7.4 (Identification of negative gaps). *For every subsequential endpoint law in Proposition 7.3, the restriction of the law of h^* to $(-\infty, 0)$ is the same. For $\psi \in C_c^\infty((-\infty, 0))$, uniformly over Gaussian ground minimizers y_N ,*

$$\frac{1}{N} \sum_{i=1}^M \psi(1 + A_i y_N) \longrightarrow \alpha \mathbb{E} \psi(h^*) \quad \text{in probability.}$$

Moreover,

$$\alpha \mathbb{E}(h^*)_-^2 = 2e. \quad (7.6)$$

Proof. We perturb the loss in a compact negative-gap interval. For either sign of the perturbation, the pressure trial gives a lower support for the ground energy. Evaluating the perturbed energy at an unperturbed minimizer gives the opposite inequality, and the two one-sided slopes identify its empirical test average. The perturbation interval must remain fixed as $\beta \rightarrow \infty$.

Let ψ be smooth with support in a compact negative-gap interval $[-R, -a]$, $a > 0$, and put $W_w(z) = V(z) + w\psi(1+z)$. Choose $w_0 > 0$ so that

$$w_0 \|\psi''\|_\infty \leq \frac{1}{2}, \quad w_0 \|\psi'\|_\infty \leq a/2, \quad w_0 \|\psi\|_\infty \leq a^2/4.$$

For $|w| \leq w_0$, W_w is convex, nonnegative, and nonincreasing, and agrees with V outside that compact interval. In particular $-\beta W_w$ is concave for every $\beta > 0$. This is a fixed interval of loss perturbations, independent of β .

Here only a trial upper bound is needed. The Gaussian interpolation with smooth concave tangent caps, followed by the spin conjugate inequality (Corollary 4.6), gives for every $q_* < 1$

$$\limsup_{N \rightarrow \infty} P_N(-\beta W_w) \leq \alpha \Phi_{-\beta W_w}(q) + E(q). \quad (7.7)$$

This uses Corollary 3.7, namely ordinary profile concavity for a concave terminal, not the small-terminal-perturbation radius in (4.11). The tangent caps have the required concavity, and removal of the cap is valid at each fixed β, w , because W_w has the same quadratic tail as V . Neither strict profile concavity nor identification of a perturbed minimizer is required for (7.7).

For a base limiting pair (γ, Δ) , let g_w solve the PDE with coefficient γ and terminal value

$$g_w(1, z) = \sup_{v \in \mathbb{R}} \left\{ -W_w(v) - \frac{(z-v)^2}{2\Delta} \right\}, \quad S(w) = -\alpha g_w(0, 0) - \frac{1}{2} \int_0^1 \frac{dt}{\bar{\chi}(t)}.$$

Write $\hat{e}_N(w) = N^{-1} \min_{\|y\|=1} \sum_i W_w(A_i y)$ and $e_N(w) = \mathbb{E} \hat{e}_N(w)$. The surface-cap pressure-energy estimate applies to W_w , whose curvature and growth bounds are uniform for $|w| \leq w_0$. Use the recovery profiles of Proposition 7.2 in (7.7), take $N \rightarrow \infty$, and then $\beta \rightarrow \infty$. For completeness, the endpoint comparison for this nonquadratic loss uses

$$U_{\beta, w}(t, z) = \sup_v \left\{ -W_w(v) - \frac{(z-v)^2}{2\bar{\chi}_\beta(t)} \right\}, \quad t < 1,$$

with continuous terminal value $U_{\beta, w}(1, z) = -W_w(z)$. Both $U_{\beta, w}$ and the scaled viscous solution $g_{\beta, w}$ have curvature in $[-C, 0]$, uniformly for $|w| \leq w_0$, and $U_{\beta, w, t} = -\gamma_\beta U_{\beta, w, z}^2/2$. The difference equation has forcing bounded by $C/2$, so

$$\sup_z |g_{\beta, w}(t, z) - U_{\beta, w}(t, z)| \leq C(1-t), \quad \sup_z |g_{\beta, w, z}(t, z) - U_{\beta, w, z}(t, z)| \leq C\sqrt{1-t}.$$

Recovery has $\int \gamma_\beta = L + o(1)$; hence its drift Lipschitz constants have bounded integrals and its diffusions have uniform moments. Compact-time L^1 coefficient stability, followed by $t \uparrow 1$, gives exactly the terminal supremum defining g_w , and the entropy converges as before. We obtain

$$\liminf_{N \rightarrow \infty} e_N(w) \geq S(w), \quad S(0) = e. \quad (7.8)$$

No existence of a perturbed thermodynamic ground-energy limit is being asserted or used.

The function S is differentiable at zero. The quadratic penalization in the endpoint supremum is strictly concave in v , so the optimizer is unique; its envelope derivative is minus ψ at the proximal gap. Linearization of the PDE propagates this bounded terminal derivative along the tilted diffusion. The coefficient stability follows from the difference equation used above. Hence

$$S'(0) = \alpha \mathbb{E} \psi(h^*). \quad (7.9)$$

For every base ground minimizer y_N , let $T_N = N^{-1} \sum_i \psi(1 + A_i y_N)$. Direct evaluation gives

$$\widehat{e}_N(w) \leq \widehat{e}_N(0) + w T_N.$$

For each fixed w , the centered ground energy $\widehat{e}_N(w) - e_N(w)$ tends to zero in probability. Indeed on $\|A\| \leq K\sqrt{N}$ its matrix Lipschitz constant is $C_w/\sqrt{N}$, uniformly over spins; extend Lipschitz and use Gaussian concentration, and then control the complement by the quadratic growth bound. Thus (7.8) and the known convergence $\widehat{e}_N(0) \rightarrow e$ imply, for fixed $w > 0$ and $v < 0$, uniformly over all base minimizers,

$$\frac{S(w) - e}{w} - o_{\mathbb{P}}(1) \leq T_N \leq \frac{S(v) - e}{v} + o_{\mathbb{P}}(1).$$

Sending $w \downarrow 0$ and $v \uparrow 0$ proves $T_N \rightarrow S'(0)$ in probability. Apply this argument to any two base limiting pairs: the same empirical tests have the two asserted limits, so their derivatives agree. Smooth compact negative tests determine the restriction of a measure to $(-\infty, 0)$, proving the asserted uniqueness and, at the same time, microscopic negative-test convergence.

For the energy identity use $W_w = (1 + w)V$, $|w| < 1/2$. The same trial-support argument applies, while now $\widehat{e}_N(w) = (1 + w)\widehat{e}_N(0)$ identically. The derivative squeeze therefore gives $e = S'(0) = \alpha \mathbb{E} V(h^* - 1)$, proving (7.6). Quadratic growth in this differentiation is justified by the uniform diffusion moments. $\square$

The first moment of the negative part also converges at microscopic ground minimizers: outside residual size R , its empirical contribution is at most $2H_{\min}/(MR)$. Together with the compact negative-test convergence just proved, this controls the far tail. For any subsequential thermal limit at fixed α , weak convergence to (7.5), atomlessness at zero, and first-moment convergence identify the force CDF as

$$F_{\alpha, \infty}(s) = \frac{1}{p} \mathbb{P} \left(0 < (h^*)_- \leq s \frac{\mu}{p} \right), \quad s > 0.$$

The conditional mean residual is μ/p ; $p, \mu > 0$ by (7.6). Proposition 7.4 makes this expression independent of the temperature subsequence. Identification with ground-minimizer force statistics additionally uses the active-mass cutoff proved in Lemma 7.7. Positive gap tests will be identified uniquely only in Section 8.

7.3 Support stability

Lemma 7.5 (Endpoint stability). *For every subsequential zero-temperature optimizer, the contact density parameter $z_c = \alpha \mathbb{P}(H < 0)$ satisfies*

$$z_c \leq (1 + 1/\Delta)^2. \quad (7.10)$$

Proof. Let Z^β denote the finite-temperature tilted diffusion. At finite temperature vary the overlap distribution, with CDF m , by moving mass to a point below 1. Linearization of the PDE and (2.21) gives, for variations supported below a cutoff $T < 1$,

$$\delta(\alpha\Phi + E) = \frac{1}{2} \int_0^T \delta m(t) J(t) dt, \quad J(t) = \mathcal{P}_\beta(t) - \int_0^t \frac{dv}{\chi_{q_\beta}(v)^2}.$$

Let ν be the minimizing overlap distribution, choose $q_{\beta,*} < T < 1$, and fix $s \in [0, T)$. The feasible direction $\nu_\varepsilon = (1 - \varepsilon)\nu + \varepsilon\delta_s$ has $m_\varepsilon(t) - m(t) = \varepsilon(\mathbf{1}_{\{s \leq t\}} - m(t))$ for $t \leq T$, and vanishes for $t > T$. With $K(s) = \int_s^T J(t) dt$, Fubini's theorem makes its directional derivative

$$\frac{1}{2} \left(K(s) - \int K(r) \nu(dr) \right) \geq 0.$$

Therefore K is at least its ν -average everywhere on $[0, T)$, and equals that average on the support, by continuity. Thus every support point in $(0, T)$ is a local minimum of K . Since $K' = -J$ and $K'' = -J'$, at each such point

$$\alpha \mathbb{E} f_{\beta,zz}(s, Z_s^\beta)^2 \chi_{q_\beta}(s)^2 \leq 1.$$

The derivatives J' and K'' are continuous before time 1, including across jumps of m . Indeed the mild heat equation and the local spatial Lipschitz bound on $f_{\beta,z}^2$ give joint continuity of $f_{\beta,zz}$ away from the terminal time; the second derivative of the heat kernel applied to a Lipschitz function has an integrable time singularity. The diffusion moment formula then gives continuity of $\mathcal{P}'_\beta$, and explicitly

$$J'(s) = \alpha \mathbb{E} f_{\beta,zz}(s, Z_s^\beta)^2 - \chi_{q_\beta}(s)^{-2}.$$

No continuity of $f_{\beta,t}$ at a jump of m is asserted.

Select continuity times $t_n \uparrow 1$ with $(1 - t_n)\gamma(t_n) \rightarrow 0$; this follows from integrability and monotonicity. A diagonal choice $t_\beta \uparrow 1$ ensures $t_\beta < q_{\beta,*}$, $\bar{\chi}_\beta(t_\beta) \rightarrow \Delta$, and $(1 - t_\beta)\gamma_\beta(t_\beta) \rightarrow 0$. Let s_β be the next overlap-support point at or above t_β . The CDF is constant between these points, so

$$\bar{\chi}_\beta(t_\beta) - \bar{\chi}_\beta(s_\beta) = \gamma_\beta(t_\beta)(s_\beta - t_\beta) \leq (1 - t_\beta)\gamma_\beta(t_\beta) \rightarrow 0.$$

Consequently $\bar{\chi}_\beta(s_\beta) \rightarrow \Delta$. Moreover, for fixed $T < 1$,

$$\int_T^{s_\beta} \gamma_\beta(t) dt \rightarrow \int_T^1 \gamma(t) dt.$$

The drift over this interval is negligible after $T \uparrow 1$, so $Z_{s_\beta}^\beta$ converges to the pre-atom endpoint Z_1 .

By (7.4), $g_{\beta,z}(s_\beta, z) \rightarrow (-1 - z)_+/(1 + \Delta)$ locally uniformly. These gradients are convex in z : differentiating the PDE three times gives a linear equation preserving the nonnegative third spatial derivative of a smoothed hinge terminal, and passage to the limit preserves convexity. Secant bounds for convex functions therefore give locally uniform convergence of their derivatives on each compact interval avoiding $z = -1$: the limit derivative is continuous there, and a finite secant mesh gives a uniform bound. Since these derivatives are bounded and Z_1 is atomless, the finite-temperature stability inequality passes to

$$\alpha \frac{\Delta^2}{(1 + \Delta)^2} \mathbb{P}(Z_1 < -1) \leq 1,$$

as asserted. □

7.4 Mechanical contacts and uniform cutoff estimates

We now relate these variational quantities to finite-dimensional minimizers. Three separate facts are needed. A positive-energy minimizer has at least N strictly violated rows and almost surely no row at exactly zero gap. A uniform small-residual bound then prevents an extensive number of those violated rows from disappearing at zero in the large- N limit. A different bound, proved on near-ground configurations later in the section, controls small positive gaps in the prescribed thermal order. Write $\Pi_y = I - yy^T$, $r_i(y) = (-1 - A_i y)_+$, and $\nabla_S H(y) = \Pi_y \nabla H(y)$ on the unit sphere.

Lemma 7.6 (Mechanical contact count). *Almost surely, every positive-energy Gaussian ground minimizer has no zero gap outside its active set $C = \{i : r_i > 0\}$, and $k = |C| \geq N$. At such a minimizer,*

$$A_C^T(A_C y + \mathbf{1}) = \lambda y, \quad \lambda = \sum_{i \in C} r_i(1 + r_i) > 0, \quad v^T(A_C^T A_C - \lambda I)v \geq 0 \quad (v \perp y). \quad (7.11)$$

Proof. For a fixed C , Gaussian genericity gives simple nonzero singular values of A_C and a nonzero component of $A_C^T \mathbf{1}$ in each corresponding right singular direction. These properties hold almost surely, since their exceptional sets are proper algebraic sets. Stationarity implies the first identity in (7.11); its scalar product with y gives $\lambda > 0$. The genericity just noted precludes λ from being an eigenvalue of $A_C^T A_C$. Thus the stationary vector is obtained by inversion. Its unit-norm equation is rational in λ , is not identically satisfied (it tends to zero at infinity), and has finitely many roots. Conditional on A_C , any row outside C almost surely avoids gap zero at all these finitely many vectors. A union over the finitely many sets C proves the first assertion, even if such a row was initially allowed to have zero gap. The energy is now smooth near the minimizer, so its constrained Hessian is nonnegative on the tangent space. Finally stationarity with $\lambda > 0$ puts y in the row span of A_C . A nontrivial kernel of A_C would be tangent and would have Hessian value $-\lambda$, a contradiction. Hence $\text{rank } A_C = N$, and $k \geq N$. $\square$

We record the common counting estimate behind the cutoff bounds. Work on $\|A\| \leq K\sqrt{N}$, whose probability tends to one for fixed sufficiently large $K = K(B_0)$. Both the residual map and the tangent gradient obey

$$\|r(y) - r(y')\|_2 \leq K\sqrt{N}\|y - y'\|, \quad \|\nabla_S H(y) - \nabla_S H(y')\| \leq CN\|y - y'\|. \quad (7.12)$$

At a deterministic unit vector y_0 , condition on all scalar projections $A_i y_0$. The tangent row components remain independent standard Gaussians, and therefore

$$\nabla_S H(y_0) \mid (A_i y_0)_i \sim \mathcal{N}(0, \|r(y_0)\|_2^2 I_{N-1}).$$

For $\|r(y_0)\|_2 \geq t\sqrt{N}/2$, its small-ball bound is

$$\mathbb{P}(\|\nabla_S H(y_0)\| \leq CaN \mid (A_i y_0)_i) \leq (Ca/t)^{N-1}. \quad (7.13)$$

A deterministic net of mesh proportional to a has at most $(C/a)^N$ points. Thus a projection event of probability at most $C^N t^k w^j$, involving specified row sets, has union probability at most

$$a^{-1} C^N t^{k-N+1} w^j \quad (7.14)$$

provided the original projection event transfers to that event at a net point, and the configurations satisfy $\|\nabla_S H\| \leq CaN$ and have residual size in $[t\sqrt{N}, 2t\sqrt{N}]$. Choose the mesh at most $t/(2K)$; then (7.12) ensures the required residual lower bound $t\sqrt{N}/2$ at the net

point. The factor t^{k-N+1} records the useful cancellation: forcing k active projections into a ball of radius $O(t\sqrt{N})$ contributes t^k , while making the tangent gradient small costs $t^{-(N-1)}$. With at least N active rows, no negative power growing with N remains as $t \downarrow 0$. Here all choices of row subsets, at most 3^M , have been absorbed into C^N . The mesh $a > 0$ is chosen after t, w and every other cutoff, and is then held fixed as $N \rightarrow \infty$. In particular the factor a^{-1} does not affect exponential decay. The operator-norm event is used only to transfer a configuration to the net, never in the conditional Gaussian probability calculation.

Lemma 7.7 (Negative cutoffs). *There are constants $c, C > 0$, independent of the density, such that $\alpha\mu \geq c\sqrt{e}$, where $\mu = \mathbb{E}(h^*)_-$. At each fixed UNSAT density, uniformly over Gaussian ground minimizers,*

$$\frac{k}{N} \rightarrow z_c \quad \text{in probability.}$$

There is $s_0 > 0$, independent of the density, such that their limiting mean-one force law obeys $F_{\alpha,\infty}(s) \leq C/|\log s|$ for all $0 < s < s_0$, uniformly as $\alpha \downarrow \alpha_J$. Furthermore $z_c \rightarrow 1$ in this limit, and every weak subsequential jamming force law has total mass one and no atom at zero.

Proof. Set $t = \sqrt{e}$. Since $H_{\min}/N \rightarrow e$, with probability tending to one,

$$\frac{3e}{4} \leq \frac{H_{\min}}{N} \leq \frac{5e}{4}, \quad t\sqrt{N} \leq \|r\|_2 \leq 2t\sqrt{N}.$$

Thus a single residual annulus suffices at every fixed density. Moreover $t \leq \sqrt{B_0}$, since $e(\alpha) \leq \alpha \leq B_0$. All exponential bases in the counting estimates can consequently be chosen independently of the density; the net mesh may still depend on its fixed value of t . If $\|r\|_1 \leq c_1 tN$, transfer to a net of sufficiently small mesh so that the same active set C has $\sum_{i \in C} |1 + A_i y_0| \leq 2c_1 tN$. The joint projection density is bounded by an absolute constant to the power k , and the k -dimensional cross-polytope volume is $2^k (2c_1 tN)^k / k!$. Since $k \geq N$, the projection probability is at most $(C c_1 t)^k$. Combining with (7.13) and the net gives $a^{-1} C^N c_1^k t^{k-N+1}$, which is exponentially small for a uniformly small c_1 , because $t \leq \sqrt{B_0}$ and $N \leq k \leq B_0 N$. Thus $\|r\|_1 \geq c_1 tN$; the previously established first-moment convergence proves the bound for μ .

Suppose at least ηN active residuals are at most st , where $0 < \eta < 1/2$. Choose $a \ll st\sqrt{\eta}/K$. By the squared projection displacement bound, at least $j = \lfloor \eta N/2 \rfloor$ corresponding net gaps have absolute value at most $2st$. The remaining coordinates of C lie in a Euclidean ball of radius $Ct\sqrt{N}$. Its volume, together with the j intervals, bounds the projection probability by $C^N t^k s^j$; here $k - j \geq N/2$ bounds the ball's dimension factor. Equation (7.14) excludes this event for $s \leq \exp(-C'/\eta)$, with C' uniform. At fixed $e > 0$, this estimate and convergence of compact negative tests show that no extensive active mass is lost at zero, proving $k/N \rightarrow \alpha \mathbb{P}(h^* < 0) = z_c$.

The mean residual among active rows is at most $\|r\|_2/\sqrt{k} \leq 2t$. A normalized force at most s therefore has residual at most Cst . The preceding estimate, divided by $k \geq N$, gives $F_{\alpha,\infty}(s) \leq C/|\log s|$. The mean-one laws are tight by Markov's inequality; in fact their second moments are uniformly bounded, since $k\|r\|_2^2/\|r\|_1^2 \leq C$. Their weak limits therefore retain mean one. The displayed logarithmic bound prevents any subsequential limit from putting mass at zero.

For the contact density, use the uniform all-spin near-gap estimate in Lemma 6.3. For each fixed $\rho > 0$, it supplies $u_0 > 0$ such that every ground minimizer satisfies

$$k \leq (1 + \rho)N + 2H_{\min}/u_0^2.$$

Take $N \rightarrow \infty$, then $\alpha \downarrow \alpha_J$, and finally $\rho \downarrow 0$. This gives $\limsup z_c \leq 1$, while Lemma 7.6 gives $z_c \geq 1$. $\square$

7.5 A uniform spectral bound

The precise Marčenko–Pastur input needed here is the following Gaussian special case [15, Theorem 1 and Example 1, pp. 459–461]: if $k_N/N \rightarrow z \geq 1$ and A_N is a k_N -by- N matrix of independent standard Gaussians, the empirical law of the eigenvalues of $A_N^T A_N/N$ converges weakly in probability to the probability measure with density

$$\frac{\sqrt{(b_z - x)(x - a_z)}}{2\pi x} \mathbf{1}_{[a_z, b_z]}(x), \quad a_z = (\sqrt{z} - 1)^2, \quad b_z = (\sqrt{z} + 1)^2.$$

In particular every interval immediately to the right of a_z has positive limiting mass. No extreme-eigenvalue theorem is needed.

Lemma 7.8 (Spectral and susceptibility bounds). *For every $\varepsilon > 0$, with probability tending to one, simultaneously for all row subsets C with $N \leq |C| \leq B_0 N$, at least two singular values of $A_C/\sqrt{N}$ are at most $\sqrt{|C|/N} - 1 + \varepsilon$. Consequently*

$$2e + \alpha\mu \leq (\sqrt{z_c} - 1)^2 \leq \Delta^{-2}, \quad \Delta \leq Ce^{-1/4}, \quad \frac{1}{\sqrt{2e}} \leq L \leq \frac{C}{\sqrt{e}}. \quad (7.15)$$

In particular $\Delta/L \rightarrow 0$ as $\alpha \downarrow \alpha_J$.

Proof. Fix $\varepsilon > 0$. For each aspect ratio in $[1, B_0]$, choose a nonnegative Lipschitz function of the singular value, supported strictly below $\sqrt{z} - 1 + \varepsilon$, with positive integral under the limiting singular-value law. By continuity of the spectral edges and density, each fixed test stays below the cutoff and retains a positive limiting integral throughout the closure of a sufficiently small ratio neighborhood. A finite collection of these neighborhoods covers the compact ratio interval, giving a common positive lower bound for the relevant limiting integrals. Convergence of their expectations is uniform within the chosen neighborhoods: otherwise a violating sequence of ratios would have a convergent subsequence and contradict the stated spectral limit. The Frobenius singular-value inequality gives

$$A \mapsto \frac{1}{N} \sum_{i=1}^N \varphi(s_i(A/\sqrt{N})) \quad \text{Lipschitz constant at most } \text{Lip}(\varphi)/N.$$

Gaussian concentration therefore bounds a fixed positive downward deviation by $2 \exp(-c_\varepsilon N^2)$. Union over at most $2^{B_0 N}$ subsets still has probability tending to zero. A positive normalized test average implies a positive multiple of N singular values below the cut, hence at least two for large N .

The span of two corresponding right singular vectors intersects $y^\perp$. The Hessian inequality (7.11) therefore yields

$$\frac{\lambda}{N} \leq (\sqrt{k/N} - 1 + \varepsilon)^2.$$

Since $\lambda/N = 2H_{\min}/N + \|r\|_1/N$, first take $N \rightarrow \infty$, then $\varepsilon \downarrow 0$, and use the contact and moment convergence. The upper bound Δ^{-2} follows from (7.10) and $z_c \geq 1$. The estimate for μ in Lemma 7.7 then gives $\Delta \leq Ce^{-1/4}$.

The second inequality in (7.2), together with $p_{q_\beta}(1-) \leq \alpha \mathbb{E} f_{\beta,z}(1, Z_1^\beta)^2$, passes to $1 \leq L^2 \alpha \mathbb{E}(h^*)_-^2 = 2eL^2$. Finally the derivative martingale identifies $g_{\beta,z}(0,0) \rightarrow \mu$, so (4.10) implies $\alpha\mu^2 L^2 \leq 1$. The lower bound for μ proves the upper bound for L . Dividing the two estimates proves the final assertion. This conclusion concerns the thermal endpoint atom; it does not preclude additional layers concentrating at the endpoint in the subsequent density limit. $\square$

Corollary 7.9 (Cutoffs after susceptibility normalization). *For every subsequential endpoint law, put $R = L(h^*)_-$ and $p = \mathbb{P}(h^* < 0)$. There are constants $c, C, u_0 > 0$, independent of the density, such that throughout the fixed UNSAT interval and for $0 < u < u_0$,*

$$c \leq \mathbb{E}R \leq C, \quad \mathbb{P}(0 < R \leq u) \leq \frac{C}{|\log u|} \quad \alpha p \longrightarrow 1 \quad (\alpha \downarrow \alpha_J).$$

Proof. The bounds $L \geq (2e)^{-1/2}$, $\alpha\mu \geq c\sqrt{e}$, and $\alpha\mu^2 L^2 \leq 1$ give the two bounds for $\mathbb{E}R = L\mu$. Moreover $p = z_c/\alpha$ is bounded away from zero because $z_c \geq 1$ and $\alpha \leq B_0$. Thus the conditional mean $\mathbb{E}[R \mid R > 0] = L\mu/p$ is bounded above and below. Apply Lemma 7.7 to the corresponding mean-one force to obtain the displayed small-value bound. The final assertion is $z_c \rightarrow 1$. $\square$

7.6 Positive gaps in the prescribed temperature order

Proposition 7.10 (Positive-gap cutoffs). *Constants $c, C, u_0 > 0$, independent of the density, exist such that, for $0 < \delta < u < 1$, $G_{N,\beta,\alpha}(\delta, u) \geq c(u - \delta)$. For every fixed UNSAT density in the interval under consideration and every $0 < u < u_0$,*

$$\limsup_{\beta \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \left\langle \frac{1}{M} \sum_{i=1}^M \mathbf{1}_{\{0 < h_i \leq u\}} \right\rangle \leq \frac{C}{|\log u|} \quad (7.16)$$

with the same constants C, u_0 at every such density. In particular every subsequential endpoint law in (7.5) satisfies the direct bound $\mathbb{P}(0 < h^ \leq u) \leq C/|\log u|$. The same estimate bounds the ordered zero-temperature and density limsups of $G_{N,\beta,\alpha}(\delta, u)$, uniformly in $0 < \delta < u$. The upper bound requires no identification of positive gap tests at exact ground minimizers.*

Proof. Remove a tested row. Its insertion denominator is at most one, whereas its penalty vanishes on the tested positive interval. Conditional on the cavity spin, the removed projection is standard Gaussian. Its density is bounded below on $[-1, 0]$, which proves the lower bound. The same proof works for the small mixtures, conditional on their variances, with a uniform lower density bound.

For the upper bound, exact ground minimizers alone are insufficient: the observable first takes $N \rightarrow \infty$ at positive temperature. We therefore prove the counting bound throughout a thin energy sublevel and only then concentrate the Gibbs measure on it. The needed replacement for the exact contact count $k \geq N$ is a lower bound $k \geq (1 - 2\eta')N$ on that whole sublevel. We derive it before counting the positive gaps.

Fix the UNSAT density and a tolerance $\rho > 0$. On the good operator-norm event the surface-cap argument implies that, for all sufficiently large β , Gibbs mass outside

$$\mathcal{L}_\rho = \{y : H(y) \leq H_{\min} + \rho N\}$$

is exponentially small in N . Along any great circle the energy derivative has Lipschitz constant CN . Descending in the negative tangent-gradient direction therefore gives

$$\|\nabla_S H(y)\| \leq C\sqrt{\rho} N \quad (y \in \mathcal{L}_\rho). \quad (7.17)$$

Indeed a derivative of magnitude d permits an energy decrease at least cd^2/N , contradicting near-minimality if $d > C\sqrt{\rho}N$.

Set $t = \sqrt{e}$ and require $\rho \leq e/4$. On the event $H_{\min}/N \in [3e/4, 5e/4]$, every point of $\mathcal{L}_\rho$ then satisfies $t\sqrt{N} \leq \|r(y)\|_2 \leq 2t\sqrt{N}$. Choose from the outset

$$0 < \eta' \leq \min \left\{ \frac{1}{4}, \frac{1}{2(1 + |\log t|)} \right\}.$$

At a sufficiently small fixed cutoff $v = v(e, \eta') > 0$, and then sufficiently small $\rho > 0$, no point of $\mathcal{L}_\rho$ has $\eta'N$ absolute gaps at most v . To see this, use the net and conditional-gradient argument with projection factor $(Cv)^{\lfloor \eta'N/2 \rfloor}$ and residual lower bound $t\sqrt{N}$. Its union probability is at most $a^{-1}C^N t^{-(N-1)v^{\lfloor \eta'N/2 \rfloor}}$. Choose v small, then $a \ll \min\{t, v\sqrt{\eta'}\}$, then $\sqrt{\rho} \ll a$. This is exponentially small by (7.17).

Every point of $\mathcal{L}_\rho$ must consequently have $k \geq (1 - 2\eta')N$ active rows, after decreasing ρ if necessary. Otherwise impose zero image on all active and all absolute-near-zero rows and intersect their kernel with $y^\perp$. This subspace has dimension at least $\eta'N - 1$. Lemma 6.4 gives a unit vector v_0 there with $\|Av_0\|_\infty \leq C/\sqrt{\eta'}$. For small fixed $b > 0$, depending on v and η' , set $y_b = \sqrt{1 - b^2}y + bv_0$. With $c_b = \sqrt{1 - b^2}$,

$$h_i(y_b) = c_b h_i(y) + (1 - c_b) + bA_i v_0.$$

Choose b so small that $c_b \geq 1/2$ and $bC/\sqrt{\eta'} < v/4$. Positive gaps above v then remain positive. All rows with $|h_i(y)| \leq v$ have $A_i v_0 = 0$, so their nonnegative gaps remain nonnegative. The same vanishing projection on every active row gives $r_i(y_b) \leq c_b r_i(y)$. Therefore $H(y_b) \leq (1 - b^2)H(y)$, an improvement of at least $b^2 eN/2$ with high probability. This contradicts membership in $\mathcal{L}_\rho$ when $\rho < b^2 e/2$.

Now suppose that ηN positive gaps are at most u . Transfer to a sufficiently fine net, retaining at least $j = \lfloor \eta N/2 \rfloor$ of these rows with absolute gap at most $2u$; these row indices are disjoint from the active set of the original point. The active coordinates at the net point lie in a Euclidean ball of radius $Ct\sqrt{N}$, with the same $t = \sqrt{e}$ fixed above. Since $k \geq N/2$, the joint projection probability is at most $C^N t^k u^j$. The net estimate (7.14) still applies, by choosing ρ after the mesh. Since $t = \sqrt{e} \leq \sqrt{B_0}$, $k \geq (1 - 2\eta')N$, and $2\eta'|\log t| \leq 1$,

$$t^{k-N} \leq \max \left\{ \max(1, \sqrt{B_0})^{B_0 N}, \exp(2\eta'N|\log t|) \right\} \leq C^N,$$

with a uniform C , even as $e \downarrow 0$. The union probability is consequently bounded by $a^{-1}C^N u^{\lfloor \eta N/2 \rfloor}$, and tends to zero for $u \leq \exp(-C'/\eta)$, where C' is uniform in the density.

The sequence of choices is: density and η ; then η' , v , and the corresponding descent step; then the uniform exponential cutoff u ; then the net mesh and ρ ; finally β large enough for Gibbs concentration, and $N \rightarrow \infty$. Thus no interchange of the temperature and size limits is used. The row count bound ηN , divided by M , is at most a uniform multiple of η , since the densities stay bounded away from zero. Taking $\eta = C/|\log u|$ proves (7.16), since the counting event included all positive gaps in $(0, u]$. At fixed density, the finite-temperature row marginal and Proposition 7.3 converge along the chosen subsequence at both endpoints 0 and u , which are atomless. Thus the bound passes directly to every subsequential law of h^* , without removing a positive lower cutoff or identifying positive tests at exact ground minimizers. $\square$

7.7 A variational characterization of the threshold

Corollary 7.11. *With g and $\bar{\chi}$ as in Proposition 7.2,*

$$\alpha_J = \inf_{\substack{\gamma \geq 0 \text{ nondecreasing, } \gamma \in L^1 \\ \Delta > 0}} \frac{\int_0^1 \bar{\chi}(t)^{-1} dt}{-2g(0, 0)}. \quad (7.18)$$

Equivalently one may impose $\bar{\chi}(0) = 1$ and replace the terminal condition by $g(1, z) = -V(z)/\Delta$.

Proof. The denominator is positive. The diffusion representation writes $g(0, 0)$ as a negative terminal expectation minus $\frac{1}{2}\mathbb{E} \int \gamma g_z^2$; atomlessness with a positive density on negative intervals makes the terminal expectation strictly negative. Recovery trials in Proposition 7.2 remain valid when $e = 0$. In that regime all trial functionals are nonnegative, so α is at most every ratio in (7.18). For $e > 0$, an attained negative functional gives a ratio strictly below α . Approaching the threshold from the two sides proves the formula.

To verify the normalization assertion, scale a fixed pair by $c > 0$. If g_c is the solution for $(c\gamma, c\Delta)$, then cg_c solves the equation with coefficient γ and terminal value $-V/(\Delta + 1/c)$. The ratio for the scaled pair is therefore the ratio for this rescaled equation. It decreases as $c \uparrow \infty$ and converges to the ratio with terminal $-V/\Delta$, by the PDE comparison and stability already proved. Every shape can be normalized to $\bar{\chi}(0) = 1$, and every normalized limiting ratio is approached by its finite scalings. The two infima are equal. $\square$

The next section uses the susceptibility bounds of Lemma 7.8 and the two direct cutoff estimates to construct the normalized critical law. Joint convexity will first supply the fixed-density uniqueness still missing for positive gaps.

8 A convex critical problem and the limiting microscopic laws

At a fixed UNSAT density, Section 7 describes subsequential zero-temperature laws by a susceptibility and its PDE. We first prove that this description is unique. We then approach the threshold, retaining the notation $\alpha_c = \alpha_J$, and construct a unique normalized pair consisting of a susceptibility χ and a nonnegative terminal force variable R . The limiting state H_1 will be the gap variable: $H_1 \geq 0$, and $RH_1 = 0$ separates positive gaps from contacts.

There are two different convergence questions. Convexity on one Wiener space identifies every weak subsequential limit, but does not preserve contact probabilities or force normalization. To obtain those facts we prove that the critical susceptibility vanishes at the endpoint, upgrade the terminal convergence to L^2 , and apply the uniform small-gap and small-force estimates from the preceding section. The resulting laws, with the limits taken in the prescribed order, are stated in Theorem 8.12.

The stochastic variational viewpoint is related to Boué–Dupuis [4] and Auffinger–Chen [2]. More specifically, the use of terminal duality to obtain a variational formula over Brownian martingales, with the optimizer identified by the derivative of the Parisi PDE along its diffusion, has a close analogue in Mourrat [17, Lemma 2.2 and Theorem 2 in the preprint]; see also Chen, Issa and Mourrat [7, Proposition 8.1 and Theorem 1.5]. Proposition 8.1 establishes the representation needed here for a concave quadratic terminal function and a nonnegative terminal variable in L^2 . Rewriting its quadratic terms using $s = 1/\chi$ gives the joint convexity in Lemma 8.2. The possibility that χ vanishes at the endpoint requires the additional compactness and equality arguments below; the critical uniqueness statement is not a direct consequence of the cited results.

8.1 The probability space and the admissible class

Fix canonical Wiener space, with its completed natural filtration $(\mathcal{F}_t)_{0 \leq t \leq 1}$ and coordinate Brownian motion B . Realize each fixed-density zero-temperature optimizer's forward diffusion as the strong solution driven by this same B . Thus the terminal variables for different densities will belong to one Hilbert space, rather than merely having laws that can be compared. Expectations in this section are on that space. For every $R \in L^2(\mathcal{F}_1)$, martingale representation gives a unique predictable $b \in L^2(\Omega \times [0, 1])$ such that

$$r_t := \mathbb{E}[R \mid \mathcal{F}_t] = c + \int_0^t b_v dB_v, \quad c := \mathbb{E}R.$$

The maps $R \mapsto c$ and $R \mapsto b$ are bounded linear maps, and $\mathbb{E}R^2 = c^2 + \int_0^1 \mathbb{E}b_t^2 dt$.

The susceptibility class extends the positive-energy class in (7.3) to allow a zero endpoint value. An admissible susceptibility is

$$\chi(t) = \Delta + \int_t^1 \gamma(v) dv, \quad \Delta \geq 0, \quad \chi(t) > 0 \quad (t < 1),$$

where $\gamma \geq 0$ is nondecreasing and belongs to $L^1(0, 1)$. Thus χ is concave, nonincreasing, and continuous on $[0, 1]$; the endpoint value is $\chi(1) = \Delta$. Set $s = 1/\chi$ on $[0, 1)$ and require $\int_0^1 s < \infty$. The value of s at the single endpoint is immaterial. We explicitly require $s \in L^2(0, 1)$ whenever that stronger condition is needed. A pair (R, s) is admissible when its susceptibility is admissible and $R \geq 0$ almost surely belongs to $L^2(\mathcal{F}_1)$.

For $\lambda \geq 0$ define the joint functional

$$J_{\alpha, \lambda}(R, s) = \alpha \mathbb{E} \left[R(1 + B_1) + \frac{\lambda}{2} R^2 \right] + \frac{\alpha}{2} \left\{ \frac{c^2}{s(0)} + \int_0^1 \frac{\mathbb{E}b_t^2}{s(t)} dt \right\} + \frac{1}{2} \int_0^1 s(t) dt. \quad (8.1)$$

All terms are finite. In particular $\chi \leq \chi(0)$ bounds the quadratic martingale term. The elementary identity

$$c^2 \chi(0) + \int_0^1 \chi(t) \mathbb{E}b_t^2 dt = \Delta \mathbb{E}R^2 + \int_0^1 \gamma(t) \mathbb{E}r_t^2 dt \quad (8.2)$$

follows from $\mathbb{E}r_t^2 = c^2 + \int_0^t \mathbb{E}b_v^2 dv$ and Tonelli's theorem.

8.2 Duality and positive-energy uniqueness

For fixed χ , quadratic duality replaces the PDE value by a minimum over terminal force variables. Passing from χ to its reciprocal s then makes the susceptibility and terminal variable jointly convex. This change of variables is what allows two physical optimizers to be compared on the Wiener space just fixed.

Proposition 8.1 (Joint martingale formulation). *For an original susceptibility χ with $\Delta > 0$, the functional in (7.3) equals the minimum over $R \geq 0$ in $L^2(\mathcal{F}_1)$ of*

$$J_{\alpha, 1}(R, s) = \alpha \mathbb{E} \left[R(1 + B_1) + \frac{1}{2} R^2 \right] + \frac{\alpha}{2} \left[\frac{c^2}{s(0)} + \int_0^1 \frac{\mathbb{E}b_t^2}{s(t)} dt \right] + \frac{1}{2} \int_0^1 s(t) dt. \quad (8.3)$$

This minimizer is unique for fixed s .

Proof. Use gap coordinates $h = 1 + z$ and write $f(t, h) = g(t, h - 1)$ and $a = f_h$. The zero-temperature equation and forward diffusion are

$$f_t + \frac{1}{2} f_{hh} + \frac{1}{2} \gamma f_h^2 = 0, \quad f(1, h) = -\frac{h_-^2}{2(1 + \Delta)}, \quad H_t = 1 + B_t + \int_0^t \gamma(v) a(v, H_v) dv,$$

where $h_- = \max\{-h, 0\}$. The coefficient is time integrable; a has linear growth and an integrable Lipschitz constant after multiplication by γ . The diffusion therefore has finite second moments. Itô's formula, first on $[0, T]$ and then with $T \uparrow 1$, gives

$$f(0, 1) = \mathbb{E} \left[-\frac{(H_1)_-^2}{2(1 + \Delta)} - \frac{1}{2} \int_0^1 \gamma(t) a(t, H_t)^2 dt \right].$$

For every nonnegative R , the pointwise quadratic dual inequality is

$$-\frac{h_-^2}{2(1 + \Delta)} \leq Rh + \frac{1 + \Delta}{2} R^2.$$

Insert $h = H_1$, condition R against $\mathcal{F}_t$ in the drift term, and use $r_t a - a^2/2 \leq r_t^2/2$. This proves

$$f(0, 1) \leq \mathbb{E} \left[R(1 + B_1) + \frac{1 + \Delta}{2} R^2 \right] + \frac{1}{2} \int_0^1 \gamma(t) \mathbb{E} r_t^2 dt.$$

For $R = a(1, H_1) = (H_1)_-/(1 + \Delta)$ the differentiated equation makes $a(t, H_t)$ a square-integrable martingale with terminal value R . Consequently $r_t = a(t, H_t)$, and both inequalities are equalities. Equation (8.2) proves the claimed formula. Strict convexity of $\mathbb{E} R^2/2$ proves uniqueness for fixed s . All uses of Itô's formula can be justified with smooth terminal data and bounded smooth coefficients; the curvature and growth bounds above dominate the terms when those approximations are removed. $\square$

Lemma 8.2 (Convexity and its equality cases). *The set of admissible s is convex, as is its normalized part $s(0) = 1$. The functional $J_{\alpha, \lambda}$ is jointly convex in (R, s) . If two pairs attain equality in this convexity inequality at a nontrivial convex combination, then*

$$\frac{c_1}{s_1(0)} = \frac{c_2}{s_2(0)}, \quad \frac{b_{1,t}}{s_1(t)} = \frac{b_{2,t}}{s_2(t)} \quad \text{for } dt \otimes \mathbb{P}\text{-almost every } (t, \omega). \quad (8.4)$$

If $\lambda > 0$, equality also forces $R_1 = R_2$ in L^2 .

Proof. For $0 < q < 1$, the reciprocal of $qs_1 + (1 - q)s_2$ is the weighted harmonic mean $H(x, y) = (q/x + (1 - q)/y)^{-1}$ evaluated at (χ_1, χ_2) . This function is increasing in each coordinate and concave on the positive quadrant: differentiation gives a negative semidefinite Hessian. Hence $H(\chi_1(t), \chi_2(t))$ is concave and nonincreasing in t . It extends continuously to the endpoint, and its negative derivative is nonnegative, nondecreasing, and integrable. Integrability of the new s is immediate from its definition as a convex combination. The same argument preserves L^2 integrability and the normalization.

For a real Hilbert space $\mathcal{H}$, the perspective $(x, u) \mapsto \|x\|_{\mathcal{H}}^2/u$, $u > 0$, is convex. Its convexity defect vanishes exactly when $x_1/u_1 = x_2/u_2$. Apply this once with $\mathcal{H} = \mathbb{R}$ to c , and for almost every t with $\mathcal{H} = L^2(\Omega)$ to b_t . Integration preserves nonnegative defects, so equality in the sum implies the asserted almost-everywhere equalities. The remaining terms are linear, except for $\alpha \lambda \mathbb{E} R^2/2$, whose strict convexity gives the last assertion. $\square$

Corollary 8.3 (Uniqueness in the UNSAT phase). *At each fixed $\alpha > \alpha_c$ the minimizing physical susceptibility in (7.3) is unique. Thus the entire zero-temperature endpoint law in (7.5), including its restriction to positive gaps, is unique.*

Proof. Two minimizing pairs have the same R and therefore the same c and b . For a PDE optimizer, $b_t = f_{hh}(t, H_t) < 0$ almost surely for $t < 1$. To see the strict inequality, the curvature is nonpositive, and its differentiated diffusion representation propagates the strictly negative terminal curvature on $(-\infty, 0)$. On every interval ending before time 1 the drift is Lipschitz, its flow derivative is positive, and the diffusion has positive probability of reaching that interval. Equivalently the strong maximum principle applied after smoothing gives strict negativity; the representation retains strictness on removing the smoothing. Equality in (8.4) therefore gives $s_1 = s_2$ almost everywhere, hence everywhere before the endpoint by continuity. Their continuous susceptibilities have the same endpoint value as well. The PDE and its forward diffusion are then identical. The subsequential description in (7.5) consequently has only one possible limit. $\square$

8.3 Normalization and compactness at the threshold

The initial physical susceptibility $\bar{\chi}(0)$ diverges as the density decreases to the threshold. Normalize that value before taking the limit. Use bars for the fixed-density quantities

in (7.3): $\bar{\gamma} = -\bar{\chi}'$, $\bar{\Delta} = \bar{\chi}(1)$, $\bar{f}(t, h) = g(t, h - 1)$, and $\bar{a} = \bar{f}_h$. The dual optimizer is $\bar{R} = \bar{a}(1, H_1) = (h^*)_-$ by (7.5). Put $L = \bar{\chi}(0)$ and define

$$\begin{aligned}\lambda &= L^{-1}, & \chi &= L^{-1}\bar{\chi}, & s &= L/\bar{\chi}, \\ \gamma &= L^{-1}\bar{\gamma}, & \Delta &= L^{-1}\bar{\Delta}, & R &= L\bar{R}, \\ f &= L\bar{f}, & a &= L\bar{a}.\end{aligned}$$

Hence $\chi(0) = s(0) = 1$, the terminal condition is $f(1, h) = -h_-^2/[2(\lambda + \Delta)]$, and the physical residual is λR . The state process H is unchanged on each Brownian path: its drift satisfies $\gamma a = \bar{\gamma}\bar{a}$. Below $\chi, \gamma, \Delta, f, a, R$ denote these normalized quantities.

Multiplying the original functional by L gives $J_{\alpha, \lambda}$. Thus the normalized physical pair minimizes this functional among normalized pairs at the fixed value $\lambda = L^{-1}$. Write $\bar{s} = 1/\bar{\chi}$. Moreover

$$J_{\alpha, \lambda}(R, s) = -\frac{e(\alpha)}{\lambda}, \quad \int_0^1 s(t)^2 dt = \alpha \mathbb{E}R^2 = \frac{2e(\alpha)}{\lambda^2}. \quad (8.5)$$

Indeed, variation of the original $\bar{\Delta} > 0$ in both directions, with $\bar{R}$ fixed in (8.3), yields $\int \bar{s}^2 = \alpha \mathbb{E}\bar{R}^2$. The second-moment identity for the physical residual in Section 7 says $\alpha \mathbb{E}\bar{R}^2 = 2e(\alpha)$. These identities give (8.5) after normalization. Here optimality is initially over $\Delta > 0$. It extends to the admissible normalized class with $\Delta = 0$ by replacing χ with $(\chi + \eta)/(1 + \eta)$ and keeping R fixed: the quadratic terms converge by domination and the entropy by monotone convergence followed by the factor $1 + \eta$.

The spectral and residual estimates in Lemma 7.8 and Corollary 7.9 give, uniformly as $\alpha \downarrow \alpha_c$:

$$\lambda \asymp \sqrt{e(\alpha)} \longrightarrow 0, \quad \int_0^1 s^2 + \mathbb{E}R^2 \leq C, \quad 0 < c_0 \leq \mathbb{E}R \leq C_0. \quad (8.6)$$

The contact-count limit and the two logarithmic cutoff bounds will enter only when we identify the endpoint masses. For compactness we need only (8.6). In particular we do not assume a critical power law or an endpoint boundary condition for χ .

Proposition 8.4 (Existence of a normalized critical pair). *The functional $J_{\alpha_c, 0}$ is nonnegative on the whole admissible class, without a normalization constraint. Every sequence of normalized physical pairs with $\alpha_n \downarrow \alpha_c$ has a subsequence such that*

$$\chi_n \rightarrow \chi \text{ uniformly on every } [0, T], \quad T < 1, \quad R_n \rightharpoonup R \text{ in } L^2(\mathcal{F}_1).$$

Its limit is normalized, satisfies $s \in L^2(0, 1)$ and $c \geq c_0$, and attains $J_{\alpha_c, 0}(R, s) = 0$.

We will prove uniqueness for the limits supplied by this proposition. That conclusion concerns physical limits; it will not require a classification of all admissible zero-value pairs.

Proof. Nonnegativity of the limiting functional. First suppose $\Delta > 0$. For $A > 0$, direct substitution gives the exact homogeneity identity

$$AJ_{\alpha, 1}(R/A, s/A) = J_{\alpha, 0}(R, s) + \frac{\alpha}{2A} \mathbb{E}R^2. \quad (8.7)$$

The reciprocal of s/A is $A\chi$, so this is an admissible zero-temperature trial. At α_c , zero ground energy and Proposition 8.1 make its value nonnegative. Letting $A \rightarrow \infty$ proves the assertion. If $\Delta = 0$, replace χ by $\chi + \eta$. The quadratic term converges by dominated convergence, while $1/(\chi + \eta) \uparrow 1/\chi$; let $\eta \downarrow 0$.

Compactness on the common space. Normalization and concavity give $1 \geq \chi_n(t) \geq 1 - t$. On each $[0, T]$ their slopes are bounded by a constant depending only on T : monotonicity of γ_n gives $\gamma_n(t) \leq (1 - t)^{-1}$. Arzelà–Ascoli and a diagonal selection give the stated local

uniform convergence. The limit is concave and nonincreasing; its continuous endpoint value is $\Delta = \lim_{t \uparrow 1} \chi(t)$, which is not initially identified with $\lim_n \chi_n(1)$. Its derivative gives an admissible integrable $\gamma = -\chi'$. At differentiability points of χ , the concavity inequalities imply $\gamma_n \rightarrow \gamma$; the same local bound then gives $L^1(0, T)$ convergence for every $T < 1$. Fatou's lemma gives $s \in L^2$.

The bound on R_n gives a weakly convergent subsequence on the fixed Wiener space. The cone of nonnegative L^2 variables is weakly closed. Martingale representation, being a bounded linear isometry after the constant component is included, gives weak convergence of b_n and convergence of c_n . In particular $c \geq c_0$.

Passage to the minimum. For each $T < 1$, uniform convergence of the positive weights χ_n on $[0, T]$ and weak convergence of b_n imply

$$\int_0^T \chi(t) \mathbb{E} b_t^2 dt \leq \liminf_n \int_0^T \chi_n(t) \mathbb{E} b_{n,t}^2 dt.$$

Let $T \uparrow 1$ for the corresponding global lower semicontinuity. In contrast the entropy term actually converges: local convergence of s_n combines with $\int_T^1 s_n \leq C\sqrt{1-T}$ from (8.6). The linear term converges weakly, and $\lambda_n \mathbb{E} R_n^2 \rightarrow 0$. Equations (8.5) and (8.6) therefore give

$$J_{\alpha_c, 0}(R, s) \leq \liminf_n J_{\alpha_n, \lambda_n}(R_n, s_n) = 0.$$

Nonnegativity proves equality. □

8.4 Interior PDE representation

A weak terminal limit need not by itself retain the PDE representation. We now recover that representation before comparing two critical limits. The strict curvature and the time regularity proved here will make the common-noise uniqueness argument possible.

For the normalized physical pairs use gap coordinates and denote the scaled PDE by f_n . Its equation and terminal data are

$$(f_n)_t + \frac{1}{2}(f_n)_{hh} + \frac{1}{2}\gamma_n(t)(f_n)_h^2 = 0, \quad f_n(1, h) = -\frac{h_-^2}{2(\lambda_n + \chi_n(1))}.$$

Write $a_n = (f_n)_h$. The comparison estimates below control the terminal error uniformly as the density changes. They will also specify the boundary condition of the critical equation if $\chi(1) = 0$:

$$\begin{aligned} -s_n(t) &\leq -\frac{1}{\lambda_n + \chi_n(t)} \leq (a_n)_h(t, h) \leq 0, \quad a_n \geq 0, \quad (a_n)_{hh} \geq 0, \\ \left| f_n(t, h) + \frac{h_-^2}{2(\lambda_n + \chi_n(t))} \right| &\leq \frac{1}{2} \int_t^1 s_n(v) dv, \\ \left| a_n(t, h) - \frac{h_-}{\lambda_n + \chi_n(t)} \right| &\leq C \sqrt{s_n(t) \int_t^1 s_n(v) dv}. \end{aligned} \tag{8.8}$$

Lemma 8.5 (Interior limits and strict curvature). *For each subsequence in Proposition 8.4, f_n and a_n converge locally uniformly on $[0, 1) \times \mathbb{R}$ to functions f and $a = f_h$ determined by χ . They satisfy (8.8) with $\lambda_n + \chi_n$ replaced by χ . They are smooth in the spatial variable in the interior and locally Lipschitz in time there, together with each spatial derivative used below. For $t < 1$,*

$$f_{hh}(t, h) < 0, \quad f_{hhh}(t, h) > 0, \quad \lim_{h \rightarrow +\infty} a(t, h) = 0.$$

On the common Wiener space the limiting martingale and diffusion obey

$$H_t = 1 + B_t + \int_0^t \gamma(v) r_v dv, \quad r_t = a(t, H_t), \quad b_t = f_{hh}(t, H_t), \quad t < 1. \quad (8.9)$$

Both H_t and r_t extend continuously and in L^2 to $t = 1$, with $r_1 = R$.

Proof. Comparison estimates. The curvature bounds are parabolic comparison bounds. Specifically, if $b = (f_n)_{hh}$, then $b_t + \frac{1}{2}b_{hh} + \gamma_n a_n b_h + \gamma_n b^2 = 0$. The spatially constant function $-(\lambda_n + \chi_n(t))^{-1}$ solves this equation, and zero is the upper comparison solution. Differentiating once more gives the linear equation $d_t + \frac{1}{2}d_{hh} + \gamma_n a_n d_h + 3\gamma_n b d = 0$ for $d = (f_n)_{hhh}$, which preserves nonnegativity backwards from the convex terminal derivative. These comparisons follow first for smooth terminal data and smooth coefficients; the bounds survive approximation. The first-order solution is $q_n(t, h) = -h_-^2 / (2(\lambda_n + \chi_n(t)))$. Its missing diffusion term is bounded in absolute value by $s_n(t)/2$. Comparison with $q_n \pm \frac{1}{2} \int_t^1 s_n$ proves the value bound. The derivative bound follows by taking secants of length $\sqrt{(\int_t^1 s_n)/s_n(t)}$ and using the curvature bound; the zero-length limiting case is immediate.

Convergence with a terminal cutoff. At a fixed cutoff $T < 1$ the value estimates compare $f_n(T, \cdot)$ and $f_m(T, \cdot)$ with quadratic functions whose coefficients converge. The error left after this quadratic comparison is bounded uniformly in space by $C\sqrt{1-T}$, uniformly in n, m . On $[0, T]$, the difference $w = f_n - f_m$ solves a linear equation with drift $\gamma_n(a_n + a_m)/2$ and source $(\gamma_n - \gamma_m)a_m^2/2$. The drift has uniformly bounded Lipschitz and linear-growth constants on that interval, and its diffusion has uniformly bounded second moments on bounded starting sets. Its Feynman–Kac representation propagates a constant terminal error without amplification. The quadratic terminal error tends to zero at fixed T , as does the source error because $\gamma_n \rightarrow \gamma$ in $L^1(0, T)$. For each fixed T , first let $n, m \rightarrow \infty$: only the constant terminal error remains, bounded on every earlier compact set by $C\sqrt{1-T}$. Then let $T \uparrow 1$. This proves the Cauchy property and shows that the limit depends only on χ . Curvature bounds and secants give convergence of the first derivatives.

Regularity for a measurable time coefficient. On a compact time interval ending strictly before 1, γ is bounded. We need spatial smoothness and local time-Lipschitz bounds, including at jumps of γ , for the inverse-curvature map used below. We give the regularity argument in a form that does not require continuity of this coefficient in time. Reverse time and multiply a by smooth space and time cutoffs supported in a larger interior rectangle, with the time cutoff zero at its initial edge. The resulting inhomogeneous heat equation has bounded source: the nonlinear term is $\gamma a a_h$, and the cutoff terms involve only a and a_h , already bounded on that rectangle. For the heat semigroup P_u , differentiation of the Gaussian kernel gives

$$\|\partial_h P_u F\|_\infty \leq C u^{-1/2} \|F\|_\infty, \quad \|\partial_{hh} P_u F\|_\infty \leq C_\eta u^{-1+\eta/2} [F]_{C_h^\eta}, \quad 0 < \eta < 1.$$

The second estimate uses the zero integral of the differentiated kernel to subtract $F(h)$. The first estimate and its spatial increment version in Duhamel’s formula give a uniform spatial $C^{1,\eta}$ bound for every $\eta < 1$: splitting the time integral at the square of the spatial increment bounds the gradient increment by a constant times $|z|(1 + |\log |z||)$. The source is therefore spatially C^η . The second displayed estimate is integrable at $u = 0$ and yields $C^{2,\eta'}$ on a smaller rectangle for $0 < \eta' < \eta$. More generally, if a is already bounded in $C_h^{m,\eta}$, $m \geq 1$, then the localized source is bounded in $C_h^{m-1,\eta}$: its derivatives of order $m-1$ use derivatives of a of order at most m . Apply the second heat-kernel estimate to $\partial_h^{m-1} F$ to obtain $C_h^{m+1,\eta'}$ on a smaller rectangle. This induction uses no derivative of a that it is trying to establish. It gives every spatial derivative on nested interior rectangles. Multiplication by the bounded function $\gamma(t)$ preserves all spatial Hölder bounds, so this bootstrap is valid for measurable

time coefficients. Smooth coefficient approximation justifies the intermediate differentiations with these same bounds. Finally the PDE bounds the almost-everywhere time derivative of every spatial derivative on smaller rectangles. This gives the claimed local time-Lipschitz property, including across jumps of γ . The uniform spatial Hölder and time-Lipschitz bounds make these derivatives jointly continuous. At the start of this argument the source uses the bounded weak derivative a_h supplied by the curvature estimate; the distributional PDE passes from a_n by local uniform convergence and L^1 convergence of γ_n . Thus the initial Duhamel representation does not presume smoothness.

Strict curvature and identification of the martingale. The value bound on $h > 0$ and $a \geq 0$ imply $a(t, h) \rightarrow 0$ as $h \rightarrow +\infty$. On the negative half-line, (8.8) makes a unbounded. Thus a is not affine, its nonnegative second derivative is nontrivial at every later interior time, and its nonpositive first derivative is nontrivial there. The strong maximum principle, on an interval between two strictly interior times, gives $f_{hhh} > 0$ and $f_{hh} < 0$ at the earlier time.

The coupled forward diffusions converge in L^2 uniformly on each compact time interval, by the local gradient convergence, L^1 convergence of γ_n , and the common linear-growth bounds. For precision, those deterministic growth bounds and Gronwall's inequality give uniformly bounded moments of every fixed finite order for $\sup_{t \leq T} |H_{n,t}|$ and $\sup_{t \leq T} |a_n(t, H_{n,t})|$, for each $T < 1$. The fourth-moment bound permits spatial localization to be removed in L^2 after the coupled convergence is proved on bounded sets. Consequently $a_n(t, H_{n,t}) \rightarrow a(t, H_t)$ strongly in L^2 for every $t < 1$. But $a_n(t, H_{n,t}) = \mathbb{E}[R_n | \mathcal{F}_t]$ converges weakly to $\mathbb{E}[R | \mathcal{F}_t]$. This identifies r_t ; Itô's formula identifies $b_t = f_{hh}(t, H_t)$ and proves (8.9). Finally Doob's inequality gives $\mathbb{E} \sup_{t \leq 1} |r_t|^2 \leq 4\mathbb{E}R^2$. Since $\gamma \in L^1$, the drift integral in (8.9) converges both almost surely and in L^2 at 1. The martingale itself has its continuous L^2 extension with terminal value R . $\square$

8.5 Uniqueness at zero energy

Proposition 8.6 (Uniqueness of the normalized critical limit). *Any two subsequential limits supplied by Proposition 8.4 coincide on the common Wiener space. In particular the normalized susceptibility and the law of (H_1, R) are independent of the density sequence.*

Proof. Equality identifies the normalized curvatures along the states. Both pairs minimize the globally nonnegative critical functional. Their midpoint is admissible and also has value zero. Lemma 8.2 and $s_i(0) = 1$ yield $c_1 = c_2 =: c > 0$ and $\chi_1 b_1 = \chi_2 b_2$ almost everywhere. Put $k_i(t, h) = -\chi_i(t)(f_i)_{hh}(t, h)$. The proof uses this matched normalized curvature to express one state process as a deterministic transform of the other. Their common Brownian noise forces the transform to have spatial derivative one; the common martingale mean then identifies the susceptibilities. Interior continuity upgrades the curvature identity to

$$k_1(t, H_{1,t}) = k_2(t, H_{2,t}), \quad 0 < t < 1,$$

on a single event of probability one. Each $k_i(t, \cdot)$ is strictly decreasing with strictly negative derivative by Lemma 8.5. Each $H_{i,t}$ has full support in $\mathbb{R}$: on every interior compact time interval its additive-noise diffusion has locally bounded Lipschitz drift, and localized Girsanov equivalence, or the Brownian support property followed by continuity of the solution map, reaches every open interval. Thus the support of $k_i(t, H_{i,t})$ is the closure of the range of $k_i(t, \cdot)$. Equality of their laws identifies the two ranges, which are open intervals. We may therefore define

$$\Psi(t, h) = k_2(t, \cdot)^{-1}(k_1(t, h)), \quad H_{2,t} = \Psi(t, H_{1,t}).$$

These inverse functions have uniform local bounds. At a fixed (t_0, h_0) put $y_0 = \Psi(t_0, h_0)$. Strict decrease gives

$$k_2(t_0, y_0 - 1) > k_1(t_0, h_0) > k_2(t_0, y_0 + 1).$$

Joint continuity preserves these inequalities on a neighborhood of (t_0, h_0) , trapping Ψ between $y_0 - 1$ and $y_0 + 1$. Compactness and uniqueness of the inverse give continuity there. The negative derivative of k_2 is bounded away from zero on a slightly smaller compact rectangle. The implicit derivative $\Psi_h(t, h) = (k_1)_h(t, h)/(k_2)_h(t, \Psi(t, h))$ is therefore jointly continuous, all needed spatial derivatives are bounded locally, and the mean-value theorem with the time-Lipschitz bounds on k_1, k_2 proves a uniform local time-Lipschitz bound on Ψ . Finite covers give the same assertions on every compact subset of $(0, 1) \times \mathbb{R}$.

The common noise forces a spatial translation. Taking quadratic covariations with the driving Brownian motion on an arbitrary interval $[u, t] \subset (0, 1)$ gives

$$t - u = [H_2, B]_t - [H_2, B]_u = \int_u^t \Psi_h(v, H_{1,v}) dv.$$

This chain rule needs no continuous time derivative. After localization, the temporal increments in a partition contribute at most $C \sum \Delta t |\Delta B|$, which tends to zero. The spatial Taylor remainders contribute at most $C \max |\Delta B| \sum |\Delta H_1|^2$, which tends to zero in probability, since quadratic-variation sums of a continuous semimartingale are tight. The first-order sums give the displayed integral. Rational interval endpoints and continuity give all these identities on one event of probability one. It follows that $\Psi_h(t, H_{1,t}) = 1$ for almost every time, almost surely. Full support gives $\Psi_h(t, h) = 1$ for every h at almost every t ; continuity extends it to every interior t . Hence $\Psi(t, h) = h + d(t)$.

The martingale mean identifies the susceptibility. The equality of k 's now holds for all h . Integrate the matched curvatures from h to $+\infty$, using gradient decay there, to obtain

$$\chi_1(t) a_1(t, h) = \chi_2(t) a_2(t, h + d(t)).$$

Along the states this gives $\chi_1 r_{1,t} = \chi_2 r_{2,t}$. Taking expectations yields $c\chi_1 = c\chi_2$, so $\chi_1 = \chi_2$. The PDE is determined by χ , and the two forward diffusions have the same initial condition and Brownian motion. They agree, and their martingales, including their L^2 terminal values, agree as well. $\square$

8.6 Variational conditions and the vanishing endpoint mass

From now on, “the critical pair” means the unique limit of the normalized physical optimizers just constructed. To reach the endpoint we must show that $\Delta = \chi(1)$ vanishes. The convergence so far is local in time: even a vanishing sequence of endpoint values $\chi_n(1)$ would not exclude mass of γ_n concentrating near 1. We instead use optimality of the critical pair in the full admissible class.

The first-order conditions are expressed by the stationarity defect $J(t)$ and its integral $K(t)$, defined by

$$P(t) = \alpha_c \mathbb{E} r_t^2, \quad I(t) = \int_0^t s(v)^2 dv, \quad J(t) = P(t) - I(t), \quad K(t) = \int_t^1 J(v) dv.$$

The quantity $K(t)/2$ is the directional derivative for the admissible positive variation $\gamma \mapsto \gamma + \varepsilon \mathbf{1}_{[t,1]}$, as proved below. Here $J(t)$ is a function of time, distinct from the joint functional $J_{\alpha,\lambda}$, and P, I, J extend continuously to 1. In the interior, $P' = \alpha_c \mathbb{E} b_t^2$ is continuous. Choose the right-continuous nondecreasing representative of γ . Let $\nu = d\gamma$ denote the nonnegative Stieltjes measure on $[0, 1)$ with $\nu([0, t]) = \gamma(t)$; this convention includes a possible initial value of γ as an atom at zero.

Lemma 8.7 (Complete first-order conditions). *The critical pair satisfies*

$$H_1 + \Delta R \geq 0, \quad R(H_1 + \Delta R) = 0 \quad \text{almost surely}, \quad (8.10)$$

$$K(t) \geq 0 \quad (0 \leq t \leq 1), \quad \int_0^1 \gamma(t) J(t) dt = \int_{[0,1)} K(t) \nu(dt) = 0. \quad (8.11)$$

The profile γ vanishes on an initial interval. At every interior point of $\text{supp } \nu$ one has $J = 0$ and $J' \leq 0$; at every non-isolated such point, $J' = 0$. Finally $J(1) = 0$.

Proof. The critical pair minimizes without normalization, since its value is the global lower bound zero. By (8.2), the Fréchet derivative in R is the L^2 variable $\alpha_c(H_1 + \Delta R)$. Explicitly the quadratic operator is

$$\mathcal{A}Q = \Delta Q + \int_0^1 \gamma(t) \mathbb{E}[Q \mid \mathcal{F}_t] dt, \quad \|\mathcal{A}\|_{L^2 \rightarrow L^2} \leq \chi(0).$$

Conditional expectations are self-adjoint contractions, so $\mathcal{A}$ is bounded and self-adjoint; the functional's gradient is $\alpha_c(1 + B_1 + \mathcal{A}R)$, as asserted. In particular $\|H_1\|_2 \leq 1 + \|B_1\|_2 + \|\gamma\|_1 \|R\|_2$ also when $\Delta = 0$. Minimization on the nonnegative cone means that this variable is nonnegative and has zero pairing with R ; these are exactly (8.10).

Adding $\varepsilon \mathbf{1}_{[t,1)}$ to γ is an admissible positive variation. Its susceptibility increment at v is $\varepsilon(1 - \max\{t, v\})$, bounded by $\varepsilon \chi(v)/\chi(0)$ by concavity. Differentiating the entropy term is therefore justified by domination: if $q(v) = 1 - \max\{t, v\}$, its difference quotient is bounded by $s(v)/\chi(0)$ for the positive variations used here. For scaling γ , the susceptibility increment is $q = \chi - \Delta \leq \chi$, and the difference quotient is bounded by $2s$ for $|\varepsilon| < 1/2$. Both bounds require only $s \in L^1$. Tonelli's theorem in the quadratic term gives the directional derivative $K(t)/2$. This proves $K \geq 0$. Scaling γ by $1 + \varepsilon$ permits either sign for small ε , and gives $\int \gamma J = 0$. Fubini gives $\int \gamma J = \int K d\nu$: its absolute version is bounded by $\|J\|_\infty \int \gamma < \infty$. Equivalently the Stieltjes boundary term at 1 vanishes because $K(t) = O(1-t)$ and $(1-t)\gamma(t) \rightarrow 0$. Thus K vanishes on $\text{supp } \nu$.

Since $J(0) = \alpha_c c^2 > 0$ and $K' = -J$, nonnegativity of K forces $K(0) > 0$. Hence ν vanishes on an initial interval, including its possible atom at zero. An interior support point is a local minimum of the nonnegative C^2 function K , so $J = 0$ and $J' \leq 0$. At a non-isolated support point, differentiability of J and nearby zeros give $J' = 0$.

If $\Delta > 0$, two-sided variation of Δ gives $0 = (P(1) - I(1))/2$. If $\Delta = 0$ and the support did not accumulate at 1, then γ would be constant on a final interval and $\chi(t)$ would be a positive constant times $1 - t$. This contradicts $s \in L^2$. Thus in this case interior support points approach 1, and continuity of J again gives $J(1) = 0$. $\square$

Proposition 8.8 (No critical endpoint susceptibility). *The critical susceptibility satisfies $\Delta = 0$. Its increase measure ν accumulates at 1, and with $k = -\chi f_{hh}$ one has, at every interior support point,*

$$\alpha_c \mathbb{E}k(t, H_t) = 1 + \alpha_c c \chi(t), \quad \alpha_c \mathbb{E}k(t, H_t)^2 \leq 1. \quad (8.12)$$

The second inequality is an equality at non-isolated interior support points.

Proof. Introduce $W(t) = s(t) + \alpha_c \mathbb{E}b_t$. The curvature equation along the forward diffusion gives $\frac{d}{dt} \mathbb{E}b_t = -\gamma(t) \mathbb{E}b_t^2$ almost everywhere. Since $s' = \gamma s^2$, we have $W' = -\gamma J'$. The Stieltjes product rule and $J d\gamma = 0$ now give

$$W(t) = W(0) - \gamma(t) J(t) \quad \text{almost everywhere}, \quad \int_0^1 W(t) dt = W(0). \quad (8.13)$$

The initial constant uses the initial zero interval of γ .

Itô's formula for $H_t r_t$ and (8.10) yield

$$-\Delta P(1) = \alpha_c c + \int_0^1 (\gamma(t)P(t) + \alpha_c \mathbb{E} b_t) dt.$$

Every term is integrable: P is bounded, $\gamma \in L^1$, and $|\mathbb{E} b_t| \leq s(t) \in L^1$. The stochastic products are justified by localization and the L^2 maximal bounds for H and r . Moreover (8.11) and integration by parts give

$$\int_0^1 \gamma P = \int_0^1 \gamma I = \int_0^1 s - \Delta I(1).$$

Since $P(1) = I(1)$, these identities show

$$W(0) = -\alpha_c c < 0. \tag{8.14}$$

Suppose for contradiction that $\Delta > 0$. Then $s \leq 1/\Delta$, and the value and derivative bounds in (8.8) give $a(t, h) \rightarrow h_-/\Delta$ as $t \uparrow 1$, uniformly on bounded sets. Convex secant bounds give $f_{hh}(t, h) \rightarrow -\mathbf{1}_{\{h < 0\}}/\Delta$ locally away from zero. The endpoint H_1 has no atom at zero. Here is a direct justification: condition on the Brownian bridge and write $B_t = tZ + \mathcal{B}_t$, where Z has a Gaussian density independent of the bridge. The map $Z \mapsto H_1$ is increasing, and its derivative is

$$\int_0^1 \exp \left\{ \int_u^1 \gamma(v) a_h(v, H_v) dv \right\} du \geq \exp \left\{ -\Delta^{-1} \int_0^1 \gamma \right\} > 0.$$

The estimate follows first by smoothing and then by the Lipschitz flow comparison. Thus the conditional endpoint law, and hence the unconditional one, is atom-free. With $p = \mathbb{P}(H_1 < 0)$, bounded convergence gives

$$W(1-) = \frac{1 - \alpha_c p}{\Delta}, \quad J'(1-) = \frac{\alpha_c p - 1}{\Delta^2}.$$

Because $J(1) = 0$ and J' is bounded when $\Delta > 0$, $|J(t)| \leq C(1 - t)$. Integrability and monotonicity of γ imply $\gamma(t)J(t) \rightarrow 0$. Hence (8.13)–(8.14) give $W(1-) = -\alpha_c c < 0$, so $\alpha_c p > 1$ and $J'(1-) > 0$. It follows that $J(t) < 0$, and therefore $K(t) < 0$, for t sufficiently close to 1. This contradicts Lemma 8.7.

Thus $\Delta = 0$, and the support accumulation follows from that lemma. At a support point $J = 0$, (8.13) and (8.14) give the first identity in (8.12). Although (8.13) was initially obtained almost everywhere, it applies here by continuity of W : approach the support point through times where that identity holds, and use local boundedness of γ and $J(t) \rightarrow 0$. Multiplying $J' = \alpha_c \mathbb{E} b_t^2 - s^2 \leq 0$ by χ^2 gives the second. The equality statement follows from $J' = 0$ at non-isolated support points. $\square$

8.7 Strong convergence through the endpoint

We have now identified the only possible weak limit and proved $\chi(1) = 0$. The remaining analytic task is to strengthen convergence of the terminal force. A trial obtained by rescaling the critical minimizer bounds the physical normalization in terms of $\alpha - \alpha_c$; comparison at that normalization then yields convergence of the full L^2 norm. Positivity probabilities will be treated separately using the cutoff bounds.

Proposition 8.9 (Strong critical convergence). *As $\alpha_n \downarrow \alpha_c$, the normalized physical variables satisfy*

$$R_n \rightarrow R \text{ in } L^2, \quad \gamma_n \rightarrow \gamma \text{ in } L^1(0, 1), \quad H_{n,1} \rightarrow H_1 \text{ in } L^2.$$

The physical endpoint gap is $h_n^ = H_{n,1} + \chi_n(1)R_n$, its residual is $(h_n^*)_- = \lambda_n R_n$, and $h_n^* \rightarrow H_1 \geq 0$ in L^2 .*

Proof. Use stars temporarily for the unique critical pair and put $S(s) = \frac{1}{2} \int_0^1 s$, $d_n = \alpha_n - \alpha_c$. The critical pair is a legitimate limiting normalized trial at any $\lambda > 0$: replace χ_* by $(\chi_* + \eta)/(1 + \eta)$, keep R_* fixed, and send $\eta \downarrow 0$. Its entropy and quadratic terms converge by domination and monotone convergence. Since its critical value is zero, its value at (α_n, λ) is

$$-\frac{d_n}{\alpha_c} S_* + \frac{\alpha_n \lambda}{2} \mathbb{E} R_*^2.$$

The original unnormalized trial has an additional factor λ . Choosing $\lambda = d_n S_*/(\alpha_c \alpha_n \mathbb{E} R_*^2)$ gives

$$e(\alpha_n) \geq \frac{d_n^2 S_*^2}{2 \alpha_c^2 \alpha_n \mathbb{E} R_*^2}.$$

Here $S_* > 0$ and $\mathbb{E} R_*^2 > 0$, so (8.6) implies $\lambda_n \geq c_1 d_n$.

Compare now at the physical value λ_n . Optimality and the same trial give

$$\left(1 + \frac{d_n}{\alpha_c}\right) J_{\alpha_c, 0}(R_n, s_n) + \frac{\alpha_n \lambda_n}{2} (\mathbb{E} R_n^2 - \mathbb{E} R_*^2) \leq \frac{d_n}{\alpha_c} (S_n - S_*). \quad (8.15)$$

The first term is nonnegative, $S_n \rightarrow S_*$ by uniform integrability, and d_n/λ_n is bounded. Therefore $\limsup \mathbb{E} R_n^2 \leq \mathbb{E} R_*^2$. Weak convergence and lower semicontinuity give the reverse inequality. The Hilbert-space identity for $\|R_n - R_*\|_2^2$ proves strong convergence. Uniqueness of all weak subsequential limits makes this conclusion valid for the original sequence.

Local L^1 convergence of γ_n becomes global because $\int_T^1 \gamma_n \leq \chi_n(T) \rightarrow \chi(T)$, and $\chi(T) \downarrow 0$ by Proposition 8.8. The same estimate gives $\chi_n(1) \rightarrow 0$. Doob's inequality applied to $\mathbb{E}[R_n - R \mid \mathcal{F}_t]$ gives L^2 convergence uniformly in time of the martingales. In (8.9), split the drift difference into $\gamma_n(r_n - r) + (\gamma_n - \gamma)r$. Conditional-expectation contraction and Minkowski's inequality give the explicit bound

$$\left\| \int_0^1 (\gamma_n r_n - \gamma r) dt \right\|_2 \leq \|\gamma_n\|_1 \|R_n - R\|_2 + \|\gamma_n - \gamma\|_1 \|R\|_2 \rightarrow 0.$$

Hence $H_{n,1} \rightarrow H_1$ in L^2 .

The terminal dual optimizer satisfies $R_n = (H_{n,1})_-/(\lambda_n + \chi_n(1))$. On its positive half-line the physical gap equals $H_{n,1}$; on its negative half-line the proximal formula (7.5) gives $h_n^* = -\lambda_n R_n$. These two cases are exactly $h_n^* = H_{n,1} + \chi_n(1)R_n$. The asserted limits follow. Finally (8.10) with $\Delta = 0$ gives $H_1 \geq 0$ and $RH_1 = 0$. $\square$

8.8 Contacts, atoms, and the prescribed order of limits

Strong L^2 convergence preserves the first two moments, but a positive variable may still converge to zero. To identify contacts we therefore use both uniform cutoff estimates from Section 7: the force estimate preserves $\mathbb{P}(R > 0)$, and the positive-gap estimate identifies $\mathbb{P}(H_1 = 0)$. Only after these masses are known can we normalize the limiting force law.

Lemma 8.10 (Contact and noncontact mass). *At the critical pair,*

$$\mathbb{P}(R > 0) = \mathbb{P}(H_1 = 0) = \frac{1}{\alpha_c}, \quad \{R > 0\} = \{H_1 = 0\} \quad \text{modulo null sets.}$$

Proof. Let $p_n = \mathbb{P}(R_n > 0)$. Corollary 7.9 gives $\alpha_n p_n \rightarrow 1$ and the direct cutoff bound

$$\lim_{u \downarrow 0} \sup_n \mathbb{P}(0 < R_n \leq u) = 0$$

after discarding a finite initial part of the density sequence. Apply strong convergence at positive continuity cutoffs and then send the cutoff to zero. This gives $\mathbb{P}(R > 0) = \lim_n p_n = 1/\alpha_c$.

Similarly the direct bound in (7.16), passed to the atomless endpoint law at each fixed UNSAT density, gives $\lim_{u \downarrow 0} \sup_n \mathbb{P}(0 < h_n^* \leq u) = 0$. Its constants and its upper cutoff are uniform in the density. This is a bound on the whole interval $(0, u]$ at fixed density; no interchange between a lower gap cutoff and the density limit is used. At every positive continuity point u of the law of H_1 , the L^2 convergence $h_n^* \rightarrow H_1$ gives convergence of the probabilities of $(-\infty, u]$. Subtracting the negative mass p_n and then sending $u \downarrow 0$ proves $\mathbb{P}(H_1 = 0) = 1/\alpha_c$; fixed-density endpoint laws have no atom at zero, as established in (7.5). Complementarity gives the inclusion $\{R > 0\} \subset \{H_1 = 0\}$. Equality of their probabilities proves the final assertion. $\square$

The contact atom at zero is now identified. To pass the microscopic observables at *every* positive cutoff, we still need to exclude atoms away from zero. The two variables require different estimates: nondegeneracy in deterministic variance time for R , and a bounded state drift away from zero for H_1 .

Lemma 8.11 (No positive atoms). *Neither R nor H_1 has an atom at any strictly positive point.*

Proof. The force variable. Monotonicity of s gives

$$s(t) \int_t^1 s(v) dv \leq \int_t^1 s(v)^2 dv \rightarrow 0.$$

Thus (8.8) approximates $a(t, h)$ by $s(t)h_-$ with an error $\varepsilon_t \rightarrow 0$ uniformly in h . Fix $x > 0$. When $x/2 \leq a(t, h) \leq 2x$ and t is sufficiently close to 1, one has $h < 0$ and $-h \geq (x/2 - \varepsilon_t)/s(t)$. A forward secant of length $x/(8s(t))$ stays negative. Convexity of a gives

$$a_h(t, h) \leq -s(t) + 16\varepsilon_t s(t)/x \leq -s(t)/2.$$

Consequently $1/2 \leq k(t, h) \leq 1$ in that region, whereas $0 \leq k \leq 1$ globally.

Change deterministic time to $v = I(t) = \int_0^t s^2$ and put $T = I(1) < \infty$. The martingale r then has Brownian coefficient $-k$: indeed, with $t(v) = I^{-1}(v)$,

$$\tilde{B}_v = \int_0^{t(v)} s(q) dB_q, \quad dr_{t(v)} = \frac{b_{t(v)}}{s(t(v))} d\tilde{B}_v = -k(t(v), H_{t(v)}) d\tilde{B}_v.$$

The process $\tilde{B}$ is Brownian in the filtration $\mathcal{F}_{t(v)}$, since its integrand and time change are deterministic and its quadratic variation is v . The coefficient has absolute value at most one and at least $1/2$ whenever $|r - x| \leq x/2$ in a sufficiently short final time interval. For $\eta > 0$ write $\ell = T - v + \eta^2$ and consider

$$\Phi(v, r) = \ell^{-\zeta/2} \exp\{-a_0(r - x)^2/\ell\}, \quad a_0 = \frac{1}{4}, \quad \zeta = \frac{1}{16}.$$

Its drift divided by Φ is

$$\frac{\zeta/2 - a_0 k^2}{\ell} + \frac{(-a_0 + 2a_0^2 k^2)(r - x)^2}{\ell^2}.$$

On $|r-x| \leq x/2$ both terms are nonpositive. Off this interval the second is at most $-x^2/(32\ell^2)$ and the first at most $1/(32\ell)$, so their sum is nonpositive when $\ell \leq x^2$. Choose $\delta > 0$ so that $v_0 = T - \delta$ lies in that final region and $2\delta \leq x^2$. For $\eta^2 \leq \delta$ one has $\ell \leq 2\delta$. At this fixed η , Φ and its stochastic derivative are bounded, so Itô's formula gives

$$\mathbb{E}\Phi(T, R) \leq \mathbb{E}\Phi(v_0, r_{t(v_0)}) \leq \delta^{-\zeta/2},$$

uniformly in η . Since $\Phi(T, R) \geq e^{-a_0}\eta^{-\zeta}$ on $|R-x| \leq \eta$,

$$\mathbb{P}(|R-x| \leq \eta) \leq C_x \eta^\zeta.$$

This proves the first assertion.

The gap variable. We show that the state drift is bounded on spatial intervals away from the contact point. Positivity of the drift and monotonicity of a imply, for $t < t' < 1$,

$$a(t, h) = \mathbb{E}a(t', H_{t'}^{t, h}) \leq \mathbb{E}a(t', h + B_{t'-t}).$$

Fix $x > 0$, write $\tau = 1 - t$, and set $t_j = 1 - 2^{-j}\tau$ and $x_j = x/2 + (x/2)2^{-j/4}$. Split the last expectation according to whether the Brownian increment is below $-(x_j - x_{j+1})$. The complement costs at most $a(t_{j+1}, x_{j+1})$. On the exceptional event, the global linear-growth bound in (8.8) and the Gaussian tail estimate bound the cost by

$$C_x(1 + s(t_{j+1})) \exp\{-c_x 2^{j/2}/\tau\}.$$

Concavity and $\chi(1) = 0$ give $s(t_{j+1}) \leq 2^{j+1}s(t)$. The remaining term $a(t_j, x_j)$ tends to zero by the uniform error ε_{t_j} . Summing the rapidly decreasing tail series proves

$$\sup_{h \geq x} a(t, h) \leq C_x s(t) \exp\{-c_x/(1-t)\}$$

for t close to 1. Since $\chi(t) = \int_t^1 \gamma(v) dv \geq (1-t)\gamma(t)$, the drift $\gamma(t)a(t, h)$ is bounded uniformly near time 1 on every compact interval contained in $(0, \infty)$; explicitly it is at most $C_x(1-t)^{-1} \exp\{-c_x/(1-t)\}$ for $h \geq x$.

Start at a fixed late time, stop on leaving such an interval, and set the drift to zero after exit. The resulting diffusion has bounded adapted drift up to time 1. The exponential martingale in Girsanov's theorem is integrable because that drift is bounded; its law is equivalent to Brownian motion with its initial value. Its terminal law therefore has no atoms. The original diffusion agrees with it on the event of no exit. Continuity of H implies that every event $\{H_1 = x\}$ with $x > 0$ is covered by a countable union of these no-exit events, using rational interval endpoints and rational starting times. Thus $\mathbb{P}(H_1 = x) = 0$. $\square$

Theorem 8.12 (The ordered limiting observables). *The Gaussian observables in the problem have all their stated ordered limits, at every $u > 0$ and $s > 0$, and are given by*

$$G_J(u) = \mathbb{P}(0 < H_1 \leq u), \quad F_J(s) = \mathbb{P}\left(\frac{R}{\alpha_c c} \leq s \mid R > 0\right). \quad (8.16)$$

The force law has mean one and no atom at zero. These limiting laws are also the laws for all Gaussian scale mixtures in the fixed-temperature comparison range. More explicitly, for every $0 < \delta < u$ and $s > 0$, the individual limits at each level of

$$\begin{aligned} \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \rightarrow \infty} \lim_{N \rightarrow \infty} G_{N, \beta, \alpha}(\delta, u) &= \mathbb{P}(\delta < H_1 \leq u), \\ \lim_{\alpha \downarrow \alpha_c} \lim_{\beta \rightarrow \infty} \lim_{N \rightarrow \infty} F_{N, \beta, \alpha}(s) &= F_J(s) \end{aligned}$$

exist; the additional limit $\delta \downarrow 0$ in the first line gives $G_J(u)$.

Proof. Thermodynamic and zero-temperature limits at fixed density. First take $N \rightarrow \infty$ at fixed (β, α) , using the inner empirical-marginal result proved earlier. At fixed UNSAT density, Corollary 8.3 and (7.5) give the zero-temperature endpoint law, with no atoms. For completeness, let $h_{\beta, \alpha}$ denote the inner limiting gap variable and put $p_{\beta, \alpha} = \mathbb{P}(h_{\beta, \alpha} < 0)$ and $\mu_{\beta, \alpha} = \mathbb{E}(h_{\beta, \alpha})_-$. Proposition 7.3 supplies uniform integrability of the negative first moment, and the absence of an atom at zero gives

$$p_{\beta, \alpha} \longrightarrow p_\alpha > 0, \quad \mu_{\beta, \alpha} \longrightarrow \mu_\alpha > 0.$$

Positivity follows from $\alpha \mathbb{E}(h_\alpha^*)_-^2 = 2e(\alpha) > 0$. The inner limiting force CDF at $s > 0$ is

$$\frac{1}{p_{\beta, \alpha}} \mathbb{P} \left(-s \frac{\mu_{\beta, \alpha}}{p_{\beta, \alpha}} \leq h_{\beta, \alpha} < 0 \right).$$

Both limiting interval endpoints are atomless under the law of h_α^* , so this expression converges at every $s > 0$. Thus the full $\beta \rightarrow \infty$ limit exists for all the required gap and force arguments, and its conditioning mass is preserved before the density is changed. Uniqueness of every subsequential endpoint law yields convergence along the full temperature limit.

The critical-density limit and the final gap cutoff. Now approach α_c from above. Proposition 8.9 gives convergence of the physical endpoint gaps and the scaled residuals, and Lemma 8.11 permits every positive cutoff, rather than just a selected continuity sequence.

For gaps keep $0 < \delta < u$ fixed throughout these three limits; the answer is $\mathbb{P}(\delta < H_1 \leq u)$. Only then let $\delta \downarrow 0$, which removes the contact atom and gives the first formula in (8.16). For forces the conditional mean at density α_n is c_n/p_n , which converges to $\alpha_c c$. Lemma 8.10 preserves the conditioning mass, and its cutoff argument loses no positive residual mass to zero. Subtracting the noncontact mass and applying the positive-atom assertion gives the second formula at every $s > 0$. Since $\mathbb{E}R = c$ and $\mathbb{P}(R > 0) = 1/\alpha_c$, its mean is one. Its conditional probability of $\{0\}$ is zero; the uniform cutoff bound also explicitly prevents an atom from appearing there in the limit of the physical conditional laws.

Finally the Gaussian and mixture inner-limit observables agree at each fixed temperature and density in a neighborhood above the common threshold, by Section 5. Equality of these functions passes through exactly the same ordered limits. The limits $u \downarrow 0$ and $s \downarrow 0$, which determine their exponents, are taken only in the subsequent sections. $\square$

9 Uniform control of the critical layer

Throughout this section, (χ, R) is the normalized critical optimizer, $\alpha = \alpha_c$, $c = \mathbb{E}R > 0$, and $Y = H_1$. The inputs from Section 8 are the PDE and diffusion representation of Lemma 8.5, the first-order conditions of Lemma 8.7, and the endpoint conclusions of Proposition 8.8 and Lemma 8.10. In particular,

$$\begin{aligned} \chi(1) &= 0, & \int_0^1 \chi^{-2} &< \infty, \\ \{R > 0\} &= \{Y = 0\} \quad \text{modulo null sets}, & \mathbb{P}(R > 0) &= 1/\alpha. \end{aligned}$$

The increase support of γ accumulates at 1, and $\gamma = 0$ on an initial interval. At its interior support points we have $J = 0$, $J' \leq 0$, and (8.12), where $J = P - I$ is the variational defect defined in Section 8. All constants below may depend on this fixed optimizer and on a fixed interior starting time; they do not depend on the distance to the endpoint. Constants with a subscript may also depend on that parameter.

Write $\tau = 1 - t$. Whenever τ is the time argument, $\chi(\tau)$ means the original susceptibility at physical time $1 - \tau$; the same convention applies to the other deterministic time-dependent

quantities. Thus $\chi_\tau = \gamma \geq 0$ and $\chi(\tau) \rightarrow 0$ as $\tau \downarrow 0$; the susceptibility is concave in this clock. The physical-time normalization remains $\chi(t=0) = 1$. In this clock write $\mathcal{S} = \{\tau \in (0,1) : 1 - \tau \in \text{supp } d\gamma\}$. It has points arbitrarily close to zero. A support gap means a component (τ_1, τ_2) of its complement, on which χ is affine. When extending estimates across gaps, we work below a fixed point of $\mathcal{S}$; every intervening gap then has both endpoints in $\mathcal{S}$. Choosing this point sufficiently small puts both endpoints in the range of any support-point estimate under consideration. Set

$$\begin{aligned} s &= \chi^{-1}, & m(t, h) &= \chi(t)a(t, h), \\ k &= -m_h, & l &= 1 - k, \\ X(t, h) &= h + m(t, h), & w(t) &= \int_t^1 s(v)^2 dv, \\ D &= \chi \sqrt{w/\tau}. \end{aligned}$$

The same symbol X_t denotes $X(t, H_t)$. In particular,

$$dr_t = -s(t)k(t, H_t) dB_t, \quad dX_t = l(t, H_t) dB_t, \quad r_1 = R, \quad X_1 = Y.$$

The curvature bounds give $0 \leq k, l \leq 1$ and $k_h < 0$, hence $0 < k < 1$ at every finite spatial point. Since $|l| \leq 1$, X_t is a square-integrable martingale. Its L^2 endpoint limit is Y , because $H_t \rightarrow Y$, $\chi \rightarrow 0$, and $r_t \rightarrow R$ in L^2 . Thus $X_t = \mathbb{E}[Y \mid \mathcal{F}_t] \geq 0$, just as $r_t = \mathbb{E}[R \mid \mathcal{F}_t] \geq 0$. Finally $w(\tau) \geq \tau/\chi(\tau)^2$ gives $D \geq 1$. Here $m = \chi a$ is the rescaled force field; the earlier finite-temperature CDF is m_q .

The identity $H_t = X_t - \chi r_t$ explains the susceptibility factor in D . We first prove that D stays bounded, making the spatial scales $\chi\sqrt{w}$ and $\sqrt{\tau}$ comparable, and derive the endpoint mass estimates (9.20). We then use the inverse-slope bounds (9.21) to exclude terminal gaps in the increase support and obtain (9.26). These are the inputs for Section 10.

The equation for m in remaining time is

$$m_\tau = \frac{1}{2}m_{hh} + dm(1 + m_h), \quad d = \gamma/\chi, \quad 0 \leq d \leq \tau^{-1}. \quad (9.1)$$

The last inequality follows from $\tau\gamma(\tau) \leq \chi(\tau)$.

9.1 Value barriers and inverse flows

Let ϕ and Φ be the standard normal density and distribution function. The following estimates do not require any information about D beyond $D \geq 1$.

Lemma 9.1 (Pointwise barriers). *For $z = h/\sqrt{\tau}$,*

$$\sqrt{\tau} [\phi(z) - z\Phi(-z)] \leq m(t, h) \leq \sqrt{\tau} \frac{\phi(z)}{\Phi(z)}. \quad (9.2)$$

Consequently, with fixed positive constants C, c_0 ,

$$k(t, h) \leq Ce^{-c_0 z^2} \quad (z \geq 1), \quad l(t, h) \leq C|z|^{-2} \quad (z \leq -1), \quad \int_{\mathbb{R}} kl dh \leq C\sqrt{\tau}. \quad (9.3)$$

Proof. First $m \geq h_-$. Indeed $X_h = l \geq 0$, and the bounded difference between $f(t, h)$ and $-h_-^2/(2\chi)$ in (8.8) forces $X(t, h) \rightarrow 0$ as $h \rightarrow -\infty$: a nonzero limit of the monotone function X would make the integral of $f_h + h/\chi = X/\chi$ unbounded on that half-line. Thus $X \geq 0$; on the other half-line use $m \geq 0$.

The heat evolution of the hinge is the lower function in (9.2). It is a subsolution of (9.1), since $m(1 + m_h) \geq 0$. For the upper function U , direct differentiation of $\rho(z) = \phi(z)/\Phi(z)$,

using $\rho' = -z\rho - \rho^2$, gives $U_\tau = U_{hh}/2 + \tau^{-1}U(1 + U_h)$ and $-1 \leq U_h \leq 0$. It is therefore a supersolution for $d \leq \tau^{-1}$.

Here comparison must start before the singular endpoint. At $\tau' > 0$, (8.8) gives

$$0 \leq m(\tau', h) - h_- \leq C\chi(\tau')\sqrt{w(\tau')} \quad \text{uniformly in } h.$$

For the difference between two solutions, the zeroth-order coefficient in the linear comparison equation is at most d . Thus the positive error at τ' is amplified by at most $\exp(\int_{\tau'}^{\tau} d) = \chi(\tau)/\chi(\tau')$. The resulting error is at most $C\chi(\tau)\sqrt{w(\tau')}$, which tends to zero as $\tau' \downarrow 0$. Comparison may first be made with smooth coefficients and on bounded spatial intervals; the uniform linear growth and curvature bounds permit the limits.

Finally, for a convex decreasing function m , its negative derivative at h is bounded above by the backward secant over a distance $\sqrt{\tau}$. Applying the upper value barrier gives the positive Gaussian tail for k . Apply the forward secant to the convex increasing function X on the negative half-line, using $X(t, h) \leq C\tau/|h|$ there; this gives the stated bound for l . The integral estimate follows by using $kl \leq 1$ on $|h| \leq \sqrt{\tau}$ and integrating the two tails. $\square$

We next turn these pointwise bounds into estimates under the law of H_t . The inverse flow retains the Gaussian density at an early time; its Jacobian records how much the critical drift compresses that law.

Choose $t_0 > 0$ in the interval on which $\gamma = 0$. Then $\chi(t_0) = 1$ and H_{t_0} has a strictly positive Gaussian density p_0 , independently of subsequent Brownian increments. Conditional on those increments, the flow F_t taking H_{t_0} to H_t is increasing and convex: its first derivative solves an equation with coefficient $-dk \in [-d, 0]$, and its second derivative has nonnegative forcing dm_{hh} . If $\mathcal{I}_t = F_t^{-1}$ and $\mathcal{J}_t = \mathcal{I}'_t$, then

$$1 \leq \mathcal{J}_t \leq \chi(t)^{-1}, \quad \mathcal{J}_t \text{ is nonincreasing}, \quad p_t(h) = \mathbb{E}[p_0(\mathcal{I}_t(h))\mathcal{J}_t(h)] \leq C/\chi(t). \quad (9.4)$$

In particular the critical state has a density at every interior time.

We shall also use the lower bound

$$\inf_{|h| \leq K\chi(t)} p_t(h) \geq c_K > 0. \quad (9.5)$$

Here is a uniform moment justification. The inverse flow, read in increasing remaining time u , solves the additive-noise equation $d\tilde{H}_u = -d(u)m(u, \tilde{H}_u)du + d\tilde{B}_u$. Its positive part is bounded by its initial positive part and twice the Brownian supremum. For $N = -\tilde{H}/\chi$,

$$dN = -s d\tilde{B} + \frac{\gamma X(u, \tilde{H})}{\chi^2} du.$$

The value barrier gives $(-h)_+ X(u, h) \leq Cu$. Applying Itô's formula to N_+^2 therefore bounds its expected growth by $Cs^2 du$, since $u\gamma/\chi \leq 1$. Its initial second moment is at most K^2 . The inverse position at t_0 consequently has a uniformly bounded second moment. On a fixed interval its Gaussian density is bounded below, and Markov's inequality puts a fixed positive fraction of the inverse positions in that interval. Equation (9.4), with $\mathcal{J}_t \geq 1$, proves (9.5). All these inverse statements can also be obtained pathwise from the additive-noise integral equation, so no independence between the inverse position and its Jacobian is used.

At an interior point of the increase support of γ , (8.12) gives $\mathbb{E}kl \geq c\chi$. Combining this with (9.3) and $p_t \leq C/\chi$ gives

$$\chi(\tau) \leq C\tau^{1/4}. \quad (9.6)$$

This holds throughout a terminal interval. Indeed on a complementary support interval χ is affine, whereas $\tau^{1/4}$ is concave, so the endpoint inequalities imply the inequality in the interval.

9.2 The ratio of the two natural scales

Let $\pi(t, h) = \mathbb{P}(R > 0 \mid H_t = h)$ and $q_g = 1 - \pi$. The Markov property supplies these conditional probabilities as measurable functions of the state. The curvature equation gives

$$dk(t, H_t) = k_h(t, H_t) dB_t - d(t)k(t, H_t)l(t, H_t) dt.$$

Its endpoint limit is $\mathbf{1}_{\{R>0\}}$. On $Y > 0$ this follows from (9.3); on $R > 0$, the uniform derivative error in (8.8) is $o(1)$, so $H_t = -\chi(t)R + o(\chi(t))$. A secant of length a fixed fraction of $\chi(t)R$ on the negative side then gives $k(t, H_t) \rightarrow 1$. The two events exhaust the space. Conditional bounded-supermartingale convergence thus gives

$$k \geq \pi, \quad q_g(t, h) \geq \Phi\left(\frac{h}{\chi\sqrt{w}}\right) \quad \text{for almost every } h. \quad (9.7)$$

For the second inequality, run the inverse flow from position zero at $t' > t$. The variable N in the preceding argument has nonnegative drift, and hence dominates, under the same noise, a centered Gaussian of variance $w(t) - w(t')$. The monotonicity of the flow shows

$$\mathbb{P}(H_{t'} > 0 \mid H_t = h) \geq \Phi\left(\frac{h}{\chi(t)\sqrt{w(t) - w(t')}}\right).$$

Now let $t' \uparrow 1$. On the force event the sign is eventually negative by the just established secant argument; on the gap event it is eventually positive. Conditional dominated convergence proves the claim.

A second useful consequence of complementarity is, for $v = r/\sqrt{w} \geq 2$,

$$X \leq C\sqrt{\tau}v\Phi(-v), \quad l \leq C\frac{v^2}{D}\Phi(-v). \quad (9.8)$$

Indeed the event $Y > 0$ requires the martingale r to hit zero. Its remaining quadratic variation is at most w , so the reflection principle after Brownian time change bounds this event's conditional probability by $q \leq 2\Phi(-v)$. The martingale X has remaining quadratic variation at most τ . Integration of its Gaussian tail gives, for any event A of probability q ,

$$\mathbb{E}[|Y - X|\mathbf{1}_A \mid H_t = h] \leq C\sqrt{\tau}q\sqrt{\log(2/q)}.$$

As Y vanishes off $A = \{Y > 0\}$, this bounds $X(1 - q)$. The normal tail estimates give the first claim, also with adjusted constants for all $v \geq 1$. To bound $l = X_h$, take its forward secant over $\delta h = \chi\sqrt{w}/v$. The variable $r/\sqrt{w}$ at the new position is at least $v - v^{-1}$, because $|r_h| \leq 1/\chi$. The ratio of the corresponding normal tails is bounded by an absolute constant. Dividing the value bound by δh proves the second claim. Continuity extends estimates derived at almost every state.

Lemma 9.2 (Subpower scale ratio). *As $\tau \downarrow 0$,*

$$\log D = o(|\log \tau|). \quad (9.9)$$

Proof. At a support point, stability says $\mathbb{E}(k^2 - \pi) \leq 0$. Suppose instead that along support points $D \geq \tau^{-\varepsilon}$ for some $\varepsilon > 0$. Since $h = X - \chi r \geq -\chi r$, (9.7) gives $q_g \geq \Phi(-v)$. Thus (9.8) implies $k^2 - \pi \geq q_g/2$ when $2 \leq v \leq a\sqrt{D}$, for a fixed sufficiently small $a > 0$. When $v > a\sqrt{D}$ its negative part is bounded by $kl \leq C\exp(-c_0D)$. When $v < 2$ and $h \leq -K\sqrt{\tau}$, take K sufficiently large in (9.3); again $k^2 - \pi \geq q_g/2$. In particular, on

$$I_- = \left[-\frac{1}{2}\chi\sqrt{w}, -\frac{1}{4}\chi\sqrt{w}\right]$$

there is a fixed positive lower bound for $k^2 - \pi$ once D is large. This follows from $q_g \geq \Phi(-1/2)$, the negative-tail slope bound, and the upper value barrier, which gives $v \leq 1$ on this interval. Writing $p_- := \inf_{I_-} p_t$, the positive contribution is at least $c_0 \chi \sqrt{w} p_-$. Equation (9.5) gives $p_- \geq c_0$ for late times.

For clarity, the required density comparison has the following quantitative form. If $h_2 > h_1$, $\delta = (h_2 - h_1)/\chi$, and P is fixed, then

$$p_t(h_2) \leq \exp\{C_P[\delta\sqrt{|\log \tau|} + \delta^2 + 1]\} p_t(h_1) + \tau^P. \quad (9.10)$$

To prove it, use the decreasing inverse Jacobian, the bound $|\mathcal{I}_t(h_2) - \mathcal{I}_t(h_1)| \leq \delta$, and the Gaussian density ratio on $|\mathcal{I}_t(h_2)| \leq C_P \sqrt{|\log \tau|}$. On the complement use $\mathcal{J}_t \leq \chi^{-1} \leq C\tau^{-1/2}$; the last inequality follows from $\tau/\chi^2 \leq w \leq w(t_0)$. Increasing C_P makes the discarded Gaussian density at most τ^P .

Apply this with $h_1 \in I_-$ and $-K\sqrt{\tau} \leq h_2 \leq C_P\sqrt{\tau|\log \tau|}$. Here $\delta \leq C_P\sqrt{w(1+|\log \tau|)}$, so the exponential factor is $\tau^{-o(1)}$. The negative contribution in this layer is at most $C\sqrt{\tau}\tau^{-o(1)}p_-$ by $(k^2 - \pi)_- \leq kl$ and (9.3). Above the layer the positive Gaussian tail of k makes the contribution an arbitrarily small power of τ . Below it the only remaining negative contribution is $C\exp(-c_0D)$. Both are negligible compared with $\chi\sqrt{w}p_- = D\sqrt{\tau}p_-$, a contradiction.

It remains to cross a support gap $\tau_1 < \tau < \tau_2$. Affineness gives the exact identity

$$w(\tau) = w_1 + \frac{\tau - \tau_1}{\chi_1 \chi(\tau)}. \quad (9.11)$$

If $\tau_2/\tau_1 \leq 2$, concavity gives $\chi/\chi_1 \leq 2$, so the assertion transfers immediately. Otherwise $\chi_2/\chi_1 \leq 2D_2^2$, by applying (9.11) at τ_2 . Hence

$$D(\tau)^2 \leq D_1^2 \frac{\tau_1}{\tau} (2D_2^2)^2 + 2D_2^2.$$

For every fixed $\eta > 0$ the endpoint estimates bound D_i by $C_\eta \tau_i^{-\eta}$. The factor τ_1/τ absorbs the possible disparity between τ_1 and τ ; since $\tau_2 \geq \tau$, for $0 < \eta < 1/2$ we obtain explicitly

$$D(\tau)^2 \leq C_\eta \frac{\tau_1^{1-2\eta}}{\tau} \tau_2^{-4\eta} + C_\eta \tau_2^{-2\eta} \leq C_\eta (\tau^{-6\eta} + \tau^{-2\eta}).$$

As η is arbitrary, this proves the same subpower conclusion in the gap. $\square$

Lemma 9.3 (Strict improvement of the endpoint power). *There exist $\eta > 0$ and $C < \infty$ such that $s(\tau) \leq C\tau^{-1/2+\eta}$. In particular $w(\tau) \leq C\tau^{2\eta}$.*

Proof. In the inverse formula, $\log \mathcal{J}_t = \int d(u)k(u, \tilde{H}_u) du$. Starting at $h \in [-a\sqrt{\tau}, 0]$, the inverse drift is nonpositive, so at every reverse time its probability of a negative position is at least $1/2$. The barriers and secants give a uniform lower bound $k \geq \kappa > 0$ for negative positions. Moreover the inverse moment bound proved above gives $\mathbb{E} \log p_0(\mathcal{I}_t(h)) \geq -C$ uniformly: its initial normalized negative position has square at most $a^2\tau/\chi^2 \leq a^2w$. Jensen's inequality therefore yields $p_t(h) \geq C^{-1}\chi^{-\kappa/2}$. Here both k and l have fixed positive lower bounds on $[-a\sqrt{\tau}, 0]$ if $a > 0$ is sufficiently small. To see this uniformly in time, the heat barrier gives $m(0) \geq \phi(0)\sqrt{\tau}$, while the upper barrier makes $m(K\sqrt{\tau})/\sqrt{\tau}$ arbitrarily small for fixed large K . The secant from 0 to $K\sqrt{\tau}$ bounds $k(0)$ below, and $k(h) \geq k(0)$ for $h \leq 0$. Likewise $X(h) \geq X(0) - |h| \geq (\phi(0) - a)\sqrt{\tau}$ on that interval, whereas $X(-K\sqrt{\tau}) \leq C\sqrt{\tau}/K$. The backward secant from $-K\sqrt{\tau}$ to h therefore gives

$$l(h) \geq \frac{X(h) - X(-K\sqrt{\tau})}{h + K\sqrt{\tau}} \geq \frac{\phi(0) - a - C/K}{K} > 0$$

after fixing $a < \phi(0)/2$ and then a sufficiently large K . Together with (9.6) this gives

$$\mathbb{E}kl \geq C^{-1}\sqrt{\tau}\chi^{-\kappa/2} \geq C^{-1}\tau^{b_0}, \quad b_0 = \frac{1}{2} - \frac{\kappa}{8} < \frac{1}{2}.$$

The drift identity for k , including its endpoint limit, gives $\mathbb{E}k - 1/\alpha = \int_t^1 d\mathbb{E}kl$. Thus, if τ_2 is a late support point and $0 < \tau_1 < \tau_2$,

$$\chi(\tau_2) \geq C^{-1}\tau_1^{b_0} \log \frac{\chi(\tau_2)}{\chi(\tau_1)}. \quad (9.12)$$

Put $b = \limsup_{\tau \downarrow 0} \log \chi(\tau) / \log \tau$. The bounds already proved imply $1/4 \leq b \leq 1/2$. Fix $0 < \varepsilon < 1$ and choose $\tau_n \downarrow 0$ realizing this limsup. For sufficiently small $\delta > 0$, eventually $\chi(u) \geq u^{b+\delta}$ at all small u , whereas along a subsequence $\chi(\tau_n) \leq \tau_n^{b-\delta}$. At $u_n = \tau_n^{1-\varepsilon}$, choosing $(2-\varepsilon)\delta < b\varepsilon/4$ gives

$$\frac{\chi(u_n)}{\chi(\tau_n)} \geq \tau_n^{-b\varepsilon+(2-\varepsilon)\delta} \geq \tau_n^{-3b\varepsilon/4}.$$

If u_n lies in a gap, move to its smaller endpoint $\tau_{2,n}$; otherwise put $\tau_{2,n} = u_n$. The loss in susceptibility is subpower in $1/u_n$. Indeed it is at most two on a gap of time ratio at most two; on any other gap it is at most $2D(\tau_2)^2$, by (9.11), and $\tau_2 \geq u_n$. Lemma 9.2 therefore gives, for all large n ,

$$\frac{\chi(\tau_{2,n})}{\chi(\tau_n)} \geq \tau_n^{-b\varepsilon/2}.$$

In particular $\tau_n < \tau_{2,n} \leq u_n$. This is the ordering needed to apply (9.12) with lower endpoint τ_n . Concavity then gives

$$\chi(\tau_n) \geq \frac{\tau_n}{\tau_{2,n}} \chi(\tau_{2,n}) \geq C^{-1}\tau_n^{b_0+\varepsilon} |\log \tau_n|.$$

Therefore $b \leq b_0 + \varepsilon$. Letting $\varepsilon \downarrow 0$ gives $b \leq b_0 < 1/2$. Choose any exponent strictly between b_0 and $1/2$ and use the defining limsup to obtain the asserted bound on s ; integration gives the bound on w . $\square$

9.3 A martingale bin comparison

We give the probability estimate needed to replace subpower control by a uniform bound. Let $(Z_v)_{0 \leq v \leq T}$ be a nonnegative continuous martingale on a finite deterministic horizon, with $dZ_v = \sigma_v dW_v$ and $|\sigma_v| \leq 1$. Write $u = T - v$ for its remaining time. Suppose that, for each fixed $a > 0$, there is $c_a > 0$, independent of v and the sample path, such that

$$Z \geq a\sqrt{u} \implies |\sigma| \geq c_a > 0 \quad (a > 0). \quad (9.13)$$

We also assume that its law has a positive continuous density on $(0, \infty)$ at each strictly interior time $0 < v < T$. Continuity of its paths and absence of positive atoms then make each moving-bin probability continuous for $0 < u < T$. For r take variance time $\int_0^t s^2$, remaining time $u = w$, and $|\sigma| = k$. Condition (9.13) follows from $m/\sqrt{\tau} \geq aD \geq a$, the upper barrier, and a secant: the corresponding z is bounded above, and k is bounded below on that half-line. No upper bound on D is needed here.

Lemma 9.4 (Bin comparison). *Under these assumptions there exist $0 < u_0 < T$ and finite constants C, P , depending on the process and the constants c_a , such that $M_j(u) := \mathbb{P}(j\sqrt{u} \leq Z_{T-u} \leq (j+1)\sqrt{u})$ satisfies*

$$M_j(u) \leq Cj^P M_1(u), \quad j \geq 2, \quad 0 < u \leq u_0. \quad (9.14)$$

Proof. We first prove a conditional steering estimate; the martingale need not itself have the Markov property. After any stopping time S , apply Brownian time change to the shifted continuous martingale $Z_{S+v} - Z_S$ in the filtration $\mathcal{F}_{S+v}$. Extending it with independent Brownian increments after the terminal time if necessary, this gives $Z_{S+v} = Z_S + W_{Q_v}$, where $Q_v = \int_S^{S+v} \sigma^2 dt$ and W is Brownian with respect to a filtration containing $\mathcal{F}_S$ at time zero. Its Brownian law thus holds conditionally on $\mathcal{F}_S$. All estimates below are uniform in that initial information. Conditional spatial and time scaling permits us to set the remaining time at S equal to one.

Consider first a single step with time ratio $\rho \in [\sqrt{2}, 2]$, elapsed time $L = 1 - \rho^{-1}$, and starting position $x \in [1, 2]$. Put

$$L_{\min} = 1 - 1/\sqrt{2}, \quad \kappa = c_{1/2}, \quad q_* = \kappa^2 L_{\min}/2, \quad y_\rho = \frac{3}{2\sqrt{\rho}}.$$

The terminal first bin is $[\rho^{-1/2}, 2\rho^{-1/2}]$. Take the fixed spatial corridor $(1/2, 3)$. Up to its first exit, (9.13) gives $\kappa^2 \leq \dot{Q} \leq 1$, since the remaining time is at most one. Let $g_{x,\rho}(q)$ move linearly from x to y_ρ on $[0, q_*]$, and stay at y_ρ on $[q_*, 1/2]$. It lies in $[1, 2]$. On the Brownian event

$$\sup_{0 \leq q \leq 1/2} |x + W_q - g_{x,\rho}(q)| < 1/16,$$

no first corridor exit can occur before elapsed time L : at such an exit, $Q_v \leq v \leq L \leq 1/2$, whereas the displayed event places the exit position strictly inside the corridor. This first-exit argument establishes the lower clock bound without assuming it on the event being proved. In particular $Q_L \geq \kappa^2 L \geq 2q_*$, so Z_{S+L} is within $1/16$ of y_ρ and lies in the terminal first bin. The paths $g_{x,\rho} - x$ form a compact family of piecewise linear paths starting at zero. Brownian motion has positive probability to stay in a fixed uniform neighborhood of each such path; a finite cover by neighborhoods of radius $1/32$ gives one positive lower bound for the displayed event, independent of x and ρ . This elementary tube fact follows, for example, by requiring finitely many Gaussian increments to lie in prescribed open intervals and the intervening Brownian bridges to stay in prescribed open strips.

For any total ratio $u'/u \geq \sqrt{2}$, partition it into factors in $[\sqrt{2}, 2]$ and iterate the conditional estimate. The number of steps is at most $1 + \log_2(u'/u)$, so the resulting probability is at least $c(u/u')^{P_1/2}$ for a fixed P_1 . On the first step the starting interval may instead be $[1.5, J]$: use corridor $(1/2, J+1)$ and the same piecewise linear path, obtaining a first-step constant depending only on fixed J , with the same subsequent power. There is also a uniform bound when starting at exactly 1.5 and $u'/u \in [1, \sqrt{2}]$. In normalized units the event $\sup_{q \leq L_{\min}} |W_q| < 1/16$ suffices, since $[1.5 - 1/16, 1.5 + 1/16]$ lies inside every corresponding terminal first bin. Here only $Q_L \leq L \leq L_{\min}$ is used; no lower clock bound is necessary.

The steering estimates now give a polynomial lower bound for reaching the first bin across the scale ratios used below. We use that bound in an induction over decreasing dyadic slabs of remaining time, comparing the other bins to the first and controlling distant starting bins by Gaussian tails. On $[u_0/2, u_0]$, M_1 has a positive minimum, so the assertion follows from $M_j \leq 1$. Suppose the assertion has been established on earlier slabs and let u lie in the next slab. For $j \geq 2$, set $u_a = \min(j^2 u, u_0) \geq 2u$. If $u_a = u_0$, steering from its first bin gives $M_1(u) \geq c(u/u_0)^{P_1/2} M_1(u_0)$. Since $j^2 u \geq u_0$, the trivial bound $M_j(u) \leq 1$ proves the desired inequality with power P_1 and a fixed constant.

In the uncapped case $u_a = j^2 u$, condition at u_a . Any path that starts below $1.5\sqrt{u_a}$ and ends in bin j at time u must hit the *moving* boundary $Z = 1.5\sqrt{\text{remaining time}}$. Let A be the event of starting below this boundary and hitting it before the terminal observation, and let S be the first hitting time. On A its remaining time u_h belongs to $[u, u_a]$. The conditional

steering estimate just proved, restarted at S , gives a probability at least cj^{-P_1} of ending in the first bin, since $u_h/u \leq u_a/u = j^2$; its short-ratio case covers $u_h/u \in [1, \sqrt{2}]$. Therefore

$$M_1(u) \geq \mathbb{E}[\mathbf{1}_A \mathbb{P}(Z \text{ ends in the first bin} \mid \mathcal{F}_S)] \geq cj^{-P_1} \mathbb{P}(A).$$

The target contribution from starting positions below the boundary is at most $\mathbb{P}(A)$, because every such target path must hit it. This proves the desired bound $Cj^{P_1}M_1(u)$ without conditioning on later target success. Starting positions between $1.5\sqrt{u_a}$ and $J\sqrt{u_a}$ obey the same bound, with C_J , by direct steering.

For an ancestor in bin $i \geq J$ at u_a , ending in the target bin requires a martingale displacement at least a constant times $i\sqrt{u_a}$, once J is a fixed sufficiently large integer: the upper target endpoint divided by $\sqrt{u_a}$ is $(j+1)/j \leq 3/2$. The exponential martingale bound therefore gives probability at most $Ce^{-c_0i^2}$. The induction hypothesis at u_a bounds the ancestor mass by $C_*i^P M_1(u_a)$, and steering gives $M_1(u_a) \leq Cj^{P_1}M_1(u)$. Thus the remaining contribution is at most

$$C_*Cj^{P_1}M_1(u) \sum_{i \geq J} i^P e^{-c_0i^2}.$$

Choose $P \geq P_1$, then J making the coefficient of C_* at most $1/2$, and finally C_* large enough to absorb the nonrecursive terms and the initial slab. These choices are independent of the slab and close the induction. $\square$

Proposition 9.5 (Bounded scale ratio). *There is a constant C such that $1 \leq D \leq C$ near the endpoint. Consequently $w \asymp \tau/\chi^2$ and $\int_t^1 s \asymp \tau/\chi$.*

Proof. Repeat the stability argument in Lemma 9.2, now along an arbitrary support sequence on which $D \rightarrow \infty$. Use the force bin $1 \leq r/\sqrt{w} \leq 2$ for the positive contribution. The first estimate in (9.8), valid also for $v \geq 1$, places its state interval below $-K\sqrt{\tau}$ when D is large. There $q_g \geq \Phi(-2)$ and $l \rightarrow 0$ uniformly, so its contribution is at least $c_0M_1(w)$. Its interval has length at least $\chi\sqrt{w}$, because $|r_h| \leq 1/\chi$.

Lemma 9.3 makes the exponential factor in (9.10) uniformly bounded when comparing this bin to $[-K\sqrt{\tau}, C_P\sqrt{\tau}|\log \tau|]$. Using the infimum density on the bin, (9.3) bounds the negative layer contribution by $CM_1(w)/D$. Errors τ^P are negligible compared with $M_1(w)$: the density lower bound and the bin length give $M_1(w) \geq c_0\chi\sqrt{w} \geq c_0\sqrt{\tau}$. For $v > a\sqrt{D}$, sum (9.8) over the force bins and use Lemma 9.4. This contribution is at most $CM_1(w) \sum_{j \geq a\sqrt{D}-1} j^{P+2} e^{-c_0j^2} = o(M_1(w))$. The other regions have nonnegative contribution as before. Stability is contradicted. Formula (9.11) transfers the result across gaps, with the same two cases as in the subpower proof. The estimate for $\int s$ follows from Cauchy–Schwarz for the upper bound and monotonicity for the lower bound. $\square$

9.4 Density comparison and endpoint masses

Bounded D implies $d \log w / d \log \tau = D^{-2} \geq \delta_0 > 0$. Combining this with $w \asymp \tau/\chi^2$ gives, for $0 < \tau \leq u \leq 1 - t_0$,

$$\frac{\chi(u)}{\chi(\tau)} \leq C(u/\tau)^{(1-\delta_0)/2}, \quad \int_{\tau}^{1-t_0} \frac{\gamma(u)}{\sqrt{u}} du \leq C \frac{\chi(\tau)}{\sqrt{\tau}}. \quad (9.15)$$

For the integral use $\gamma(u) \leq \chi(u)/u$ and integrate the first bound; the power in the resulting integral is $-1 - \delta_0/2$.

We need the additional gradient estimate $|k_h| \leq C/\sqrt{\tau}$. Differentiating (9.1) gives $k_{\tau} = k_{hh}/2 + dm k_h + dk l$. On $[\tau/2, \tau]$, the drift has spatial Lipschitz constant at most $2/\tau$,

and the source is bounded by $2/\tau$. For completeness, the semigroup estimate used here is

$$|\partial_x \mathbb{E}_x g(Z_T)| \leq \frac{e^{LT}}{\sqrt{T}} \|g\|_\infty$$

for $dZ = b(v, Z) dv + dB$, $|b_x| \leq L$. If $J_v = \partial_x Z_v$, Brownian integration by parts represents the derivative by $\mathbb{E}[g(Z_T) T^{-1} \int_0^T J_v dB_v]$. Indeed contraction of the noise derivative with the adapted process J_v/T produces the endpoint variation J_T ; its divergence is its Itô integral. For smooth simple adapted integrands this is Gaussian integration by parts, and the general statement follows by L^2 approximation. Cauchy–Schwarz and $|J_v| \leq e^{LT}$ give the estimate. Apply it to the bounded datum $k(\tau/2, \cdot)$ and to Duhamel’s source integral; $\int_0^{\tau/2} v^{-1/2} dv$ gives $|k_h| \leq C/\sqrt{\tau}$. Smooth approximation and localization justify the argument for the time-measurable coefficient and linear-growth drift.

Couple inverse flows from h and 0 . At reverse time u , their separation is at most $|h|\chi(u)/\chi(\tau)$. The logarithmic Jacobian formula, the preceding gradient estimate, and (9.15) consequently give

$$|\log \mathcal{J}_t(h) - \log \mathcal{J}_t(0)| \leq C|h|/\sqrt{\tau}, \quad |\mathcal{I}_t(h) - \mathcal{I}_t(0)| \leq |h|/\chi.$$

These imply

$$p_t(h) \asymp_K p_t(0) \quad (|h| \leq K\sqrt{\tau}), \quad p_t(h) \leq Cp_t(0) \exp\{C|z| + \varepsilon(\tau)z^2\}, \quad \varepsilon(\tau) \rightarrow 0. \quad (9.16)$$

Here and subsequently $\asymp_K$ denotes two positive constants depending only on K and the fixed optimizer. The first comparison is uniform in all sufficiently late times for each fixed K ; the second holds for every $z \in \mathbb{R}$ at each such time, with a single function $\varepsilon(\tau)$ independent of z . To check the global assertion, write $\delta = |h|/\chi \leq C\sqrt{w}|z|$. The Gaussian density satisfies, for $0 < \eta \leq 1/2$,

$$p_0(\mathcal{I}_t(h)) \leq Cp_0(\mathcal{I}_t(0))^{1-\eta} \exp(C\delta^2/\eta).$$

Hölder’s inequality with measure $\mathcal{J}_t(0) d\mathbb{P}$ bounds its expectation by $p_t(0)^{1-\eta} \chi^{-\eta}$. Take $\eta = \sqrt{w}$; Lemma 9.3 makes $\chi^{-\eta}$ bounded, while (9.5) bounds $p_t(0)$ below. This proves the global claim with $\varepsilon(\tau) = C\sqrt{w}$ after enlarging constants. Apply the same argument with h and 0 interchanged for the compact lower bound; the Jacobian estimate is symmetric, and powers $p_t(0)^{\pm\eta}$ are uniformly bounded because $c_0 \leq p_t(0) \leq C/\chi$.

Put $M_{\text{lay}} = p_t(0)\sqrt{\tau}$. The first force bin has mass comparable to M_{lay} . Indeed D is bounded, the barriers put its endpoints in a fixed scaled spatial interval, and k is bounded below there. Thus its spatial length is comparable to $\chi\sqrt{w} \asymp \sqrt{\tau}$, and (9.16) applies. The same bin lemma applies to X with remaining time τ and coefficient l . If $X \geq a\sqrt{\tau}$, its scaled position is bounded below by the upper barrier for X on the negative half-line; a secant then bounds l below. Its first bin also lies in a fixed scaled spatial interval. Therefore both families of bins have mass at most $Cj^P M_{\text{lay}}$, and

$$\mathbb{E}kl \asymp M_{\text{lay}}, \quad \mathbb{E}(rX) \asymp (\tau/\chi)M_{\text{lay}}. \quad (9.17)$$

For the lower bounds integrate over a fixed central interval: both $m/\sqrt{\tau}$ and $X/\sqrt{\tau}$ are bounded below there by the heat barrier, and so are k and l by secants. For the upper bounds split into large force bins, large gap bins, and their complement. In force bin j , (9.8) bounds kl and $\chi rX/\tau$ by a polynomial in j times $e^{-c_0 j^2}$; bounded D is used here. In gap bin j , the positive barrier places z above a fixed positive multiple of j for large j , and (9.2)–(9.3) give the same estimate. The polynomial bin masses are summable against these tails. The complement is in a fixed scaled spatial interval and has mass $O(M_{\text{lay}})$.

Recall $J = P - I$ from the critical variational conditions, where $P(t) = \alpha \mathbb{E} r_t^2$ and $I(t) = \int_0^t s^2$. The identities for W and J' give globally

$$\frac{\alpha \mathbb{E} k l}{\chi} = \alpha c + \gamma J - \chi J' = \alpha c - (\chi J)'.$$

Itô's formula for rX , using its covariation $-skl \, dt$ and $RY = 0$, now gives

$$\mathbb{E}(r_t X_t) = c\tau + \chi(t)J(t)/\alpha. \quad (9.18)$$

The terminal boundary term is zero: J is bounded by $\alpha \mathbb{E} R^2 + \int_0^1 s^2$ and $\chi \rightarrow 0$. At support points $J = 0$, so (9.17) implies $M_{\text{lay}} \asymp \chi$ there.

These comparisons hold throughout a terminal interval. First the ratios τ_2/τ_1 across support gaps are bounded. Otherwise (9.11) and bounded D give $\chi_2/\chi_1 \leq C$ along a sequence of gaps with unbounded time ratio. Their affine slope satisfies $d \leq C/\tau_2$. The inverse flow from 0 at τ_1 to τ_2 therefore has a bounded Jacobian and a position whose ratio to $\sqrt{\tau_2}$ has a uniform Gaussian tail. To see the latter directly, the barrier gives $|m(u, h)| \leq C(\sqrt{\tau_2} + |h|)$ for $u \leq \tau_2$; Gronwall's inequality bounds the inverse position by a constant times $\sqrt{\tau_2}$ plus the Brownian supremum. Use the inverse density formula on this interval and (9.16) at τ_2 . The linear exponential integrates against this Gaussian tail, as does the quadratic exponential once $\varepsilon(\tau_2)$ is sufficiently small. Thus $p_{1-\tau_1}(0) \leq C p_{1-\tau_2}(0)$. But the support comparisons make their ratio comparable to $(\chi_1/\chi_2)\sqrt{\tau_2/\tau_1}$, which tends to infinity. This contradiction bounds the time ratios.

For a point inside a gap, compare to its larger endpoint by the same inverse formula. The bounded time ratio and $d \leq 1/\tau$ give bounded Jacobians and uniformly tight scaled inverse positions. Equation (9.16) gives the upper comparison. For the lower comparison restrict to a fixed compact containing a positive probability of inverse positions and use its compact lower bound and $\mathcal{J}_t \geq 1$. As χ and $\sqrt{\tau}$ also have bounded ratios,

$$p_t(0)\sqrt{\tau} \asymp \chi(t) \quad (9.19)$$

throughout a terminal interval.

The two endpoint windows and the atoms they exclude are shown in Figure 1.

Proposition 9.6 (Two-sided endpoint mass estimates). *There exist $t_1 < 1$ and constants $0 < c_1 < C_1 < \infty$ such that, for every $t_1 < t < 1$,*

$$\mathbb{P}(0 < R \leq \sqrt{w}) \asymp \chi(t), \quad \mathbb{P}(0 < Y \leq \sqrt{\tau}) \asymp \chi(t). \quad (9.20)$$

Both comparisons use the same fixed constants c_1, C_1 , after enlarging their range if necessary; the time t need not be a support point.

Proof. For a lower bound, start the relevant martingale in the scaled interval $[0.4, 0.6]$. The barriers, bounded D , and (9.16)–(9.19) put at least $c_0\chi$ mass in this interval. Its remaining quadratic variation is at most the square of the corresponding scale. Brownian time change shows that its displacement stays within 0.2 of that scale with a fixed positive probability: it suffices that standard Brownian motion stay in $[-0.2, 0.2]$ up to time one. Its endpoint is then strictly positive and below the desired cutoff.

For the force upper bound, paths starting in force bin $j \geq 2$ must make a displacement of at least $(j-1)\sqrt{w}$ to finish below $\sqrt{w}$, and hence pay a Gaussian tail. Paths starting in gap bin $j \geq 2$ must make a displacement at least $j\sqrt{\tau}$ to have $R > 0$, since that event forces $Y = 0$. Sum the polynomial bin estimates against these Gaussian tails. The remaining region $\{r \leq 2\sqrt{w}, X \leq 2\sqrt{\tau}\}$ is in a fixed scaled spatial interval and has mass $O(\chi)$. For the gap upper bound interchange the two martingales, using $Y > 0 \Rightarrow R = 0$. This proves both statements. $\square$



Figure 1: The unconditional endpoint laws of Y and R and the windows in (9.20). Each bracket denotes an interval open at zero, so its indicated mass excludes the atom. These are probability windows, with no density assumption or common horizontal scale. The force law F_J conditions on $R > 0$ and rescales R to conditional mean one. The martingales X_t and r_t have remaining quadratic variations at most τ and w , respectively. In the identity $H_t = X_t - \chi r_t$, the corresponding spatial scales are $\sqrt{\tau}$ and $\chi\sqrt{w}$. Proposition 9.5 makes them comparable.

9.5 Quantitative inverse-slope barriers

For the rest of this section put $V = -\sqrt{\tau} k_h > 0$. Since k decreases strictly from 1 to 0, we may also regard V as a function of $k \in (0, 1)$ at a fixed time. Subscripts z below mean differentiation in $z = h/\sqrt{\tau}$, so $V = -k_z$.

Lemma 9.7 (Inverse-slope bounds). *At every interior critical time,*

$$\frac{9}{7}k(1-k)^{3/2} \leq V \leq \frac{6}{5}[k(1-k)]^{3/4}. \quad (9.21)$$

Proof. We first establish the claim for a regularization that makes all endpoint comparisons classical. In the approximants of (8.8), replace m by $(\lambda_n + \chi_n)a_n$. Its terminal value is the hinge, and it solves (9.1) with $d = \gamma_n/(\lambda_n + \chi_n)$, $0 \leq \tau d \leq 1$. At fixed n this coefficient is integrable. Approximate it in L^1 by bounded coefficients satisfying the same inequality. Next replace the initial hinge by its heat smoothing of variance $e' > 0$ and replace the bounded coefficient $d(\tau)$ by $d(\tau)\tau/(\tau + e')$. Put $T = \tau + e'$, and denote the replaced coefficient again by d . Then $D_0 = Td \in [0, 1]$. These operations converge on interior compact sets, first as $e' \downarrow 0$, then in the coefficient approximation, then as $n \rightarrow \infty$: the linear difference equation and the uniform curvature and value bounds give local uniform convergence, and the semigroup gradient estimates give convergence of spatial derivatives. It is therefore enough to prove the bounds uniformly for this regularization, with $\sqrt{T}$ in the definition of V .

The slopes remain in $[-1, 0]$ and are strictly monotone. Their endpoint values are -1 and 0 . Write $v = -k_h$, and let $h = h(\tau, k)$ be the inverse position. From the equation for k ,

$$h_\tau = \frac{1}{2}v_k - dm + \frac{dk(1-k)}{v}, \quad h_k = -1/v.$$

Differentiate the first identity in k , use $m_k = k/v$, and compare with the time derivative of h_k . For $V = \sqrt{T}v$ the result is

$$\partial_{\log T} V = \frac{1}{2}V^2 V_{kk} + [\frac{1}{2} + D_0(1-3k)]V - D_0k(1-k)V_k. \quad (9.22)$$

The endpoint values are $V(0) = V(1) = 0$, uniformly on compact time intervals in this regularization. To justify this at spatial infinity, v solves $v_\tau = v_{hh}/2 + dm v_h + d(1-3k)v$, with Gaussian initial datum, Lipschitz drift of at most linear growth, and bounded potential. Its reverse diffusion representation shows that v vanishes uniformly at spatial infinity on a compact time interval: starting positions tending to infinity remain there in probability, by

the bounded integrated Lipschitz norm. The corresponding equations for k and $1 - k$ show that their fixed intermediate levels stay in bounded spatial intervals. These facts give the asserted endpoint behavior in inverse coordinates.

Let $t_k = k(1 - k)$ and $b_+ = (6/5)t_k^{3/4}$. Direct differentiation gives

$$-b_+b_+'' = \frac{36}{25} \left(\frac{3}{4}\sqrt{t_k} + \frac{3}{16\sqrt{t_k}} \right), \quad 1 - 3k - t_k b_+' / b_+ = \frac{1}{4} - \frac{3}{2}k.$$

For $k \geq 1/6$, $-b_+b_+'' \geq 27/25 > 1$ suffices to make the right-hand side of (9.22) nonpositive at b_+ . For $k < 1/6$, the required inequality is

$$\frac{27}{100} \frac{1 + 4t_k}{\sqrt{t_k}} \geq \frac{3}{2} \sqrt{1 - 4t_k}.$$

After squaring, it follows from $x(1 - x)/(1 + x)^2 \leq 1/8$ for $0 \leq x \leq 1$, with $x = 4t_k$. Thus b_+ is a stationary supersolution for every $D_0 \in [0, 1]$.

For $b_- = Ck(1 - k)^{3/2}$, the right-hand side divided by b_- is bounded below by

$$\frac{1 - k}{2} \left[1 - C^2 \left(3k(1 - k) - \frac{3}{4}k^2 \right) \right].$$

The quadratic in parentheses has maximum $3/5$; with $C = 9/7$, its product with C^2 is $243/245 < 1$. This proves the subsolution inequality.

At the heat-smoothed initial time, the inverse profile is $V_0(k) = \phi(\Phi^{-1}(k))$ and satisfies $V_0'' = -1/V_0$. Both barriers have zero endpoint values. Their stationary inequalities at $D_0 = 0$ give $b_-b_-'' \geq -1$ and $b_+b_+'' \leq -1$. If $b_- - V_0$ had a positive interior maximum, then $b_-'' \leq V_0'' = -1/V_0 < -1/b_-$, a contradiction. The same argument at a positive maximum of $V_0 - b_+$ proves $V_0 \leq b_+$. Parabolic comparison in (9.22) now yields the bounds at all later times. One may apply the maximum principle on $[\delta, 1 - \delta]$ and then let $\delta \downarrow 0$. The zeroth-order coefficient in the linearized difference is bounded above: it involves $(V + b)b''$, which is bounded above for the supersolution and, for the subsolution, on the only relevant region $V < b_-$. Multiplying the difference by a sufficiently decreasing exponential in time deals with this bound. The uniform endpoint vanishing deals with the spatial boundary. Smooth approximation handles measurable time coefficients. Passing through the regularizations proves (9.21). $\square$

We record the compactness and tail information used below. Uniformly near the endpoint,

$$|V(z)| + |V_z(z)| \leq Ce^{-c_0 z^2}. \quad (9.23)$$

For V use (9.21), the positive Gaussian tail in (9.3), and the negative tail in (9.8); bounded D makes $r/\sqrt{w}$ comparable to $|z|$ on the negative half-line. For V_z apply the semigroup gradient argument given above to the linear equation $v_\tau = v_{hh}/2 + dm v_h + d(1 - 3k)v$ for $v = -k_h$ on $[\tau/2, \tau]$, using spatial unit $\sqrt{\tau}$ and time unit τ . In these units the drift has uniformly bounded Lipschitz constant and at most linear growth, and the potential is bounded. The bound just proved for V holds at every time in this interval, so both the initial datum and the Duhamel source (the potential times $\sqrt{\tau}v$) have uniform Gaussian decay in the scaled position. A reverse diffusion started at z stays at distance at least $c|z| - C$ from zero except on an event of probability $Ce^{-c_0 z^2}$, by the flow separation estimate and the Brownian supremum bound. Thus the second moment of either decaying datum evaluated along that diffusion is at most $Ce^{-c_0 z^2}$. Cauchy-Schwarz in the integration-by-parts gradient formula bounds its gradient by $CT^{-1/2}e^{-c_0 z^2}$, with a possibly smaller c_0 . The source cost is integrable because $\int_0^{1/2} T^{-1/2} dT < \infty$. This proves (9.23). Repeating the spatial derivative estimates on smaller

time intervals gives local bounds and equicontinuity of k, V, V_z . In particular sequences of scaled spatial profiles have subsequences converging locally with these derivatives. Time differentiability of the coefficient is not needed in these semigroup estimates.

The limiting density ratios have a useful order property. For $h_1 < h_2$ with bounded scaled positions, the inverse displacement is $O(\sqrt{w})$. On $|\mathcal{I}_t(h_2)| \leq w^{-1/4}$, the Gaussian density ratio $p_0(\mathcal{I}_t(h_2))/p_0(\mathcal{I}_t(h_1))$ is at most $1 + o(1)$, uniformly on the chosen scaled compact. The complement contributes at most $C\chi^{-1}e^{-c_0/\sqrt{w}} = o(p_t(0))$, using the strict power bound. The inverse Jacobian is decreasing. Hence

$$\frac{p_t(h_2)}{p_t(0)} \leq \frac{p_t(h_1)}{p_t(0)} + o(1).$$

The compact two-sided bounds in (9.16) and the elementary selection theorem for monotone functions now give subsequential limits

$$\frac{p_t(z\sqrt{\tau})}{p_t(0)} \longrightarrow P_\infty(z) \quad (9.24)$$

at continuity points, where P_∞ is positive, nonincreasing, and bounded above and below on every compact interval. One can apply the selection theorem to the decreasing upper envelopes on each compact, whose difference from the original functions tends uniformly to zero, and then use a diagonal subsequence.

9.6 A strict curvature inequality

Lemma 9.8 (Weighted curvature inequality). *There is $c_1 > 0$ such that, at all sufficiently late times, with $A_0 = \tau\gamma/\chi \in [0, 1]$,*

$$\mathbb{E}\left[V_z^2 + 6A_0(2k-1)V^2 + 6A_0^2k^2(1-k)^2\right] \geq c_1p_t(0)\sqrt{\tau}. \quad (9.25)$$

Proof. We first prove an unweighted inequality for every limiting profile satisfying (9.21). Put

$$F_A(z) = V_z^2 + 6A(2k-1)V^2 + 6A^2k^2(1-k)^2, \quad 0 \leq A \leq 1.$$

Its cumulative integral from $-\infty$ is nonnegative until $k = 1/2$, since every term is then pointwise nonnegative. For an endpoint with $k = k_0 \leq 1/2$, change variables from z to k to obtain

$$\int_{-\infty}^{z(k_0)} F_A(z) dz = H + 6AC_1 + 6A^2B_1,$$

where all following integrals are over $[k_0, 1]$ and

$$\begin{aligned} H &= \int VV_k^2, \quad C_1 = \int (2k-1)V, \quad B_1 = \int t_k^2/V, \\ T_1 &= \int t_k, \quad U_b = \frac{6}{5} \int t_k^{3/4}, \quad B_b = \frac{7}{9} \int k\sqrt{1-k}. \end{aligned}$$

Cauchy–Schwarz applied to $(V^{3/2})'$ on $[1/2, 1]$ gives

$$H \geq \frac{8}{9}V(1/2)^3 \geq \frac{81}{5488\sqrt{2}} > \frac{1}{100}.$$

Put $b = T_1/U_b$. Integration by parts and weighted Cauchy–Schwarz give

$$\begin{aligned} C_1 &= V(k_0)\{t_{k_0} - bV(k_0)/2\} + \int V(t_k/V - b)V_k \\ &\geq V(k_0)\{t_{k_0} - bV(k_0)/2\} - \sqrt{HB_1 \left(1 - \frac{T_1^2}{U_b B_b}\right)}. \end{aligned}$$

Indeed the square integral in this application is $B_1 - 2bT_1 + b^2 \int V \leq B_1 - T_1^2/U_b$; (9.21) gives $\int V \leq U_b$ and $B_1 \leq B_b$.

Here are explicit bounds for the ratio appearing in this estimate. Under probability weight t_k/T_1 , the mean of $t_k^{-1/4}$ on $[k_0, 1]$, $0 \leq k_0 \leq 1/2$, is at most its full-interval mean. This follows by symmetry and monotonic decrease on $[0, 1/2]$: the integral of t_k times its deviation from the full mean over $[0, k_0]$ is nonnegative. Cauchy–Schwarz bounds the full mean by $\sqrt{3\pi/4}$, since $\int_0^1 \sqrt{k(1-k)} = \pi/8$ and $\int_0^1 k(1-k) = 1/6$. The mean of $(1-k)^{-1/2}$ increases with the lower cutoff; at $k_0 = 1/2$ it is $7\sqrt{2}/5 < 2$. If $k_0 < 1/50$, its value is less than $17/10$: integrate $k\sqrt{1-k}$ explicitly and evaluate at $k_0 = 1/50$. Consequently, using $\pi < 22/7$,

$$\frac{U_b B_b}{T_1^2} < \frac{28}{15} \sqrt{\frac{33}{14}} < \frac{287}{100}, \quad \frac{U_b B_b}{T_1^2} < \frac{119}{75} \sqrt{\frac{33}{14}} < \frac{61}{25} \quad (k_0 < 1/50).$$

The endpoint term is nonnegative unless $t_{k_0} < 1/64$. Indeed $b \leq 5/(6\sqrt{2})$ and $V(k_0) \leq (6/5)t_{k_0}^{3/4}$. When $t_{k_0} < 1/64$, necessarily $k_0 < 1/50$, and the endpoint term is still greater than $-1/20000$. To verify the latter, its dependence on $V(k_0)$ is concave, so its minimum on the permitted interval is at an endpoint. With $x = t_{k_0}^{1/4}$, the nonzero endpoint is bounded below by $(6/5)x^6(x - 1/(2\sqrt{2}))$. The absolute value of this polynomial's minimum on $x \geq 0$ is $6^7/(5 \cdot 7^7 \cdot 1024\sqrt{2}) < 1/20000$.

Complete the square in A in $H + 6AC_1 + 6A^2B_1$ and use $A \leq 1$ for the endpoint loss. The result is at least

$$H \left[1 - \frac{3}{2} \left(1 - \frac{T_1^2}{U_b B_b} \right) \right] - \frac{6}{20000} \mathbf{1}_{\{t_{k_0} < 1/64\}}.$$

The preceding rational bounds make this greater than

$$\min \left\{ \frac{13}{57400}, \frac{7}{6100} - \frac{3}{10000} \right\} > \frac{1}{5000}.$$

Thus, if $C_A(z) = \int_{-\infty}^z F_A$, then $C_A \geq 0$ everywhere and $C_A \geq 1/5000$ after the unique point z_* at which $k = 1/2$.

Let P be any positive nonincreasing limiting density in (9.24). Integration by parts against its Stieltjes measure gives

$$\int_{\mathbb{R}} F_A P \, dz = C_A(+\infty)P(+\infty) + \int_{\mathbb{R}} C_A \, d(-P) \geq \frac{1}{5000} P(z_*) > 0.$$

The negative boundary product is zero by the Gaussian tail of C_A and the exponential bound for P ; one may first truncate both tails. The inequality includes the case $P(+\infty) > 0$, which contributes the displayed boundary term.

Finally suppose (9.25) failed along a sequence to the endpoint. Extract a limit of $A_0 \in [0, 1]$, of the spatial profiles, and of the density ratios. Equations (9.16) and (9.23) give an integrable Gaussian majorant for the normalized integrands, so dominated convergence applies. The level z_* stays in a fixed compact by the value and slope barriers, and the limiting density is uniformly bounded below there. The preceding strictly positive lower bound contradicts failure. This proves (9.25). $\square$

9.7 Full increase support near the endpoint

Proposition 9.9 (Absence of terminal support gaps). *The increase support of γ contains a terminal interval $(t_*, 1)$. On this interval $J = J' = 0$, and γ has a continuous version with*

$$A_0 = \frac{\mathbb{E}V^2}{2\mathbb{E}k^2(1-k)}. \quad (9.26)$$

Proof. On an open gap, γ is constant. Write $b = f_{hh} < 0$ and $H_b = \mathbb{E}b(t, H_t)^2$. The diffusion equations and Itô's formula give

$$H'_b = \mathbb{E}b_h^2 - 2\gamma\mathbb{E}b^3, \quad H''_b = \mathbb{E}b_{hh}^2 - 12\gamma\mathbb{E}(bb_h^2) + 6\gamma^2\mathbb{E}b^4.$$

For example $db = b_h dB - \gamma b^2 dt$ and $db_h = b_{hh} dB - 3\gamma b b_h dt$; these yield both identities by the product rule. Localization and the spatial tails justify taking expectations.

Substitute $b = -sk$, $b_h = sV/\sqrt{\tau}$, $b_{hh} = sV_z/\tau$, and $\gamma s\tau = A_0$. Then

$$\begin{aligned} \frac{2H_b H''_b - 3(H'_b)^2}{2H_b s^2/\tau^2} &= \mathbb{E}V_z^2 + 12A_0\mathbb{E}(kV^2) + 6A_0^2\mathbb{E}k^4 \\ &\quad - \frac{3}{2\mathbb{E}k^2}(\mathbb{E}V^2 + 2A_0\mathbb{E}k^3)^2. \end{aligned}$$

Use $\mathbb{E}k^3/\mathbb{E}k^2 \leq 1$ and the identity

$$\mathbb{E}k^4 - \frac{(\mathbb{E}k^3)^2}{\mathbb{E}k^2} = \mathbb{E}[k^2(1-k)^2] - \frac{(\mathbb{E}[k^2(1-k)])^2}{\mathbb{E}k^2}.$$

The last display is therefore at least the left-hand side of (9.25) minus

$$\frac{6A_0^2(\mathbb{E}[k^2(1-k)])^2}{\mathbb{E}k^2} + \frac{3(\mathbb{E}V^2)^2}{2\mathbb{E}k^2}.$$

Both errors are $O(\chi^2)$: (9.17), (9.19), and the Gaussian derivative tails bound each numerator's unsquared expectation by $C\chi$. Moreover $\mathbb{E}k^2 \rightarrow \mathbb{P}(R > 0) = 1/\alpha > 0$ by bounded convergence. The main term in (9.25) is at least $c_0\chi$. Hence $2H_b H''_b - 3(H'_b)^2 > 0$ on every sufficiently late open gap. Equivalently $\psi = (\alpha H_b)^{-1/2}$ is strictly concave there.

Choose a support point t_* beyond the time at which this determinant inequality holds uniformly. Every support gap to the right of t_* has both endpoints in the increase support: the left endpoint lies at or after t_* , and accumulation at 1 excludes a final gap. For any such gap, stability gives $\psi \geq \chi$ at those endpoints, whereas χ is affine on the gap. Strict concavity consequently gives $\psi > \chi$ at every interior point. But then $J' = \alpha H_b - s^2 < 0$ throughout the gap, contradicting $J = 0$ at both endpoints. Thus there are no complementary intervals to the right of t_* . Closedness of the support proves that it contains $(t_*, 1)$.

On this interval the variational conditions give $J = J' = 0$, $\mathbb{E}k^2 = 1/\alpha$, and $\mathbb{E}kl = c\chi$. Apply Itô's formula to $k(t, H_t)^2$:

$$0 = \frac{d}{dt}\mathbb{E}k^2 = \frac{1}{\tau}\mathbb{E}V^2 - \frac{2\gamma}{\chi}\mathbb{E}[k^2(1-k)] \quad \text{for almost every } t.$$

The denominator is strictly positive. Interior spatial regularity, the diffusion representation, and dominated convergence make the quotient on the right of (9.26) continuous in t . It supplies the continuous version of A_0 , and hence of γ , and proves the asserted identity everywhere on the terminal interval. $\square$

10 Selection of the critical similarity profile

The estimates of Section 9 leave a scalar quantity to identify: the limiting logarithmic derivative of the susceptibility. We prove a rigidity statement for every limit obtained by translating the critical layer in logarithmic time. Its quantitative part consists of finitely many inequalities for one-dimensional functions. The analytic implications of those inequalities are proved here; their finite verification and rounding-error bounds are given in Section 12. The

stationary scaling equations have their antecedents in [5, Section IX] and [10, Section V]. Here their selection is deduced from the entire limits of the critical problem constructed in Sections 8–9.

The coefficient reduction first confines A to $[0, .41]$. With this bound, stationary-reference comparisons give curvature and density barriers for every entire critical limit. Using its critical integral identities, signed tests then narrow the coefficient interval to $[\text{.285}, \text{.303}]$, where a strict contraction around the root forces $A \equiv a_*$. Compactness finally transfers this identification to the original critical layer. The opening table in Section 12 maps the finite checks to these analytic steps.

10.1 Entire limits and their identities

We regard χ , γ , and m as functions of remaining time: their value at τ in this section is their previous value at physical time $t = 1 - \tau$. The density p_t retains its physical-time index. Write $v = \log \tau$, $u = -v$, and $q = 1/2$. Throughout this section the letters M, X, P refer to rescaled quantities:

$$\begin{aligned} M(v, z) &= \frac{m(\tau, z\sqrt{\tau})}{\sqrt{\tau}}, & X(v, z) &= z + M(v, z), \\ P(u, z) &= \frac{\sqrt{\tau}}{\chi(\tau)} p_{1-\tau}(z\sqrt{\tau}), & A &= \frac{\tau\gamma}{\chi}. \end{aligned}$$

Thus P is a positive density divided by a mass scale and need not integrate to one. Set $k = -M_z$, $l = 1 - k$, and $V = -k_z = M_{zz} > 0$. Spatial derivatives in this section are derivatives in z , except where inverse-slope coordinates are explicitly specified. Direct change of variables gives

$$\begin{aligned} M_v &= \tfrac{1}{2}M_{zz} + BM_z + (A - q)M, \\ B &= qz + AM, \\ P_u &= \tfrac{1}{2}P_{zz} - (BP)_z + AP. \end{aligned} \tag{10.1}$$

The opposite time directions are essential: M is parabolic in v , whereas P is parabolic in u . The last term in the density equation comes from $\chi_u = -A\chi$.

Lemma 10.1 (Compactness in logarithmic time). *Every sequence $u_n \rightarrow \infty$ has a subsequence along which the translated rescaled profiles converge locally to a solution of (10.1) for all $u \in \mathbb{R}$. The limiting coefficient A is continuous and takes values in $[0, 1]$. The value barriers (9.2), the inverse-curvature bounds (9.21), and the Gaussian bounds for k on the positive half-line, l on the negative half-line, and V, V_z on both half-lines hold uniformly in entire time. Moreover $P > 0$, $P \leq C \exp(C|z|)$, and on every bounded spatial interval P has a positive lower bound independent of entire time. The limits satisfy*

$$\int_{\mathbb{R}} MXP \, dz = \int_{\mathbb{R}} klP \, dz = c, \quad A = \frac{\int_{\mathbb{R}} V^2 P \, dz}{2 \int_{\mathbb{R}} k^2 l P \, dz}, \tag{10.2}$$

where $c = \mathbb{E}R > 0$ is the critical normalization constant.

Proof. The density estimate (9.16), together with $p_{1-\tau}(0)\sqrt{\tau}/\chi \asymp 1$, bounds P above and below on spatial compact sets. The same estimate bounds it globally by $C \exp(C|z| + o(1)z^2)$, which is integrable against the Gaussian tail bounds already proved. The value and curvature barriers bound the coefficients of (10.1) locally, uniformly after sufficiently late translation.

Interior estimates for the localized heat equation, applied forward in v to M and forward in u to P , give local bounds and equicontinuity for their spatial derivatives. Applying the equations on a slightly smaller cylinder gives time equicontinuity. For precision, one may first differentiate the localized heat-kernel formula in space: the initial C^1 bound controls the drift term, and iteration gives the derivatives needed below. Bounded, initially merely measurable, time coefficients suffice in this argument.

On the full-support endpoint interval, (9.26) expresses A as a quotient of spatial integrals. Its denominator, after division by χ , is bounded below uniformly. Indeed the value barriers place an interior interval of slope values in a bounded spatial interval; inversion and (9.21) give positive spatial width, and central positivity of P gives a positive integral of $k^2 l P$. Gaussian tails and the density bound permit uniform truncation of both integrals. Consequently the same compactness gives compactness of A itself. Pass first in the integral form of the equations; the quotient then shows that the limiting A is continuous.

The critical identities in physical units are $\mathbb{E}[r_t X_t] = c\tau$ and $\mathbb{E}[kl] = c\chi$. They become the first two integrals in (10.2). Their passage to the limit, and passage in the quotient, use the value barriers, the negative-tail estimate (9.8), and the Gaussian domination just described. The constants in all these arguments are uniform under translation, so the conclusions hold on the whole time axis. $\square$

We now use Q for the logarithmic spatial slope of P throughout this section. There is a global, translation-uniform bound:

$$Q = (\log P)_z \in [-C, 0], \quad Q_u = \frac{1}{2}Q_{zz} + (Q - B)Q_z - (q - Ak)Q - AV. \quad (10.3)$$

Here nonincrease of P follows from the inverse-flow comparison in Section 9. To obtain the lower bound for Q , compare two inverse positions corresponding to fixed rescaled positions z_1, z_2 . Their displacement is $O(\sqrt{w})$, so the ratio of the initial Gaussian densities tends to one after truncating the inverse positions as in that comparison; the complement is negligible. The inverse-Jacobian logarithmic ratio is bounded by $C|z_1 - z_2|$, with C independent of the chosen compact set. Hence in the limit $|\log P(z_1) - \log P(z_2)| \leq C|z_1 - z_2|$ for all z_1, z_2 . Differentiating the density equation gives (10.3).

All subsequent integrations by parts can be made first with compact cutoffs. To justify removal of the cutoffs also after differentiation, use cylinders of spatial radius $(1 + |z|)^{-1}$ and time length $(1 + |z|)^{-2}$. Rescaling bounds the drift and its required spatial derivatives there. Interior derivative estimates thus introduce only polynomial losses into the Gaussian bounds for V and its derivatives and into the exponential bounds for P . These losses remain integrable in every expression below.

10.2 An absolute improvement of the coefficient bound

Let $\mathcal{L} = \frac{1}{2}\partial_z^2 + B\partial_z$. Differentiating the M equation gives

$$(\partial_u + \mathcal{L})k = -Akl, \quad (\partial_u + \mathcal{L})V = -(q + A - 3Ak)V.$$

For a smooth integrable function $F = F(u, z)$, the density equation therefore supplies the useful differentiation rule

$$\frac{d}{du} \int F P = \int (F_u + \mathcal{L}F + AF) P.$$

Apply it to V^2 and $k^2 l$. The quotient in (10.2) is C^1 and obeys

$$2 \left(\int_{\mathbb{R}} k^2 l P \, dz \right) A_u = \int_{\mathbb{R}} [(V_z)^2 + (12Ak - 5A - 1)V^2 + 6A^2 k^2 l^2] P \, dz. \quad (10.4)$$

For example the derivative of the denominator without its factor two is

$$\int [A(-1 + 3k)k^2l + (1 - 3k)V^2]P.$$

In simplifying the quotient derivative one uses $\int (-2AV^2 + 4A^2k^2l)P = 0$, which is precisely (10.2).

Lemma 10.2 (Coefficient reduction). *Every entire limit in Lemma 10.1 has $0 \leq A \leq 0.41$.*

Proof. We first show that the right side of (10.4) is strictly positive whenever $0.41 \leq A \leq 1$. Denote $t_k = k(1 - k)$. Its unweighted integrand is positive while $k \geq 5/8$. For a cumulative integral ending at a slope $k_0 \leq 5/8$, change variables from z to k , so the cumulative runs from k_0 to 1. Put

$$b = 2(0.6 - k), \quad E = -0.4t_k, \quad K = 12Ak - 5A - 1 - b^2 + 0.8(1 - 2k).$$

Integrating the inequality $V(V_k - b - E/V)^2 \geq 0$ gives the following lower bound for that cumulative:

$$-(bV^2 + 2EV)(k_0) + \int_{k_0}^1 \left[2V^2 + KV - 2bE + (6A^2 - 0.16)\frac{t_k^2}{V} \right] dk. \quad (10.5)$$

The boundary term at $k = 1$ vanishes by (9.21).

The finite check `checkA` proves strict positivity of (10.5), uniformly for $A \in [0.41, 1]$ and $k_0 \in [0, 5/8]$. Its input is only the two explicit bounds in (9.21). It partitions A and k , subdivides the interval between the lower and upper allowed values of V , and integrates cellwise lower bounds. For the boundary quadratic its maximum is at an endpoint when $b \geq 0$; when $b < 0$ it is decreasing for $V \geq 0$. The two omitted endpoint cells each cost at most 9 times their width, and on the first cell the boundary expression before its negation is less than 0.001. A possible positive partial cell is subtracted rather than included in full. These are exactly the corrections in `checkA`; Section 12 verifies the arithmetic.

For completeness, positive cumulatives are preserved under the weight P . If f is the unweighted integrand and $F(z) = \int_{-\infty}^z f$, then $F > 0$ at every finite point and its terminal value is positive. Since P is positive and nonincreasing, Stieltjes integration by parts gives $\int fP = F(\infty)P(\infty) + \int F d(-P) > 0$; limits of truncated expressions give the same formula when a tail value is infinite. The Gaussian/exponential bounds make the left endpoint product zero and justify the limits. Strictness follows either from $P(\infty) > 0$ or from the nonzero positive measure $d(-P)$.

Finally let L be the endpoint limsup of the original rescaled A . Choose translates realizing L at time zero and take an entire limit. In that limit $A(0) = L$ and $A(u) \leq L$ for all u . If $L \geq 0.41$, (10.4) makes $A_u(0) > 0$, contradicting this maximum. Thus $L < 0.41$, which implies the claimed, slightly weaker bound for every entire limit. $\square$

10.3 Stationary reference profiles

For $a \in [0, 0.41]$, let M_a denote the entire solution of the first equation of (10.1) with $A \equiv a$ and the value barriers (9.2), in the inherited admissible class $-1 \leq M_z \leq 0$, $M_{zz} > 0$. This solution exists and is unique. For existence, evolve the heat-smoothed hinge of variance one with coefficient $a/(1 + \tau)$ in the original m equation, and rescale by $T = 1 + \tau$. The resulting equation is autonomous in $\log T$; the heat and Mills-ratio barriers, together with (9.21), give compactness of translates. For uniqueness, subtract two entire solutions. Their bounded difference solves a linear equation whose zeroth-order coefficient is $al_2 - q \leq a - q < 0$ and

whose drift is $qz + aM_1$. Comparison from time $v - T$ to v bounds its supremum by $Ce^{-(q-a)T}$; let $T \rightarrow \infty$. Time translation and uniqueness then imply stationarity. The same compactness and uniqueness argument proves continuity of M_a in a on compact spatial sets, with its needed spatial derivatives.

The maximum principles used here and below are valid on the whole line. For bounded differences, localize to $[-R, R]$ and add $\varepsilon e^{Ct}(1 + z^2)$; bounded diffusivity and at most linear drift allow C independent of R . Send $R \rightarrow \infty$ and then $\varepsilon \downarrow 0$. This also proves the corresponding comparison for bounded ratios whenever their equation has bounded diffusivity and at most linear drift.

At a fixed reference, abbreviate $p = q - a > 0$ and $W = M_a''$. Its ODE and its derivative give

$$W = 2pMl + Xk = 2pM + 2Bk, \quad W_z = -2BW + 2at_k, \quad W_k = 2B - 2a\frac{t_k}{W}. \quad (10.6)$$

Here and subsequently W_k is taken at the inverse position $z_a(k)$. In particular, for $x > 0$,

$$\begin{aligned} z = x : \quad k &\leq e^{-qx^2}, \quad M \leq k/x, \\ z = -x : \quad l &\leq e^{-px^2}, \quad X \leq l/(2px). \end{aligned}$$

Indeed $W \geq xk$ on the positive side, and $W \geq 2pxl$ on the negative side; integrate the logarithmic derivatives of k or l , and then their tail integrals. These formulas also bound the tails of W by a polynomial times the same Gaussians.

Define the reference density up to a positive factor by

$$\frac{1}{2}P_a'' - (B_a P_a)' = -\lambda(a)P_a, \quad Q_a = (\log P_a)'.$$

The substitution $P_a = \exp(\int_0^z B_a) \psi_a$ gives the self-adjoint operator

$$\mathcal{H}_a = -\frac{1}{2}\partial_z^2 + \frac{1}{2}(B_a^2 + B_a').$$

Its potential is bounded below by $c_0 z^2 - C_0$, uniformly in $a \in [0, 0.41]$. Minimization of its quadratic form over unit vectors therefore has a minimizer: the confining bound makes the mass outside large compact intervals uniformly small, and bounded one-dimensional Sobolev norms give compactness on each such interval. Replace a minimizer by its absolute value; the ODE and uniqueness for initial values make it strictly positive. The lowest eigenvalue is simple, by the Wronskian identity for two positive square-integrable solutions. Smooth compactly supported trials give upper semicontinuity of $\lambda(a)$; compactness of minimizing sequences gives lower semicontinuity. Thus λ is continuous.

Section 12 certifies

$$\lambda(0.2936533) > 0.2936533, \quad \lambda(0.2936535) < 0.2936535.$$

Fix a root $a_* \in (0.2936533, 0.2936535)$ of $\lambda(a) = a$. For every tabulated density reference used below, and for each root in this bracket, the tail construction in that section proves polynomial-order tails for P_a and $Q_a \rightarrow 0$ at both ends. No table at the unknown root is needed: at $\lambda = a_*$ the two tail coefficients in (12.6) obey

$$\begin{aligned} c_{\text{tail},+} &= 2a_* - 1 \in (-0.4126934, -0.4126930), \\ c_{\text{tail},-} &= \frac{a_*}{1/2 - a_*} - 1 \in (0.4231063, 0.4231088). \end{aligned}$$

These intervals lie inside the sided hypotheses of the analytic tail construction, and (10.6) supplies its coefficient estimates uniformly for the root. Also $Q_a \leq 0$: its stationary equation

is (10.3), since differentiating the eigenvalue equation removes the constant eigenvalue term. A positive interior maximum would have right side at most $-(q - ak)Q_a - aW < 0$. Together with the zero tail limits this excludes positive values (also at $a = 0$).

At the root the two integral identities in (10.2) hold, after a common positive normalization of P_* . In fact

$$\mathcal{L}(MX) = (1 - a)MX - kl, \quad \mathcal{L}(kl) = -a(1 - 2k)kl - W^2.$$

Integration against P_a replaces $\mathcal{L}$ by $-\lambda(a)$. At $\lambda = a$, the first formula gives $\int (MX - kl)P_a = 0$ and the second gives $\int W^2 P_a = 2a \int k^2 l P_a$. Boundary terms vanish by the reference tail estimates. Normalize the positive first integral to c .

10.4 Comparisons with the references

For the comparisons and integral tests, an *input* consists of a continuous coefficient $A : \mathbb{R} \rightarrow [0, 0.41]$ and a solution $M_i(v, z)$ of the first equation in (10.1), defined for all $v, z \in \mathbb{R}$. We require $0 < k_i = -M_{i,z} < 1$, $V_i = M_{i,zz} > 0$, the scaled value barriers (9.2), and the inverse-curvature bounds (9.21), uniformly in time. A density input also includes a positive function $P_i(u, z)$ whose logarithmic slope $Q_i = (\log P_i)_z$ is bounded on $\mathbb{R}^2$, nonpositive, and satisfies (10.3), with $u = -v$. The regularity is that of the entire limits above, so the equations and comparisons hold classically on interior space-time cylinders.

Every critical entire limit is an input by Lemmas 10.1 and 10.2. Every stationary M_a is an input. The stationary density references used below, including every root in the certified bracket, are density inputs by the tail construction and the bound $Q_a \leq 0$. The density comparison class requires neither the full density equation in (10.1) nor the integral identities (10.2). Indeed a stationary eigenreference with $\lambda(a) \neq a$ need not solve that density equation, but its logarithmic derivative solves (10.3): spatial differentiation removes the constant eigenvalue term. Consequently all the comparisons and the resulting **Env** enclosures apply to such references. Later, the reference at 0.293650 enters only through direct functional evaluation and interpolation. Converting a functional sign into a coefficient bound will additionally require the homogeneous consequences of (10.2); a reference satisfies them at a root. If $a \leq A \leq b$, then at the same spatial position

$$M_a \leq M_i \leq M_b. \quad (10.7)$$

For example $\delta = M_i - M_a$ solves, in v , a linear equation with drift B_i , damping $q - Al_a \geq q - b > 0$, and source $(A - a)M_a l_a \geq 0$. Its negative part vanishes by comparison from the arbitrarily remote past. The upper comparison is identical with the inequality reversed.

Curvatures must instead be compared at the same slope k . Fix a reference a , put

$$T_a = 1 - 3k - t_k \frac{W_k}{W} = \frac{2pMl}{W} - 2k + 2a \left(\frac{t_k}{W} \right)^2, \quad D_a = q + aT_a,$$

and write $R = V_i/W$ as a function of the reference position $z = z_a(k)$. Substitution into the inverse-curvature equation of Section 9 gives

$$R_v = \frac{1}{2}R^2 R_{zz} + \left[-R^2 B + (aR^2 + A) \frac{t_k}{W} \right] R_z + R[D_a(1 - R^2) + (A - a)T_a]. \quad (10.8)$$

Equivalently, its derivative terms in k are $\frac{1}{2}R^2 W^2 R_{kk} + (R^2 W W_k - A t_k) R_k$.

Lemma 10.3 (A bounded curvature ratio). *For every input, at the same $k \in (0, 1)$,*

$$\sqrt{1 - 0.82 \cdot 1.12} \leq \frac{V_i}{W_0} \leq \sqrt{1 + 0.82 \cdot 0.13}, \quad W_0(k) = \phi(\Phi^{-1}(k)). \quad (10.9)$$

Consequently the ratio of any two input curvatures is less than 3.8.

Proof. The central reference check and the tail estimates proved below give $-1.12 \leq T_0 \leq 0.13$. Let $\alpha = q + 0.41 \cdot 0.13$ and solve $C' = C(\alpha - qC^2)$ with $C(v) \rightarrow \infty$ as $v \downarrow v_0$. Equation (10.8) says that CW_0 is a supersolution for V_i in the k equation. This comparison does not initially assume a bounded ratio. Indeed in the equation for $V_i - CW_0$ the extra zeroth-order coefficient is $\frac{1}{2}(V_i + CW_0)CW_{0,kk} \leq 0$, because $W_{0,kk} = -1/W_0$. The remaining positive coefficient is bounded independently of C . At time $v_0 + h$, the positive part of $V_i - CW_0$ has supremum tending to zero as $h \downarrow 0$: on interior k -intervals $C \rightarrow \infty$, and near both endpoints the uniform upper bound (9.21) tends to zero. Comparison, followed by $h \downarrow 0$ and $v_0 \rightarrow -\infty$, gives $R \leq \sqrt{\alpha/q}$.

To obtain a positive lower ratio without assuming one, set $U = 1/R$ in heat-reference position. Differentiating (10.8) and dropping its nonpositive gradient-square term gives

$$U_v \leq \frac{1}{2}R^2U_{zz} + \left[-qR^2z + A\frac{t_k}{W_0}\right]U_z - \delta_0U + qR, \quad \delta_0 = q - 0.41 \cdot 1.12 > 0.$$

Here $0 \leq At_k/W_0 \leq C$ and the already proved upper bound controls R . The lower bound in (9.21) and elementary normal-tail estimates give, uniformly in entire time,

$$U(z) \leq C(1 + z_+^2 + e^{z_-^2/3}).$$

The diffusion representation for this subsolution has no surviving boundary term at infinity. Here is a direct justification. For $r < q$ and an even integer $m \geq 2$, use a C^1 , piecewise C^2 Lyapunov function comparable to $1 + z_+^m + e^{rz_-^2}$. On the negative half-line its generator is bounded above by a constant: the quadratic exponential contribution has coefficient $2r(r - q)R^2z^2 \leq 0$, and the additional drift is nonnegative, hence contributes nonpositively there. On the positive half-line its generator is at most $C(1 + z_+^{m-1})$. Stopped Itô and Jensen therefore bound its expectation over duration T by $C_z(1 + T)^m$. Taking $m = 4$, $r = 0.45$ supplies uniform integrability for the smaller growth bound on U , permitting removal of stops at each finite duration. The representation thus yields

$$U(v, z) \leq e^{-\delta_0 T} \mathbb{E}[U(v - T, Z_T)] + q \sup R \int_0^T e^{-\delta_0 s} ds.$$

The first term tends to zero polynomially times an exponential. We obtain $U \leq q \sup R/\delta_0$, a uniform positive lower bound for R .

Now compare R to the spatially constant solution of $r' = r(\delta_0 - qr^2)$ starting below that positive lower bound. The coefficients in (10.8) are bounded in diffusion and at most linear in drift, so the whole-line comparison stated above applies. Let its starting time tend to $-\infty$; this gives $R \geq \sqrt{\delta_0/q}$. The quotient of the upper and lower constants in (10.9) is less than 3.8. $\square$

10.5 Quantitative curvature and density barriers

Suppose that $a \leq A(v) \leq b$ for every $v \in \mathbb{R}$, where $[a, b]$ is one of the intervals used in the coefficient reductions or the tight root enclosure below. Their finite checks are verified in Section 12. Put $g = b - a$ and define

$$e(z) = \frac{0.13}{(1 + (2 - z)_+^2/2)^{1/4}}.$$

The following inequalities hold at equal slope values:

$$V_i(k) \leq W_a(k)(1 + ge(z_a(k))), \quad V_i(k) \geq W_b(k)(1 - 1.1ge(z_b(k))). \quad (10.10)$$

We spell out both the finite tests and their implication for an entire solution, since simply testing a stationary barrier would not by itself prove an entire comparison.

For either endpoint temporarily call the reference a , and let $R_0 = 1 + fge$, with $f = 1$ at the lower reference and $f = -1.1$ at the upper reference. Set $C_0 = e''/2 + (-B + at_k/W)e'$. The sufficient inequalities are

$$\begin{aligned} R_0^2 C_0 + A \frac{t_k}{W} e' + R_0 \left[-(R_0 + 1) D_a e + \frac{(T_a)_+}{|f|} \right] &< 0, \\ f g C_0 - R_0 D_a &< -0.02. \end{aligned} \quad (10.11)$$

The first line bounds the right side of (10.8) after division by fg ; for $f < 0$ the reversal gives the required subsolution inequality. On replacing R_0 by CR_0 , the right side divided by C changes by $(C^2 - 1)R_0^2(fgC_0 - R_0 D_a)$. Start with $C > 1$ for the upper barrier or $0 < C < 1$ for the lower barrier, chosen to dominate the bounded ratios of Lemma 10.3. For a sufficiently small fixed $c_0 > 0$, the solution $C' = c_0 C(1 - C^2)$ preserves the corresponding super- or subsolution inequality. Entire comparison and passage of the starting time to $-\infty$ give $C \rightarrow 1$ and prove (10.10). The products of drift differences with barrier derivatives are bounded, including Be' . To see explicitly why whole-line comparison is legitimate, write

$$\beta = -B + a \frac{t_k}{W}, \quad F(r) = r[D_a(1 - r^2) + (A - a)T_a].$$

For the difference between an input R and a trial $S = CR_0$, the diffusion and drift coefficients are $R^2/2$ and $R^2\beta + At_k/W$. The zeroth-order coefficient is

$$\frac{1}{2}(R + S)S_{zz} + (R + S)\beta S_z + \frac{F(R) - F(S)}{R - S},$$

where the quotient is interpreted as $F'(S)$ at equality. Both R and S stay in fixed positive bounded intervals; $D_a, T_a, t_k/W$ are bounded, S_{zz} is bounded, and βS_z is bounded because Be' is bounded. Hence this coefficient is bounded above uniformly, the diffusion is bounded, and the drift grows at most linearly. Multiplying the difference by a decaying time exponential and using the polynomial localization already stated proves comparison. The barrier is C^1 and piecewise C^2 at $z = 2$; comparison applies on both sides, or follows by smoothing inside the strict margins.

The central inequalities (10.11) are the test **checkV**. The following tail estimates complete their whole-line verification and also supply the heat bound used above. For $z = x \geq 6$, (10.6) gives

$$\frac{t_k}{W} \leq \frac{1}{x}, \quad -2k \leq T_a \leq \frac{1}{x^2};$$

e is constant there, so both inequalities in (10.11) follow directly. For $z = -x$, $x \geq \sqrt{13/p}$, put $U_0 = (4p^2x^2)^{-1} \leq (52p)^{-1}$. Write $W = 2pMl(1 + y)$. Then

$$0 \leq y \leq U_0, \quad k \geq 1 - 3 \cdot 10^{-6}, \quad \frac{x}{M} \geq 1 - 3 \cdot 10^{-6},$$

and, more sharply, $l \leq e^{-13} < 2.261 \cdot 10^{-6}$ and $X/x \leq e^{-13}/26 < 8.694 \cdot 10^{-8}$. Thus $(kx/M)^2 > 1 - 4.7 \cdot 10^{-6} > 1 - 10^{-5}$, and hence

$$-1 - U_0 + \frac{2a(1 - 10^{-5})U_0}{(1 + U_0)^2} \leq T_a \leq 1 - 2k + 2aU_0 < 0.$$

In particular $D_a > 0.56p$: maximize the quadratic in a in $a[1 - 2a(1 - 10^{-5})/(1 + U_0)^2] < 0.185$, using $U_0 < 0.214$. Also

$$\frac{t_k}{W} \leq \frac{1}{2px}, \quad 0 \leq -B \leq px, \quad 0 \leq \frac{e'}{e} \leq \frac{1}{2(x+2)}, \quad \frac{|e''|}{e} \leq \frac{1}{(x+2)^2}.$$

In the first line of (10.11), divided by e , the positive terms are at most

$$1.12 \left[\frac{p}{2} + \frac{0.5p}{13} + \frac{0.41}{52} \right] + \frac{0.41}{52} < 0.8p,$$

whereas the negative term has magnitude at least $0.94 \cdot 1.94 \cdot 0.56p > p$. For the second line, $|fgC_0| < 0.059(0.5 + 0.039 + 0.088)p$, while $R_0D_a > 0.94 \cdot 0.56p$; since $p \geq 0.09$, its value is below -0.02 . At $a = 0$ these tail bounds and the central check give $-1.12 \leq T_0 \leq 0.13$.

For density comparison use equal spatial positions and the trial $\hat{Q} = Q_a + a\delta$, where $\delta = M_i - M_a$. Substitution into (10.3), using the difference equation for δ with time reversed from v to u , gives the exact residual

$$H = a(1 - a)\delta + (A - a) [M_i(Q'_a - al_a) - k_iQ_a + V_i - ak_i\delta]. \quad (10.12)$$

In particular the second spatial derivatives of δ cancel. The difference $\hat{Q} - Q_i$ solves a linear equation with drift $Q_i - B_i$, source H , and damping

$$DQ = q - Q'_a - ak_a - (A - a)k_i.$$

At either endpoint the certificate proves $DQ > 0.06$ and that the square bracket in (10.12) is at least $-0.12DQ$. Together with (10.7) and entire comparison in u , these inequalities give

$$Q_b - b(M_b - M_a) - 0.12g \leq Q_i \leq Q_a + a(M_b - M_a) + 0.12g. \quad (10.13)$$

For example at the lower reference $\delta \geq 0$ and $H \geq -0.12gDQ$, so adding $0.12g$ to $\hat{Q}$ gives a supersolution. At the upper reference $\delta \leq 0$ and $A - b \leq 0$, so subtracting $0.12g$ gives a subsolution.

On $|z| \leq 12$ the test is `checkH`. It uses (10.7), $V_i \geq (9/7)k_i l_i^{3/2}$, and the possible same-position slope interval obtained by inverting (10.10); the inversion is described below. The term $-ak_i\delta$ is discarded only at the upper reference, where it is nonnegative. The damping uses the worst possible positive part of $A - a$. For $x \geq 12$, the reference tail certification gives, with $S = \partial_x \log P_a(\pm x)$,

$$|S| < 0.071, \quad |xS' + S| < 0.0055.$$

On the positive side $M_i \leq M_b$ is Gaussian-small, and so is k_i by a secant from $z - 1$; also $Q_a \leq 0$ and $|Q'_a| < 0.0064$. On the negative side

$$M_i = x + X_i, \quad X_i \leq X_b < 1.2 \cdot 10^{-6}, \quad l_i \leq xX_b(-x + 1/x) < 2 \cdot 10^{-5}.$$

There $Q_a = -S$, $Q'_a = S'$, and the leading terms of the bracket are $xS' + S$. The adverse remainder terms X_iS' , $-l_iS$, $-aM_i l_a$, and $-ak_i\delta$ have total absolute size below 0.0001, using (10.6) and decrease of xe^{-px^2} in this range. Since $DQ \geq 0.09 - 0.0064$, these estimates prove the required tail inequalities.

10.6 Inverse-coordinate functionals and their enclosures

The comparisons now give bounds for the fields at every entire time. We use them to evaluate a signed integral whose sign determines whether A lies above or below a tested value. Curvature is compared at equal slope, so we first express the other fields in that same coordinate.

The primitive reconstruction formulas are

$$M(k) = \int_0^k \frac{j}{V(j)} dj, \quad X(k) = \int_k^1 \frac{1-j}{V(j)} dj, \quad z(k) = X(k) - M(k). \quad (10.14)$$

Indeed $z_k = -1/V$, $M_k = k/V$, and $X_k = -(1 - k)/V$; the zero constants at the appropriate tails follow from the value barriers. These formulas fix the spatial origin, so no translation parameter is suppressed when comparing curvatures at equal k .

For a tested coefficient ξ and a nonnegative number c_+ define

$$N_\xi(V, P) = \int_0^1 \frac{P}{V} [V^2 + c_+(MX - t_k) - 2\xi kt_k] dk.$$

The critical identities have the following homogeneous form:

$$\int_{\mathbb{R}} (MX - kl)P dz = 0, \quad \int_{\mathbb{R}} V^2 P dz = 2A \int_{\mathbb{R}} k^2 l P dz. \quad (10.15)$$

They hold for each critical entire limit by (10.2), and for each stationary reference at a root of $\lambda(a) = a$. For every input satisfying these identities, change of variables gives exactly

$$N_\xi(V_i, P_i) = 2(A - \xi) \int_0^1 \frac{P_i}{V_i} kt_k dk.$$

The value comparisons give Gaussian tails for MX and kl , and bounded Q_i gives at most exponential growth of P_i . The integrals are therefore finite; the one on the right is strictly positive. Thus a sign enclosure for N_ξ bounds A at every time; no assumption of stationarity of the input is involved.

The remaining construction turns the value, curvature and log-slope comparisons into enclosures of this functional. All comparison bounds apply to inputs before the identities (10.15) are imposed; only the conversion of a functional sign into a coefficient bound uses those identities.

Here we specify the mathematical meaning of the interval operations used in the certificate. Its middle grid covers $[\kappa, 1 - \kappa]$, where $\kappa = 2^{-38}$, and contains $1/2$ as a cell boundary. A *box* is an interval containing every value of a function throughout one closed grid cell. A sum of boxes times cell widths encloses the corresponding integral. The operation **partialsum** bounds a primitive ending anywhere in a cell: from the inclusive full-cell sum it subtracts that cell's box multiplied by the interval from zero to its width. The operation **anchored** forms the analogous signed primitive from $1/2$. Thus endpoints and partial cells are included, not replaced by point sampling.

The object **Env** applies (10.10) with

$$r_h = 1 + ge_a, \quad r_l = 1 - 1.1ge_b, \quad V_h = r_h W_a, \quad V_l = r_l W_b.$$

To bound M and X below, integrate $1/V_h$ in (10.14), writing it as the known a -reference primitive plus the correction $(1/r_h - 1)/W_a$. Use the b -reference and $1/r_l$ for upper bounds. Since $g \leq 0.41$ and $0 \leq e \leq 0.13$,

$$|1/r_h - 1| \leq 0.16g, \quad |1/r_l - 1| \leq 0.16g.$$

Hence the omitted correction in each primitive tail costs at most $0.16g$ times the corresponding reference tail primitive. The coarser direct factor bound in **adjusted** is a second valid enclosure, and intersection improves the result.

Subtracting the primitive bounds encloses $z_i(k)$. A spatial range query in (10.13), with this position box as input, then bounds $Q_i(z_i(k))$. Together with (10.10) and $Q_i \leq 0$ it yields cellwise bounds s_l, s_h for

$$\frac{d}{dk} \log P_i(k) = -\frac{Q_i(z_i(k))}{V_i(k)}.$$

Except in the final local comparison, normalize $P_i(1/2) = 1$ in inverse coordinates. The primitive of these slope bounds is the **logp** enclosure. This normalization is made separately at each fixed time, after the evolution comparisons. It preserves Q_i and the homogeneous identities $\int(MX - kl)P_i = 0$ and $\int V_i^2 P_i = 2A \int k_i^2 l_i P_i$. No evolution equation for this renormalized density is used in the functional tests.

Conversely, the test of (10.12) needs possible slope values at a given spatial position. The lower position bounds are replaced by their prefix minima and the upper bounds by their suffix maxima, which preserves enclosure and makes inversion monotone. Invert the resulting position intervals against the given spatial cell. At a clipped endpoint include the corresponding entire k tail. The checked edge positions straddle the central region: the positive edge is above 4, and the negative edge below -5.5 . A spatial cell beyond an edge uses that extreme slope cell and its tail. This is **possiblek**. Thus the density comparison has no circular dependence on an assumed bound for Q_i in its own residual test: **checkH** uses only M and V enclosures to bound its input slopes.

Several uniform tail bounds will be used in the integral tests. Equation (10.13) gives $Q_i \geq -0.4$ for $z \leq -5.5$: the central reference table checks $Q_b > -0.28$ and $X_b < 0.03$ there, and the reference S estimates continue the bounds outside the table. Also **logp** verifies $P_i(1 - \kappa) \leq e^5$. The two curvature factors lie in $[0.94, 1.054]$. For either reference let x_c be the absolute position of its grid edge in the tail under consideration. The checked bounds give $x_c < 20$, $x_c > 6$ on the positive side, and $2px_c > 2$ on the negative side. Integration of the reference slope bounds gives

$$\begin{aligned} k_a(x) &\leq \kappa e^{-q(x^2 - x_c^2)}, & x \geq x_c & \text{ (positive tail),} \\ l_a(-x) &\leq \kappa e^{-p(x^2 - x_c^2)}, & x \geq x_c & \text{ (negative tail).} \end{aligned}$$

The same statements hold at b . In the negative tails $W \leq (x+1)l$, by (10.6). On the positive tails the bound $W \leq (x+2)k$ suffices.

10.7 From signed integral tests to a narrow interval

First take $c_+ = 0$. To prove an upper bound, use the monotonicity in $V > 0$ of $V - 2\xi kt_k/V$ and replace V_i by V_h . Define p_h on the middle grid by its piecewise constant logarithmic slope s_l , anchored by $p_h(1/2) = 1$. Then P_i/p_h is nondecreasing. The upper **coarse** test verifies that the right cumulatives of

$$f_h = p_h \left(V_h - 2\xi \frac{kt_k}{V_h} \right)$$

are negative, with margin 10^{-7} until the far region $k > 0.95$, where the integrand itself is checked negative. If $F_h(k) = \int_k^{1-\kappa} f_h$, integration by parts gives

$$\int_{\kappa}^{1-\kappa} \frac{P_i}{p_h} f_h = \left(\frac{P_i}{p_h} \right) (\kappa) F_h(\kappa) + \int_{\kappa}^{1-\kappa} F_h d(P_i/p_h) \leq -10^{-7} \left(\frac{P_i}{p_h} \right) (\kappa).$$

In the negative spatial tail the actual integrand remains negative:

$$\frac{V_i^2}{k^2 l} \leq 1.054^2 \frac{(x+1)^2 l_a}{k^2},$$

which is Gaussian-small from the edge onward. In the positive spatial tail P_i is increasing in k , $p_h(\kappa) \leq 1$, and the only harmful term has $\int_0^{\kappa} P_i V_i dk < \kappa (P_i/p_h)(\kappa)$ by (9.21). The strict margin therefore survives both tails.

For a lower bound use V_l and p_l with logarithmic slope s_h . Then P_i/p_l is nonincreasing. The lower **coarse** test checks positive left cumulatives with the same margin, except in

$k < 0.05$, where the integrand itself is positive. Integration by parts with left cumulatives gives a lower bound $10^{-7}(P_i/p_l)(1 - \kappa)$ on the middle integral. Beyond the positive edge actual positivity follows from $V_l \geq 0.94xk$ in b -reference position. On the negative side the possible negative contribution is at most

$$\left(\frac{P_i}{p_l}\right)(1 - \kappa)\frac{2\xi e^5}{0.94} \int_{x_c}^{\infty} l_b(-x)e^{0.43(x-x_c)} dx < 10^{-7} \left(\frac{P_i}{p_l}\right)(1 - \kappa).$$

Here $p_l(1 - \kappa) \leq e^5$, and $x_i - x_{i,c} \leq (x_b - x_{b,c})/0.94$, by the lower curvature bound. The Gaussian rate $2px_c > 2$ proves the last inequality (even the bound $\kappa/(2 - 0.43)$ for the integral suffices). This establishes the analytic validity of every **coarse** sign test.

The sharper test uses $c_+ = 0.3$. Fix the reference r with $a_r = 0.293650$, which lies in every interval to which this test is applied. At fixed inverse coordinate k , set

$$\epsilon = \log(V_i/V_r), \quad \pi = \log(P_i/P_r), \quad v_\sigma = V_r e^{\sigma\epsilon}, \quad p_\sigma = P_r e^{\sigma\pi}, \quad 0 \leq \sigma \leq 1.$$

This is interpolation in the functional only. Define M_σ, X_σ by (10.14); no interpolated PDE is claimed or needed. Both endpoints, and hence their geometric interpolants and interpolated logarithmic density slopes, lie in **Env**'s envelopes. It bounds ϵ and π' by **eps** and **pis**, and bounds π by the anchored primitive of **pis**. The interpolated primitives are obtained by correcting the r primitives. On their omitted tails $|e^{-\sigma\epsilon} - 1| < 3$ by Lemma 10.3; the code uses the harmless larger constant 3.01. Intersecting with the static envelopes remains valid.

Let v, p, M, X denote these interpolated profiles, and put $S = c_+(MX - t_k) - 2\xi kt_k$. Differentiating the middle-grid contribution gives $\int(g_p\pi + g_v\epsilon)$, up to the explicitly bounded primitive-tail terms, where

$$\begin{aligned} g_p &= \frac{p}{v}(v^2 + S), \\ g_v &= \frac{p}{v}(v^2 - S) - c_+ \frac{k}{v} \int_k^{1-\kappa} \frac{p}{v} X dj - c_+ \frac{l}{v} \int_\kappa^k \frac{p}{v} M dj. \end{aligned} \tag{10.16}$$

The integration variable in the inner integrals is j , with every profile there evaluated at j . To verify the formula, differentiate (10.14):

$$\partial_\sigma M(k) = - \int_0^k \frac{j\epsilon(j)}{v(j)} dj, \quad \partial_\sigma X(k) = - \int_k^1 \frac{(1-j)\epsilon(j)}{v(j)} dj.$$

The local factors p/v and v^2 give the first terms in (10.16); Fubini for the two primitive derivatives gives its last terms. The omitted Fubini regions are bounded in absolute value by

$$\mathcal{E}_{\text{prim}} = 6M_r(\kappa) \int_\kappa^{1-\kappa} c_+ \frac{p}{v} X dk + 6X_r(1 - \kappa) \int_\kappa^{1-\kappa} c_+ \frac{p}{v} M dk.$$

Indeed $|\epsilon|V_r/v \leq 3.8 \log 3.8 < 6$. This is **terror**, checked below $3 \cdot 10^{-7}$ uniformly on each interpolation-parameter box.

Since $\pi(1/2) = 0$, the density term can instead be expressed against π' : its coefficient is $-\int_\kappa^k g_p$ for $k < 1/2$ and $\int_k^{1-\kappa} g_p$ for $k > 1/2$. This is just Fubini applied to the anchored primitive; there is no endpoint term. The code intersects this enclosure with the direct enclosure for $\int g_p\pi$. It integrates the resulting derivative enclosures in σ and adds them to the directly enclosed reference middle integral, called **base**.

To recover a sign for the full functional, add an absolute bound for the actual tails. On the positive side $P_i \leq 1$, $V_i \leq 1$; moreover

$$\frac{dk}{V_i} \leq \frac{dx_b}{0.94}, \quad M_i X_i \leq \frac{M_b X_b}{0.94^2} \leq 1.2k.$$

On the negative side the corresponding estimates are $M_i X_i \leq 7l$ and $P_i \leq e^5 e^{0.43(x_b - x_{b,c})}$. For the term $P_i V_i dk$, use the upper reference instead:

$$P_i V_i dk \leq 1.054 e^5 (x_a + 1)^2 l_a^2 e^{1.52(x_a - x_{a,c})} dx_a.$$

The exponential factor uses the position-change bound 3.8 from (10.9) and $3.8 \cdot 0.4 = 1.52$. Integrating with the tail rates above, and $2\xi < 1.1$, bounds the total actual tail by $3 \cdot 10^{-7}$. In fact a much smaller explicit allowance suffices. With $s = x - x_c$ and $e^5 < 149$, the positive-tail contribution is at most

$$\kappa + \frac{1.76}{0.94} \frac{\kappa}{2} < 7.05 \cdot 10^{-12}.$$

Here $|c_+(MX - t_k) - 2\xi kt_k| \leq 1.76k$ and the tail rate is at least 2. The negative-tail contribution excluding $P_i V_i$ is at most

$$\frac{3.5 \cdot 149}{0.94} \frac{\kappa}{2 - 0.43} < 1.30 \cdot 10^{-9},$$

since the corresponding numerator is at most $3.5l$. Its remaining $P_i V_i$ contribution is at most

$$1.054 \cdot 149 \kappa^2 \int_0^\infty (21 + s)^2 e^{-2.48s} ds < 4 \cdot 10^{-19}.$$

These bounds follow from $x_c < 20$ and the displayed Gaussian rates; their sum is below $1.31 \cdot 10^{-9}$. Thus the allowance 10^{-6} in `nonlineartest` covers both the primitive-tail error and the actual tails with slack. This proves the analytic validity of that sign test as well.

The certificate applies the following interval reductions. Entries are the resulting allowed intervals for A , in units of 10^{-3} :

test	successive intervals
coarse	[167, 410], [183, 410], [185, 400], [193, 397], [197, 390], [203, 386], [207, 381], [211, 377], [215, 373], [218, 369], [220, 370],
nonlineartest	[228, 370], [231, 368], [234, 363], [239, 358], [244, 350], [251, 341], [258, 332], [266, 324], [272, 316], [277, 311], [281, 307], [283, 304], [285, 303].

The initial interval is $[0, 0.410]$. Widening the upper endpoint from 0.369 to 0.370 before the last coarse step is harmless; only the improved lower endpoint requires a sign test there. At each step, the input satisfies (10.15) and its coefficient lies in the incoming interval for every entire time. The comparisons and sign test therefore apply at every time, giving the resulting interval on the whole time axis. That whole-time enclosure is the hypothesis for the next round of comparisons. The resulting conclusion is

$$0.285 \leq A(v) \leq 0.303 \quad \text{for all entire times } v, \quad \sup_v |A(v) - a_*| < 0.01.$$

10.8 A local barrier at the root

The signed tests have placed every input satisfying (10.15) within distance 0.01 of the root. To turn this interval bound into equality, we need comparison errors that are proportional to the remaining deviation. Put $d = \sup_v |A(v) - a_*| \leq 0.01$. At the root reference,

$$1 - dL(k) \leq \frac{V_i(k)}{W_*(k)} \leq 1 + dL(k), \quad L(k) = 0.26 + 1.30k - 1.86k^2 + 3.25k^3. \quad (10.17)$$

For $d > 0$ let $C_L = W^2 L_{kk}/2 + WW_k L_k$. The two required stationary inequalities, for $R_0 = 1 \pm dL$, are

$$R_0^2 C_L - At_k L_k + R_0[-(R_0 + 1)D_a L + |T_a|] < 0, \quad \pm d C_L - R_0 D_a < 0.$$

Scaling and entire comparison work exactly as in (10.11). In the linearization just given, the local trial $S = C(1 \pm dL(k(z)))$ has $S_z = \mp CdWL_k$ and $S_{zz} = \pm Cd(W^2 L_{kk} - W_z L_k)$. The Gaussian derivative bounds make S_{zz} and βS_z bounded here too. The test `localcheck` verifies these inequalities on its central k grid, enclosing the root by the reference interval $[0.293650, 0.293655]$ and (10.7), (10.10), (10.13). In particular, the root is an admissible stationary input for (10.13); this encloses its unknown Q_* between the two tabulated endpoint expressions without using rigidity. The omitted positive and negative tails have respectively $z > 7$ and $x = -z > 10$, as checked by those enclosures; both root edge positions also have absolute value below 20. The reference Gaussian tail bounds stated above therefore apply to the root. On the positive side the preceding bound for T_a applies and $|C_L| < 0.001$, using $W \leq (x + 2)e^{-qx^2}$ and $|WW_k| \leq 2|B|W + 2at_k$. On the negative side $U_0 < 0.059$, so

$$T_a \in [-1 - U_0, 0], \quad D_a \geq p - aU_0 > 0.1889, \quad L > 2.94.$$

Again $|C_L| < 0.001$, now using $W \leq (x + 1)e^{-px^2}$. In both tails $t_k |L_k| < 0.001$. These bounds give the displayed strict inequalities, since $0 < L < 3$ and $1.97 \cdot 0.1889 \cdot 2.94 > 1.09$.

At the same spatial position one also has

$$|M_i - M_*| \leq 0.65d, \quad |Q_i - Q_*| \leq 2d. \quad (10.18)$$

For the first, the difference equation used in (10.7) is controlled by the certified bound

$$\sup_z \frac{M_* l_*}{q - (a_* + 0.01)l_*} < 0.65.$$

The finite test covers the inverse middle grid. Outside that grid, $z > 7$ or $x = -z > 10$, while the denominator exceeds 0.196. On the positive tail $M_* l_* \leq e^{-z^2/2}/z$; on the negative tail $M_* l_* \leq (x + 1)e^{-0.2063465x^2}$, by (10.6) and $X_* < 1$. These decreasing envelopes make the ratio less than 10^{-6} on both omitted tails. Thus the displayed supremum bound holds on the whole line, and the constants $\pm 0.65d$ are barriers. For the second, use (10.12) in absolute value. The central test checks positive damping and, cell by cell,

$$a_* 0.65 + \frac{a_*(1 - a_*) 0.65 + \sup |\text{bracket in (10.12)}|}{D_{Q,\min}} < 2.$$

It obtains the possible k_i range by the outward inversion described above, using the larger position boxes from (10.17) and (10.14); at a clipped tail cell the extra 0.0001 in its upper curvature bound is justified by (9.21): outside the inverse grid, $t_k \leq \kappa = 2^{-38}$, so $W_* \leq (6/5)\kappa^{3/4} < 3.162 \cdot 10^{-9}$. Equation (10.17) increases this by a factor at most 1.03; the overlapping grid cell is already enclosed by the range query. For the spatial tails, $a_* + 0.01 < 0.31$ gives $M_i \leq M_{0.31}$ by (10.7), and also $X_i \leq X_{0.31}$. Convex secants and (10.6) give, for $x \geq 12$,

$$k_i(x) \leq M_{0.31}(x - 1) \leq \frac{e^{-(x-1)^2/2}}{x - 1} < 4.829 \cdot 10^{-28},$$

$$l_i(-x) \leq x X_{0.31}(-x + 1/x) \leq \frac{x e^{-0.19(x-1/x)^2}}{0.38(x - 1/x)} < 5.075 \cdot 10^{-12}.$$

The bounding functions decrease on this range. Substitution in (9.21), rather than the secants alone, yields $V_i < 4.059 \cdot 10^{-9} < 0.0001$ on both tails. The same S bounds and cancellation used above give bracket absolute value below 0.08 and $D_Q > 0.18$; with $|\delta| \leq 0.65d$ these imply the second inequality of (10.18) there also.

10.9 Strict contraction and rigidity

Proposition 10.4 (Rigidity of the logarithmic-time limit). *Let (M_i, P_i, A) be a density input as defined in Subsection 10.4, and suppose that the two identities (10.15) hold at every entire time. Then $A \equiv a_*$. In particular every entire limit of the critical rescaled profiles has this coefficient, and*

$$\lim_{\tau \downarrow 0} \frac{\tau \gamma(\tau)}{\chi(\tau)} = a_*, \quad \lim_{\tau \downarrow 0} \frac{\log \chi(\tau)}{\log \tau} = a_*.$$

The root of $\lambda(a) = a$ in the certified interval is unique.

Proof. Let $d = \sup |A - a_*| < 0.01$. If $d = 0$ there is nothing to prove; assume $d > 0$. Interpolate from the root to the input as above, with $c_+ = 0.3$ and $\xi = a_*$. Normalize the two densities at the *same spatial position* $z_c = z_*(1/2)$ to have value one. This normalization preserves both identities (10.15). Unlike the previous normalization, it does not require the input density at its own median slope to be one.

Our target is $|A(v) - a_*| \leq c_{\text{loc}} d$ at every entire time, with $c_{\text{loc}} < 1$. We obtain it by bounding the change in the signed functional from its zero value at the root and dividing by the positive denominator in the identity for N_{a_*} . The estimates below control that one quotient, including its endpoint tails, by a constant times d .

Write $W = W_*$. Equation (10.17) implies

$$\frac{|\epsilon|}{d} \leq E_m := \frac{L}{1 - 0.01L}, \quad |(V_i/W)^{-\sigma} - 1| \leq \sigma d E_m, \quad E_m < 3.1.$$

For the second inequality, if $V_i/W < 1$, convexity of $t \mapsto (V_i/W)^{-t}$ puts its value below the chord on $[0, 1]$; if $V_i/W \geq 1$, use $1 - e^{-x} \leq x$ and the logarithmic bound. Integrating E_m/W with weights k and l in (10.14) gives bounds L_M, L_X for primitive changes divided by σd . Their sum L_z bounds the position change:

$$|z_i(k) - z_*(k)| \leq d L_z(k).$$

The code arrays `1m`, `1x`, `1z` enclose these functions; the global bounds $3.11M_*$ and $3.11X_*$ give convenient caps and tail allowances.

The logarithmic density difference splits as

$$\begin{aligned} \pi(k) &= \log P_i(z_*(k)) - \log P_*(z_*(k)) \\ &\quad + \log P_i(z_i(k)) - \log P_i(z_*(k)). \end{aligned}$$

The first term is zero at $k = 1/2$ and has k -derivative bounded by $2d/W$, by (10.18). The absolute value of the second is bounded by d times `shifts`: use the position interval $z_* \pm dL_z$ and (10.18) to bound Q_i there by a range query for Q_* plus $2d$. Hence

$$\frac{|\pi(k)|}{d} \leq 2|z_*(k) - z_c| + \text{shifts}(k).$$

These estimates enclose the interpolated v, p, M, X on each of the four parameter subintervals used by `localcheck`.

Apply (10.16) to the middle contribution, divided by d . The curvature term has the enclosure `cv`; the density term has the enclosure `cp`. For the latter, integrate by parts only the first, anchored, part of π just displayed, and bound the shift part directly. The primitive-tail error now uses 4 instead of 6: indeed

$$\frac{|\epsilon|}{d} \frac{W}{v} \leq \frac{3/0.97}{0.97} < 4.$$

This holds on the full open interval $(0, 1)$, including the omitted tails, because (10.17) was proved there analytically. For example the omitted lower primitive derivative obeys

$$\frac{1}{d} \int_0^\kappa \frac{j|\epsilon(j)|}{v(j)} dj \leq 4 \int_0^\kappa \frac{j}{W(j)} dj = 4M_*(\kappa),$$

and the upper primitive has the corresponding bound $4X_*(1 - \kappa)$. These are precisely the two factors in the local **terror** bound. The certificate again checks this error below $3 \cdot 10^{-7}$. It also gives a positive lower bound, called **den**, for

$$D_i = 2 \int_0^1 \frac{P_i}{V_i} kt_k dk,$$

by integrating its lower enclosure on the middle grid only. The last parameter box includes $\sigma = 1$, so the denominator enclosure applies to the actual input.

The full reference functional is exactly zero by the stationary identities at a_* . Therefore its tail contribution must be subtracted from the input tail contribution before applying the local bound. We bound this difference by differentiating along the interpolation, not by an error independent of d . Here are explicit bounds for that step. Equations (10.13) and (10.18), with the certified tight reference interval, give $|Q_i| \leq 1.5$. The value barriers give $M_* + X_* \leq x + 1.6$ and the certificate gives $|z_c| < 2$. Thus

$$\frac{|\pi|}{d} \leq 2(x + 2) + 1.5 \cdot 3.11(x + 1.6) < 7x + 12.$$

With $s = x - x_c \geq 0$, the interpolated density satisfies

$$p \leq 5e^5 e^{0.47s} \quad \text{on the negative spatial tail,} \quad p \leq 5e^{0.07s} \quad \text{on the positive spatial tail.}$$

At the edge the normalization estimate above, $d < 0.01$, and $x_c < 20$ permit the factor 5. Along a tail one may use the reference density bound together with $p = P_* \exp(\sigma\pi)$ and the $7s$ part of the last display. On the negative side the reference has logarithmic growth bounded by 0.4; the positive-side reference density is at most its edge value. This gives exactly the displayed exponents.

Every primitive derivative has absolute value at most $3.1d$ times the primitive itself: the full-interval bound $|\epsilon(j)| < 3.1d$ and positivity imply

$$|\partial_\sigma M_\sigma(k)| \leq 3.1d \int_0^k \frac{j}{v_\sigma(j)} dj = 3.1d M_\sigma(k),$$

and likewise for X_σ . The reference tail bounds and $v_\sigma \geq 0.97W$ dominate these differentiated integrals, so differentiation under their endpoint integrals is justified. Therefore the pointwise absolute functional derivative, divided by d , is at most

$$\frac{p}{v} \left[\left(\frac{|\pi|}{d} + 3.1 \right) (v^2 + 0.3(MX + t_k) + 2a_* kt_k) + 0.3 \cdot 6.2MX \right].$$

Use $dk/v \leq dx/0.97$. On the negative side, $MX \leq 2.6l$ and $v^2 \leq 1.061(x + 1)^2 l^2$; on the positive side, $MX \leq 1.2k$ and $v^2 \leq 1.061(x + 2)^2 k^2$. For the product bound on the negative side, $p_* := q - a_* \geq 0.2063465$ and $x \geq 10$ give explicitly

$$\frac{MX}{l} \leq \frac{1/(2p_*) + \kappa/(400p_*^2)}{0.97^2} < 2.576 < 2.6;$$

on the positive side, $x \geq 7$ gives $MX/k \leq (1 + \kappa/49)/0.97^2 < 1.063 < 1.2$. The reference tail bounds give l or k at most κe^{-2s} , and $x_c < 20$. Integrating proves a total tail error below 10^{-6} after division by d . More explicitly, the non- v^2 part on the negative side is bounded by

$$\frac{5e^5}{0.97} \kappa \int_0^\infty [(156 + 7s)1.68 + 4.84] e^{-1.53s} ds.$$

The positive-side non- v^2 counterpart is

$$\frac{5}{0.97} \kappa \int_0^\infty [(156 + 7s)1.26 + 2.232] e^{-1.93s} ds.$$

The two v^2 contributions are bounded respectively by

$$\begin{aligned} & \frac{5e^5}{0.97} 1.061 \kappa^2 \int_0^\infty (156 + 7s)(21 + s)^2 e^{-3.53s} ds, \\ & \frac{5}{0.97} 1.061 \kappa^2 \int_0^\infty (156 + 7s)(22 + s)^2 e^{-3.93s} ds. \end{aligned}$$

Using $e^5 < 149$ and $\int_0^\infty s^n e^{-bs} ds = n!/b^{n+1}$ reduces their sum to a rational expression below $5.035 \cdot 10^{-7}$. This proves the stated 10^{-6} allowance. Including the primitive-tail error, the $2 \cdot 10^{-6}$ used by `localcheck` is sufficient.

The certified middle derivative bound plus these errors, divided by the positive denominator enclosure, gives a number $c_{\text{loc}} < 0.95$. Hence at each entire time

$$|A(v) - a_*| = \frac{|N_{a_*}(V_i, P_i)|}{D_i} \leq c_{\text{loc}} d.$$

Taking the supremum yields $d \leq c_{\text{loc}} d$, contradicting $d > 0$. Thus every such density input has $A \equiv a_*$. In particular this holds for every critical entire limit. Compactness now implies convergence of the original A to a_* : otherwise a sequence with a fixed nonzero deviation would produce an entire limit with that deviation at time zero. Finally $d \log \chi / d \log \tau = A$; integrate between a fixed positive τ_0 and τ , divide by $\log \tau$, and use the convergence of A to obtain the logarithmic susceptibility limit.

A stationary reference at any other root in the certified interval is a density input: its value and curvature bounds were proved above, and its positive eigenfunction has bounded nonpositive logarithmic slope satisfying (10.3). The stationary integral identities are exactly (10.15). The implication just proved therefore forces its constant coefficient to equal a_* . This proves uniqueness in the interval without assuming that the other reference arises as a critical translation limit. $\square$

11 The cumulative exponents

We now deduce the cumulative exponents from the selected susceptibility slope. This completes the main theorem using the finite inequalities invoked in Section 10; their verification, including rounding and exterior error bounds, follows in Section 12.

Completion of Theorem 1.1. Sections 5 and 6 establish the common threshold and mixture comparison. Theorem 8.12 identifies the limits in the prescribed order (1.4), followed by removal of the gap cutoff, using a nonnegative terminal gap $Y = H_1$ and a nonnegative terminal force variable R on the common Brownian space. If $c = \mathbb{E}R$, then

$$G_J(u) = \mathbb{P}(0 < Y \leq u), \quad F_J(s) = \mathbb{P}\left(\frac{R}{\alpha_c c} \leq s \mid R > 0\right). \quad (11.1)$$

In particular $\mathbb{P}(Y = 0) = \mathbb{P}(R > 0) = 1/\alpha_c$. The gap convention removes exactly the first of these masses; conditioning in the second formula removes the noncontact mass of R . The small-force cutoff estimate and strong L^2 convergence give a probability law with no atom at zero, and

$$\mathbb{E} \left[\frac{R}{\alpha_c c} \mid R > 0 \right] = \frac{c}{\alpha_c c \mathbb{P}(R > 0)} = 1.$$

We now restore physical time as the argument of χ and w , and write $\tau = 1 - t$. Proposition 10.4 gives

$$\lim_{\tau \downarrow 0} \frac{\log \chi(1 - \tau)}{\log \tau} = a_*.$$

Equivalently $\chi(1 - \tau) = \tau^{a_* + o(1)}$. The scale estimate in Section 9 states

$$w(1 - \tau) = \int_{1-\tau}^1 \chi(t)^{-2} dt \asymp \frac{\tau}{\chi(1 - \tau)^2} = \tau^{1-2a_*+o(1)}.$$

Together with (9.20), this yields

$$\begin{aligned} \mathbb{P}(0 < Y \leq \sqrt{\tau}) &= \tau^{a_*+o(1)}, \\ \mathbb{P}(0 < R \leq \sqrt{w(1 - \tau)}) &= \tau^{a_*+o(1)}. \end{aligned}$$

For the gap distribution set $u = \sqrt{\tau}$ to obtain $G_J(u) = u^{2a_*+o(1)}$. For the force distribution, $w(1 - \tau)$ is continuous and strictly increasing in τ , and tends to zero with τ . Its square root therefore covers all sufficiently small positive cutoffs. Taking logarithms of the second display gives the force exponent $2a_*/(1 - 2a_*)$. Conditioning and the fixed mean normalization in (11.1) change no logarithmic exponent. Thus (1.11) holds.

Returning to the gap-exponent notation of Theorem 1.1, $\gamma = 1 - 2a_*$, the exponent relation is exact:

$$2 + \theta = 1 + \frac{2a_*}{1 - 2a_*} = \frac{1}{1 - 2a_*} = \frac{1}{\gamma}.$$

Both maps $a \mapsto 1 - 2a$ and $a \mapsto 2a/(1 - 2a) - 1$ are strictly monotone on the certified interval, in opposite directions. The endpoint arithmetic gives

$$0.4126930 < \gamma < 0.4126934$$

and

$$0.4231063544994904 \dots < \theta < 0.4231087030795289 \dots,$$

which implies the stated rational decimal bounds. Finally, the mixture comparison identifies the fixed-temperature row marginals on an entire density neighborhood above the common threshold. Each subsequent ordered limit and each exponent is therefore the same for every fixed $\varepsilon \in [0, \varepsilon_0)$. $\square$

12 Verification of the finite certificate

The computer-assisted part of the argument has three logically distinct components: integer certificates for the stationary ODEs, analytic estimates that turn small residuals into errors relative to the exact profiles, and interval bounds for the remaining integrals. This section describes the interface between these components. In particular, a successful floating-point calculation alone is not the assertion being used: every computed box must contain the corresponding exact mathematical quantity. We first certify the reference profiles and eigenfunction logarithmic derivatives using integer residuals and analytic error estimates; we

then enclose the barrier and integral tests by interval arithmetic and record the complete execution.

Throughout this section $q = 1/2$, $0 \leq a \leq .41$, $p = q - a \geq .09$, and

$$B(z) = qz + aM_a(z), \quad k = -M'_a, \quad l = 1 - k, \quad X = z + M_a.$$

The existence, uniqueness, and monotonicity of M_a , its ODE and tail bounds (10.6), and the confining ground-state construction of P_a are analytic inputs from Section 10. All decimal constants in inequalities below denote terminating decimals, hence rational numbers. Machine padding constants are discussed separately in Subsection 12.4.

The table maps the finite checks to their uses in Section 10. The enclosure proofs and recorded execution below establish the finite inequalities; the comparison principles and exterior estimates of that section supply their analytic consequences. The tail estimates used with each line have been established in Section 10; a finite grid by itself would not justify any statement over an infinite spatial domain.

Routine	Role in the analytic argument
<code>checkA</code>	Positivity of the cumulative lower bound (10.5), excluding an entire limit with $A \geq .41$.
<code>checkV</code>	Central strict inequalities (10.11), which validate the curvature barriers (10.10).
<code>Env.checkH</code>	Central positive damping and source bounds for (10.12), hence the log-slope comparison (10.13), and the density bounds needed for integration tails.
<code>Env.coarse</code>	Signs of the left or right signed cumulatives for $c_+ = 0$, with a 10^{-7} allowance for the adverse tail.
<code>nonlineartest</code>	Signs of the full calibrated functional at $c_+ = .3$, using geometric interpolation and (10.16), with a 10^{-6} allowance.
<code>localcheck</code>	The local barriers (10.17), (10.18), a positive denominator, and the strict contraction coefficient, with a $2 \cdot 10^{-6}$ allowance.
<code>certified</code>	Integer residual and shooting tests, including the opposite signs of $\lambda(a) - a$ at the final two parameter values.

12.1 Integer polynomial certificates

Set $D = 10^{36}$ and $h = 1/100$. The program uses integer coefficients c_n for the polynomial

$$\widetilde{M}(z) = D^{-1} \sum_{n=0}^{18} c_n s^n, \quad z = \varepsilon(j + s)h, \quad 0 \leq s \leq 1, \quad \varepsilon \in \{-1, 1\}.$$

The input parameter is the rational number $a = A/\text{AD}$; initially $\text{AD} = 10^6$, and the two final parameter tests use $\text{AD} = 10^7$. The spatial equation is

$$M'' + 2(qz + aM)M' - 2pM = 0. \quad (12.1)$$

After substitution of the polynomial, multiplication by the integer common denominator gives exactly the coefficient expression `rhs(n)` in `mstep`. Successive coefficients are obtained by integer division. Their manner of construction is only a way to find a good trial: certification uses the residual of the completed polynomial, not the accuracy of this recurrence as an ODE solver.

Specifically, if r_n are the integer residual coefficients after the multiplication by ADD^2 , the assertion in `mstep` is

$$10^{23} \sum_n |r_n| < \text{ADD}^2.$$

Because $0 \leq s \leq 1$, it proves, throughout the closed cell,

$$|\widetilde{M}'' + 2(qz + a\widetilde{M})\widetilde{M}' - 2p\widetilde{M}| < 10^{-23}. \quad (12.2)$$

At the next cell the initial value and the derivative in s are the integer sums $\sum c_n$ and $\sum nc_n$. Thus joins are exactly C^1 , including the join at zero: the two first derivative coefficients are chosen with opposite signs. Second derivatives may jump at joins; the comparison arguments below apply separately on cells and use continuity of the first derivative.

The routine `mint`, with `save=True`, additionally certifies

$$|\widetilde{M}(24)| < 10^{-23}, \quad |\widetilde{M}(-24) - 24| < 10^{-23}. \quad (12.3)$$

The derivative coefficient tests in `certified` give $-1.01 \leq \widetilde{M}' \leq .01$ on every cell. Indeed, for $k = -\widetilde{M}'$, the constant coefficient is the integer $-\varepsilon 100c_1$, and the sum of the absolute values of the remaining coefficients is at most $100 \sum_{n \geq 2} n|c_n|$; these are precisely `kp` and `rad`.

Lemma 12.1 (Error of the stationary profile). *For every certified rational parameter,*

$$\|M_a - \widetilde{M}\|_{L^\infty([-24, 24])} < 10^{-22}, \quad \|M'_a - \widetilde{M}'\|_{L^\infty([-24, 24])} < 4 \cdot 10^{-20}.$$

Proof. Let $E = M_a - \widetilde{M}$ and let r denote the residual in (12.2). Subtraction gives

$$E'' + 2BE' - (2p - 2a\widetilde{M}')E = -r. \quad (12.4)$$

The zero-order damping satisfies $2p - 2a\widetilde{M}' \geq .18 - .0082 > .17$. The true deviations from the hinge at ± 24 are bounded, respectively, by $e^{-q24^2}/24$ and $e^{-p24^2}/(2p24)$, by (10.6). Combining these with (12.3), the maximum principle applied to the constant barriers $\pm 10^{-22}$ yields the first claim. At an interior join a strict positive maximum of the difference is excluded by the same one-sided argument, since its first derivative is continuous.

On either half-line write $x = |z|$, $\beta(x) = \varepsilon B(\varepsilon x)$, and $F = \partial_x E(\varepsilon x)$. Equation (12.4) reads

$$F' + 2\beta F = f, \quad |f| < 2.1 \cdot 10^{-22}.$$

On the positive half-line $\beta \geq qx$; on the negative half-line $\beta = px - aX \geq px - .34$, using $X \leq .8$. Consequently the total magnitude of the negative part of 2β is at most $.34^2/p < 1.29$. On $[0, .1]$ we also have $\beta \leq .38$. The integrating-factor formula for F , integrated once more between zero and $.1$, and $|E(.1) - E(0)| < 2 \cdot 10^{-22}$ now imply $|F(0)| < 2.4 \cdot 10^{-21}$. Explicitly, the coefficient of $F(0)$ in this second integration is at least $.1e^{-.076}$, whereas the source contribution has absolute value at most $.1^2 e^{1.29} (2.1 \cdot 10^{-22})/2$. Finally, on $[0, 24]$ the same formula gives

$$|F(x)| \leq e^{1.29} (2.4 \cdot 10^{-21} + 24 \cdot 2.1 \cdot 10^{-22}) < 4 \cdot 10^{-20}.$$

□

12.2 The decaying eigenfunction branch at infinity

For the polynomial-order branch of P_a , put $S_\varepsilon(x) = \partial_x \log P_a(\varepsilon x)$. The eigenvalue equation gives the Riccati equation

$$S' = 2\beta S - S^2 + 2(q - ak - \lambda). \quad (12.5)$$

Let $p_+ = q$ and $p_- = p$. For either sign set

$$\begin{aligned} c &= \lambda/p_\varepsilon - 1, & d &= \frac{c(c-1)}{2p_\varepsilon}, & e &= \frac{(2c-3)d}{2p_\varepsilon}, \\ T(x) &= \frac{c}{x} + \frac{d}{x^3} + \frac{e}{x^5}, & K &= \frac{10|e| + 3d^2 + 30}{p_\varepsilon}. \end{aligned} \quad (12.6)$$

The rational assertions in **tailinit**, applied at all three tested values of λ , verify $-.71 < c < .81$ and, on the negative side, $c > -.01$. The same bounds hold at $\lambda = a_*$ in the final root bracket. On the positive side $p_+ = 1/2$ and $|c(c-1)| < 1.215$; on the negative side $|c(c-1)| \leq 1/4$ and $p_- \geq .09$. These sided bounds imply

$$|d| < 1.4, \quad |e| < 24, \quad K < 3066. \quad (12.7)$$

Lemma 12.2 (Riccati tail enclosure). *For $x \geq 12$, the polynomial-order branch is uniquely determined by*

$$T(x) - Kx^{-7} \leq S_\varepsilon(x) \leq T(x) + Kx^{-7}.$$

It depends continuously on λ within each trial window. Moreover,

$$|S_\varepsilon(x)| < .071, \quad |xS'_\varepsilon(x) + S_\varepsilon(x)| < .0055. \quad (12.8)$$

Proof. First replace the coefficients in (12.5) by their limiting values $\beta = p_\varepsilon x$ and $q - ak = p_\varepsilon$. For the residual $\mathcal{R}(S) = S' - 2\beta S + S^2 - 2(q - ak - \lambda)$, direct expansion gives

$$\mathcal{R}(T) = \frac{(2c-5)e + d^2}{x^6} + \frac{2de}{x^8} + \frac{e^2}{x^{10}}.$$

Thus $x^6|\mathcal{R}(T)| \leq 6.62|e| + d^2 + 1$ for $x \geq 12$. Replacing T by $T \pm Kx^{-7}$ adds the leading term $\mp 2p_\varepsilon Kx^{-6}$ and a remainder whose absolute value, after multiplication by x^6 , is at most

$$\frac{|-7 + 2c + 2d/x^2 + 2e/x^4|K}{x^2} + \frac{K^2}{x^8} \leq .7p_\varepsilon K.$$

The true coefficients differ from their limiting values by at most

$$|\beta - p_\varepsilon x| \leq \frac{.41e^{-p_\varepsilon x^2}}{2p_\varepsilon x}, \quad |q - ak - p_\varepsilon| \leq .41e^{-p_\varepsilon x^2}.$$

Their additional residual cost, multiplied by x^6 , is less than 8. For example, the largest exponential envelope is obtained with $p_\varepsilon = .09$ at $x = 12$, since $x^6e^{-p_\varepsilon x^2}$ is decreasing for these values; inserting $|T \pm Kx^{-7}| < .071$ proves the stated bound. The definition of K leaves strict room:

$$1.3p_\varepsilon K - (6.62|e| + d^2 + 1 + 8) = 6.38|e| + 2.9d^2 + 30 > 0.$$

Accordingly the lower barrier has positive residual and the upper barrier negative residual. These are the required signs for integration from larger to smaller x .

Start at $T(R)$ at a finite $R > 12$ and integrate inward. Comparison keeps the solution within the barriers. Compactness on finite intervals supplies a limit as $R \rightarrow \infty$. If two such limits are compared, their difference satisfies a homogeneous first-order equation with coefficient $2\beta - S_1 - S_2 \geq p_\varepsilon x$ for $x \geq 12$. Inward integration therefore forces their difference to vanish as the comparison point tends to infinity. The same formula with the additional source $-2(\lambda_1 - \lambda_2)$ proves continuity in λ on compact subintervals.

The first estimate in (12.8) follows immediately from (12.7). For the second, substitute the enclosure into (12.5). The resulting upper bound is

$$\begin{aligned} \frac{2|d|}{x^3} + \frac{1}{x^5} \left(4|e| + 6.62|e| + d^2 + 1 + 8 + 2p_\varepsilon K \right. \\ \left. + \frac{(1 + 2 \cdot .82)K}{x^2} + \frac{K^2}{x^8} \right). \end{aligned}$$

To justify the constant .82 in this expression, observe that d and e have opposite signs and $|e|/(|d|x^2) = |2c - 3|/(2p_\varepsilon x^2) < 1$ when $d \neq 0$. Thus xT has the sign of c and magnitude at most $|c| < .81$; the case $d = 0$ is immediate. The displayed bound, using (12.7), is less than .00514. Here one must retain the identity $p_\varepsilon K = 10|e| + 3d^2 + 30$; replacing the two factors by separate worst-case bounds would lose the required estimate.

Finally, integrating $S = c/x + O(x^{-3})$ gives $P(\varepsilon x) = Cx^c(1 + O(x^{-2}))$ with $C > 0$. After the conjugation $P = e^{\int B\psi}$, this is the square-integrable branch of the Schrödinger equation. Reduction of order gives a second independent solution growing by the reciprocal Gaussian factor; hence no other square-integrable branch is available at this endpoint. $\square$

12.3 Inward shooting and eigenvalue certification

The shooting routine begins at $x = 20$. The value `tailinit` is an integer multiple of D^{-1} within D^{-1} of the exact rational number $T(20)$. The routine `revpoly` expresses the stationary polynomial in the reversed cell coordinate. In `qcoefs`, the resulting coefficients represent $2\tilde{\beta}$ with denominator `ADD`, and $\tilde{k}$ with denominator D . The inward cell step is $-1/100$; this explains the sign in `qint`'s recurrence.

For the completed degree-18 Riccati trial $\tilde{S}$, the integer residual test is again a sum of absolute coefficients on $0 \leq s \leq 1$. It proves a residual smaller than 10^{-20} in the spatial Riccati equation with the approximate $\tilde{\beta}, \tilde{k}$. The trial is exactly continuous between cells. No numerical differentiation is used.

Let $E_S = S - \tilde{S}$. With λ equal to the trial value,

$$E'_S = (2\beta - 2\tilde{S} - E_S)E_S + f_S, \quad |f_S| < 3 \cdot 10^{-19}, \quad (12.9)$$

where the source bound includes the residual and both stationary-profile errors from Lemma 12.1. When λ differs from the middle trial value by at most 10^{-9} , add $2 \cdot 10^{-9}$ to this source bound.

Under the provisional condition $|E_S| \leq .001$, each integer `gs` is a lower bound for the drift in (12.9), with denominator `ADD`. The subtraction .002 in its definition covers $|E_S|$ and the much smaller profile error. The tests on `gs` imply:

1. every inward integrating factor, including factors with endpoints inside cells, is at most e^2 ;
2. the integrating factor from 20 to 0 is smaller than e^{-20} ;
3. the factor from 20 to any $x < 12.02$ is smaller than e^{-6} .

For the last assertion, the cells indexed from 1203 upward have total integral greater than 8, while the sum of all negative cell contributions is greater than -2 . This also covers the possible partial cell.

The initial error at 20, including variation of $T(20)$ over the eigenvalue window, is less than $2.41 \cdot 10^{-6}$. Variation of constants gives the following deliberately loose enclosures:

$$|E_S| < 2.9 \cdot 10^{-5} \quad (0 \leq x \leq 20), \quad |E_S| < 4.5 \cdot 10^{-7} \quad (0 \leq x < 12.02). \quad (12.10)$$

For example, the global estimate is bounded by $e^2[2.41 \cdot 10^{-6} + 20(2 \cdot 10^{-9} + 3 \cdot 10^{-19})]$; the central estimate replaces the first factor e^2 on the initial error by e^{-6} . Both are strictly below the provisional .001, which closes the bootstrap and proves that the positive branch extends to the center. For an endpoint trial compared at its own eigenvalue, the error at zero is bounded instead by

$$(2.41 \cdot 10^{-6})e^{-20} + 20e^2(3 \cdot 10^{-19}) < 10^{-10}.$$

The matching condition is $S_-(0) + S_+(0) = 0$, because derivatives in z acquire opposite signs on the two half-lines. At the lower and upper shooting values the integer tests give, respectively, a trial sum smaller than -10^{-9} and greater than 10^{-9} . The total error of the two halves is less than $2 \cdot 10^{-10}$, so these signs also hold for the true shoots. Continuity therefore supplies a match. It produces a positive square-integrable eigenfunction after conjugation, and consequently the lowest eigenvalue. We have proved

$$\frac{\mathbf{lam} - 10}{10^{10}} < \lambda(A/AD) < \frac{\mathbf{lam} + 10}{10^{10}} \quad (12.11)$$

for every call to `certified`. This also justifies the log-slope errors assigned to the middle shoot in subsequent tables.

12.4 Floating-point enclosures

The interval stage uses IEEE binary64 round-to-nearest arithmetic, including square root. Integer polynomial residual tests precede every tabulation. Let $u = 2^{-53}$ be unit roundoff. Away from underflow each elementary operation has relative error at most u ; at underflow an absolute error of one subnormal unit suffices. The functions `down` and `up` subtract or add

$$(|x| + 10^{-280})2^{-43}$$

to a rounded endpoint. This is more than one thousand unit roundoffs in relative scale, and its absolute component is much larger than a subnormal unit. Rounding in forming and applying the padding consumes less than this allowance. Therefore the operations of class I, including the four-product rule for multiplication, reciprocal on intervals of fixed sign, squaring, and square root, enclose the exact operations. Assertions exclude division by a zero-containing interval, invalid square roots, overflow, and invalid floating-point operations.

The class constructor itself does not enlarge a literal. Integer and dyadic inputs are exact. A terminating decimal that defines a mathematical barrier or functional is consequently entered by `rat(n,d)`, rather than by an unpadded decimal literal. The other non-dyadic literals in the code are allowances or coarse caps with strict slack; they may be interpreted as their binary64 values. In particular the caps `.1431` and `.1301` in `adjusted` are strictly weaker than the analytic bounds they represent. Integer-coefficient conversion and node evaluation require the separate allowance below.

No library exponential or logarithm is used for interval evaluation. For $|x| \leq 100$, `iexp` evaluates a degree-20 Taylor polynomial at $x/256$, adds the interval remainder $[-2^{-80}, 2^{-80}]$, and squares eight times. The remainder bound follows from $e^4 \cdot 4^{21}/21! < 2^{-80}$. For $1/8 \leq x \leq 8$, `ilog` takes three interval square roots and uses

$$\log x = 16 \left(t + \frac{t^3}{3} + \cdots + \frac{t^{21}}{21} \right) + \text{remainder}, \quad t = \frac{x^{1/8} - 1}{x^{1/8} + 1}.$$

Here $|t| < .14$ and $16|t|^{23}/[23(1-t^2)] < 2^{-60}$, the remainder interval used by the code. All polynomial operations in these two routines are interval operations.

For summation of at most n real terms, the usual induction on rounded addition gives

$$\left| \text{fl} \left(\sum_{j=1}^n y_j \right) - \sum_{j=1}^n y_j \right| \leq \frac{nu}{1 - nu} \sum_{j=1}^n |y_j|$$

up to the negligible absolute underflow contribution. The quadrature arrays have $n < 2 \cdot 10^5$, so the coefficient is less than $3 \cdot 10^{-11}$. The routine `sums` uses the much larger allowance

10^{-9} times the accumulated absolute sum, plus `tiny`. This also covers rounding in that accumulated sum and in the final correction. The shorter summation in `checkA` has at most 4096 terms of absolute value below 9; its error after division by 4096 is less than $\gamma_{4096} 9 < 4.1 \cdot 10^{-12}$. Its remaining scalar arithmetic costs less than 10^{-13} , since all terms have magnitude less than 10. The subtraction of $2 \cdot 10^{-9}$ therefore covers both, including the unpadded bound arithmetic after the summation.

The last scalar addition or division in the nonlinear and local tests is also unpadded. The analytic allowances retain sufficient slack to cover it. In the recorded run all nonlinear returned values have magnitude below .00204, so the final addition of the signed 10^{-6} changes its exact value by less than 10^{-18} ; the two substantive errors total less than $6 \cdot 10^{-7}$, leaving nearly $4 \cdot 10^{-7}$ unused. For the local test, an additional diagnostic evaluation of the unchanged function gave

$$O = \text{out.hi} = .7290900413708653, \quad D_0 = \text{den.lo} = .7928198472818054.$$

With $O < 1$ and $.7 < D_0 < 1$, rounding the final expression $(O + 2 \cdot 10^{-6})/D_0$ costs less than $6 \cdot 10^{-16}$. The exterior and primitive-tail estimates cost less than $1.3 \cdot 10^{-6}$ in the numerator, leaving $.7 \cdot 10^{-6}$ unused there. Small rounding in the positive scalar calculation of `terror` is absorbed in the same slack. Thus the returned local number is an upper bound for the exact contraction coefficient, and the returned signed nonlinear numbers retain the required mathematical signs.

12.5 From polynomial tables to spatial and inverse boxes

Each step of length .01 is sampled at the 50 fractions $0, 1/50, \dots, 49/50$. The resulting spacing is $1/5000$. The coefficients for X , k , and l are formed by exact integer operations from those of M ; they are evaluated independently. Here is a quantitative bound for the conversion and evaluation. CPython's integer true division computes the correctly rounded ratio c_n/D : it forms an integer quotient with guard bits, records the discarded remainder, rounds ties to even, and then converts the rounded integer and rescales exactly.¹ Every nonzero ratio here is at least 10^{-36} in magnitude, so it is normal and has relative conversion error at most u . Both an ordinary sample $j/50$ and a reversed sample $1 - j/50$ have absolute node error at most $2u$, and the exact and rounded nodes lie in $[0, 1]$. Writing $C = \sum |c_n|/D$, the node perturbation therefore costs at most $36uC$. The 36 rounded operations in degree-18 Horner evaluation cost at most $\gamma_{36}(1 + u)C$, where $\gamma_n = nu/(1 - nu)$; coefficient conversion costs at most uC . The total is less than $74uC < 8.22 \cdot 10^{-15}C$, and in particular smaller than $2 \cdot 10^{-14}C$. The node allowance used for each of M, X, k, l is

$$3 \cdot 10^{-14} \sum_n |c_n|/D + 3 \cdot 10^{-18}. \quad (12.12)$$

It covers this evaluation error, Lemma 12.1, and the additions and subtractions that form node bounds. The code checks that the coefficient sum is less than $30D$. The separate Q allowances contain the evaluation error and (12.10); the global node radius is $2.95 \cdot 10^{-5}$ and the central refinement uses $5 \cdot 10^{-7}$.

Between nodes, monotonicity gives boxes for M, X, k, l . For $Q = (\log P)'$, `zcells` first bootstraps $|Q| \leq 1$: the trial nodes have magnitude below .951, and the Riccati equation bounds $|Q'| < 30$ on a fine cell under this provisional bound. Its length is $1/5000$, so enlargement by .006 is sufficient and is strictly inside the provisional bound. Twice applying the derivative equation then sharpens the cell box. The cap $Q \leq 0$ uses the analytic

¹The implementation and its rounding argument are in CPython 3.12.13, `long_true_divide`, `Objects/longobject.c`.

maximum-principle argument for the exact eigenprofile; the cap is not an additional numerical assumption.

Inverse lookup must also include uncertainty in node values. Since k is decreasing, suffix maxima of its node lower bounds remain valid lower bounds, and prefix minima of its upper bounds remain valid upper bounds. The analogous operations for increasing l give its monotone guaranteed bounds. A binary search through these outward-monotonized arrays supplies outer nodes for the inverse position. The code uses k or l according to which is better separated from 1; the two schemes are mathematically equivalent. The inverse curvature is then evaluated through the exact identity

$$W = 2pMl + Xk,$$

rather than by differentiating an inverse interpolant.

The routines `Rmq.query` and `kquery` return minima and maxima over all cells between two indices. For a range of length n , take the two blocks of length $2^{\lceil \log_2 n \rceil}$ abutting its endpoints. Their union covers the range, and their individual extrema have been computed by successive doubling. Thus the returned box is valid even when these blocks overlap. Spatial coordinate bounds and index calculations are enlarged by 10^{-12} or 10^{-10} before multiplication and flooring; these allowances are larger than the binary64 errors on the bounded coordinate range. This padding also validates the scalar negative-edge test $2px_c > 2$, even though its code uses the rounded parameter `af`. If x_f is the negation of the padded upper position bound used in that test, then $x_f < x_c - 0.99 \cdot 10^{-12}$. Since $p \geq 0.09$ and $x_c < 20$, replacing the rounded parameter and product by their exact values costs less than $3 \cdot 10^{-14}$, whereas the position padding leaves more than $1.7 \cdot 10^{-13}$ in $2px_c$. Thus a successful scalar comparison implies the exact tail-rate inequality.

The inversion of the envelope $z_i(k)$ in `possiblek` and `localcheck` uses the same outward monotonicity principle. At a clipped edge it explicitly restores the missing k -tail; a point outside the central range is never silently treated as an interior grid point.

More explicitly, let L_j, U_j be the outward-monotonized bounds in inverse cell j . If an admissible inverse value belongs to the queried spatial interval $[z_-, z_+]$, necessarily $L_j \leq z_+$ and $U_j \geq z_-$. The binary searches give the first and last cells satisfying these necessary inequalities. Clipping either index to the available grid and restoring $k = 0$ or $k = 1$ therefore retains all admissible slopes, including those in the omitted tails. For the local curvature query the retained cells obey $V_i(k) \leq 1.03W(k)$ by (10.17). If an admissible k is in an omitted tail, $k(1 - k) \leq \kappa$ and (9.21) gives instead

$$V_i(k) \leq 1.2\kappa^{3/4} = 1.2 \cdot 2^{-28.5} < 3.2 \cdot 10^{-9}.$$

Consequently the code's bound `1.031 wi.hi + .0001` contains the curvature in both cases; the clipped query cannot discard a larger tail curvature.

12.6 Integral enclosures and their logical use

The integration grid covers $[\kappa, 1 - \kappa]$ with $\kappa = 2^{-38}$, and has 2048 subdivisions per dyadic scale. It splits at $1/2$ and all edges and widths are exactly representable binary numbers: the finest required denominator is 2^{49} . A box on a closed cell, multiplied by its width, encloses its integral. The routine `partialsum` removes an arbitrary fraction of the endpoint cell by subtracting its box multiplied by `[0, width]`. This is valid for either sign of the integrand. The routine `anchored` integrates separately to the left and right of $1/2$, avoiding unnecessary cancellation.

The opening table records the analytic uses of these enclosures.

Several implementation details are important for this correspondence. The ratio caps in **Env** apply to the exact profiles by the analytic barriers; primitive corrections use (10.14), including the omitted primitive tail. In **nonlinear**test, the intermediate pairs are defined by geometric interpolation solely inside the functional. No assertion that they solve a parabolic equation is made or needed. Their log slopes lie in the same envelope because these slopes interpolate affinely. The two forms of the density variation—direct integration against π , and integration by parts against its anchored derivative—each enclose the same quantity; taking their interval intersection is therefore justified. The quantity **terror** bounds the missing primitive terms in that integration by parts, rather than the full exterior integral. These are distinct errors: **nonlinear**test’s 10^{-6} contains both, and **local**check’s $2 \cdot 10^{-6}$ contains their local analogues after division by d .

These primitive-tail allowances require derivative bounds, which can be checked directly. Put $r = V_i/W$ and $v_\sigma = Wr^\sigma$. Then

$$\partial_\sigma(v_\sigma^{-1}) = -(\log r)v_\sigma^{-1}.$$

In the local regime, $|\log r|/d \leq L/(1 - .01L) < 3.1$, so differentiating the positive integrals in (10.14) gives $|\partial_\sigma M_\sigma| \leq 3.1dM_\sigma$ and the identical estimate for X_σ . Relative to the reference integrands, $|\log r|r^{-\sigma}/d \leq 3.1/.97 < 4$, proving the primitive-tail coefficient 4 after division by d . For the nonlinear refinement, $1/3.8 < r < 3.8$ gives instead $|\log r|r^{-\sigma} \leq 3.8 \log 3.8 < 6$. This proves its coefficient 6. The distinct finite-difference bound $|r^{-\sigma} - 1| < 2.8 < 3.01$ justifies its primitive correction. These arguments use dominated differentiation with the reference primitive as an integrable majorant; no derivative estimate is inferred merely from a finite-difference estimate.

The omitted endpoint cells of **checkA** admit particularly crude bounds. In the notation of (10.5), $|b| \leq 1.2$, $b \geq -.8$, $E = -.4t_k$, $t_k \leq 1/4$, and $K \geq -8.24$. Since $V < .5$ and $6A^2 - .16 > 0$ for $.41 \leq A \leq 1$, its integral density satisfies

$$2V^2 + KV - 2bE + (6A^2 - .16)t_k^2/V \geq -8.24 \cdot .5 - .16 = -4.28 > -9.$$

This holds throughout the interval, hence on both omitted cells. At $0 \leq k_0 \leq 1/4096$, the endpoint expression obeys

$$bV^2 + 2EV \leq 1.2V^2 \leq 1.728t_k^{3/2} \leq 1.728/262144 < 6.6 \cdot 10^{-6} < .001.$$

Thus its endpoint allowances also have analytic coverage.

The driver follows the successive output intervals listed with the signed tests in Section 10: eleven coarse updates from $[0, 0.410]$, then thirteen nonlinear updates, give $[0.285, 0.303]$. Each printed checkpoint is the *input* interval for the checks about to run. In particular, the last printed interval is $[0.283, 0.304]$; successful completion of its two sign tests proves the final interval $[0.285, 0.303]$. The slight enlargement from 0.369 to 0.370 in the coarse sequence keeps a valid coefficient enclosure. All sign tests include the error allowances described above. Their analytic consequence is the input enclosure required by the local contraction, which then forces $A \equiv a_*$.

12.7 Reproduction and numerical output

The accompanying file `certificate/original_certificate.py` is the unaltered code printed in the appendix. From the manuscript directory, run

```
python3 certificate/original_certificate.py
```

with NumPy installed and assertions enabled; do not use `-O`. The certificate requires IEEE binary64 arithmetic with round-to-nearest operations as described above. Its interval checkpoints are printed to standard output. The accompanying reproducibility record identifies the software versions, source hashes, and saved output of a complete run.

The complete unchanged driver passed on October 4, 2026, under Python 3.12.13 and NumPy 2.3.5, with assertions enabled, in 338.52 seconds. It printed all 24 checkpoints and had empty standard error; the actual output is retained with the certificate. The separate return-observer record from September 24, which leaves the source, function arguments and results unchanged, gives

$$\text{checkA}() = 0.002977541856187091 > 0, \quad \text{localcheck}() = 0.9196188060510444 < .95.$$

The displayed decimals are a returned lower bound and upper bound, respectively. A fresh targeted evaluation of the unchanged local routine reproduces the second value and the numerator and denominator reported in Subsection 12.4. The complete unchanged execution and this diagnostic evaluation are recorded separately. Their interpretation as rigorous bounds is supplied by the error analysis and comparison arguments above. The exact source hash is

$$7\text{bd}5\text{dbd}918\text{d}9\text{a}48\text{fe}4\text{da}62044\text{e}295239\text{b}74\text{f}7\text{a}1607\text{e}2643337\text{ad}4\text{d}073\text{a}85\text{dc}7\text{f}.$$

The integer seeds for the final two spectral tests are 2936535033 and 2936533047, with denominator 10^{10} . Formula (12.11) therefore yields

$$\begin{aligned} .2936535023 &< \lambda(.2936533) < .2936535043, \\ .2936533037 &< \lambda(.2936535) < .2936533057. \end{aligned}$$

In particular the first eigenvalue is strictly larger than its parameter and the second strictly smaller. Continuity supplies

$$.2936533 < a_* < .2936535.$$

The contraction argument applies to every root in this interval and proves its uniqueness there. Combined with the analytic identification of the critical exponents, these rational inequalities give the exponent bounds stated in the conclusion.

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A Executable certificate

The following listing is the complete certificate distributed with the manuscript. Run it with Python 3 and NumPy, with assertions enabled; see Section 12 for the relation between the finite checks and the analytic estimates. An arrow at the start of a displayed line marks a continuation of the same source line.

```

1 import math, numpy as np
2 D=10**36
3 HN=100 # step reciprocal
4 L=24
5 N=L*HN
6 DEG=18
7 AD=10**6 # a denominator
8 LD=10**10
9 # all stored coefficients are integers /D
10 def mstep(A, side, j, c0,c1, check=False):
11     # z0=side*j/HN, h=side/HN, q=.5, a=A/AD
12     c=[c0,c1]
13     # recurrence common h^2 terms: 2(q-a-q*n)c_n -2q*(z0/h)*(n+1)c_{n+1}; nonlinear -2*a/h ...
14     # denominator HN^2 AD D
15     def rhs(n):
16         val=0
17         if n<len(c): val+=(AD*(1-n)-2*A)*D*c[n]
18         if n+1<len(c): val-=j*(n+1)*AD*D*c[n+1]
19         conv=sum(c[i]*(n-i+1)*c[n-i+1] for i in range(max(0,n+2-len(c)),min(n+1,len(c))))
20         val=-2*A*side*HN*conv
21         return val
22     den=HN**2*AD*D
23     for n in range(DEG-1):
24         c.append(rhs(n)/(den*(n+2)*(n+1)))
25     if check:
26         # equation residual in M_zz - rhs/h^2; scale D^2 AD
27         err=sum(abs(((c[n+2]*(n+2)*(n+1)*den) if n+2<len(c) else 0)-rhs(n)) for n in range(2*DEG+1))
28         assert err*10**23 < AD*D**2, (side,j,err/(AD*D**2))
29     return c
30 def mint(A,m0,k0,save=False):
31     data=[]; ends=[]
32     for side in [-1,1]:
33         arr=[]
34         c0=m0
35         # to ensure initial derivative exactly same with sides define k0 here as k0/HN scaled!
36         c1=-side*k0
37         for j in range(N):
38             c=mstep(A,side,j,c0,c1,save)
39             c0=sum(c);c1=sum(n*x for n,x in enumerate(c))
40             if save: arr.append(c)
41             ends.append(c0-(L*D if side<0 else 0));data.append(arr)
42         if save:
43             assert max(abs(x) for x in ends)*10**23 < D
44             return data
45     return np.array([x/D for x in ends])
46 LQ=20
47 def revpoly(c):
48     return [(-1)**j*sum(c[i]*math.comb(i,j) for i in range(j,len(c))) for j in range(len(c))]
49 def qcoefs(A,data):
50     out=[]

```



```

123 304000: [474830062508701471123549605498961791, 5236950205867338402676604016732928, 2832226234],
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132 363000: [493947768520916359508306619953883636, 5303695170717058167276507568238026, 2171028129],
133 368000: [495646114469944087456537737737225530, 5309735201830213364679958592968569, 2109122819],
134 369000: [49598728111702052425538529154406594, 5310950548984813632995589189134366, 2096618989],
135 370000: [496328947871629374756867582623647432, 5312168346853911202387504058544337, 2084073726],
136 373000: [497356952751536597848406417862848528, 5315836459606920685663169284212728, 2046187111],
137 377000: [498734651097943921003750035530256643, 5320761672756152026272462715404391, 1995077904],
138 381000: [500120399939983459354541920108240900, 5325726268072088182877928567867189, 1943276576],
139 386000: [501863941040929695618157081757711250, 5331987493849022591294812731983611, 1877528371],
140 390000: [503267883372610202334955761574692406, 5337040922801031344612478693686727, 1824113384],
141 397000: [505744327598534384034341787833659036, 5345979603431905326263929284728143, 1728838531],
142 400000: [506813294676182538273397955089170882, 5349847568311569582933277586355971, 1687287204],
143 410000: [510409698904365400089610218268197555, 5362901520459144207633059993983021, 1545546607],
144 2936533: [471647910495218087096951828814828891, 5226079904138646272764052074508064, 2936535033],
145 2936535: [471647971528437792087151567621746964, 5226080111933846901160172613080866, 2936533047],
146 }
147 def certified(A):
148     m,k,lam=db[A]; data=mint(A,m,k,save=True); co=qcoefs(A,data)
149     for lr in [lam-10,lam,lam+10]:
150         ends,out=qint(A,lr,co,save=True)
151         for side,arr,(barr,_),sarr in zip([-1,1],data,co,out):
152             for c in arr:
153                 kp=-side*HN*c[1]; rad=HN*sum(n*abs(x) for n,x in enumerate(c) if n>=2)
154                 assert (kp-rad)*100>=-D and (kp+rad)*100<=101*D
155                 # integer lower g scaled AD*D
156                 gs=[b[0]-sum(abs(x) for x in b[1:])-2*AD*(c[0]+sum(abs(x) for x in c[1:]))-AD*D//500 for b,c in zip(
157                     ↪ barr,sarr)]
158                 assert all(sum(abs(v) for v in c)*100<95*D for c in sarr)
159                 assert sum(gs[1203:])>8*HN*AD*D
160                 assert sum(min(g,0) for g in gs)>-2*HN*AD*D and sum(gs)>20*HN*AD*D
161                 if lr<lam:assert sum(ends)*10**9 < -D
162                 if lr>lam:assert sum(ends)*10**9 > D
163                 if lr==lam:qdata=out
164             return data,qdata,lam
165
166 # float interval library; padding assumes IEEE
167 np.seterr(over="raise",invalid="raise",divide="raise")
168 tiny=1e-280
169 def down(x):return x-(abs(x)+tiny)*2**-43
170 def up(x):return x+(abs(x)+tiny)*2**-43
171 class I:
172     def __init__(self,lo,hi=None):
173         self.lo=lo;self.hi=lo if hi is None else hi
174     def __add__(x,y):
175         y=iv(y);return I(down(x.lo+y.lo),up(x.hi+y.hi))
176     __radd__=__add__
177     def __neg__(x):return I(-x.hi,-x.lo)
178     def __sub__(x,y):return x+-iv(y)
179     def __rsub__(x,y):return iv(y)+-x
180     def __mul__(x,y):
181         y=iv(y)
182         a=x.lo*y.lo;b=x.lo*y.hi;c=x.hi*y.lo;d=x.hi*y.hi
183         return I(down(np.minimum(np.minimum(a,b),np.minimum(c,d))),up(np.maximum(np.maximum(a,b),np.maximum(c
184             ↪ ,d))))
185     __rmul__=__mul__
186     def inv(x):
187         assert np.all(x.lo*x.hi>0)
188         return I(down(1/x.hi),up(1/x.lo))
189     def __truediv__(x,y):return x*iv(y).inv()
190     def __rtruediv__(x,y):return iv(y)*x.inv()
191     def sq(x):
192         a=x.lo*x.lo;b=x.hi*x.hi
193         return I(np.where((x.lo<=0)&(x.hi>=0),0,down(np.minimum(a,b))),up(np.maximum(a,b)))
194     def sqrt(x):

```

```

193     assert np.all(x.lo>=0)
194     return I(np.maximum(0,down(np.sqrt(x.lo))),up(np.sqrt(x.hi)))
195 def cap(x,lo,hi):
196     z=I(np.maximum(x.lo,lo),np.minimum(x.hi,hi));assert np.all(z.lo<=z.hi)
197     return z
198 def sel(x,s):return I(x.lo[s],x.hi[s])
199 def maxabs(x):return np.maximum(abs(x.lo),abs(x.hi))
200 def iv(x):return x if isinstance(x,I) else I(x)
201 def rat(n,d):return I(n)/d
202 def choose(cond,x,y):
203     x=iv(x);y=iv(y);return I(np.where(cond,x.lo,y.lo),np.where(cond,x.hi,y.hi))
204 # exp interval Taylor scale /256, require |x|<=100
205 def iexp(x):
206     x=iv(x);assert np.all(x.maxabs())<=100
207     def fn(y,upper):
208         # Taylor point using interval class
209         w=I(y)/256; t=I(1.)
210         for i in range(20,0,-1):t=1+w*t/i
211         t=I(-2**-80,2**-80) # exp(.4) *.4^21/21! < 2^-80
212         for i in range(8):t=t.sq()
213         return t.hi if upper else t.lo
214     return I(fn(x.lo,False),fn(x.hi,True))
215 # log for args [1/8,8] via repeated sqrt and atanh
216 def ilog(x):
217     x=iv(x); assert np.all(x.lo>=.125) and np.all(x.hi<=8)
218     for _ in range(3):x=x.sqrt()
219     t=(x-1)/(x+1); ts=t.sq()
220     p=rat(1,21)
221     for i in range(19,0,-2):p=rat(1,i)+ts*p
222     return 16*t*p+I(-2**-60,2**-60) # |t|<.14
223
224 ZF=50 # fine per HN =.0002
225 ZN=HN*ZF
226 def polytab(intarr,reverse=False):
227     # table each cell at s=0,...,(ZF-1)/ZF; q reversed variable use 1-s
228     co=np.array([[float(v/D) for v in c] for c in intarr])
229     frac=np.arange(ZF)/ZF
230     if reverse:frac=1-frac
231     val=co[:,-1]
232     for j in range(co.shape[1]-2,-1,-1): val=val*frac+co[:,j]]
233     # final endpoint omitted
234     return val.reshape(-1)
235 class Profile:
236     def __init__(self,A):
237         data,qdata,lam=certified(A)
238         self.A=A; self.a=rat(A,AD); self.af=A/AD
239         self.lam=rat(lam,LD)+I(-1.01e-9,1.01e-9) # use pads later
240         sides=[]
241         for side,arr,qs in zip([-1,1],data,qdata):
242             arr=arr[:LQ*HN]
243             mx=[];ks=[];ls=[]
244             for j,c in enumerate(arr):
245                 xx=c.copy();xx[0]+=side*j*D//HN;xx[1]+=side*D//HN;mx.append(xx)
246                 kk=[-side*HN*n*v for n,v in enumerate(c) if n>0];ks.append(kk)
247                 ll=[-v for v in kk];ll[0]+=D;ls.append(ll)
248             # error bound for polytab rounding using sumabs
249             for coeff in [c,xx,kk,ll]:
250                 # absolute eval error <=2e-14 sumabs; want 1e-14 when small; adaptive use? store separately
251                 assert sum(abs(x) for x in coeff)<30*D
252             sides.append([polytab(arr),polytab(mx),polytab(ks),polytab(ls),side*polytab(qs,True)])
253 # join z indices -(20-1/ZN)..(20-1/ZN)
254 self.offset=LQ*ZN-1
255 self.M,self.X,self.k,self.l,self.Q=[np.concatenate([x[:0:-1],y]) for x,y in zip(*sides)]
256 # adaptive rounding error: for M,X,k,l near small sumabs maybe bound globally tighter by cell checking
257 # compute arrays of err coefficients for each
258 errrsides=[]
259 for side,arr in zip([-1,1],data):
260     lists=[[] for _ in range(4)]
261     for j,c in enumerate(arr[:LQ*HN]):
262         xx=c.copy();xx[0]+=side*j*D//HN;xx[1]+=side*D//HN
263         kk=[-side*HN*n*v for n,v in enumerate(c) if n>0]
264         ll=[-v for v in kk];ll[0]+=D

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265     for li,coef in zip(lists,[c,xx,kk,ll]):
266         li.append(sum(abs(x) for x in coef)/D*3e-14+3e-18)
267     errsides.append([np.repeat(li,ZF) for li in lists])
268     self.err=[np.concatenate([x[:0:-1],y]) for x,y in zip(*errsides)]
269     # monotone search approximate k and l enforce using certified thresholds cumulative
270     # exact k decreasing. guaranteed lower k and upper l prefix if node bound; take suffix max lower k?
271     self.klo=np.maximum.accumulate((self.k-self.err[2])[:-1])[:-1] # suffix nodes prove lower
272     self.khi=np.minimum.accumulate(self.k+self.err[2]) # prefix upper
273     self.llo=np.maximum.accumulate(self.l-self.err[3]) # prefix lower (l inc)
274     self.lhi=np.minimum.accumulate((self.l+self.err[3])[:-1])[:-1]
275     def at(self,name,idx):
276         vals=getattr(self,name)[idx]
277         er=(2.95e-5 if name=='Q' else self.err[['M','X','k','l'].index(name)][idx])
278         return I(vals)-0+I(-er,er) # padded
279     def inverse_indices(self,k,l): # intervals arrays, find z interval outer nodes
280         use=k.hi<=.6
281         # lower idx: want node k_lower >= khi, choose last such; equivalently search on -klo
282         l1=np.searchsorted(-self.klo,-k.hi,side='right')-1
283         u1=np.searchsorted(-self.khi,-k.lo,side='left')
284         # l increasing: lower node upper l <= llo
285         l2=np.searchsorted(self.lhi,l.lo,side='right')-1
286         u2=np.searchsorted(self.llo,l.hi,side='left')
287         low=np.where(use,l1,l2);hi=np.where(use,u1,u2)
288         assert np.all(low>=0) and np.all(hi<len(self.k))
289         return low,hi
290     def inverse(self,k,l):
291         low,hi=self.inverse_indices(k,l)
292         z=I((low-self.offset)/ZN-1e-12,(hi-self.offset)/ZN+1e-12)
293         M=I(self.at('M',hi).lo,self.at('M',low).hi).cap(0,100)
294         X=I(self.at('X',low).lo,self.at('X',hi).hi).cap(0,100)
295         W=2*(.5-self.a)*M*1+X*k
296         return z,M,X,W
297
298     from functools import lru_cache
299     profiles=lru_cache(maxsize=6)(Profile)
300
301     MGRID=2048
302     CUT=38
303     def grid():
304         ed=np.concatenate([(1+np.arange(MGRID)/MGRID)*2.**(-i) for i in range(CUT,1,-1)]+[np.array([.5])])
305         ed=np.concatenate([ed,1-ed[-2::-1]]) # exact dyadic? smallest subdivisions require CUT+11=49 <53 yes
306         w=np.diff(ed); assert len(w)%2==0
307         return I(ed[:-1],ed[1:]),I(1-ed[1:],1-ed[:-1]),w,len(w)//2
308     kg,lg,width,mid=grid()
309     def sums(x,rev=False): # intervals of weighted cumulative over cells inclusive full endpoint
310         v=x*I(width)
311         def side(a,sgn):
312             if rev:a=a[::-1]
313             cs=np.cumsum(a)
314             err=np.cumsum(abs(a))*1e-9+tiny # n<1e6
315             out=cs+sgn*err
316             return out[:-1] if rev else out
317         return I(side(v.lo,-1),side(v.hi,1))
318     def partialsum(x,rev=False): # integral to any point in current cell from relevant tail (within grid)
319         v=x # allow either sign
320         full=sums(v,rev)
321         # remove optional part
322         amt=x*I(0,width)
323         return full-amt
324     def total(x):
325         sm=sums(x);return I(sm.lo[-1],sm.hi[-1])
326     def anchored(x): # integral from .5 to any cell; avoid total cancellation
327         left=choose(np.arange(mid*2)<mid,x,0)
328         right=choose(np.arange(mid*2)>=mid,x,0)
329         return choose(np.arange(mid*2)<mid,-partialsum(left,True),partialsum(right))
330     # RMQ for cell bounds on z table adjacent nodes
331     class Rmq:
332         def __init__(self,g,values):
333             self.g=g;self.los=[values.lo];self.his=[values.hi]
334             while 2**len(self.los)<=len(values.lo):
335                 step=2**(len(self.los)-1)
336                 self.los.append(np.minimum(self.los[-1][:-step],self.los[-1][step:]))

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337     self.his.append(np.maximum(self.his[-1][:-step],self.his[-1][step:]))
338 def query(self,z):
339     low=np.floor((z.lo-1e-10)*ZN).astype(int)+self.g.offset
340     high=np.floor((z.hi+1e-10)*ZN).astype(int)+self.g.offset
341     assert np.all(low>=0) and np.all(high<len(self.los[0]))
342     lens=high-low+1
343     olo=np.empty_like(z.lo);ohi=np.empty_like(z.lo)
344     for j,(los,his) in enumerate(zip(self.los,self.his)):
345         sel=(lens>=2**j)&(lens<2**(j+1)); u=low[sel];v=high[sel]-(2**j-1)
346         olo[sel]=np.minimum(los[u],los[v]);ohi[sel]=np.maximum(his[u],his[v])
347     return I(olo,ohi)
348 def zcells(g):
349     idx=np.arange(len(g.M)-1)
350     def monot(name,inc):
351         vals=g.at(name,np.arange(len(g.M)))
352         return I(vals.lo[:-1] if inc else vals.lo[1:],vals.hi[1:] if inc else vals.hi[:-1])
353     M=monot('M',False).cap(0,30);X=monot('X',True).cap(0,30)
354     k=monot('k',False).cap(0,1);l=monot('l',True).cap(0,1)
355     z=I((idx-g.offset)/ZN-1e-12,(idx+1-g.offset)/ZN+1e-12)
356     beta=.5*z+g.a*M
357     # bootstrap derivative bound: start Q node error, assume |Q|<=1 within length => |Q'|<25, enlarge,
358     #   ↪ refine
359     qnodes=g.at('Q',idx)
360     # refine error central (cert proof)
361     qnodes=qnodes.cap(g.Q[idx]-np.where(abs(z.lo)<12.01,5e-7,3e-5),g.Q[idx]+np.where(abs(z.lo)<12.01,5e
362     #   ↪ -7,3e-5))
363     Q=qnodes+I(-.006,.006)
364     for _ in range(2):
365         qp=2*beta*Q-Q.sq()+2*(.5-g.a*k-g.lam)
366         Q=(qnodes+I(-1,1)*I(qp.maxabs())/ZN).cap(-1,0)
367         qp=2*beta*Q-Q.sq()+2*(.5-g.a*k-g.lam)
368         V=2*(.5-g.a)*M*1+X*k
369     return locals()
370 def eloc_z(z,cp): # .25 power
371     t=2-z
372     t=I(np.maximum(t.lo,0),np.maximum(t.hi,0))
373     den=1+t.sq()/2
374     e=cp/2/den.sqrt().sqrt()
375     ep=e*t/(4*den)
376     epp=e*(5*t.sq()/(16*den.sq()))-1/(4*den)
377     epp=I(np.where(z.hi>=2,np.minimum(0,epp.lo),epp.lo),np.where(z.hi>=2,np.maximum(0,epp.hi),epp.hi))
378     return e,ep,epp
379 def checkV(g,gapA,upper,cp):
380     d=zcells(g);a=g.a
381     p=.5-g.af
382     sel=(d['z'].lo<6)&(d['z'].hi>-math.sqrt(13/p)-.01)
383     z,M,k,l,W,B=[d[x].sel(sel) for x in ['z','M','k','l','V','beta']]
384     t=k*1
385     T=2*(.5-a)*M*1/W-2*k+2*a*(t/W).sq()
386     if g.A==0: assert np.min(T.lo)>-1.12 and np.max(T.hi)<.13
387     e,ep,epp=eloc_z(z,cp)
388     fac=rat(1,1) if upper else -rat(11,10)
389     R=1+fac*e*gapA
390     Ar=(a+I(0,1)*gapA) if upper else (a-I(0,1)*gapA)
391     core=.5*epp+(-B+a*t/W)*ep
392     rhs=R.sq()*core+Ar*t/W*ep+R*(-(R+1)*(.5+a*T)*e+I(np.maximum(T.lo,0),np.maximum(T.hi,0)))/(fac if upper
393     #   ↪ else -fac))
394     core0=fac*gapA*core-(.5+a*T)*R
395     return np.max(rhs.hi),np.max(core0.hi)
396 CP=rat(26,100); ERR=rat(12,100)
397 def tail_mx_bounds(g): # cut primitives station first/last cell from inverse already include true cutoff
398     z,m,x,v=g
399     return m.hi[0],x.hi[-1]
400 class Env:
401     def __init__(self,A,B,check=True):
402         self.a=profiles(A);self.b=profiles(B); gap=self.gap=self.b.a-self.a.a
403         if check:
404             for g,u in [(self.a,1),(self.b,0)]:
405                 rr,co=checkV(g,gap,u,CP); assert rr<0 and co<-.02,(g.A,rr,co)
406         self.va0=self.a.inverse(kg,lg); self.vb0=self.b.inverse(kg,lg)
407         za,ma,xa,wa=self.va0; zb,mb,xb,wb=self.vb0
408         rh=1+gap*eloc_z(za,CP)[0]; rl=1-rat(11,10)*gap*eloc_z(zb,CP)[0]

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```

406 self.vhi=rh*wa; self.vlo=rl*wb
407 # envelopes primitive correction
408 def adjusted(box,w,R,weight,rev):
409     corr=1/R-1
410     tail=I(-.16,.16)*gap*(box.sel(-1 if rev else 0))
411     out=box+tail+partialsum(weight/w*corr,rev)
412     direct=box/(1+I(-.1431,.1301)*gap)
413     return out.cap(direct.lo,direct.hi)
414 ml=adjusted(ma,wa,rh,kg,False);xl=adjusted(xa,wa,rh,lg,True)
415 mh=adjusted(mb,wb,rl,kg,False);xh=adjusted(xb,wb,rl,lg,True)
416 self.m=I(ml.lo,mh.hi);self.x=I(xl.lo,xh.hi)
417 self.z=self.x-self.m
418 # Q combo cell rmq
419 da=zcells(self.a);db=zcells(self.b)
420 delta=(db['M']-da['M']).cap(0,30) # may fail over tiny equal? intervals overlap
421 upperq=db['Q']+self.a.a*delta
422 lowerq=db['Q']-self.b.a*delta
423 qhi=Rmq(self.a,upperq).query(self.z)+ERR*gap
424 qlow=Rmq(self.b,lowerq).query(self.z)-ERR*gap
425 qhi=I(np.minimum(0,qhi.lo),np.minimum(0,qhi.hi))
426 self.sl=I(np.maximum(0,(-qhi/self.vhi).lo)); self.sh=I((-qlow/self.vlo).hi)
427 # actual slope between sl.lo and sh.hi (qlo.lo negative)
428 self.slope=I(self.sl.lo,self.sh.hi)
429 self.logp=anchored(self.slope)
430 if check:self.checkH(da,db,delta)
431 def possiblek(self,z): # central away grid tails
432     # high k where zhi >= z.lo; low k where zlo <= z.hi
433     low=np.minimum.accumulate(self.z.lo) # enforce decreasing lower? prefix min remains lower
434     hi=np.maximum.accumulate(self.z.hi[::-1])[:-1]
435     i1=np.searchsorted(-low,-z.hi,side='left')
436     i2=np.searchsorted(-hi,-z.lo,side='right')-1
437     ok=(i1<len(kg.lo))&(i2>=0)
438     i1=np.clip(i1,0,len(low)-1);i2=np.clip(i2,0,len(low)-1)
439     klow=np.where(ok,kg.lo[i1],np.where(z.lo>0,0,kg.lo[-1]))
440     khigh=np.where(ok,kg.hi[i2],np.where(z.lo>0,kg.hi[0],1))
441     # even if ok at extreme include tail if i1==0 or i2==last
442     klow=np.where(i1==0,0,klow);khigh=np.where(i2==len(low)-1,1,khigh)
443     assert np.all(klow<=khigh)
444     k=I(klow,khigh)
445     l=1-k;l=I(np.maximum(0,l.lo),np.minimum(1,l.hi))
446     # lower V via 32
447     vl=rat(9,7)*k*1*.sqrt()
448     return k,vl.lo
449 def checkH(self,da,db,delta):
450     sel=(da['z'].lo<12)&(da['z'].hi>-12)
451     z=da['z'].sel(sel)
452     mi=I(da['M'].lo,db['M'].hi).sel(sel)
453     ki,vl=self.possiblek(z)
454     for d,g,upper in [(da,self.a,True),(db,self.b,False)]:
455         q,qp,k,l=[d[x].sel(sel) for x in ['Q','qp','k','l']]
456         damp=.5-qp-g.a*k-(self.gap*ki if upper else 0)
457         br=mi*(qp-g.a*1)-ki*q+I(vl)
458         if upper:br=g.a*ki*delta.sel(sel)
459         val=br+ERR*damp
460         assert np.min(damp.lo)>.06,np.min(damp.lo)
461         assert np.min(val.lo)>0,(g.A,np.min(val.lo),z.lo[np.argmin(val.lo)])
462     # tail q @ z <= -5.5 numerical
463     for d in [da,db]:
464         s=(d['z'].lo<-5.5)
465         assert np.min(d['Q'].lo[s])>-.28
466     assert self.z.hi[-1]<-5.5 and self.z.lo[0]>4
467     # delta tail <= X_b(-5.5) for negative z; numerical
468     s=db['z'].lo<-5.5
469     assert np.max(db['X'].hi[s])<.03
470     assert self.logp.hi[-1]<5
471     for g,arr in [(self.a,self.va0),(self.b,self.vb0)]:
472         assert -arr[0].hi[-1]*2*(.5-g.af)>2 and arr[0].lo[0]>6
473 def coarse(self,test,upper,detail=False):
474     test=iv(test)
475     slope=self.sl if upper else self.sh
476     p=iexp(anchored(slope))
477     v=self.vhi if upper else self.vlo

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478 integrand=p*(v-2*test*kg*kg*lg/v)
479 # sign cumulative, allow tail error absolute 1e-7, only when away far endpoint
480 if upper:
481     # g negative near k=1 verify central tail say k>.95, analytical beyond cut
482     far=kg.lo>.95
483     if np.max(integrand.hi[far])>=0:return False
484     cum=partialsum(integrand,True)
485     val=np.max(cum.hi[~far])+1e-7
486     # positive tail k->0 total error
487     return val if detail else val<0
488 else:
489     far=kg.hi<.05
490     if np.min(integrand.lo[far])<=0:return False
491     cum=partialsum(integrand)
492     val=np.min(cum.lo[~far])-1e-7
493     return val if detail else val>0
494 @lru_cache(maxsize=3)
495 def refdata(g):
496     z,m,x,v=g.inverse(kg,lg)
497     q=Rmq(g,zcells(g)['Q']).query(z)
498     slope=-q/v; lp=anchored(slope); p=iexp(lp)
499     return z,m,x,v,q,slope,lp,p
500 def nonlineartest(env,test,upper,nt=8):
501     test=iv(test);c=rat(3,10)
502     gr=profiles(293650)
503     z,mr,xr,vr,q,sr,lpr,pr=refdata(gr)
504     tk=kg*lg
505     base=total(pr/vr*(vr.sq()+c*(mr*xr-tk)-2*test*kg*tk))
506     # actual slopes bounded, reference slope cell intervals
507     eps=ilog(I(env.vlo.lo,env.vhi.hi)/vr)
508     pis=env.slope-sr
509     pi=anchored(pis) # tighter than logp - lpr
510     out=base
511     for it in range(nt):
512         t=I((it)/nt,(it+1)/nt) # nt power2
513         vf=iexp(t*eps);pf=iexp(t*pi)
514         v=vr*vf;p=pr*pf
515         # cap v,p actual envelope geometric? both ref within => yes slope convex for interpolated log
516         v=v.cap(env.vlo.lo,env.vhi.hi)
517         pbox=iexp(env.logp);p=p.cap(pbox.lo,pbox.hi)
518         pv=p/v
519         inv=1/vf-1
520     def prim(ref,weight,rev):
521         # omitted tail 1/vfac-1/<=3 from heat ratio <4
522         tail=I(-3.01,3.01)*ref.sel(-1 if rev else 0)
523         y=ref+tail+partialsum(weight/vr*inv,rev)
524         static=(env.x if rev else env.m)
525         return y.cap(static.lo,static.hi)
526     m=prim(mr,kg,False);x=prim(xr,lg,True)
527     signed=c*(m*x-tk)-2*test*kg*tk
528     gp=pv*(v.sq()+signed)
529     gv=pv*(v.sq()-signed)-c*kg/v*partialsum(pv*x,True)-c*lg/v*partialsum(pv*m)
530     varv=total(gv*eps)
531     # omitted M epsilon derivative tails <=6 M_ref(cut), heat/log ratio
532     terror=6*(mr.hi[0]*total(c*pv*x).hi+xr.hi[-1]*total(c*pv*m).hi)
533     assert terror<3e-7,terror # may pv*m divergent near tail but finite cut logs polynomial!
534     coeff=choose(np.arange(len(width))<mid,-partialsum(gp),partialsum(gp,True))
535     varp=total(coeff*pis)
536     alt=total(gp*pi)
537     varp=varp.cap(alt.lo,alt.hi)
538     out+=(varv+varp)/nt
539     val=out.hi+1e-6 if upper else out.lo-1e-6
540     return val
541 def kquery(vals,i1,i2):
542     # range extrema sparse using generic
543     los=vals.lo;his=vals.hi
544     n=i2-i1+1
545     outlo=np.empty(len(n));outhi=np.empty(len(n))
546     step=1
547     while step<=max(n):
548         sel=(n>=step)&(n<2*step)
549         a=i1[sel];b=i2[sel]-step+1

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550 outlo[sel]=np.minimum(los[a],los[b]);outhi[sel]=np.maximum(his[a],his[b])
551 los=np.minimum(los[:-step],los[step:]);his=np.maximum(his[:-step],his[step:])
552 step*=2
553 return I(outlo,outhi)
554 def localcheck():
555     E=Env(293650,293655)
556     a=I(E.a.a.lo,E.b.a.hi); d=rat(1,100)
557     k,l=kg,lg; z=E.z;m=E.m;x=E.x
558     assert -20<z.lo[-1] and z.hi[-1]<-10 and z.lo[0]>7 and z.hi[0]<20
559     W=I(E.vlo.lo,E.vhi.hi)
560     c0,c1,c2,c3=[rat(t,100) for t in [26,130,-186,325]]
561     L=c0+k*(c1+k*(c2+k*c3));Lp=c1+2*k*c2+3*k*k*c3;Lpp=2*c2+6*k*c3
562     assert np.max(L.hi)<3
563     t=k*l
564     T=2*(.5-a)*m*l/W-2*k+2*a*(t/W).sq()
565     B=.5*z+a*m
566     Wk=2*B-2*a*t/W
567     R=1+I(-1,1)*d*L
568     core=.5*W.sq()*Lpp+W*Wk*Lp
569     rhs=R.sq()*core-(a*I(-1,1)*d)*t*Lp+R*(-(R+1)*( .5+a*T)*L+I(0,T.maxabs())) # absolute adverse source
570     core0=I(-1,1)*d*core-( .5+a*T)*R
571     assert np.max(rhs.hi)<0,np.max(rhs.hi)
572     assert np.max(core0.hi)<0
573     epsmax=L/(1-d*L)
574     def lprim(weight,ref,rev):
575         val=I(0,3.11)*ref.sel(-1 if rev else 0)+partialsum(weight/W*epsmx,rev)
576         return I(0,np.minimum(val.hi,3.11*ref.hi))
577     lm=lprim(k,m,False);lx=lprim(1,x,True);lz=lm+lx
578     zi=z+I(-1,1)*d*lz
579     da=zcells(E.a);db=zcells(E.b)
580     delta=(db['M']-da['M']).cap(0,30)
581     Qh=da['Q']+E.a.a*delta+ERR*E.gap;Ql=db['Q']-E.b.a*delta-ERR*E.gap
582     Qsp=I(Ql.lo,np.minimum(0,Qh.hi))
583     rq=Rmq(E.a,Qsp)
584     Qstar=rq.query(z)
585     shifts=lz*I(rq.query(zi).maxabs()+2*.011) # pad
586     # H same z
587     sel=(da['z'].lo<12)&(da['z'].hi>-12); zz=da['z'].sel(sel)
588     def poss(box,useW=False):
589         low=np.minimum.accumulate(box.lo);hi=np.maximum.accumulate(box.hi[:,:-1])[:-1]
590         i1=np.searchsorted(-low,-zz.hi,side='left');i2=np.searchsorted(-hi,-zz.lo,side='right')-1
591         i1=np.clip(i1,0,len(low)-1);i2=np.clip(i2,0,len(low)-1)
592         kk=I(np.where(i1==0,0,k.lo[i1]),np.where(i2==len(low)-1,1,k.hi[i2]))
593         assert np.all(i2>=i1)
594         ww=kquery(W,i1,i2) # tail max <= .0001 below from (32)
595         return kk,ww
596     ks,_=poss(z);ki,wi=poss(zi)
597     Ms=I(da['M'].lo[sel],db['M'].hi[sel]);ls=1-ks
598     assert np.max((m*l/(.5-(a+d)*l)).hi)<.65
599     dm=rat(65,100)
600     Mi=Ms+I(-1,1)*dm*d
601     qq=Qsp.sel(sel)
602     qp=2*(.5*zz+a*Ms)*qq-qq.sq()+2*(.5-a*ks-a)
603     damp=.5-qp-a*ks-d*ki
604     # wi max tails use endpoint or simple .001 extra
605     br=Mi*(qp-a*ls)-ki*qq+I(0,(1.031*wi.hi+.0001))+I(-1,1)*a*ki*dm*d
606     assert np.min(damp.lo)>.1
607     gain=a*dm+(a*(1-a)*dm+I(br.maxabs()))/damp
608     assert np.max(gain.hi)<2,np.max(gain.hi)
609     # functional
610     cadd=rat(3,10)
611     sr=-Qstar/W;lpr=anchored(sr);pr=iexp(lpr)
612     zc=z.sel(mid)
613     assert zc.maxabs()<2
614     pimax=2*I((z-zc).maxabs())+shifts
615     out=0
616     for it in range(4):
617         td=d*(it+1)/4
618         pvf=iexp(I(-1,1)*td*pimax)
619         vf=iexp(I(-1,1)*td*epsmx)
620         v=W*vf;p=pr*pvf;pv=p/v
621         mt=(m+I(-1,1)*td*lm).cap(.968*m.lo,1.033*m.hi)

```

```

622 xt=(x+I(-1,1)*td*lx).cap(.968*x.lo,1.033*x.hi)
623 signed=cadd*(mt*xt-t)-2*a*k*t
624 gp=pv*(v.sq()+signed)
625 gv=pv*(v.sq()-signed)-cadd*k/v*partialsum(pv*xt,True)-cadd*l/v*partialsum(pv*mt)
626 cv=total(I(gv.maxabs()))*epsmax)
627 terror=4*(m.hi[0]*total(cadd*pv*xt).hi+x.hi[-1]*total(cadd*pv*mt).hi)
628 assert terror<3e-7
629 coeff=choose(np.arange(len(width))<mid,partialsum(gp),partialsum(gp,True))
630 cp=total(I(coeff.maxabs()))*2/W)+total(I(gp.maxabs()))*shifts)
631 out+=(cv+cp)/4
632 den=2*total(pv*k*t)
633 assert den.lo>0
634 return (out.hi+2e-6)/den.lo
635 def checkA():
636     n=4096
637     ind=np.arange(1,n-1)
638     k=I(ind/n,(ind+1)/n);l=1-k;t=k*1
639     low=rat(9,7)*k*1*1.sqrt(); high=rat(6,5)*(t*t.sqrt()).sqrt()
640     b=2*(rat(6,10)-k); B=-rat(4,10)*t
641     # max endpoint term convex when b>=0, decreasing if b<0
642     edge=(b*low.sq()+2*B*low).hi
643     edge=np.maximum(edge,(b*high.sq()+2*B*high).hi)
644     ws=[low+(high-low)*I(i/128,(i+1)/128) for i in range(128)]
645     mini=100.
646     for aa in range(419,1024): # covers [.41,1]
647         a=I(aa/1024,(aa+1)/1024)
648         K=12*a*k-5*a-1-b.sq()+rat(8,10)*(1-2*k)
649         pos=(6*a.sq()-rat(16,100))*t.sq()
650         mins=np.minimum.reduce([(2*w.sq()+K*w-2*b*B+pos/w).lo for w in ws])
651         # last cell crude integrand >=-9; first likewise and endpoint [...] upper<.001
652         assert np.max(abs(mins))<9
653         cs=np.cumsum(mins[:-1])[:-1]/n-2e-9
654         bound=cs-np.maximum(0,mins)/n-9/n-edge
655         val=min(np.min(bound[ind<=2560]),cs[0]-18/n-.001)
656         mini=min(mini,val)
657         assert val>0,(aa,val,cs[0])
658     return mini
659
660 def driver():
661     global AD
662     assert checkA()>0
663     A,B=(0, 410000)
664     for an,bn in [(167000, 410000), (183000, 410000), (185000, 400000), (193000, 397000), (197000, 390000)
        ↪ , (203000, 386000), (207000, 381000), (211000, 377000), (215000, 373000), (218000, 369000),
        ↪ (220000, 370000)]:
665         e=Env(A,B); print(A,B,flush=True)
666         if an>A: assert e.coarse(rat(an,AD),False)
667         if bn<B: assert e.coarse(rat(bn,AD),True)
668         A,B=an,bn
669     A,B=(220000, 370000)
670     for an,bn in [(228000, 370000), (231000, 368000), (234000, 363000), (239000, 358000), (244000, 350000)
        ↪ , (251000, 341000), (258000, 332000), (266000, 324000), (272000, 316000), (277000, 311000),
        ↪ (281000, 307000), (283000, 304000), (285000, 303000)]:
671         e=Env(A,B); print(A,B,flush=True)
672         if an>A: assert nonlineartest(e,rat(an,AD),False)>0
673         if bn<B: assert nonlineartest(e,rat(bn,AD),True)<0
674         A,B=an,bn
675
676     assert localcheck()<.95
677     AD=10**7
678     for A in [2936533,2936535]:
679         _,_,lam=certified(A)
680         assert (lam-10>A*1000 if A==2936533 else lam+10<A*1000)
681 driver()

```